MATHEMATICS

STUDENT'S TEXTBOOK GRADE 12

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FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY OF EDUCATION

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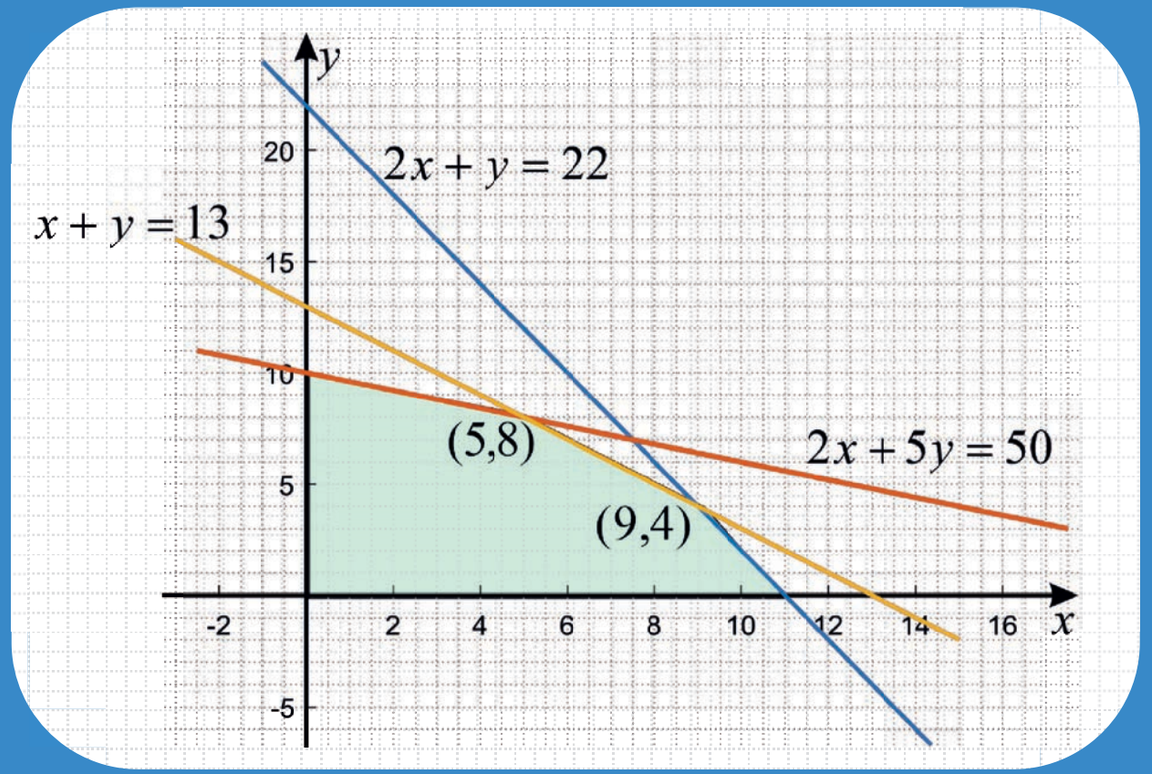

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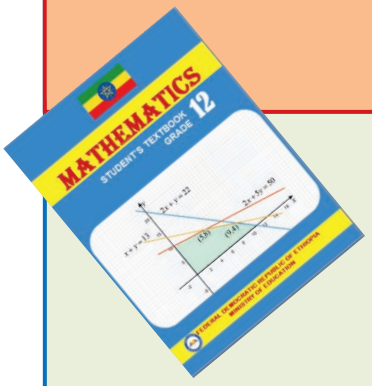

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STUDENT'S TEXTBOOK GRADE 12

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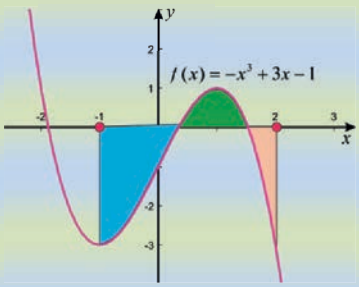

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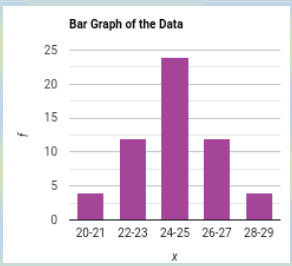

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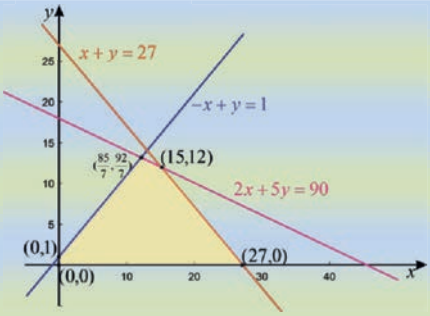

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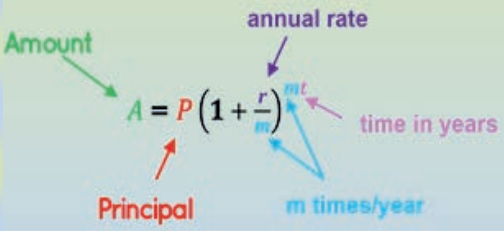

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Welcoming Message to Grade 12 Students

Dear students, you are welcome to Grade 12 Mathematics educations. This is a golden opportunity in your academic career. Joining this grade level is a new experience and a transition to tertiary level mathematics educations. At this grade level, you are expected to receive new and advanced opportunities that can help you learn and grow in the field of Mathematics, in life and work. Enjoy it!

Introduction about the Students' Textbook

Dear student, this textbook is organized in such a way that there is an introductory remark about how to use the textbook, some suggestions about how to care the textbook so that it will serve for many beneficiaries for a long period of time, and basic units are addressed after the introduction section.

The Grade 12 textbook is organized having 5 units. The units include: Sequence and Series, Introduction to Pre-calculus, Statistics, Introduction to Linear Programming, and Mathematical Applications in Business**.** Each unit is composed of an introduction, learning outcome, lessons, key words, and summary and review exercise.

In a unit, a brief introduction about the unit is provided. Following a brief introduction of a unit, the expected learning outcomes are presented. Once the objectives are communicated, the different lessons of the unit are presented. In the end, summary and review exercise are packed, respectively.

As mentioned earlier, a unit in the textbook is divided into different lessons. In every lesson of each unit, you learn about mathematics frequently through five components. That is, structurally, the lessons in this textbook usually

have five components; namely, Activity, Definitions/Theorems/Notes, Examples and Solutions, and Exercises.

The most important part in this process is to practice problems by yourself based on what your teacher shows and explains. Your teacher will also give you feedback and assistance, and facilitate further learning. In such a way, you will be able not only acquire new knowledge and skills but also develop them further. Herewith, a brief explanation of each sub-component of a lesson is forwarded.

Activity

This part of the lesson requires you to revise what you have learnt at different stages of your mathematics education and reflect on the topic under discussion by using this background knowledge. The activity also introduces you to what you are going to learn in the respective new lesson or topic.

Definition/Theorem/Note

This part of the textbook presents and explains new concepts/ definitions/theorems.

Example and Solution

Here, the textbook provides you specific examples and thereby helps you to improve your understanding of the new content. In this part, your teacher will give you explanations and/or solutions and you are, therefore, advised to listen to your teacher's explanations very carefully and participate actively in the process. You are also required to refer to the solution part of the textbook for your review and self-learning. In this sub-section, a student is

further expected to study on the remaining examples that a classroom teacher will not explain during his/her classroom instruction.

Exercise

Under this part of the textbook, you will be required to solve the problems or questions that are given as exercise individually, in pairs or groups so as to further practice what you have learnt in the examples. That is, a teacher may provide selected exercise questions for you as class work, homework or project work. When you are doing the exercises either individually, in pairs or groups, you are expected to share your opinions with your friends, listen to others' ideas carefully and compare your ideas with others. However, it is always advisable to make some efforts individually before working in pair or groups in order to have a better understanding and input in the process.

In some of the cases, some exercises may not be covered by a classroom teacher. In that case, a student is expected to study the remaining items of the exercise a teacher will not cover during classroom instruction. A teacher will provide the possible answers for those items in the exercises after completing your study, and you are expected to check what you will do against the respective teacher's answers.

Checking in every step

Throughout a lesson, students shall check their progress and the respective teacher should do the same. It is also true that students are required to check the correct answers and solutions for the exercises at the end of each unit. If you want to practice more, therefore, you need to go to the review exercises. Doing the Review Exercise is always recommended since they help you develop a concrete view of the lesson contents.

Generally, students are advised to practice, drill and exercise each activity, example, and exercise on a daily basis as mastering mathematical skills take a long period of time and frequent practice to be second nature. You are also requested to read the notes with deep understanding.

UNIT 1

SEQUENCES AND SERIES

Unit Outcomes

By the end of this unit, you will be able to:

- Understand sequence and series.
- Compute terms of a sequence from a given rule.
- Use given terms to develop a formula that represent the sequence.
- Identify different types of sequences and series.
- Compute the partial and infinite sum of some sequences.
- Apply your understanding the knowledge of sequences and series to real-life problems.

Unit Contents

1.1 Sequence

1.2 Arithmetic and Geometric Sequences

1.3 The Sigma Notation and Partial Sums

1.4 Infinite Series

1.5 Applications of Sequence and Series in Daily Life

Summary

Review Exercise

- arithmetic sequence
- finite sequence
- partial sums
- common difference
- general term
- sequence
- common ratio
- geometric sequence
- series
- convergent series
- infinite sequence
- terms of a sequence
- divergent series
- infinite series
- recursion formula

Introduction

Mathematics has an enormous number of uses in our daily life. There is, in fact, no area of life that is not affected by mathematics. For instance, civil engineers use mathematics to determine how to best design new structures; economists use mathematics to describe and predict how the economy will react to certain changes; investors use mathematics to price certain types of shares or calculate how risky particular investments are; software developers use mathematics for many of the algorithms (such as Google searches and data security) that make programs useful. Mathematics is present everywhere from distance, time and money to art, design and music.

A sequence is an arrangement of numbers in a definite order according to some rule. Sequences and Series play an important role in various aspects of our lives. They help us to predict, evaluate and monitor the outcome of a situation or event and help us in decision making.

1.1 Sequences

Activity 1.1

1. People wants to plant trees in a certain pattern in the green area of a community like 20 plants in the first row, 34 plants in the second row and 48 plants in the third row, and so on. How many trees people will plant in the 5^{th} and 6^{th} row?
2. Find the next two terms of the following sequence.

   a. $\{50,\ 47,\ 44,\ 41,...\}$ b. $\{2,\ 12,\ 22,\ 32,...\}$

Definition 1.1

A sequence is a function whose domain is the collection of all integers greater than or equal to a given integer m (usually 0 or 1). A sequence is usually denoted by a_n. The functional values: $a_1, a_2, a_3, ..., a_n, ...$ are called the terms of a sequence, and a_n is called the general term, or the n^{th} term of the sequence. There are two types of sequences depending on its last term.

Finite Sequence: A sequence that has a last term. The domain of a finite sequence is $1, 2, 3, ..., n$.

Infinite sequence: A sequence that does not have a last term. The domain of an infinite sequence is the set of natural numbers $(\mathbb{N})$.

Example 1

List the first five terms of each of the sequences whose general terms are given below where n is a positive integer.

$(a)\ a_n = 2n-1 \qquad (b)\ a_n = \left(-\frac{1}{3}\right)^{n-1} \qquad (c)\ a_n = \frac{1}{n}$

Solution

$(a)\ a_n = 2n-1$:

$a_1 = 2(1)-1 = 1, \quad a_2 = 2(2)-1 = 3, \quad a_3 = 2(3)-1 = 5,$

$a_4 = 2(4)-1 = 7, \quad a_5 = 2(5)-1 = 9.$

Therefore, the first five terms are $1, 3, 5, 7$ and 9.

$(b)\ a_n = \left(-\frac{1}{3}\right)^{n-1}$

$a_1 = \left(-\frac{1}{3}\right)^{1-1} = 1, \quad a_2 = \left(-\frac{1}{3}\right)^{2-1} = -\frac{1}{3}, \quad a_3 = \left(-\frac{1}{3}\right)^{3-1} = \frac{1}{9},$

$a_4 = \left(-\frac{1}{3}\right)^{4-1} = -\frac{1}{27}, \quad a_5 = \left(-\frac{1}{3}\right)^{5-1} = \frac{1}{81}.$

Therefore, the first five terms are 1, $-\frac{1}{3}$, $\frac{1}{9}$, $-\frac{1}{27}$ and $\frac{1}{81}$.

(c) $a_n = \frac{1}{n}$:

$a_1 = \frac{1}{1} = 1$, $a_2 = \frac{1}{2}$, $a_3 = \frac{1}{3}$, $a_4 = \frac{1}{4}$, $a_5 = \frac{1}{5}$.

Therefore, the first five terms are 1, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$.

Example 2

Draw the graph of sequence $a_n = 2n + 1$.

Solution

Make a table with n and a_n, as follows. Then plot each ordered pair (n, a_n).

n	1	2	3	4	5
a_n	3	5	7	9	11

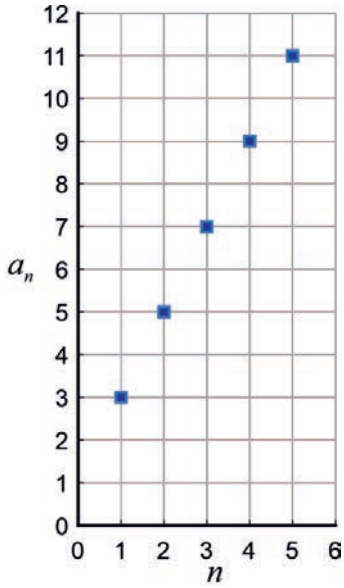


Figure 1.1

Note

From the Figure 1.1, we can observe that the graph of the sequence follows the pattern of a linear equation.

Exercise 1.1

1. List the first five terms of each of the sequences whose general terms are given below where n is a positive integer:

   a. $a_n = 2n$ b. $a_n = \left(\frac{1}{2}\right)^{n-1}$ c. $a_n = \frac{n}{n+1}$ d. $a_n = n^3$

2. Draw the graph of the following sequences and observe the pattern of the sequence.

   a. $a_n = 3n-1$, b. $a_n = \frac{1}{n}$, c. $a_n = (-1)^n$.

3. Bontu's uncle gave 130 Ethiopian birr to her in January, in the next month she saves money and has 210 Ethiopian birr and in the third month she has 290 Ethiopian birr. How much money will she have in the fourth, fifth, sixth and seventh month respectively.

Fibonacci and Mulatu sequences

Activity 1.2

Two squares with a side length of 1 are arranged side by side. The rectangle made has a vertical length of 1 and a horizontal length of 2. Place a square with a side length of 2 next to it. Then, the rectangle made has vertical length is 3, horizontal length is 2. After this, arrange a square with a side length of 3, a square with a length of 5, and a square with a length of 8 to make a large rectangle.

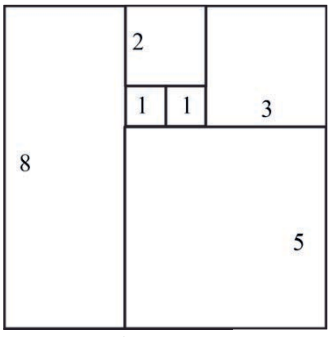


Figure 1.2

Based on Figure 1.2 above, answer the following questions:

a. What is the sum of the area of all the inner squares?

b. What is the area of the outer rectangle?

c. What is the relation between your answer to parts a and b?

Fibonacci's sequence

HISTORICAL NOTE

Leonardo Fibonacci (circa 1170, 1240)

Italian mathematician Leonardo Fibonacci made advances in number theory and algebra. Fibonacci, also called Leonardo of Pisa, produced numbers that have many interesting properties such as the birth rates of rabbits and the spiral growth of leaves on some trees.

He is especially known for his work on series of numbers, including the Fibonacci series. Each number in a Fibonacci series is equal to the sum of the two numbers that came before it. Fibonacci sequence arose when he was trying to solve a problem of the following kind concerning the breeding of rabbits.

"Suppose that rabbits live forever and that every month each pair produces a new pair which becomes productive at the age of two months. If we start with one new born pair, how many pairs of rabbits will we have in the n^{th} month?"

Fibonacci sequence is defined as: $F_n = \begin{cases} 1 & \text{if } n = 0 \\ 1 & \text{if } n = 1 \\ F_{n-1} + F_{n-2} & \text{if } n \geq 2. \end{cases}$

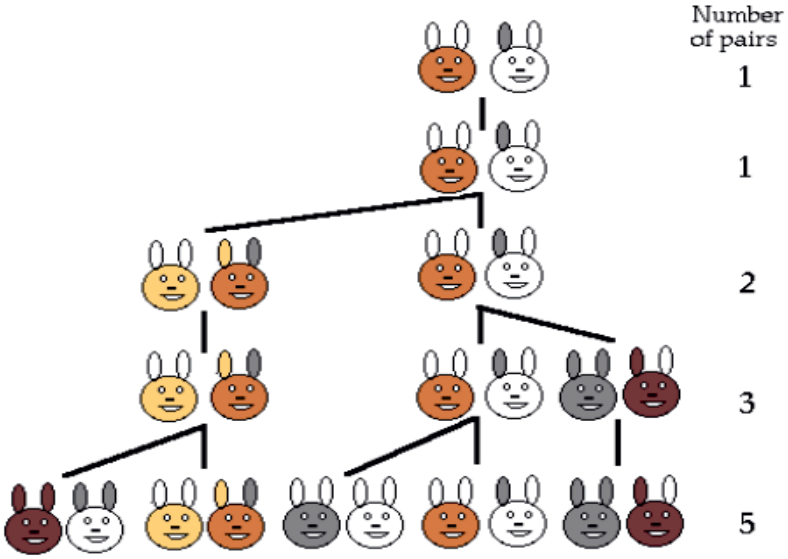


Figure 1.3

Example

List the first eight terms of the Fibonacci sequence and draw its graph.

Solution

Make a table with n and a_n, as follows. Then plot each ordered pair (n, F_n).

n	0	1	2	3	4	5	6	7	8	9
F_n	1	1	2	3	5	8	13	21	34	55

So, the graph becomes

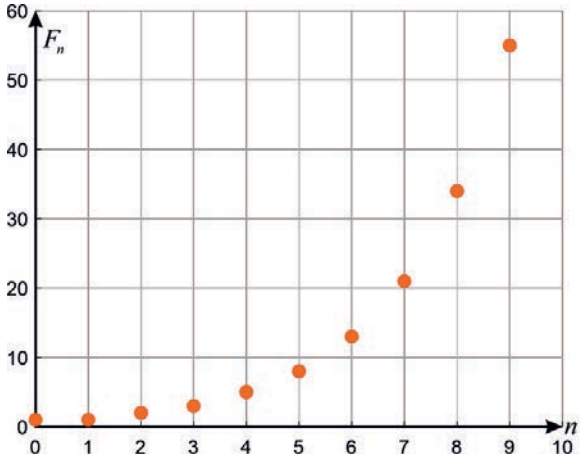


Figure 1.4

Mulatu sequence

HISTORICAL NOTE

Mulatu Lemma

Professor Mulatu Lemma is an Ethiopian Mathematician.
He pursued an education through a university in Ethiopia, earning a Bachelor of Science in 1977 from Addis Ababa University.
He continued his studies at the aforementioned university and obtained a Master of Science in applied mathematics in 1982. Following these accomplishments, he was enrolled at Kent State University, where he pursued a Master of Arts in pure mathematics in 1993. Prof. Mulatu completed his academic journey with a Doctor of Philosophy from Kent State University in 1994. In 2011, he introduced the Mulatu's Number (named after him) to the mathematical community and to the world.

Professor Mulatu introduced a sequence of the form:

$$M_n = \begin{cases} 4 & \text{if } n = 0 \\ 1 & \text{if } n = 1 \\ M_{n-1} + M_{n-2} & \text{if } n \geq 2. \end{cases}$$

Recursive Sequence

A sequence that relates to the general term a_n of a sequence where one or more of the terms that comes before it is said to be defined recursively. The domain of recursive sequence can be the set of whole numbers. For Example, Mulatu and Fibonacci sequences are some of the examples of recursion formula.

Exercise 1.2

1. Find the 12^{th} term of the Fibonacci's sequence {1, 1, 2, 3, 5, 8,...}.
2. List the first eight terms of Mulatu's sequences and draw its graph.

1.2 Arithmetic and Geometric Sequences

1.2.1 Arithmetic Sequences

Activity 1.3

Find the difference between consecutive terms for each of the following sequences.

a. 3, 7,11,15, ... b. 3, −1,−5,−9,... c. $2,\frac{7}{3},\frac{8}{3},3,...$

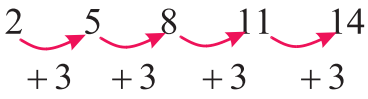
Definition 1.2

Arithmetic sequence or arithmetic progression is a sequence in which each term except the first is obtained by adding a fixed number (positive or negative) to the preceding term. The fixed number is called common difference of the sequence.

Example 1

For the following arithmetic sequence 2, 5, 8, 11, 14,…, what is the first term, third term and common difference? Find the 6^{th} term.

Solution



1^{st} term: 2, 3^{rd} term: 8

The common difference is 3.

6^{th} term: 5^{th} term + 3 = 14 + 3 = 17.

Example 2

Given that the 1^{st} term of an arithmetic sequence is 10, and the common difference is -4, find the terms from 2^{nd} to 5^{th} term.

Solution

2^{nd} term: 1^{st} term + (-4) = 10 + (-4) = 6.

3^{rd} term: 2^{nd} term + (-4) = 6 + (-4) = 2.

4^{th} term: 3^{rd} term + (-4) = 2 + (-4) = -2.

5^{th} term: 4^{th} term + (-4) = -2+ (-4) = -6.

Therefore, the terms from 2^{nd} to 5^{th} term are 6, 2, -2, -6.

Exercise 1.3

1. For the following arithmetic sequences, what are the first term, third term and common difference? Find the 6^{th} term.

   a. {3, 5, 7, 9, 11...} b. {9, 6, 3, 0, -3…}

2. Find the terms from 2^{nd} to 5^{th} of an arithmetic sequence when the

   a. 1st term is 1, and the common difference is 5.

   b. 1st term is 2, and the common difference is $-\frac{1}{2}$.

General term of arithmetic sequence

From the Activity 1.3, (a), 3, 7, 11, 15, 19,… the first term is 3, and the common difference is 4. Let the common difference be d, and let A_n be the n^{th} term of the sequence, then,

$$A_1 = 3$$
$$A_2 = 3+4 = A_1 + d$$
$$A_3 = 3+4+4 = A_1 + 2d$$
$$A_4 = 3+4+4+4 = A_1 + 3d$$
$$\cdots$$
$$A_n = 3+4+4+4+\ldots+4 = A_1 + (n-1)d$$

Suppose $A_n : A_1, A_2, A_3, A_4, A_5, A_6, \ldots$ is an arithmetic sequence.

$$A_2 = A_1 + d$$
$$A_3 = A_2 + d = A_1 + d + d = A_1 + 2d$$
$$A_4 = A_3 + d = A_1 + 2d + d = A_1 + 3d$$

$$A_5 = A_4 + d = A_1 + 3d + d = A_1 + 4d$$
$$A_6 = A_5 + d = A_1 + 4d + d = A_1 + 5d$$
$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$$
$$A_n = A_{n-1} + d = A_1 + (n-2)d + d = A_1 + (n-1)d$$

Thus, the following formula is derived:

Theorem 1.1

If A_n is an arithmetic sequence with the first term A_1 and a common difference d, then the n^{th} term of the sequence is given by $A_n = A_1 + (n-1)d$.

Example 1

Find the general term of the sequence A_n, when the first term is 5, and the common difference is 4.

Solution

The given sequence is arithmetic sequence with $A_1 = 5$ and $d = 4$.

From the theorem 1.1, we have the formula $A_n = A_1 + (n-1)d$. So,

$$A_n = A_1 + (n-1)d = 5 + (n-1)(4) = 5 + 4n - 4 = 4n + 1.$$

Example 2

What is the 31^{st} term of the sequence $1, 4, 7, 10, \ldots$?

Solution

The given sequence is arithmetic sequence with $A_1 = 1$ and $d = 3$

$$A_n = A_1 + (n-1)d = 1 + 3(n-1) = 3n - 2.$$

Then,

$$A_{31} = 3(31) - 2 = 93 - 2 = 91.$$

Exercise 1.4

1. Find the general term of the sequence A_n, when

   a. $A_1 = 2,\ d = 3.$ b. $A_1 = 10,\ d = -5.$

2. What is the 10^{th} term of the sequence, 10, 6, 2, -2, …?

Further on arithmetic sequences

Example 1

When the third term is 10 and the sixth term is 1,

a. Find the general term of sequence A_n.

b. Find A_8.

Solution

a. Applying arithmetic sequence formula and substituting existing values yields:

$$A_3 = A_1 + (3-1)d, \qquad 10 = A_1 + 2d$$
$$A_6 = A_1 + (6-1)d, \qquad 1 = A_1 + 5d$$
$$\begin{cases} A_1 + 2d = 10 \\ A_1 + 5d = 1 \end{cases}$$

Subtracting these two equations,

$$3d = -9 \Rightarrow d = -3\ ,\ A_1 + 2d = 10,\ A_1 + 2(-3) = 10,\ A_1 = 10 + 6 = 16.$$

Therefore, the general term becomes

$$A_n = A_1 + (n-1)d = 16 + (n-1)(-3) = 16 - 3n + 3 = 19 - 3n.$$

b. $A_n = 19 - 3n \Rightarrow A_8 = 19 - 3(8) = 19 - 24 = -5.$

Example 2

Determine whether or not the sequences with the following general terms are arithmetic. a. $a_n = 3n - 2$ b. $a_n = 3n^2 - 2$

Solution

a. To solve such types of problem, we have to show the difference between successive terms is constant.

$a_n = 3n-2$

$a_{n+1} = 3(n+1)-2 = 3n+1$

$a_{n+1} - a_n = 3n+1-(3n-2) = 3.$

Since, the difference between successive term is constant, it arithmetic sequence

b. $a_n = 3n^2 - 2$

$a_n = 3n^2 - 2$

$a_{n+1} = 3(n+1)^2 - 2 = 3(n^2+2n+1) - 2 = 3n^2 + 6n + 1$

$a_{n+1} - a_n = 3n^2 + 6n + 1 - (3n^2 - 2) = 6n + 3.$

Since, the difference between successive terms is not constant, it is not arithmetic sequence.

Exercise 1.5

1. Find the general term of the arithmetic sequence A_n, when

   a. $A_4 = 15,\ A_8 = 27.$ b. $A_5 = 20,\ A_{10} = 0.$

2. Given arithmetic sequence with $A_2 = 3$ and $A_5 = 24$. Find A_n and A_{11}.

3. Determine whether or not the sequences with the following general terms are arithmetic.

   a. $a_n = 7n-3$ b. $a_n = 5n-3$ c. $a_n = n^2+n+1$ d. $a_n = 3^n$

Arithmetic mean between two numbers

The term(s) of arithmetic sequence that lie between two given terms are called the arithmetic mean.

Example 1

Given that $1, x, 8$ is an arithmetic sequence, find x.

Solution

Since it is arithmetic sequence, the difference between two consecutive terms is constant.

$$x-1=8-x$$
$$2x=9$$
$$x=\frac{9}{2}$$

Example 2

The first and sixth terms of an arithmetic sequence are 4 and 29. Find the values of terms 2, 3, 4 and 5.

Solution

Let the four terms be A_2, A_3, A_4, A_5.

So, $4, A_2, A_3, A_4, A_5, 29$ form an arithmetic sequence. Then, $A_n = A_1 + (n-1)d$.

$$A_1 = 4$$
$$A_6 = A_1 + (6-1)d$$
$$29 = 4 + 5d$$
$$d = 5$$

Thus, $A_n = 4 + (n-1)5 = 5n-1$

$A_2 = 4 + 5 \times 1 = 9$

$A_3 = 4 + 5 \times 2 = 14$

$A_4 = 4 + 5 \times 3 = 19$

$A_5 = 4 + 5 \times 4 = 24.$

Exercise 1.6

1. Given that the sequence $3, x, 7$ is an arithmetic sequence, find x.
2. Given that the sequence $\frac{1}{12}, \frac{1}{x}, \frac{1}{6}$ is an arithmetic sequence, find x.
3. Find the arithmetic mean of 4 and 14.
4. Insert four arithmetic means between 4 and 14 to create an arithmetic sequence.

1.2.2 Geometric Sequences

Activity 1.4

Find the ratio between the consecutive terms of each of the following sequences.

a. 2 ,6, 18, 54,... *b.* $2, -2, 2, -2, \ldots$ *c.* $3, -\frac{3}{2}, \frac{3}{4}, -\frac{3}{8}, \ldots$

Definition 1.3

A geometric sequence or geometric progression is one in which the ratio between consecutive terms is a non-zero constant. This constant is called the common ratio.

$\{G_n\}$ is geometric sequence if and only if $r = \frac{G_{n+1}}{G_n}$, $r \in \mathbb{R}$ & $r \neq 0$ where r is the common ratio.

Example 1

For the following geometric sequence 3, 6, 12, 24, 48,…, find the common ratio, r, and the 6^{th} term.

Solution

The common ratio:

$$\frac{G_2}{G_1}=\frac{6}{3}=2,\quad \frac{G_3}{G_2}=\frac{12}{6}=2,\quad \frac{G_4}{G_3}=\frac{24}{12}=2,\ldots$$

$$3 \xrightarrow{\times 2} 6 \xrightarrow{\times 2} 12 \xrightarrow{\times 2} 24 \cdots$$

Thus, $r=2$.

The 6^{th} term: $G_6=G_5r=48\times 2=96$.

Example 2

The 1^{st} term of a geometric sequence is $-\frac{1}{2}$, and its common ratio is $\frac{1}{2}$, find the 2^{nd}, 3^{rd}, 4^{th} and 5^{th} term.

Solution

We are given $G_1=-\frac{1}{2}$ and $r=\frac{1}{2}$, then

$$G_2=G_1r=-\frac{1}{2}\times\frac{1}{2}=-\frac{1}{4}$$

$$G_3=G_2r=-\frac{1}{4}\times\frac{1}{2}=-\frac{1}{8}$$

$$G_4=G_3r=-\frac{1}{8}\times\frac{1}{2}=-\frac{1}{16}$$

$$G_5=G_4r=-\frac{1}{16}\times\frac{1}{2}=-\frac{1}{32}.$$

Exercise 1.7

1. For the geometric sequence 1, 2, 4, 8, 16…, find the common ratio, r, and the 6^{th} term.
2. Given that the 1^{st} term of a geometric sequence is 1, and its common ratio is 3, find the 2^{nd}, 3^{rd}, 4^{th} and 5^{th} term.

Determining the n^{th} term of geometric sequence

From the activity 1.4, (a), we have the geometric sequence 2, 6, 18, 54,… which has a common ratio of 3.

$$G_1 = 2$$
$$G_2 = 6 = 2\times 3 = G_1 \times 3^1$$
$$G_3 = 18 = 6\times 3 = G_2 \times 3 = G_1 \times 3 \times 3 = G_1 \times 3^2$$
$$G_4 = 54 = 18\times 3 = G_3 \times 3 = G_1 \times 3 \times 3 \times 3 = G_1 \times 3^3$$
$$\cdots$$
$$G_n = G_1 \times 3 \times 3 \times 3 \times \ldots \times 3 = G_1 \times 3^{n-1}$$

From Definition 1.3,

$$G_2 = G_1 \times r$$
$$G_3 = G_2 \times r = G_1 \times r \times r = G_1 \times r^2$$
$$G_4 = G_3 \times r = G_1 \times r^2 \times r = G_1 \times r^3$$
$$G_5 = G_4 \times r = G_1 \times r^3 \times r = G_1 \times r^4$$
$$\cdots \qquad\qquad \cdots$$
$$G_n = G_1 \times r^{n-1}$$

Thus, the following theorem is deduced:

Theorem 1.2

If $\{G_n\}$ is a geometric sequence with the first term G_1 and common ratio r, then the n^{th} term of the sequence is given by $G_n = G_1 r^{n-1}$.

Example 1

Find G_n, when the first term is 3 and the common ratio is 2.

Solution

Given $G_1 = 3$ and $r = 2$, then

$$G_n = G_1 r^{n-1} = 3\times 2^{n-1}.$$

Example 2

Find the n^{th} term, G_n of the sequence $1, -2, 4, -8, 16, \ldots$.

Solution

$G_1 = 1$, $r = -2$

Applying the formula for the n^{th} term of a geometric sequence, $G_n = G_1 r^{n-1}$.

$$G_n = (-2)^{n-1}.$$

Example 3

Find the 6^{th} term of the geometric sequence whose first term is 1 and common ratio is 2.

Solution

Given $G_1 = 1$ and $r = 2$, then $G_n = G_1 r^{n-1} = 1 \times 2^{n-1} = 2^{n-1}$.

Therefore, $G_6 = 2^{6-1} = 2^5$ or 32.

Exercise 1.8

1. For each of the following, find the n^{th} term of the geometric sequence.
   a. $G_1 = 2,\ r = 5$ b. $G_1 = 1,\ r = -3$
   c. $G_1 = 2,\ r = -2$ d. $G_1 = -3,\ r = \frac{1}{2}$
2. Find the n^{th} term G_n of the following sequences
   a. $3, 6, 12, 24, \ldots$ b. $\frac{3}{2}, \frac{3}{4}, \frac{3}{8}, \frac{3}{16}, \ldots$ c. $27, 9, 3, 1, \ldots$
3. Find the 5^{th} term of the geometric sequence whose first term is -3 and common ratio is $-\frac{1}{2}$.

Geometric mean between two numbers

When a, m, and b are terms in a geometric sequence, then m is called the geometric mean between a and b. $(a \neq 0,\ b \neq 0,\ m \neq 0)$. In a geometric sequence, the ratio between consecutive terms is constant.

$$r = \frac{m}{a} = \frac{b}{m}$$
$$m^2 = ab$$
$$m = \pm\sqrt{ab}$$

Example 1

When $2, x, 5, \ldots$ is a geometric sequence, find $x(x \neq 0)$.

Solution

As the ratio between the consecutive terms is the same,

$$\frac{x}{2} = \frac{5}{x},\quad x^2 = 10,\quad x = \pm\sqrt{10}.$$

Example 2

Find geometric mean between 2 and 8.

Solution

Let the geometric mean be $m(m \neq 0)$, then

$$r = \frac{m}{2} = \frac{8}{m}$$
$$m^2 = 2 \times 8$$
$$m = \pm\sqrt{16} = \pm 4.$$

Therefore, the geometric mean between 2 and 8 is -4 or 4.

Example 3

Find the 8^{th} term of the geometric sequence whose 1^{st} term is 5 and 4^{th} term is $\frac{1}{25}$.

(Express the answer in the form of exponent).

Solution

$G_1 = 5$ and $G_4 = \frac{1}{25}$,

$G_4 = G_1 r^3 = 5r^3 = \frac{1}{25}$.

Then, $r^3 = \left(\frac{1}{5}\right)^3$, therefore $r = \frac{1}{5}$.

$$G_8 = G_1 r^7 = 5\left(\frac{1}{5}\right)^7 = \frac{1}{5^6}.$$

Exercise 1.9

1. Find the geometric mean between 3 and 12.
2. In a geometric sequence, the 2^{nd} term is 12 and the 6^{th} term is 192. Find the 11^{th} term.
3. If $x, 4x+3, 7x+6$ are consecutive terms of a geometric sequence, find the value(s) of x, $x \neq 0$.
4. Find three consecutive terms of a geometric sequence, such that their sum is 35 and their product is 1000. Let the terms be $\frac{a}{r}$, a and ar. $(a \neq 0, r \neq 0)$

1.3 The Sigma Notation and Partial Sums

In the previous sections, you learned about the individual terms of a sequence. In this section, you will learn how to add the terms of a sequence, i.e. find the sum of the terms.

Partial sums

Given the sequence a_n

$S_1 = a_1$, S_1 is the first term of the sequence.

$S_2 = a_1 + a_2$, S_2 is the sum of the first two terms of the sequence.

$S_3 = a_1 + a_2 + a_3$, S_3 is the sum of the first three terms of the sequence.

$S_4 = a_1 + a_2 + a_3 + a_4$, S_4 is the sum of the first four terms of the sequence,

and so on.

$S_n = a_1 + a_2 + a_3 + a_4 + ... + a_n$, S_n is the sum of the first n terms of the sequence called the partial sum.

Definition 1.4

Let $\{a_n\}_{n=1}^{n=\infty}$ be a sequence. The sum of the first n terms of the sequence, denoted by S_n is called the partial sum of the sequence. Such summation is denoted as follows.

$S_n = \sum_{k=1}^{n} a_k = a_1 + a_2 + a_3 + a_4 + ... + a_n$, where k is the index of the summation, 1 is the lower limit of summation, n is the upper limit of the summation and $\sum$ is the sigma notation or the summation notation.

Example 1

Find the sum of the first five even natural numbers.

Solution

$a_1 = 2,\ a_2 = 4,\ a_3 = 6,\ a_4 = 8,\ a_5 = 10$. Then,

$$\begin{aligned} S_5 &= a_1 + a_2 + a_3 + a_4 + a_5 \\ &= 2 + 4 + 6 + 8 + 10 \\ &= 30 \end{aligned}$$

Example 2

Let $a_n = 3n + 1$, find S_6

Solution

$$a_n = 3n + 1$$

$$a_1 = 4, a_2 = 7, a_3 = 10, a_4 = 13, a_5 = 16, a_6 = 19$$

$$\begin{aligned} S_6 &= a_1 + a_2 + a_3 + a_4 + a_5 + a_6 \\ &= 4 + 7 + 10 + 13 + 16 + 19 \\ &= 69 \end{aligned}$$

Example 3

Given the general term $a_n = \dfrac{1}{n(n+1)}$, find the sum of the first

a. 99 terms b. n terms

Solution

By using partial fraction decomposition:

$$\frac{1}{n(n+1)} = \frac{A}{n} + \frac{B}{n+1}$$

Solving for A and B gives $A = 1$ and $B = -1$

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$(a)\ \ S_{99} = (1-\frac{1}{2})+(\frac{1}{2}-\frac{1}{3})+(\frac{1}{3}-\frac{1}{4})+\ldots+(\frac{1}{99}-\frac{1}{100}) = 1-\frac{1}{100} = 0.99$

$(b)\ \ S_n = 1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\ldots+\frac{1}{n-1}-\frac{1}{n}+\frac{1}{n}-\frac{1}{n+1} = 1-\frac{1}{n+1} = \frac{n}{n+1}$

Note: Such a sequence is said to be telescoping sequence.

Exercise 1.10

1. Find the sum of:
   a. the first five odd natural numbers.
   b. the first ten odd natural numbers.
2. Find the sums of the following sequences to the term given.

   a. $a_n = 4n-3,\ S_5.$ b. $a_n = 3-5n,\ S_8.$ c. $a_n = n^2+1,\ S_6.$

3. Given the general term $a_n = \frac{2}{n^2+5n+6}$, find the sum of the first n^{th} terms.

Sigma notation

Sigma notation is a method used to write out a long sum in a concise way. We use sigma notation for writing finite and infinite numbers of terms in a sequence. The sum is denoted by the sigma notation using the Greek letter $\sum$ (sigma).

Example 1

Express the following sigma notation in the form of the sum

a. $\sum_{k=1}^{8} 3k$ b. $\sum_{k=2}^{6} k^2$

Solution

a. $\sum_{k=1}^{8} 3k = 3+6+9+12+15+18+21+24 = 108.$ b. $\sum_{k=2}^{6} k^2 = 4+9+16+25+36 = 90.$

Example 2

Which one of the following express the sum, $3^2+4^2+5^2+6^2+7^2$?

a. $\sum_{k=3}^{7} k^2$ b. $\sum_{i=3}^{7} i^2$ c. $\sum_{k=2}^{6} (k+1)^2$

Solution

All of them express the given sum.

Exercise 1.11

1. Express the following sigma notations in the form of a sum.

a. $\sum_{k=1}^{6} 2k$ b. $\sum_{k=3}^{5} k^2$ c. $\sum_{k=1}^{n} 3^k$ d. $\sum_{k=3}^{5} k^3$

2. Express the following using the sigma notation

$2^2+4^2+6^2+8^2+10^2+12^2$.

1.3.1 Sigma Notation

Properties of sigma notation

The sequence $\{a_n\}$, where all the terms are c, the sum of the first nth term is

$$\sum_{k=1}^{n} a_n = c+c+c+\ldots+c = nc$$

That is, $\sum_{k=1}^{n} a_n = nc$

In particular, $\sum_{k=1}^{n} 1 = n(1) = n.$

Properties of Sigma Notation

(1) $\sum_{k=1}^{n} ca_k = c\sum_{k=1}^{n} a_k$, c is a constant

(2) $\sum_{k=1}^{n} (a_k+b_k) = \sum_{k=1}^{n} a_i + \sum_{k=1}^{n} b_k$

(3) $\sum_{k=1}^{n} (a_k-b_k) = \sum_{k=1}^{n} a_k - \sum_{k=1}^{n} b_k$

(4) $\sum_{k=1}^{n} a_k = \sum_{k=1}^{m} a_k + \sum_{k=m+1}^{n} a_k$, $1 \le m \le n$

Example

Evaluate each of the following sigma notations:

$(a)\ \sum_{k=1}^{3} 4k$ $(b)\ \sum_{k=1}^{5} (3k\text{-}2)$ $(c)\ \sum_{k=1}^{6} 2^{k-1}$

Solution

$$(a)\ \sum_{k=1}^{3} 4k = 4+8+12 = 24$$

Using the above property (1), you can also calculate,

$$\sum_{k=1}^{3} 4k = 4\sum_{k=1}^{3} k = 4(1+2+3) = 4(6) = 24$$

$$(b)\ \sum_{k=1}^{5} (3k-2) = 1+4+7+10+13 = 35$$

Using property (1) and (3), you can also calculate,

$$\begin{aligned}\sum_{k=1}^{5} (3k-2) &= 3\sum_{k=1}^{5} k - \sum_{k=1}^{5} 2 = 3(1+2+3+4+5) - (2+2+2+2+2)\\ &= 3(1+2+3+4+5) - (2+2+2+2+2)\\ &= 3(15) - 2(5)\\ &= 35\end{aligned}$$

$$\begin{aligned}(c)\ \sum_{k=1}^{6} 2^{k-1} &= 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5\\ &= 1+2+4+8+16+32\\ &= 63\end{aligned}$$

Using the above property (4), you can also calculate,

$$\begin{aligned}\sum_{k=1}^{6} 2^{k-1} &= \sum_{k=1}^{3} 2^{k-1} + \sum_{k=4}^{6} 2^{k-1}\\ &= (2^0 + 2^1 + 2^2) + (2^3 + 2^4 + 2^5)\\ &= 63\end{aligned}$$

Exercise 1.12

Evaluate each of the following sigma notations.

a. $\sum_{k=1}^{4} 5k$ b. $\sum_{k=1}^{5} (4k-1)$ c. $\sum_{k=3}^{6} (k^2-4)$

d. $\sum_{k=2}^{5} 3$ e. $\sum_{k=1}^{8} (k^3+2k^2-3k+5)$

1.3.2 Sum of Arithmetic Sequences

Activity 1.5

Find the sum of the first ten terms of the sequence $5, 15, 25, 35, \ldots$

HISTORICAL NOTE

Carl Friedrich Gauss (1777-1855)

A teacher of Gauss, at his elementary school, asked him to add all the integers from 1 to 100. When Gauss returned with the correct answer after only a few moments, the teacher could only look at him in astounded silence. This is what Gauss did:

$$\begin{array}{r} 1+\ 2+\ 3+\ldots+100 \\ \underline{100+99+98+\ldots+\ 1} \\ 101+101+101+\ldots+101 \end{array}$$

$$\frac{100\times 101}{2} = 5050.$$

To find the sum of the first 100 natural numbers, Gauss worked as follows. Writing the sum forward and backward then adding them together yields:

$$S_{100} = 1+2+3+\ldots+98+99+100$$
$$S_{100} = 100+99+98+\ldots+3+2+1$$
$$2S_{100} = 101+101+101+\ldots+101+101+101$$
$$2S_{100} = 100\times 101$$

Therefore, $S_{100} = \frac{1}{2} \times 100 \times 101 = 5050$

The sum of the first n natural numbers can also be calculated as follows:

$$S_n = 1+2+3+\ldots+(n-2)+(n-1)+n$$

$$S_n = n+(n-1)+(n-2)+\ldots 3+2+1$$

$$2S_n = (n+1)+(n+1)+\ldots+(n+1)+(n+1)$$

Therefore, $2S_n = n(n+1)$

The sum of the first n consecutive natural numbers is $S_n = \frac{n}{2}(n+1)$.

Example

Find the sum of the first a. 40 natural numbers b. 150 natural numbers

Solution

a) Using the formula $S_n = \frac{n}{2}(n+1)$, $S_{40} = \frac{40}{2}(40+1) = 20(41) = 820$

b) Using the formula $S_n = \frac{n}{2}(n+1)$, $S_{150} = \frac{150}{2}(150+1) = 75(151) = 11,325$

Exercise 1.13

1. Find the sum of the first
   a. 30 natural numbers.
   b. 99 natural numbers.
   c. 200 natural numbers.
2. If the sum of the first n natural numbers is 3240, what is the value of n ?

Derivation of sum of arithmetic sequence

Let $\{A_n\}_{n=1}^{\infty}$ be an arithmetic sequence.

$$S_n = A_1 + A_2 + A_3 + \ldots + A_n \text{ , where } A_n = A_1 + (n-1)d$$

$$S_n = A_1 + (A_1 + d) + (A_1 + 2d) + (A_1 + 3d) + \ldots + (A_1 + (n-1)d)$$

By collecting all the A_1 terms (there are n of them) we get,

$$S_n = nA_1 + [d + 2d + 3d + ... + (n-1)d]$$

Now factoring out d from within the brackets,

$$S_n = nA_1 + d[1 + 2 + 3 + ... + (n-1)]$$

Inside the brackets you have the sum of the first $(n-1)$ positive integers. Thus by using the formula, $S_n = \frac{n}{2}(n+1)$,

$$S_n = nA_1 + d\left(\frac{n-1}{2}\right)n = \frac{2nA_1 + n(n-1)d}{2} = \frac{n}{2}[2A_1 + (n-1)d].$$

This formula can be written as:

$$S_n = \frac{n}{2}[A_1 + \{A_1 + (n-1)d\}]$$

$$= \frac{n}{2}(A_1 + A_n)$$

Hence, the following theorem is derived:

Theorem 1.3

The sum S_n of the first n terms of an arithmetic sequence with first term A_1 and common difference d is $S_n = \sum_{k=1}^{n} A_k = \frac{n}{2}[2A_1 + (n-1)d]$.

This formula can also be written as:

$S_n = n\left(\frac{A_1 + A_n}{2}\right)$, where A_n is the n^{th} term.

This alternative formula is useful when the first and the last terms are known.

Example 1

Find the sum of the first 35 terms of the sequence whose general term is $A_n = 5n$.

Solution

From the general term, we get $A_1 = 5, A_2 = 10, A_3 = 15, \ldots,$ this shows that the given sequence is an arithmetic sequence. So, $A_{35} = 5(35) = 175.$

Since we can easily identify the first and the 35^{th} terms, we use the formula

$$S_n = \frac{n}{2}(A_1 + A_n) = n\left(\frac{A_1 + A_n}{2}\right)$$

Thus, substituting $A_1 = 5$, and $A_{35} = 175$

$$S_{35} = \frac{35}{2}(5+175) = 35\left(\frac{5+175}{2}\right) = 35(90) = 3{,}150$$

Example 2

If the 1^{st} term of arithmetic sequence is 4, common difference is 5, then find the sum of the first 40 terms.

Solution

Given $A_1 = 4, \quad d = 5,$

$$S_n = \frac{n}{2}[2A_1 + (n-1)d] \Rightarrow S_{40} = \frac{40}{2}[2(4) + (40-1)(5)] = 20(8+195) = 20(203) = 4060.$$

Exercise 1.14

1. Find the partial sum of the following arithmetic sequences:
   a. $A_1 = 2,$ and last term $A_{10} = 21.$
   b. $A_1 = 40,$ and last term $A_{26} = 0.$
2. Find the sum of the following arithmetic sequences:
   a. $A_1 = 2,\ d = 3,\ n = 10$ b. $A_1 = 30,\ d = -5,\ n = 12$

Further on sum of arithmetic sequence

Example 1

Find the sum S_7 of the arithmetic sequence whose 4^{th} term is 2 and 7^{th} term is 17.

Solution

Applying the formula $A_n = A_1 + (n-1)d$

$$\begin{cases} A_4 = A_1 + 3d = 2 \\ A_7 = A_1 + 6d = 17 \end{cases}$$

Subtracting the first equation from the second equation,

$$3d = 17 - 2$$
$$d = 5$$

Thus,

$$A_1 = A_4 - 3d$$
$$A_1 = 2 - 3 \times 5 = -13$$
$$S_n = n\left(\frac{A_1 + A_n}{2}\right)$$
$$S_7 = 7\left(\frac{-13+17}{2}\right) = 7(2) = 14.$$

Example 2

For a given arithmetic sequence the sum $S_{10} = 165$ and $A_1 = 3$, find A_{10}.

Solution

Since the first term and the sum are given, applying the formula gives:

$$S_n = n\left(\frac{A_1 + A_n}{2}\right)$$

$$S_{10} = 165, A_1 = 3, A_{10} = ?$$

$$S_{10} = 10\left(\frac{A_1 + A_{10}}{2}\right)$$

$$165 = \frac{10}{2}(3 + A_{10})$$

$$\frac{165}{5} = 3 + A_{10}$$

$$A_{10} = 33 - 3 = 30.$$

Example 3

Find the sum of integers from 1 to 100 that are divisible by 10.

Solution

The number of such integer is $\frac{100}{10} = 10$

The 1^{st} term is 10, and the last term is 100. Then, $S_n = 10\left(\frac{10+100}{2}\right) = 550.$

Exercise 1.15

1. Find S_5 of the arithmetic sequence whose 3^{rd} term is 5 and 5^{th} term is 11.
2. Given the sum of an arithmetic sequence is $S_8 = 120$ and $A_1 = 1$, find A_8 and A_n.
3. Find the sum of the integers from 1 to 100 that are divisible by 2 or 5.
4. Find the sum of odd integers from 1 to 2001.

1.3.2 Sum of Geometric Sequences

Activity 1.6

1. Find the sum of the following geometric sequences:

   a. $\{2, 4, 8, 16, 32\}$ b. $\left\{1, \frac{2}{3}, \frac{4}{9}, \frac{8}{27}\right\}$

2. Find the sum of the first 5 terms of the sequence $1, \frac{2}{3}, \frac{4}{9}, \frac{8}{27}, \frac{16}{81}, \ldots$

In order to answer such types of problems, consider the following:

Let $\{G_n\}_{n=1}^{\infty}$ be a geometric sequence, then its associated geometric sum, S_n is:

$$S_n = G_1 + G_2 + G_3 + ... + G_{n-1} + G_n, \text{ where } G_n = r^{n-1}G_1$$
$$S_n = G_1 + rG_1 + r^2G_1 + ... + r^{n-2}G_1 + r^{n-1}G_1$$

Factorizing out G_1,

$$S_n = G_1(1 + r + r^2 + ... + r^{n-2} + r^{n-1}) \quad (1)$$

Multiplying both sides of equation (1) by r

$$rS_n = G_1(r + r^2 + r^3 ... + r^{n-1} + r^n) \quad (2)$$

Subtracting the equation (2) from equation (1),

$$S_n - rS_n = G_1(1 + r + r^2 + r^3 + ... + r^{n-2} + r^{n-1}) - G_1(r + r^2 + r^3 + ... + r^{n-1} + r^n)$$
$$(1-r)s_n = G_1(1-r^n),$$
$$S_n = \frac{G_1(1-r^n)}{1-r} \text{ for } r \neq 1$$

Thus, the following theorem is inferred:

Theorem 1.4

Let $\{G_n\}_{n=1}^{\infty}$ be a geometric sequence with common ratio r. Then the sum of the first n terms S_n is given by, $S_n = \begin{cases} nG_1, & \text{if } r = 1 \\ G_1\dfrac{(1-r^n)}{1-r} = G_1\dfrac{(r^n-1)}{r-1}, & \text{if } r \neq 1. \end{cases}$

Example

Find the sum of the sequences in activity 1.6 using this theorem, and confirm the results.

Solution

a. $r=2$, $G_1=2$ and $n=5$

$$S_5=2\left(\frac{2^5-1}{2-1}\right)=62$$

b. $\left\{1,\frac{2}{3},\frac{4}{9},\frac{8}{27}\right\}$

$G_1=1$, $r=\frac{2}{3}$ and $n=4$

$$S_4=1\left(\frac{\left(\frac{2}{3}\right)^4-1}{\frac{2}{3}-1}\right)=\frac{65}{27}$$

Exercise 1.16

1. Find the sum of the from 1st to nth of the following geometric sequences:
   a. $G_1=3,\ r=2$ b. $G_1=1,\ r=\frac{1}{2}$
2. Given the geometric sequence: 1, 3, 9, 27,.... find S_n.

Further on the sum of geometric sequences

Example

The sum of the first three terms of a geometric sequence is 7, and the sum from 4^{th} to 6^{th} terms is 56. Find the first term and the common ratio.

Solution

Let the first term be G_1 and common ratio r. Then,

$$G_1+G_1r+G_1r^2=7 \qquad (1)$$
$$G_1r^3+G_1r^4+G_1r^5=56 \qquad (2)$$

From (2),

$$r^3(G_1 + G_1r + G_1r^2) = 56$$

Substituting (1),

$$7r^3 = 56$$
$$r^3 = 8$$
$$r = 2$$

Then,

$$G_1 + 2G_1 + 4G_1 = 7$$
$$7G_1 = 7$$
$$G_1 = 1$$

Therefore, the common ratio is 2, and the 1^{st} term is 1.

Exercise 1.17

1. The sum of the first three terms of a geometric sequence is 9, and the sum from the 4^{th} to 6^{th} term is -18. Find the first term and common ratio.
2. How many terms of the sequence: $3, \frac{3}{2}, \frac{3}{4}, \frac{3}{8}, \ldots$ are needed to give the sum $\frac{3069}{512}$?
3. Find the sum to indicated number of terms in each of the geometric sequence in questions *a* to *d*:
   a. 0.15, 0.015, 0.0015, ... *n* terms.
   b. $\sqrt{7}, \sqrt{21}, 3\sqrt{7}, \ldots$ *n* terms.
   b. 1, $-a$, a^2, $-a^3$, ... *n* terms (if $a \neq -1$).
   d. $x^3, x^5, x^7, \ldots$ *n* terms (if $x \neq \pm 1$).

1.4 Infinite Series

Activity 1.7

Supposing that a tree grows by half its height in a year. It then grows half of the amount of the previous year. What would be the height of the tree be if it continues to grow at the same rate?

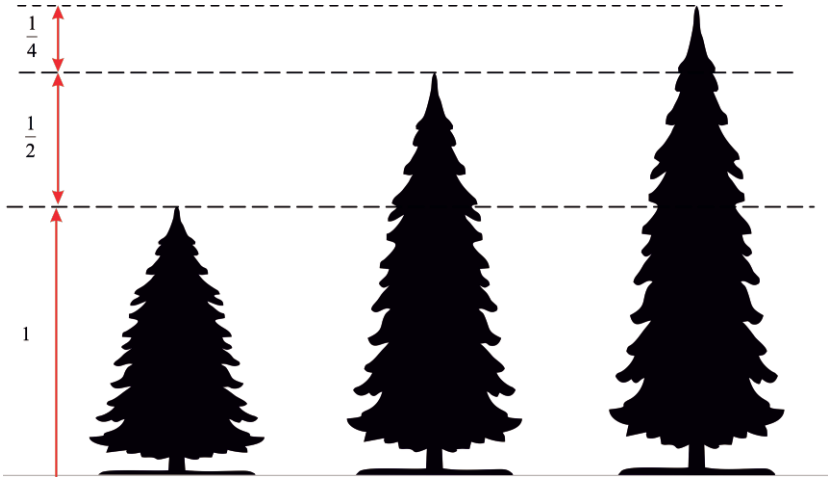


Figure 1.5

Suppose we have the sequence: $a_1, a_2, a_3, ..., a_n, ...$

An infinite sum of the form: $a_1 + a_2 + a_3 + ... + a_n + ...$ is called an infinite series and using summation, we can write as: $a_1 + a_2 + a_3 + ... + a_n + ... = \sum_{n=1}^{\infty} a_n$

1.4.1. Divergence and Convergence of Infinite Sequence

Activity 1.8

For each of the sequences i – v:

a. Write the formula for the n^{th} term.

b. Find the sum where the sequence converges.

i. 1, 2, 3, 4,...

ii. $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, ...$

iii. $-1, 1, -1, 1, ...$

iv. 1, 1.5, 2.25, 3.375,...

v. 0.12, 0.0012, 0.000012, 0.00000012,...

For an infinite sequence, there are cases of convergence and divergence, when n approaches to infinity, as follows:

Convergence

a. Sequence a_n converges to α: $n \to \infty$, $a_n \to \alpha$

Divergence

b. Sequence a_n diverges to positive infinity: $n \to \infty$, $a_n \to \infty$.

c. Sequence a_n diverges to negative infinity: $n \to \infty$, $a_n \to -\infty$.

d. Sequence a_n vibrates: a_n has no limit. i.e. the value oscillate or vibrate back and forth between numbers.

Divergence/convergence of infinite geometric sequence

Considering infinite geometric sequence $G_n = r^n$, where $G_1 = r$, and the common ratio is r, there are cases of convergence and divergence, when n approaches to infinity, as follows:

i) $r > 1$, When $n \to \infty$, then $G_n \to \infty$ (diverge)

ii) $r = 1$, When $n \to \infty$, then $G_n \to 1$ (converge)

iii) $|r| < 1$, When $n \to \infty$, then $G_n \to 0$ (converge)

iv) $r \leq -1$, When $n \to \infty$, then G_n vibrates (no limit, diverge).

Example

Find whether the given geometric sequences diverge, converge or vibrate as n approaches to infinity.

a. $G_n = (\sqrt{2})^n$ *b.* $G_n = \left(\frac{2}{3}\right)^n$ *c.* $G_n = (-3)^n$

Solution

a. $G_n = (\sqrt{2})^n$

As $\sqrt{2} > 1$, when $n \to \infty$, then $G_n \to \infty$, it diverges.

b. $G_n = \left(\frac{2}{3}\right)^n$

As $\left|\frac{2}{3}\right| < 1$, when $n \to \infty$, then $G_n \to 0$, it diverges.

c. $G_n = (-3)^n$

As $-3 \leq -1$, when $n \to \infty$, then G_n vibrates

Exercise 1.18

Find whether the given geometric sequences diverge, converge or vibrates as n approaches infinity.

a. $G_n = (\sqrt{3})^n$ b. $G_n = \left(\frac{3}{4}\right)^n$ c. $G_n = \left(-\frac{1}{2}\right)^n$

Infinite series

Definition 1.5

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence and S_n be the n^{th} partial sum such that, as n gets larger and larger, S_n tends to s, where s is a real number, then we say the infinite series $\sum_{n=1}^{\infty} a_n$ converges and is written as $\sum_{n=1}^{\infty} a_n = s$.

However, if such an s does not exist or is infinite, we say the infinite series $\sum_{n=1}^{\infty} a_n$ diverges.

Let us consider the geometric sequence $1, \frac{2}{3}, \frac{4}{9}, \ldots$ where $G_1 = 1$ and $r = \frac{2}{3}$. We have partial sum,

$$S_n = \frac{G_1(1-r^n)}{1-r} = \frac{1\left(1-\left(\frac{2}{3}\right)^n\right)}{1-\frac{2}{3}} = 3\left[1-\left(\frac{2}{3}\right)^n\right]$$

Let us study the behavior of $\left(\frac{2}{3}\right)^n$ as n becomes larger and larger, approaching to infinity.

n	1	5	10	20
$\left(\frac{2}{3}\right)^n$	0.6667	0.1316872428	0.01734152992	0.00030072866

We observe that as n becomes larger and larger, $\left(\frac{2}{3}\right)^n$ becomes closer and closer to zero. Mathematically, we say that as n becomes increasingly large, $\left(\frac{2}{3}\right)^n$ becomes increasingly small. In other words, as $n \to \infty$, $\left(\frac{2}{3}\right)^n \to 0$. Consequently, we find set on the line below the sum of infinitely many terms of the sequence above is $S = 3$.

Thus, for an infinite geometric sequence $G_1, G_1r, G_1r^2, ...,$ if numerical value of the common ratio r is between -1 and 1, then $S_n = \frac{G_1(1-r^n)}{1-r} = \frac{G_1}{1-r} - \frac{G_1r^n}{1-r}$.

In this case, $r^n \to 0$ as $n \to \infty$ since $|r| < 1$ and then $\frac{G_1r^n}{1-r} \to 0$. Therefore, $S_n \to \frac{G_1}{1-r}$ as $n \to \infty$. Symbolically, the sum to infinity of an infinite geometric series is denoted by S_∞. Thus, we have $S_\infty = \frac{G_1}{1-r}$.

Note

Recall that if $\sum_{n=1}^{\infty} G_1 r^{n-1} = G_1 + G_1 r + G_1 r^2 + \ldots$ is a geometric series with first term G_1 and common ratio r, then $S_n = \frac{G_1(1-r^n)}{1-r} = \frac{G_1}{1-r} - \frac{G_1 r^n}{1-r}$.

(i) If $|r| < 1$, as $n \to \infty$, $r^n \to 0$ so that $S_n = \frac{G_1}{1-r} - \frac{G_1 r^n}{1-r} \to \frac{G_1}{1-r}, S_\infty = \frac{G_1}{1-r}$.

(ii) If $|r| \geq 1$, then the series diverges or vibrates as follows.

a. If $r = 1$, then $\sum_{k=1}^{n} G_k = nG_1$, when $n \to \infty$, $\sum_{k=1}^{\infty} G_k = \infty \ (G_1 > 0)$

$\sum_{k=1}^{\infty} G_k = -\infty \ (G_1 < 0)$

b. If $r \neq 1$, then $S_n = \sum_{k=1}^{\infty} G_k = \frac{G_1(1-r^n)}{1-r}$

Then, If $r \leq -1$, when $n \to \infty$, then $\sum_{k=1}^{\infty} G_k$ vibrates,

If $r > 1$, when $n \to \infty$, then $\sum_{k=1}^{\infty} G_k = \infty \ (G_1 > 0)$

$\sum_{k=1}^{\infty} G_k = -\infty \ (G_1 < 0)$

Example 1

Determine whether the series $\sum_{n=1}^{\infty} 3^n$ converges or diverges.

Solution

The series $\sum_{n=1}^{\infty} 3^n = 3 + 9 + 27 + \ldots + \ldots$ is a geometric series with first term $G_1 = 3$ and common ratio $r = 3$.

Hence, the partial sum is given by $S_n = \frac{G_1(1-r^n)}{1-r}$

Substituting the values, we obtain,

$$S_n = \frac{G_1(1-r^n)}{1-r} = \frac{3[1-3^n]}{1-3} = -\frac{3}{2}[1-3^n] = -\frac{3}{2} + \frac{3}{2}3^n.$$

Thus, as $n \to \infty$, $S_n \to \infty$. Therefore, the series diverges.

Example 2

Find the sum of the geometric series $5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \ldots$

Solution

The geometric series has the first term $G_1 = 5$ and the common ratio $r = \frac{-\frac{10}{3}}{5} = -\frac{2}{3}$.

Since

$$|r| = \left|\frac{-2}{3}\right| = \frac{2}{3} < 1,$$

$$S_\infty = 5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \ldots = \frac{G_1}{1-r} = \frac{5}{1-\left(\frac{-2}{3}\right)} = \frac{5}{\frac{5}{3}} = 3.$$

then the series converges and its sum is given by

Exercise 1.19

Find the sum for each of the following, if it exists, assuming the patterns continue as in the first few terms.

a. $3 + 1 + \frac{1}{3} + \frac{1}{9} + \ldots$

b. $1 + \frac{3}{4} + \frac{9}{16} + \frac{27}{64} + \ldots$

c. $\frac{1}{5} + \frac{1}{10} + \frac{1}{20} + \ldots$

d. $\frac{1}{5} + \frac{-1}{10} + \frac{1}{20} + \frac{-1}{40} + \ldots$

e. $7 + \frac{7}{10} + \frac{7}{100} + \frac{7}{1000} + \ldots$

Further on infinite series

Example 1

Find the sum $\sum_{k=1}^{\infty} 5^{3-k}$

Solution

$$5^{3-k} = 5^3 \times 5^{-k} = 5^3\left(\frac{1}{5}\right)^k = 5^2\left(\frac{1}{5}\right)^{k-1}$$

It is a geometric series whose first term $G_1 = 25$ and the common ratio $r = \frac{1}{5}$.

Therefore, the sum becomes:

$$S_\infty = \sum_{k=1}^{\infty} 5^{3-k} = 5^2 + 5^1 + 5^0 + 5^{-1} + \ldots = \frac{G_1}{1-r} = \frac{25}{1-\frac{1}{5}} = \frac{125}{4}$$

Example 2

Find the sum $\sum_{k=1}^{\infty} (-1)^{k+1}\left(\frac{2}{3}\right)^k$

Solution

$$(-1)^{k+1}\left(\frac{2}{3}\right)^k = (-1)^k(-1)\left(\frac{2}{3}\right)^k = (-1)\left(-\frac{2}{3}\right)^k = -\left(-\frac{2}{3}\right)^k$$

It is a geometric series whose first term $G_1 = \frac{2}{3}$ and the common ratio $r = -\frac{2}{3}$.

Therefore, the sum becomes

$$S_\infty = \sum_{k=1}^{\infty} (-1)^{k+1}\left(\frac{2}{3}\right)^k = \frac{2}{3} - \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 - \left(\frac{2}{3}\right)^4 + \ldots = \frac{G_1}{1-r} = \frac{\frac{2}{3}}{1-\left(-\frac{2}{3}\right)} = \frac{2}{5}$$

Exercise 1.20

Find the sums for each of these geometric series

a. $\sum_{k=1}^{\infty} 5\left(\frac{1}{3}\right)^{k-1}$ b. $\sum_{k=1}^{\infty} 2^{1-k}$ c. $\sum_{k=1}^{\infty} 5^{k-3}$ d. $\sum_{k=1}^{\infty} \left(\frac{3}{4}\right)^{k+3}\left(\frac{2}{3}\right)^{k-2}$

1.4.2. Recurring Decimals

Recurring or repeating decimals are rational numbers (fractions) whose representations as a decimal contain a pattern of digit that repeats indefinitely after decimal point. The decimal that start their recurring cycle immediately after the decimal point are called **purely recurring decimals.**

Purely recurring decimals are converted to an irreducible fraction whose prime factors in the denominator can only be the prime numbers other than 2 or 5, i.e. the prime numbers from the sequence $\{3,7,11,13,17,19,...\}$. The decimals that have some extra digits before the repeating the sequence of digits are called the **mixed recurring decimals**.

The repeating sequence may consist of just one digit or of any finite number of digits. The number of digits in the repeating pattern is called **the period.**

Mixed requiring decimals are converted to an irreducible fraction whose denominator is a product of 2's and/or 5's besides the prime numbers from the sequence $\{3,7,11,13,17,19,...\}$.

All recurring decimals are infinite decimals.

Converting purely recurring decimals to fraction

Example 1

Convert the recurring decimal $0.\dot{3}$ to a fraction.

Solution

We can write the given decimal as the sum of the infinite converging geometric series as follows:

$$0.\dot{3} = 0.333333... = 0.3 + 0.03 + 0.003 + ...$$
$$= \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + ...$$

This is the infinite geometric series, where $G_1 = \frac{3}{10}$ and $r = \frac{\frac{3}{100}}{\frac{3}{10}} = \frac{1}{10}$

Since, $|r| < 1$ then $S_\infty = \frac{G_1}{1-r} = \frac{\frac{3}{10}}{1-\frac{1}{10}} = \frac{\frac{3}{10}}{\frac{9}{10}} = \frac{1}{3}$

Example 2

Convert the recurring decimal $0.\dot{4}\dot{7}$ to a fraction.

Solution

We can write the given recurring decimal as the sum of the infinite converging geometric series as follows:

$$0.\dot{4}\dot{7} = 0.474747... = 0.47 + 0.0047 + 0.000047 + ...$$
$$= \frac{47}{100} + \frac{47}{10000} + \frac{47}{1000000} + ...$$

Since, $r = \frac{G_n}{G_{n-1}} = \frac{1}{100}$, $|r| < 1$ then $S_\infty = \frac{G_1}{1-r}$,

Therefore, by substituting $G_1 = \frac{47}{100}$ and $r = \frac{1}{100}$ in to the formula:

$$0.\dot{4}\dot{7} = \frac{G_1}{1-r} = \frac{\frac{47}{100}}{1-\frac{1}{100}} = \frac{\frac{47}{100}}{\frac{99}{100}} = \frac{47}{99}$$

Notice that when converting a recurring decimal that is less than one to a fraction, we write the repeating digits to the numerator and in the denominator of the equivalent fraction write as many 9's as is the number of digits in the repeating pattern.

Exercise 1.21

Convert the following recurring decimals to fractions.

a. $0.\dot{4}$ b. $3.\dot{7}$ c. $0.\dot{5}\dot{6}$

1.5 Applications of Sequence and Series in Daily Life

This section is devoted to the application of arithmetic and geometric progressions or geometric series that are associated with real-life situations.

Activity 1.9

1. A man accepts a position with an initial salary of 5200 ETB per month. It is understood that he will receive an automatic increase of 320 ETB in the very next month and each month thereafter.
   a. Find his salary for the tenth month.
   b. What are his total earnings during the first year?
2. A carpenter was hired to build 192 window frames. The first day he/she made five frames and each day, thereafter he/she made two more frames than he /she made the day before. How many days did it take him/her to finish the job?

Here are some examples followed by exercises. They illustrate some useful applications.

Example 1

Observe the pattern in Figure 1.5:

a. If the pattern continues, find the number of letters in the column containing the letter M.

b. If the total number of letters in the pattern is 361, which letter will be in the last column?

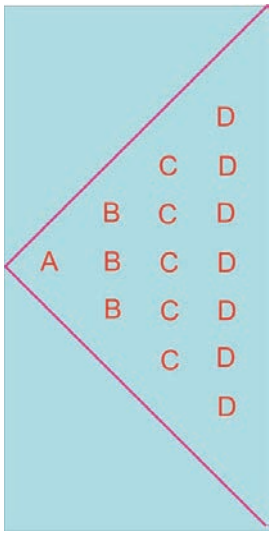


Figure 1.5

Solution

a. If you observe the structure of the letters in figure 1.5, it is of the form: $1,3,5,7,9,...$

This is an arithmetic series with first term $A_1 = 1$ and common difference $d = 2$. Therefore, the n term of the arithmetic sequence is given by

$$A_n = A_1 + (n-1)d$$

For the letter M: $n = 13$;

$$A_{13} = A_1 + (n-1)d = 1 + (13-1)(2) = 1 + (12)(2) = 25.$$

b.

$$S_n = \frac{n}{2}[2A_1 + (n-1)d]$$

$$361 = \frac{n}{2}[2(1) + (n-1)(2)]$$

$$361 = n[1 + n - 1]$$

$$361 = n^2$$

$$n = \pm 19 \text{ (as } n > 0)$$

$$\therefore n = 19$$

So, the letter "S" will be in the last column.

Example 2

A theatre is filling up at a rate of 4 people in the first minute, 6 people in the second minute and 8 people in the third minute and so on. After 6 minutes, the theatre is half full. After how many minutes will the theatre be full?

Solution

The structure of the problem has an infinite arithmetic series of the form: $4+6+8+...$

So, the common difference of the problem is calculated as

$$d = A_2 - A_1 = A_3 - A_2 = 2$$

Therefore, the sum of arithmetic series whose first term $A_1 = 4$ and common difference $d = 2,$ is written as

$$S_n = \frac{n}{2}[2A_1 + (n-1)d] \Rightarrow S_6 = \frac{6}{2}[2(4) + (6-1)(2)] = 3(18) = 54 \text{ (theatre half full).}$$

Therefore, the capacity of the theatre is $2 \times 54 = 108$

$$S_n = \frac{n}{2}[2A_1 + (n-1)d]$$

$$108 = \frac{n}{2}[2(4) + (n-1)(2)]$$

$$216 = n[8 + 2n - 2]$$

$$2n^2 + 6n - 216 = 0$$

$$n^2 + 3n - 108 = 0$$

$$(n+12)(n-9) = 0$$

$\therefore n = -12$ or $n = 9$, where n must be a positive integer.

Therefore, $n = 9.$ It takes 9 minutes for the theatre to be full.

Example 3

A job applicant finds that a firm offers a starting annual salary of Birr 32,500 with a guaranteed raise of Birr 1,400 per year.

a. What would the annual salary be in the tenth year?

b. How much would be earned at the firm over the first 10 years ?

Solution

a. The annual salary at the firm forms the arithmetic sequence;

32,500, 33,900, 35,300, … with first term $A_1 = 32,500$ and common difference $d = 1,400$. Thus, $A_n = A_1 + (n-1)d$, substituting the values we obtain;

$A_{10} = 32,500 + (10-1)(1,400) =$ Birr 45,100

b. To determine the amount that would be earned over the first 10 years, we need to add the first 10 annual salaries;

$$S_{10} = A_1 + A_2 + A_3 + \ldots + A_{10} = 10\left(\frac{A_1 + A_{10}}{2}\right)$$ (It is 10 times the average of the first and the last term.)

$$S_{10} = \frac{10}{2}(32,500 + 45,100) = \text{ Birr } 388,000$$

Therefore, a total of Birr 388,000 would be earned at the firm over the first 10 years.

Exercise 1.22

1. A person is scheduled to get a raise of Birr 250 every 6 months during his/her first 5 years on the job. If their starting salary is Birr 25,250 per year, what will his/her annual salary be at the end of the 3rd year?
2. Bontu begins a saving program in which she will save Birr 1,000 the first year. Each subsequent year, she will save 200 more than she did the previous year. How much will she save during the eighth year?

Example 4

A woman deposits Birr 3,500 in a bank account paying an annual interest at a rate of 6%. Show that whether the amounts she has in the account at the end of each year form a geometric sequence.

Solution

Let $G_1 = 3,500$. Then,

$$G_2 = G_1 + \frac{6}{100}G_1 = G_1(1+0.06) = G_1(1.06)$$

$$G_3 = G_2 + \frac{6}{100}G_2 = G_2(1+0.06) = G_1(1.06)(1.06) = G_1(1.06)^2$$

$$G_4 = G_3 + \frac{6}{100}G_3 = G_3(1+0.06) = G_1(1.06)^2(1.06) = G_1(1.06)^3$$

Continuing in this way, you get $G_n = G_1(1.06)^{n-1}$

Since the ratio of any two consecutive terms is a constant, which is 1.06, this sequence is a geometric sequence.

Example 5

Suppose a radioactive substance loses half of its mass per year. If we start with 100 grams of a radioactive substance, how much will be left after 10 years?

Solution

Let us record the amount of the radioactive substance left after each year starting with $G_0 = 100$. Note that each term is half of the previous term and hence,

$G_1 = \frac{1}{2}G_0 = 100\left(\frac{1}{2}\right)$ is the amount left at the end of year 1.

$G_2 = \frac{1}{2}G_1 = 100\left(\frac{1}{2}\right)^2$ is the amount left at the end of year 2.

$G_3 = \frac{1}{2}G_2 == 100\left(\frac{1}{2}\right)^3$ is the amount left at the end of year 3.

If you continue in this way, you see that the ratio of any two consecutive terms is a constant, which is $\frac{1}{2}$, and hence this sequence is a geometric sequence.

Therefore, after ten years, the amount of the substance left is given by:

$$G_{10} = G_0\left(\frac{1}{2}\right)^{10} = 100\left(\frac{1}{2}\right)^{10} = \frac{100}{1,024} = 0.09765625 \text{ g.}$$

Exercise 1.23

1. A certain item loses one-tenth of its value each year. If the item is worth Birr 28,000 today, how much will it worth 4 years from now?
2. A boat is now worth Birr 34,000 and loses 12% of its value each year. What will it worth after 5 years?
3. The population of a certain town is increasing at a rate of 2.5% per year. If the population is currently 100,000, what will the population be 10 years from now?
4. Sofia deposits Birr 3,500 in a bank account paying an annual interest rate of 6%.Find the amount she has at the end of the
   a. first year b. second year c. third year
   d. fourth year e. n^{th} year
   f. Do the amounts she has at the end of each year form a geometric sequence? Explain.

Example 6

A man was injured in an accident at work. He receives a disability grant of 4800 ETB in the first year. This grant increases by a fixed amount each year.

a. What is the annual rate of increase if he received a total of 143,500 ETB over 20 years?

b. His initial annual expenses are 2600 ETB, which increases at a rate of 400 ETB per year. After how many years will his expenses exceed his income?

Solution

a. Since the grant increases with a fixed amount each year, the problem is infinite arithmetic series and the sum is given as

$$4800+(4800+d)+(4800+2d)+...$$

This is an arithmetic series with first term is $A_1=4800$ and the common difference d. So, the sum of the arithmetic series is given by

$$S_n=\frac{n}{2}[2A_1+(n-1)d]$$

Therefore, the sum over 20 years is given by

$$S_{20}=\frac{20}{2}[2(4800)+(20-1)d]=143,500$$
$$143,500=10[9600+19d]$$
$$14,350=9600+19d$$
$$19d=4750$$
$$d=250.$$

Therefore, the annual increase if he received a total of 143,500 ETB over 20 years is 250 ETB.

b. The structure of the series is given by

$$2600+3000+3400+...$$

This is an arithmetic series whose first term $A_1=2600$ and common difference $d=400.$

So, the sum of the arithmetic series is given by

$$S_n = \frac{n}{2}[2A_1 + (n-1)d]$$

$$S_{expenses} = \frac{n}{2}[2(2600) + (n-1)(400)]$$

$$= \frac{n}{2}[5200 + 400n - 400] = \frac{n}{2}[4800 + 400n]$$

$$S_{income} = \frac{n}{2}[2(4800) + (n-1)(250)]$$

$$= \frac{n}{2}[9600 + 250n - 250] = \frac{n}{2}[9350 + 250n]$$

Let $S_{expenses} = S_{income}$

$$\frac{n}{2}[4800 + 400n] = \frac{n}{2}[9350 + 250n]$$

$$4800 + 400n = 9350 + 250n$$

$$150n = 4550$$

$$n = \frac{4550}{150} = 30.333...$$

Therefore, his expenses will exceed his income after 30 years.

Exercise 1.24

1. A job applicant finds that a firm A offers a starting salary of Birr 31,100 with a guaranteed raise of Birr 1,200 per year, whereas firm B offers a higher starting salary of Birr 35,100 but will guarantee a yearly raise of only Birr 900.
   a. What would the annual salary be in the 11^{th} year at firm A?
   b. What would the annual salary be in the 11^{th} year at firm B?
   c. Over the first 11 years, how much would be earned at firm A?
   d. Over the first 11 years, how much would be earned at firm B?
   e. Compare the amount earned in 11 years in firms A and B.
2. A contest offers a total of 18 prizes. The first prize is worth Birr 10,000, and each consecutive prize is worth Birr 500 less than the next higher prize. Find the value of the eighteenth prize and the total value of the prizes.

3. A contest offers 10 prizes with a total value of Birr 13,250. If the difference in value between consecutive prizes is Birr 250, what is the value of the first prize?

Example 7

A flower 110 cm high is planted in a garden. At the end of the first year, the flower is 120 cm tall. Thereafter the growth of the flower each year is half of its growth in the previous year. Show that the height of the flower will never exceed 130 cm. Draw a graph of the relationship between time and growth.

Solution

The annual growth of the flower: $10, 5, \frac{5}{2}, \frac{5}{4}, \ldots$ and the sum of the growth:

$$10 + 5 + \frac{5}{2} + \frac{5}{4} + \ldots$$

This is a geometric series with the first term $G_1 = 10$ and common ratio $r = \frac{1}{2}$.

$$S_\infty = \frac{G_1}{1-r} = \frac{10}{1-\frac{1}{2}} = 20.$$

Note: we may join the points on the graph because the growth is continuous.

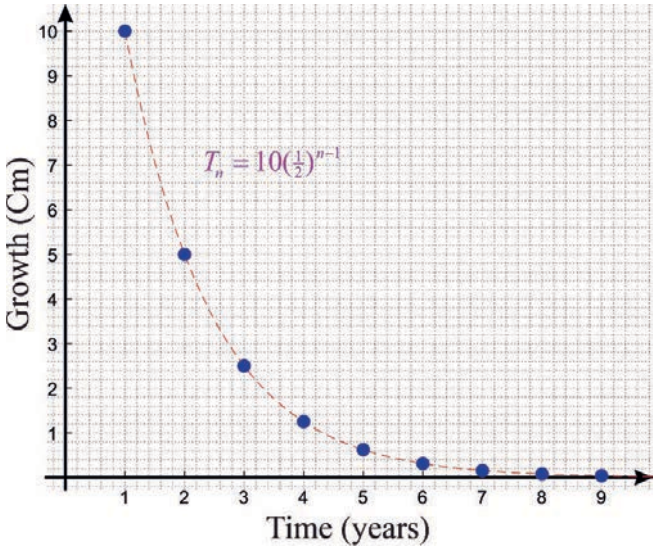


Figure 1.6

Therefore the growth of the flower is limited to 20 cm, and the maximum height of the shrub is therefore 110 cm + 20 cm = 130 cm.

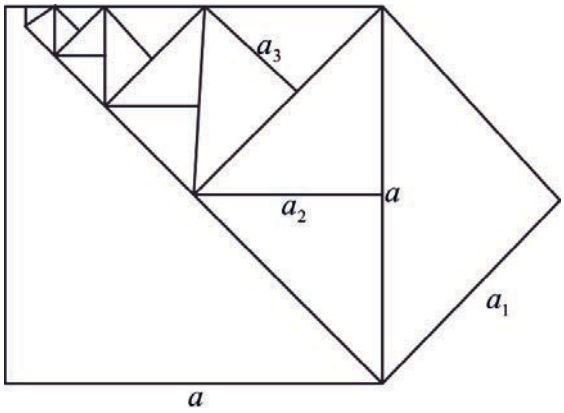
Example 8

Given a square with side length a. The side of the second square is half of its diagonal. The side of the third square is half of the diagonal of the second square and so on as shown in the figure below. Find the sum of the areas of all these squares.

Solution

Let the side of square be a_n and diagonal d_n.

Then,



$$a_0 = a$$

$$a_1 = \frac{1}{2}d_0 = \frac{1}{2}\sqrt{2}a = \frac{\sqrt{2}a}{2}$$

$$a_2 = \frac{1}{2}d_1 = \frac{1}{2}\sqrt{2}a_1 = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2}a}{2}$$

$$a_3 = \frac{1}{2}d_2 = \frac{1}{2}\sqrt{2}a_2 = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{2} \times \frac{\sqrt{2}a}{2}$$

.....

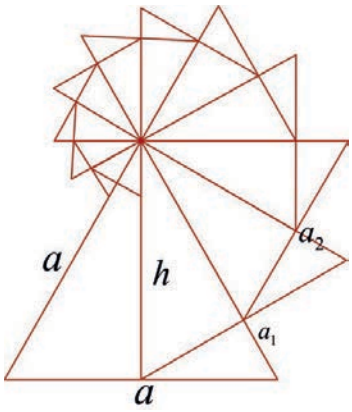
$$S_n = a_0^2 + a_1^2 + a_2^2 + a_3^2 + \ldots$$

$$= a^2 + \frac{1}{2}a^2 + \frac{1}{4}a^2 + \frac{1}{8}a^2 + \ldots$$

Since, $r = \frac{G_n}{G_{n-1}} = \frac{1}{2}$, $|r| < 1$, then $S_\infty = \frac{G_1}{1-r} = \frac{a^2}{1-\frac{1}{2}} = 2a^2$.

Exercise 1.25

1. Suppose a ball is dropped from a height of h m and always rebounds to r % of the height from which it falls. Show that the total vertical distance that could be covered by the ball is $h\left(\frac{r+1}{1-r}\right)$ m. Assume that the ball will never stop bouncing.



2. Given an equilateral triangle with side length a, its height is the side of another equilateral triangle. The height of this triangle is then the side of the third equilateral triangle and so on, as shown in the diagram. Find the sum of the areas of all these triangles.

Summary

1. **Sequence**

   - A sequence $\{a_n\}$ is a function whose domain is the set of positive integers or a subset of consecutive integers starting with 1.
   - The sequence $\{a_1, a_2, a_3, ...\}$ is denoted by $\{a_n\}$ or $\{a_n\}_{n-1}^{\infty}$.
   - A sequence that has a last term is called a **finite sequence**. Otherwise it is called **infinite sequence**.
   - **Recursion formula** is a formula that relates the general term a_n of a sequence to one or more of the terms that come before it.

2. **Arithmetic and geometric progression**

   - **An arithmetic sequence** is one in which the difference between consecutive terms is a constant, and this constant is called the common difference.
   - If $\{A_n\}$ is an arithmetic progression with the first term A_1 and the common difference d, then the nth term is given by:

$$A_n = A_1 + (n-1)d.$$

   - **A geometric progression** is one in which the ratio between consecutive terms is a constant, and this constant is called the common ratio.
   - If $\{G_n\}$ is a geometric progression with the first term G_n and a common ratio r, then the nth term is given by: $G_n = G_1 r^{n-1}$.

3. **Partial sums:**

   - The sum of the first n terms of the sequence $\{a_n\}_{n=1}^{\infty}$, denoted by S_n is called the partial sum of the sequence.
   - The sum S_n of the first n terms of an arithmetic sequence with first term A_1, and common difference d is:

$$S_n = \sum_{k=1}^{n} A_k = \frac{n}{2}[2A_1 + (n-1)d] \text{ or } S_n = \sum_{k=1}^{n} A_k = \frac{n}{2}[A_1 + A_n].$$

- In a geometric sequence, $\{G_n\}_{n=1}^{\infty}$, with common ratio r, the sum of the first n terms S_n is given by: $S_n = \begin{cases} nG_1, & \text{if } r = 1 \\ \dfrac{G_1(1-r^n)}{1-r}, & \text{if } r \neq 1 \end{cases}$

4. **Convergent series and divergent series**

- In a sequence if $\{a_n\}_{n=1}^{\infty}$, S_n is the nth partial sum such that, as $n \to \infty$, $S_n \to s$ where s is a real number, we say the infinite series $\sum_{n=1}^{\infty} a_n$ converges to s, otherwise the series diverges.

Review Exercise

1. List the first five terms of each of the sequences whose general terms are given below where n is a positive integer:

   a. $a_n = \dfrac{n^n}{n!}$ b. $a_n = (-1)^n + (-1)^n \sin(n\pi)$

   c. $a_n = Sgn(3-n)$

2. List the first five terms of each of the sequences whose general terms are given below where n is a positive integer:

   a. $a_n = 3^{\frac{1}{n}}(-1)^n$ b. $a_n = ne^{-2n}$ c. $a_n = \dfrac{(-2)^n + 6}{(n-1)!}$

   d. $a_n = (-1)^n - \dfrac{1}{n^2}$ e. $a_n = \cos\left(\dfrac{n\pi}{2}\right)$

3. Determine whether the sequences with the following general terms are arithmetic.

   a. $a_n = \dfrac{n^2 + 5n + 6}{n+2}$ b. $a_n = \sqrt{4n+1}$

4. Find the arithmetic mean between 1 and 9.

5. Find the 10^{th} and n^{th} terms of the geometric progression 5, 25, 125,…

6. Find three numbers in geometric progression such that their sum is $\dfrac{13}{3}$ and the sum of their squares is $\dfrac{91}{3}$.

7. Find the sum of

   a. $\sum_{k=1}^{8}(k^3 + 2k^2 - 3k + 5)$ b. $\sum_{k=2}^{5} 3$ c. $\sum_{k=1}^{5} \dfrac{1}{k^2 + 5k + 6}$

   d. $\sum_{m=1}^{10} \ln\left(\dfrac{m}{m+1}\right)$

8. Let $\sum_{i=1}^{5} x_i = 37$, $\sum_{i=1}^{5} y_i = 127$, $\sum_{i=1}^{5} x_i^2 = 303$, $\sum_{i=1}^{5} y_i^2 = 50$ and $\sum_{i=1}^{5} x_i y_i = 105$.

Evaluate the following

a. $\sum_{i=1}^{5}(2x_i - 3y_i)$ b. $\sum_{i=1}^{5}(2x_i + 3y_i)$ c. $\sum_{i=1}^{5}(2x_i - 3y_i)^2$

d. $\left(\sum_{i=1}^{5} x_i\right)^2$ e. $\sum_{i=1}^{5}(2x_i - 5y_i + 3)$

9. Find the sum of all the natural numbers between 100 and 1000 which are multiples of 5.

10. In an arithmetic progression the 1st term is 2 and the sum of the 1st five terms is one-quarter of the next five terms. Show that the 20^{th} term is –112.

11. How many terms of the arithmetic progression $-6, -\frac{11}{2}, -5, \ldots$ are needed to give the sum –25.

12. The sum of a certain number of terms of the arithmetic progression 25, 22, 19, … is 116. Find the last term.

13. If the sum of n terms of an arithmetic progression is $(pn + qn^2)$, where p and q are constants, find the common difference.

14. If the n^{th} partial sum of an arithmetic sequence $\{A_n\}$ is $4n^2$, find A_n and A_{10}.

15. Convert this mixed recurring decimal $0.3\overline{17}$ to fraction.

16. Convert the mixed recurring decimal to fraction.

a. $0.3\overline{7}$ b. $3.23\overline{54}$

17. The first three terms of a convergent geometric series are: $x+1; x-1; 2x-5$.

a. Find the values of $x (x \neq 1 \text{ or } -1)$.

b. Find sum to infinity of the series.

18. Find $p: \sum_{k=1}^{\infty} 27p^k = \sum_{x=1}^{12}(24 - 3x)$.

19. Find the product $4 \times 4^{\frac{1}{2}} \times 4^{\frac{1}{4}} \times 4^{\frac{1}{8}} \times \ldots \times 4^{\frac{1}{2^n}} \times \ldots$

20. If $\sum_{k=1}^{\infty} 5^{kt} = \frac{1}{24}$, find the values of t.

21. If the product $5^k \cdot 5^{k^2} \cdot 5^{k^3} \ldots = 5$, find k.

22. Evaluate $\sum_{k=1}^{11} (2 + 3^k)$.

23. The sum of the first three terms of a geometric progression is $\frac{39}{10}$ and their product is 1. Find the common ratio and the three terms.

24. How many terms of geometric progression $3, 3^2, 3^3, \ldots$ are needed to give the sum 120?

25. The sum of first three terms of a geometric progression is 16 and the sum of the next three terms is 128. Determine the first term, the common ratio and the sum to n terms of the geometric progression.

26. Given a geometric progression with $a = 729$ and 7^{th} term 64, determine S_7.

27. If the 4^{th}, 10^{th} and 16^{th} terms of a geometric progression are x, y and z, respectively. Prove that x, y, z are in geometric progression.

28. Find the sum to n terms of the sequence 8, 88, 888, 8888....

29. Show that the products of the corresponding terms of the sequences $a, ar, ar^2, \ldots ar^{n-1}$ and $A, AR, AR^2, \ldots AR^{n-1}$ form a geometric progression, and find the common ratio.

30. If the p^{th}, q^{th} and r^{th} terms of a geometric progression are a, b and c, respectively. Prove that $a^{q-r} b^{r-p} c^{p-q} = 1$

31. If the first and the n^{th} term of a geometric progression are a and b, respectively, and if P is the product of n terms, prove that $P^2 = (ab)^n$

32. If a, b, c and d are in geometric progression. Show that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (ab + bc + cd)^2$

33. Insert two numbers between 3 and 81 so that the resulting sequence is geometric progression.

34. The sum of two numbers is 6 times their geometric mean, show that numbers are in the ratio $(3+2\sqrt{2}):(3-2\sqrt{2})$.

35. If A and G are an arithmetic mean and geometric mean, respectively between two positive numbers,

    prove that the numbers are $A \pm \sqrt{(A+G)(A-G)}$.

36. 150 workers were hired to finish a job in a certain number of days. 4 workers dropped out on second day, four more workers dropped out on third day and so on. It took eight more days to finish the work. Find the number of days in which the work was completed.

UNIT 2

INTRODUCTION TO CALCULUS

Unit Outcomes

By the end of this unit, you will be able to:

- Deduce rate of change of different quantities.
- Calculate rate of change of different quantities.
- Understand gradient of functions at a point.
- Analyze the geometrical and mechanical meaning of derivative.
- Find the derivative of simple functions using gradient method.
- Find area of a region bounded by a function and x-axis on a given interval.
- Understand definite integral.
- Value the real world contributions of having skills of derivatives and integrations.
- Apply the knowledge of integral calculus to solve real life mathematical problems.
- Apply the concept of derivatives to solve basic problems in business, economics, and road taxation.

Unit Contents

2.1 Introduction to Derivatives
2.2 Applications of Derivative
2.3 Introduction to Integration
Summary
Review Exercise

- Absolute maximum
- Partial sums
- Extreme values
- Differentiation
- Absolute minimum
- Rate of change
- Anti-derivative
- Area
- Critical number
- Definite integral
- Derivative
- First derivative test
- Fundamental theorem
- Gradient
- Indefinite integral
- Partial fraction
- Relative maximum
- Relative minimum
- Secant
- Tangent
- Volume

Introduction

In our daily life, we come across things that change according to some well recognizable rules. Many physical phenomena involve changing quantities such as the speed of a rocket, the inflation of currency, the number of bacteria in a culture, the shock intensity of an earthquake, the AC voltage of an electric signal, and so on.

Calculus is one of the components of mathematics that is concerned with change and motion; it deals with quantities that approach other quantities. Thus, the knowledge of calculus is essential, especially when dealing with quantities that change at a variable rate; you need to understand the concept of a derivative, which is the mathematical tool that is used to study the rates at which physical quantities change. The derivative is the exact rate at which one quantity changes with respect to another. Geometrically, the derivative is the slope of a curve at a point on the curve, defined as the slope of the tangent to the curve at the same point. The process of finding the derivative is called differentiation. This process is central to the branch of

mathematics called differential calculus. In this unit, you are going to study the meaning and techniques of differentiation, applications of derivatives, and integration.

Historical Notes

Calculus, known in its early history as infinitesimal calculus, is a mathematical discipline focused on limits, continuity, derivatives, integrals, and infinite series. Isaac Newton and Gottfried Wilhelm Leibniz independently developed the theory of infinitesimal calculus in the late 17^{th} century. Newton was only 22 at the time.

2.1 Introduction to Derivatives

2.1.1 Understanding Rates of Change

Rate of change

Activity 2.1

1. Consider a square of length of side 4 cm.
   a. If the length of the side of the square increases by 1 cm, find the change in the perimeter of the square.
   b. If the length of the side of the square increases by 1 cm/s find the perimeter of the square when t = 1s, 2s, 3s.
   c. What is the time rate of change of the perimeter of the square when its length of side increases by 2 cm/s?
2. What does a rate of change mean?

The fundamental philosophical truth is that everything changes. In physics, the change in position is known as displacement, in economics, the price change is known as inflation, in business, the change in costs is sometimes known as a trend, in mathematics, the change in values of a function is known as the derivative. But, to understand the derivatives, which measure instantaneous change, we need to understand the concepts of change and average change over intervals.

Definition 2.1

A rate of change is a rate that describes how one quantity changes in relation to another quantity. The units on a rate of change are output units per input units.

For Example: Cost per minute, plants per hectare, kilometers per liter of benzene, salary per month, population per square kilometer, etc.

Rates of change can be positive or negative or even zero. When a quantity does not change over time, it has zero rates of change.

1. **A positive rate of change** is the case when the values of the two quantities (say x and y) increases at the same time and the graph slopes upward (See Figure 2.1).

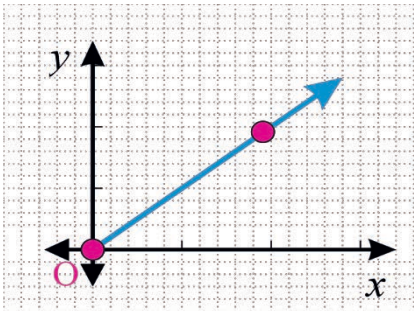


Figure 2.1

2. **Negative rate of change** is the case when the value of one quantity (say x) increases and the value of the other quantity (say y) decreases. The graph slopes downward (See Figure 2.2).

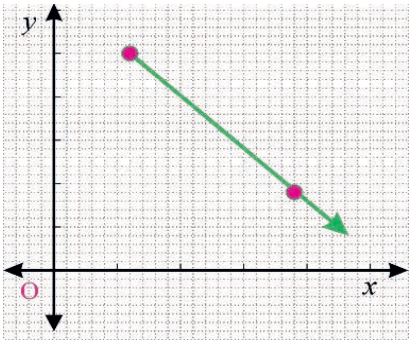


Figure 2.2

3. **Zero rate of change** is the case when the value of one quantity (say x) increases, the value of the other quantity (say y) remains constant. The graph is a horizontal line.

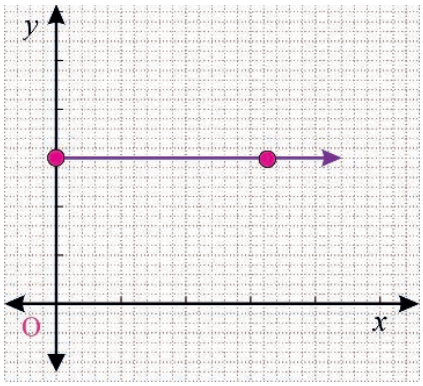


Figure 2.3

Example

Calculate the rate of change for the following pair of numbers.

a. $(1,2)$ and $(3,4)$ b. $(2,3)$ and $(3,1)$ c. $(0,2)$ and $(2,2)$ d. $(2,0)$ and $(2,2)$

Solution

a. $\text{Rate of change} = \dfrac{\text{change in } y}{\text{change in } x} = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{4-2}{3-1} = \dfrac{2}{2} = 1.$

Thus, the rate of change is positive.

b. $\text{Rate of change} = \dfrac{\text{change in } y}{\text{change in } x} = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{1-3}{3-2} = \dfrac{-2}{1} = -2.$

Thus, the rate of change is negative.

c. $\text{Rate of change} = \dfrac{\text{change in } y}{\text{change in } x} = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{2-2}{2-0} = \dfrac{0}{2} = 0.$

Thus, the rate of change is zero.

d. $\text{Rate of change} = \dfrac{\text{change in } y}{\text{change in } x} = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{2-0}{2-2} = \dfrac{2}{0}$ (undefined).

Thus, the rate of change is undefined.

Remark

A function is increasing where its rate of change is positive and decreasing where its rate of change is negative.

Exercise 2.1

Calculate the rate of change for the following pairs of numbers:

a. $(4,6)$ and $(5,9)$ b. $(4,7)$ and $(5,6)$ c. $(1,2)$ and $(3,-4)$

d. $(-2,4)$ and $(3,-6)$ e. $(0,9)$ and $(9,9)$ f. $(8,0)$ and $(8,8)$

g. $(3,-1)$ and $(1,-1)$

Note

The two kinds of rate of change are average rate of change and instantaneous rate of change.

Average rate of change

Definition 2.2

If $y = f(x)$ is a function, then the **average rate of change** of y with respect to x on the interval $[a,b]$ is given by the expression:

$$\text{Average rate of change} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x} = \frac{f(b)-f(a)}{b-a}.$$

Example 1

Find the average rate of change of a function $f(x) = 2x - 9$ beteween $x_1 = 3$ and $x_2 = 6$.

Solution

Step 1: Write the formula for the average rate of change over the interval $x_1 \le x \le x_2$

$$\frac{f(x_2)-f(x_1)}{x_2-x_1}.$$

Step 2: Solve for $f(x_1)$ and $f(x_2)$.

$$f(x_2)=f(6)=2(6)-9=12-9=3 \text{ and } f(x_1)=f(3)=2(3)-9=6-9=-3$$

Step 3: Substitute the known values into the formula and simplify.

$$\frac{f(x_2)-f(x_1)}{x_2-x_1}=\frac{f(6)-f(3)}{6-3}=\frac{3-(-3)}{3}=2.$$

Example 2

Examine the motion of a car that travels along a straight road and assume that we can measure the distance traveled by the car (in meter) at l-second intervals as in the following chart:

T*(s)*	0	1	2	3	4	5
S*(m)*	0	2	10	25	43	78

a. Find the average velocity during the time interval $2 \le t \le 4$.

b. Find the average velocity during the time interval $2 \le t \le 3$.

Solution

a. **Step 1:** Write the formula for the average rate of change over the interval $a \le x \le b$

$$\text{Average rate of change} = \frac{f(b)-f(a)}{b-a}$$

Step 2: Identify the two points corresponding to the interval

$$(a, f(a)) = (2,10) \text{ and } (b, f(b)) = (4,43)$$

Step 3: Substitute the values into the formula and simplify

$$\text{Average rate of change} = \frac{f(b)-f(a)}{b-a}=\frac{43-10}{4-2}=16.5\text{m/s}.$$

b. Similarly, Average velocity $= \dfrac{\text{distance travelled}}{\text{time elapsed}} = \dfrac{25-10}{3-2} = 15\text{m/s}.$

Example 3

The accompanying figure shows the graph of a function f. Find the average rate of change of the function f over the interval $0 \le x \le 9$:

Solution

Step 1: Write the formula for the average rate of change over the interval $0 \le x \le 9$

$\dfrac{f(x_2)-f(x_1)}{x_2-x_1}$, where $x_1 = 0$ and $x_2 = 9$.

Step 2: Observe from the graph the value of $f(0)$ and $f(9)$. We can see from the graph that $f(0) = -7$ and $f(9) = 3$.

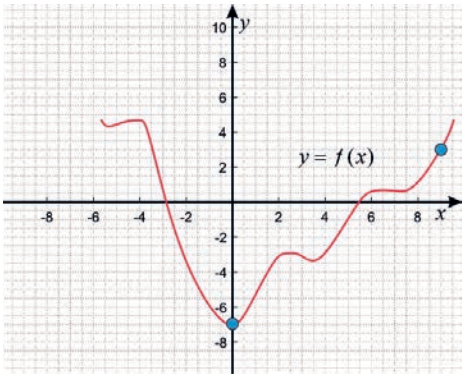


Figure 2.4

Step 3: Substitute the known values into the formula.

Average rate of change $= \dfrac{f(x_2)-f(x_1)}{x_2-x_1} = \dfrac{f(9)-f(0)}{9-0} = \dfrac{3-(-7)}{9} = \dfrac{10}{9}.$

Note: The average rate of change is a measure of how much the function changed per unit, on average, over the interval. Geometrically, the average rate of change of a function $y = f(x)$ on the interval $[a,b]$ is the slope of the **secant line** through $(a, f(a))$ and $(b, f(b))$ See Figure 2.5.

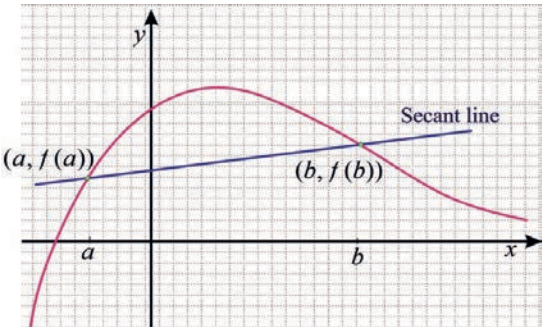


Figure 2.5

Exercise 2.2

1. Compute the average rate of change of
   a. $f(x)=3x-2$ over the interval $0\le x\le 2$
   b. $f(x)=x^3-3x$ over the interval $1\le x\le 6$
   c. $f(x)=3x-2$ on the interval $[1,5]$
   d. $f(x)=x^2+3x+1$ on the interval $[0,k]$
2. In the following table, the average cost in Birr of a gallon of gasoline for the years 2005–2013 is illustrated. Using the data in the table, find the average rate of change of the price of gasoline between 2007 and 2009.

x	2005	2006	2007	2008	2009	2010	2011	2012	2013
$c(x)$	2.31	2.62	2.84	3.30	2.41	2.84	3.58	3.68	4.21

3. A candle has a starting length of 10 cm. Thirty minutes after lighting it, the length became 7 cm. Determine the rate of change in the length of the candle as it burns. Determine how long the candle takes to completely burn out.

Instantaneous Rate of Change

For many of the functions such as polynomial, trigonometric, exponential, logarithmic, and rational functions, it is possible to find their instantaneous rate of change for values of x in their domain.

Activity 2.2

Let $y=x^3$.

i. Find the average rate of change of the given function on the following intervals.

a. $x=1$ and $x=1.1$ *b.* $x=1$ and $x=1.001$ *c.* $x=1$ and $x=1+\Delta x$

ii. Determine what happens as Δx approaches 0.

Definition 2.3

The instantaneous rate of change is the change at that particular moment.

Instantaneous rate of change $= \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$, where Δx gets closer and closer to zero.

Note

Average rate of change is the rate of change over the entire period. The **instantaneous rate** of change is the change at a certain period of time.

Therefore, when h gets closer and closer to zero, the instantaneous rate of change of a function f at $x = x_0$ is $\frac{\Delta y}{\Delta x} = \frac{f(x_0 + h) - f(x_0)}{h}$.

Example 1

Evaluate the instantaneous rate of change of $f(x) = 2x^2 + 9$ at $x = -3$.

Solution

The instantaneous rate of change of a function f at $x = x_0$ is

$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + h) - f(x_0)}{h}$, as h gets closer and closer to zero. For the given function $f(x) = 2x^2 + 9$ at $x = -3$, the instantaneous rate of change is

$$\text{Instantaneous rate of change} = \frac{f(-3+h) - f(-3)}{h}$$

$$= \frac{2(-3+h)^2 + 9 - (2(-3)^2 + 9)}{h} = \frac{2(h^2 - 6h + 9) + 9 - (18 + 9)}{h}$$

$$= \frac{2h^2 - 12h + 18 + 9 - (18 + 9)}{h} = \frac{2h^2 - 12h}{h}$$

$= 2h - 12$, as h gets closer and closer to zero

$= -12$.

Thus, the instantaneous rate of change is -12.

Example 2

Find the instantaneous rate of change of $f(x)=\sqrt{x}$ at $x=x_0>0$.

Solution

Recall that the instantaneous rate of change of a function $f(x)$ when $x=x_0$ is

$$\text{Instantaneous rate of change} = \frac{f(x_0+h)-f(x_0)}{h}$$

$$= \frac{\sqrt{x_0+h}-\sqrt{x_0}}{h}, \text{ rationalizing the numerator}$$

$$= \frac{\sqrt{x_0+h}-\sqrt{x_0}}{h}\left(\frac{\sqrt{x_0+h}+\sqrt{x_0}}{\sqrt{x_0+h}+\sqrt{x_0}}\right)$$

$$= \frac{h}{h(\sqrt{x_0+h}+\sqrt{x_0})}$$

$$= \frac{1}{\sqrt{x_0+h}+\sqrt{x_0}}, \text{ as } h \text{ gets closer and closer to zero}$$

$$= \frac{1}{2\sqrt{x_0}}.$$

Instantaneous rates of change have many real-world applications; for example, the velocity of a moving body at a particular time is the instantaneous rate of change of the displacement at that time. In the final example, we will consider a real-world problem and determine the instantaneous rate of change at a particular time for a polynomial function representing the biomass of a bacterial culture.

Example 3

The biomass of a bacterial culture in milligrams as a function of time in minutes is given by $f(t)=71t^3+63$. What is the instantaneous rate of growth of the culture when $t=2$ minutes?

Solution

In this example, we want to determine the instantaneous rate of change of a cubic function representing the biomass of a bacterial culture.

Recall that the instantaneous rate of change of a function $f(t)$ when $t = t_0$ is

$$\begin{aligned}\text{Instantineous rate of change} &= \frac{f(2+h)-f(2)}{h}\\ &= \frac{71(2+h)^3+63-(71\times 2^3+63)}{h}\\ &= \frac{71(h^3+6h^2+12h+8)-71\times 8}{h}\\ &= \frac{71h^3+426h^2+852h}{h}\\ &= 71h^2+426h+852, \text{ as } h \text{ gets closer and closer to zero}\\ &= 852.\end{aligned}$$

Since the rate of change is positive, it is equivalent to the rate of growth. Hence, the rate of growth of the biomass of a bacterial culture when $t = 2$ minutes is 852 mg/min.

The instantaneous rate of change is also related to the derivative, at an arbitrary point x. In particular, the instantaneous rate of change at $x = x_0$ of a function is the derivative of a function evaluated at $x = x_0$.

Exercise 2.3

1. Evaluate the instantaneous rate of change of $f(x) = 6x^2 - 3$ at $x = 1$.
2. Find the instantaneous rate of change of $f(x) = 2\sqrt{x}$ at $x = x_0 > 0$.
3. The biomass of a bacterial culture in milligrams as a function of time in minutes is given by $f(t) = 81t^3 + 90$. What is the instantaneous rate of growth of the culture when $t = 3$ minutes?

2.1.2 Gradient of Curves and Rate of Changes

Activity 2.3

1. What is a secant line to a curve? What is a tangent line to a curve?
2. What is slope of a line? How can we calculate the slope of a line?

Definition 2.4

a. **A tangent line** to a curve or a graph of a function $y = f(x)$ is a line that touches the curve exactly at one point, but does not cross the curve. The point where the tangent line touches the graph is said to be the **point of tangency.**

b. **Gradient (slope)** is a number that describes steepness and direction of a line. If slope is positive, then the line is increasing, which means it goes up as we move from left to right. If slope is negative then the line is decreasing, which means it goes down when we move from left to right. If the slope is zero, then the line is horizontal.

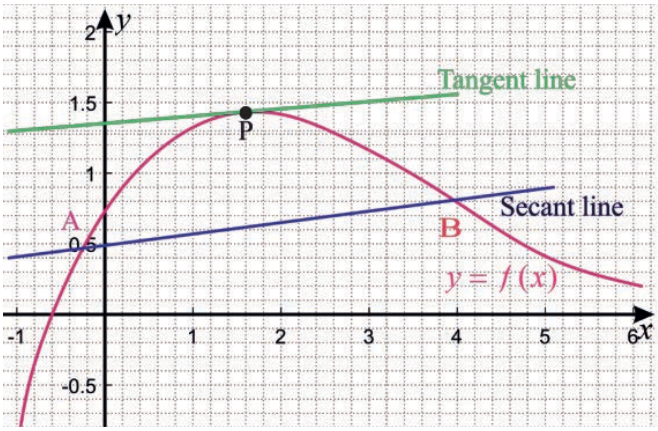


Figure 2.6

Note: If we are given two points $A(x_0, y_0)$ and $B(x_1, y_1)$ on the graph of $y = f(x)$, then using definition 2.2 and understanding that 'change' is the difference between

the two values we get the general formula for the slope of a secant line through the two points, which is given as:

$$\text{Slope} = \frac{\text{vertical change}}{\text{horizontal change}} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x} = \frac{y_1 - y_0}{x_1 - x_0}.$$

From this formula, we can see that the gradient of the secant line to the curve of a function is equal to the average rate of change.

The slope of a secant line = average rate of change.

Example 1

Find the slope of a line passing through the given two points.

a. $(-5, 7)$ and $(0, 0)$ **b.** $(3, -5)$ and $(-2, 9)$

Solution

a. $\text{Slope} = \dfrac{\Delta y}{\Delta x} = \dfrac{y_1 - y_0}{x_1 - x_0} = \dfrac{0-(7)}{0-(-5)} = \dfrac{-7}{5} = -\dfrac{7}{5}.$

b. $\text{Slope} = \dfrac{\Delta y}{\Delta x} = \dfrac{y_1 - y_0}{x_1 - x_0} = \dfrac{(9-(-5))}{(-2-3)} = \dfrac{14}{-5} = -\dfrac{14}{5}.$

Example 2

Find the slope of a secant line to the graph of the function $f(x) = 2x^3 + 3x - 5$ over the interval [1, 3].

Solution

Given $f(x) = 2x^3 + 3x - 5$

Step 1: Write the formula for the slope of a secant line over the interval $1 \le x \le 3$.

$$\text{Average rate of change} = \frac{f(3) - f(1)}{3 - 1}.$$

Step 2: Solve for $f(3)$ and $f(1)$.

$$f(3)=2(3)^3+3(3)-5=58 \text{ and } f(1)=2(1)^3+3(1)-5=0$$

Step 3: Substitute the known values into the formula and simplify.

$$\text{Average rate of change}=\frac{f(3)-f(1)}{3-1}=\frac{58-0}{2}=29.$$

Therefore, the slope of a secant line to the graph of $f(x)=2x^3+3x-5$ with respect to x over the interval $[1,3]$ is 29.

Exercise 2.4

1. Find the slope of a line passing through the two given points.
   a. $(-3,-5)$ and $(3,11)$ b. $(4,-9)$ and $(5,-7)$
2. For each of the following functions, find the slope of a secant line to the graph of the functions over the given intervals.
   a. $f(x)=x^2+1;[-1,3]$ b. $g(x)=x^3-4x+6;\ [1,2]$.

2.1.3 Gradient at a Point on a Curve

Activity 2.4

1. Let $f(x)=x^2$. Compute and simplify the algebraic expression
   a. $\dfrac{f(x+h)-f(x)}{h}$ b. $\dfrac{f(x)-f(x_0)}{x-x_0}$
2. Let $y=x^3$.
   i. Find the slope of the secant line to the curve of the function between
      a. $x=1$ and $x=1.1$ b. $x=1$ and $x=1.001$ c. $x=1$ and $x=1+h$.
   ii. Determine what happens as h approaches 0.

Suppose you are given some curve, and a point P on the curve as shown below in figures 2.7 a and b.

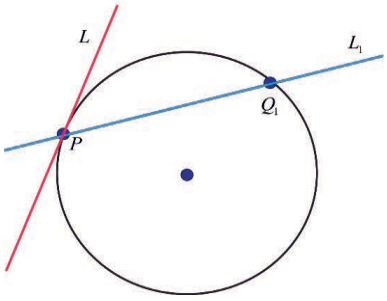


Figure 2.7a

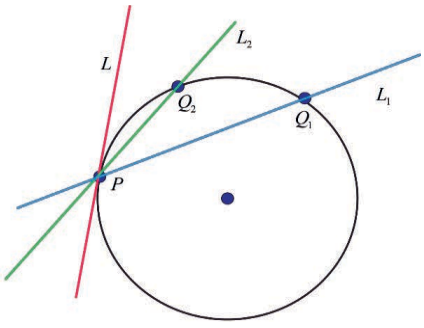


Figure 2.7b

- The line L passing through the point P is called a tangent line.
- Given another point Q_1 on the curve, let L_1 be the line passing through the points P and Q_1.
- L_1 is a secant line and is sometimes written as PQ_1.
- Pick a point Q_2 closer to P. Notice that the slope of the secant line L_2 is closer to the slope of L compared to the slope of L_1.
- As we pick points closer and closer to P, the slope of the associated secant lines will get closer and closer to the slope of the tangent line L.
- Let P (a, f (a)) be any point on the graph of a function f. Another point on the graph may be denoted by $Q\ (a + h, f\ (a + h))$, where h is the difference between the x-coordinates of Q and P (see Figure 2.8 a). By definition the slope m of the secant line through P and Q is

$$m_{PQ} = m_{\text{secant}} = \frac{f(a+h) - f(a)}{h}.$$

Note

m_{PQ} denotes the slope of the secant line through P and Q and m_{secant} represents the slope of the secant line to the graph of $f(x)$.

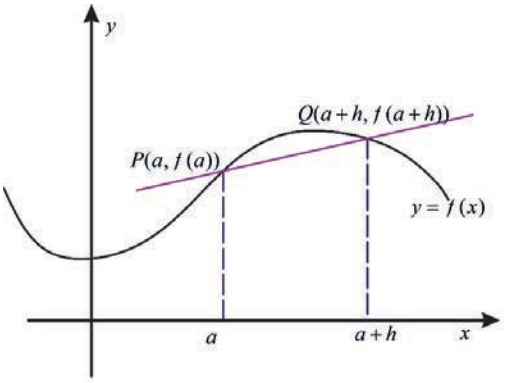


Figure 2.8 *a*

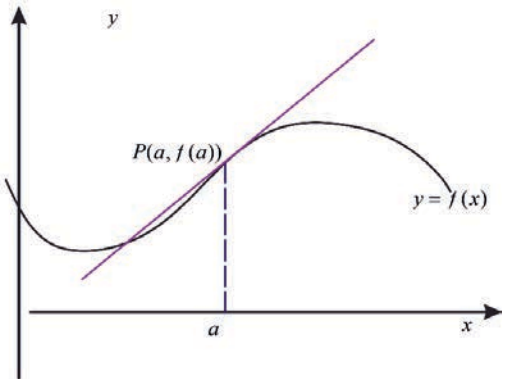


Figure 2.8 *b*

If f is a given function, then we can make Q approach P by letting h approach 0 (see figure 2.8*b*) and we get the slope of the tangent line to the curve. It is natural, therefore, to define m as follows:

Definition 2.5

If f is a function defined on an open interval containing a, then **as h gets closer and closer to zero**, the slope m of the tangent line to the graph of $y = f(x)$ at the point $P(a, f(a))$ is given by: $m = \dfrac{f(a+h) - f(a)}{h}$.

Note

1. If we allow the quantity h to get smaller and smaller, the point $(a+h, f(a+h))$ is getting closer to $(a, f(a))$, which in turn means that the slope of the secant line through $(a+h, f(a+h))$ is getting closer to the slope of the tangent line at $(a, f(a))$.

   Hence, the slope of the graph of a function at a point is the same as the slope of the tangent line at that point.
2. The slope of a tangent line represents the **instantaneous rate of change** of the function at that one point. That is:

   The slope of a tangent line = instantaneous rate of change.

Procedure to find gradient of a curve at a point

To find the gradient of a curve $y = f(x)$ at $x = a$ use the following procedure:

Step 1: Write the quotient difference $\frac{f(x)-f(a)}{x-a}$ of f at $x = a$.

Step 2: Compute the algebraic expression in step1 and simplify.

Step 3: Find the value of the result obtained in step 2 as the value of x approaches a.

The final result is the gradient of the tangent line to the curve $y = f(x)$ at $x=a$.

Example 1

Find the gradient (slope) of the functions $f(x) = x^2$ at $x = 2$.

Solution

$$\text{Slope} = \frac{f(x)-f(2)}{x-2} = \frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x+2.$$

Therefore, the slope of the function at $x=2$ becomes

$$x+2\big|_{x=2} = 2+2 = 4$$

Note: $x+2\big|_{x=2}$ means value obtained by substituting $x = 2$ to $x+2$.

Example 2

If $f(x) = x^2$ find the gradient of the tangent line to the graph of $f(x)$ at the point $P(a, a^2)$.

Solution

Given $y = f(x) = x^2$, using the above procedure, we have

i. $$\frac{f(x)-f(a)}{x-a} = \frac{x^2-a^2}{x-a} = \frac{(x-a)(x+a)}{x-a} = x+a$$

ii. As x approaches to a, the expression or the quantity $x+a$ approaches $2a$.

Therefore, the gradient of the tangent line to the curve $y = f(x) = x^2$ at the point $P(a, a^2)$ is $m_{\tan} = 2a$, where $m_{\tan}$ denotes the slope of the tangent line to the graph of the function at the specified point of tangency.

This means that the instantaneous rate of change of the function at the point P is $2a$, and that the gradient of the tangent to the graph of the curve $f(x) = x^2$ at any point is found by doubling the x-coordinate. For example, the gradient of the function $f(x) = x^2$ at the point $P(4,16)$ is equal to $m_{\tan} = 2(2) = 8$, at $P(-5,25)$ is $m_{\tan} = -10$, at the origin $P(0,0)$ is $m_{\tan} = 0$. See Figure 2.9.

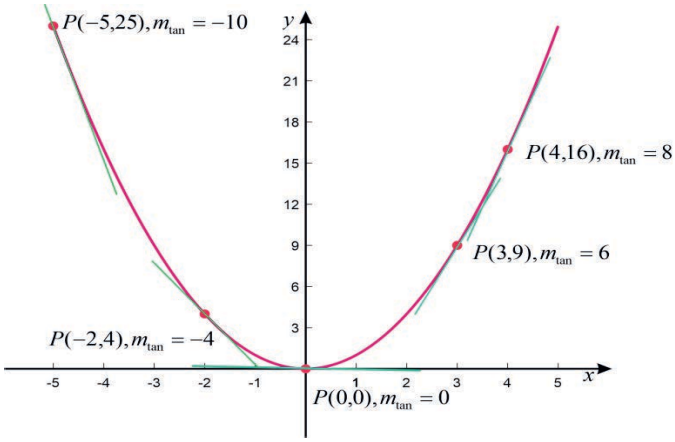


Figure 2.9

To clarify this idea condider the following table:

x	-5	-2	0	3	4
$f(x) = x^2$	25	4	0	9	16
Gradient (m)	-10	-4	0	6	8

Exercise 2.5

1. Find the gradients of the tangent lines to the graph of the following function at specified points.

   a. $f(x) = x^2$ at $x = -1$ and $x = 4$.

   b. $f(x) = 3x^2 - 5x + 4$ at $x = 2$ and $x = -\sqrt{2}$.

   c. $g(x) = \sqrt{x}$ and $x = 4$.

   d. $f(x) = \frac{1}{x}$ at the point $(2, \frac{1}{2})$.

2. If $f(x)=x^3$, find the gradient of the tangent line to the graph of $f(x)$ at the point $P(a,a^3)$.

2.1.4 Definition of Derivative

The slope concept usually pertains to straight lines. When a function is non-linear, its slope may vary from one point to the next. We must therefore introduce the notion of derivative which allows us to obtain the slope at all points of these non-linear functions.

2.1.4.1 Derivative of function at a point

Activity 2.5

Let $f(x)=\frac{1}{x^2}$. Then

a. Simplify $\frac{f(x)-f(1)}{x-1}$.

b. Simplify $\frac{f(x+h)-f(x)}{h}$.

Definition 2.6 (Derivatives of functions at a point)

The derivative of a function $f(x)$ at a number "a" in the domain of f, denoted by $f'(a)$, is the gradient of the tangent line to the graph of $y=f(x)$ at $(a,f(a))$. That is, as h gets closer and closer to zero from both directions, $\frac{f(a+h)-f(a)}{h}$ becomes closer to $f'(a)$.

Note

- $f'(a)$ is the slope of the line tangent to the graph of f at the point $(a,f(a))$.
- If $y=f'(x)$ is defined, then we say that f has a derivative at a or f is differentiable at a.

Example

Use definition 2.6, to find the derivatives of the following functions at the given value of a.

a. $f(x)=5x-3;\ a=-2$ b. $f(x)=x^2+3;\ a=5$ c. $f(x)=\sqrt{2x+4};\ a=6$

Solution

Using the definition, as h gets closer and closer to zero $f'(a)=\dfrac{f(a+h)-f(a)}{h}$, and we have the following results:

a. $f'(-2)=\dfrac{f(-2+h)-f(-2)}{h}$

$=\dfrac{[5(-2+h)-3]-(-2\times5-3)}{h}$

$=\dfrac{(-10+5h-3)-(-13)}{h}=\dfrac{5h}{h}=5.$

b. $f'(5)=\dfrac{f(5+h)-f(5)}{h}$

$=\dfrac{[(5+h)^2+3]-(5^2+3)}{h},$

$=\dfrac{25+10h+h^2+3-(25+3)}{h}$

$=\dfrac{h^2+10h}{h}$

$=h+10=10$ as h approaches to 0

c. $f'(6)=\dfrac{f(6+h)-f(6)}{h}=\dfrac{\sqrt{2(6+h)+4}-\sqrt{2(6)+4}}{h}$

$=\left(\dfrac{\sqrt{2h+16}-\sqrt{16}}{h}\right)\left(\dfrac{\sqrt{2h+16}+\sqrt{16}}{\sqrt{2h+16}+\sqrt{16}}\right)$, rationalization

$=\dfrac{2h+16-16}{h\left(\sqrt{2h+16}+\sqrt{16}\right)}$, simplification

$$= \frac{2}{\sqrt{2h+16}+\sqrt{16}} \quad \text{as } h \text{ approaches to } 0,$$

$$= \frac{2}{\sqrt{16}+\sqrt{16}} = \frac{1}{4}.$$

Exercise 2.6

Use definition 2.6, to find the derivatives of the following functions at the given value.

a. $f(x) = 9x - 7, \quad a = 2$

b. $f(x) = x^2 - 1, \quad a = 2$

c. $f(x) = x^3 + 2x, \quad a = 3$

d. $f(x) = \sqrt{3x-6}, \quad a = 3$

2.1.4.2 The derivative as a function

Activity 2.6

For each of the following functions, find the set of values of x_0 such that $f'(x_0)$ is defined and is unique.

a. $f(x) = x^2$

b. $f(x) = |x|$

c. $f(x) = \frac{1}{x}$

From Activity 2.6, you have observed that there are functions that are differentiable at all numbers in their domains and there are also functions that are not differentiable at some numbers in their domain. We are now ready to define derivatives.

Definition 2.7

The function f' whose domain consists of those values of x at which f is differentiable and whose value at any such number as h gets closer and closer to zero is given by $f'(x) = \frac{f(x+h) - f(x)}{h}$

Here, we represent the derivative of a function by a prime symbol. For example, writing $f'(x)$ represents the derivative of the function f evaluated at a point x.

Similarly, writing $(5x^2+2x+3)'$ indicates we are carrying out the derivative of the function $5x^2+2x+3$. The prime symbol disappears as soon as the derivative has been calculated. If we let $t=x+h$, as h gets closer and closer to zero, t approaches x and the definition above can be re-written as

$$f'(x)=\frac{f(t)-f(x)}{t-x} \text{ as } t \text{ gets very close to } x.$$

Note

The derivative is the exact rate at which one quantity changes with respect to another.

The different notations for the derivative

The following are some of the other notations for derivatives. If $y=f(x)$, then $f'(x)$ is denoted by either of the following notations: $\frac{dy}{dx}, \frac{d}{dx}f(x), Df(x), D_x f$

Example

Find the derivative of the following functions.

a. $f(x)=k, k$ be any constant b. $f(x)=x^2$ c. $f(x)=\sqrt{x}$

Solution

Use the definition and let h get closer and closer to 0.

Now, from $f'(x)=\frac{f(x+h)-f(x)}{h}$, we obtain:

a. For $f(x)=k, k$ be any constant, the quotient $\frac{f(x+h)-f(x)}{h}=\frac{k-k}{h}=0.$

Thus, by definition 2.6 $f'(x)=0$ or $\frac{d}{dx}(k)=0.$

b. When $f(x)=x^2$, the quotient difference

$$\frac{f(x+h)-f(x)}{h}=\frac{(x+h)^2-x^2}{h}$$
$$=\frac{(x^2+2xh+h^2)-x^2}{h}$$
$$=\frac{2xh+h^2}{h}=2x+h.$$

As h gets closer and closer to 0, the quantity $2x+h$ gets closer to $2x$. Therefore, by definition 2.6

$$f'(x)=2x \quad \text{or} \quad \frac{d}{dx}(x^2)=2x.$$

c. For $f(x)=\sqrt{x}$ the quotient

$$\frac{f(x+h)-f(x)}{h}=\frac{\sqrt{x+h}-\sqrt{x}}{h}, \text{ using rationalization}$$
$$=\left(\frac{\sqrt{x+h}-\sqrt{x}}{h}\right)\left(\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}\right)$$
$$=\frac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})}=\frac{1}{(\sqrt{x+h}+\sqrt{x})}$$

As h tends to 0, this quantity approaches $\frac{1}{2\sqrt{x}}$.

Hence, by definition 2.7,

$$f'(x)=(\sqrt{x})'=\frac{1}{2\sqrt{x}} \quad \text{or} \quad \frac{d}{dx}(\sqrt{x})=\frac{1}{2\sqrt{x}}.$$

Exercise 2.7

Use definition 2.7, find the derivative of the function with respect to x.

a. $f(x)=x$ b. $f(x)=x^3$ c. $f(x)=\frac{1}{x}$ d. $f(x)=2\sqrt{x}$

2.1.4.3 Derivatives of various functions

i. Differentiation of power functions

Activity 2.7

Using definition 2.7, find the derivative of each of the following power functions.

1. a. $f(x)=x$ b. $f(x)=x^2$ c. $f(x)=x^{-2}$.
2. What is the derivative of $f(x)=x^n$?

Theorem 2.1

Let $f(x)=x^n$, where n is a positive integer. Then $f'(x)=nx^{n-1}$.

Proof

Let $f(x)=x^n$. The quotient difference

$$\frac{f(t)-f(x)}{t-x}=\frac{t^n-x^n}{t-x}$$

$$=\frac{(t-x)(t^{n-1}+xt^{n-2}+\ldots+tx^{n-2}+x^{n-1})}{t-x}$$

$$=t^{n-1}+xt^{n-2}+\ldots+tx^{n-2}+x^{n-1}, t\neq x$$

Thus, by definition of the derivatives as t gets very close to x

$$f'(x)=\underbrace{x^{n-1}+x^{n-1}+\ldots+x^{n-1}+x^{n-1}}_{\text{n-times}}=nx^{n-1}$$

Note: By long division, we have

$$t^n-x^n=(t-x)(t^{n-1}+xt^{n-2}+x^2t^{n-3}+\ldots+x^{n-2}t+x^{n-1}).$$

i.e.,

$$
\begin{array}{r|l}
t-x & t^{n-1}+xt^{n-2}+x^2t^{n-3}+\ldots+x^{n-2}t+x^{n-1} \\
\hline
 & t^n \qquad\qquad\qquad\qquad -x^n \\
 & \underline{-(t^n-xt^{n-1})} \\
 & \qquad xt^{n-1}-x^n \\
 & \qquad \underline{-(xt^{n-1}-x^2t^{n-2})} \\
 & \qquad\qquad x^2t^{n-2}-x^n \\
 & \qquad\qquad \underline{-(x^2t^{n-2}-x^3t^{n-3})} \\
 & \qquad\qquad\quad \vdots \\
 & \qquad\qquad x^{n-1}t-x^n \\
 & \qquad\qquad \underline{-(x^{n-1}t-x^n)} \\
 & \qquad\qquad\qquad 0
\end{array}
$$

Example 1

Find the derivatives of each of the following functions.

a. $f(x)=x^{15}$ b. $f(x)=x^{24}$

Solution

a. $f'(x)=15x^{15-1}=15x^{14}$ b $f'(x)=24x^{24-1}=24x^{23}$

Corollary 2.1

If $f(x)=x^{-n}$, where n is a positive integer, then $f'(x)=-nx^{-(n+1)}$.

Example 2

Find the derivatives of each of the following functions.

a. $f(x)=x^{-5}$ b. $f(x)=x^{-17}$

Solution

a. $f'(x)=-5x^{-5-1}=-5x^{-6}$ b. $f'(x)=-17x^{-17-1}=-17x^{-18}$

Example 3

Let $f(x)=x^{-6}$, find

a. $f'(1)$ b. $f'(\frac{1}{3})$ c. $f'(k),\ k\neq 0.$

Solution

$f'(x)=-6x^{-7} \Rightarrow a)\ f'(1)=-6(1)^{-7}=-6$

$b)\ f'(\frac{1}{3})=-6(\frac{1}{3})^{-7}=-6(3)^{7}=-13{,}122$

$c)\ f'(k)=-6(k)^{-7}=-6k^{-7}$

Corollary 2.2

Let $f(x)=kx^n$, then $f'(x)=knx^{n-1}$, where n is any non-zero integer and k is a constant number.

Example 4

Find the derivative of each of the following functions.

a. $f(x)=-2x^{11}$ b. $f(x)=4x^{20}$ c. $f(x)=-\frac{3}{x^7}$ d. $f(x)=\frac{1}{3x^{15}}$

Solution

Using corollary 2.2, we obtain

a. $f'(x)=-2(11)x^{10}=-22x^{10}$

b. $f'(x)=4(20)x^{19}=80x^{19}$

c. $f'(x)=-3(-7)x^{-8}=21x^{-8}=\frac{21}{x^8}$

d. $f'(x)=\frac{1}{3}(-15)x^{-16}=-\frac{5}{x^{16}}$

Exercise 2.8

1. Find the derivative of each of the following functions.
   a. $f(x)=x^8$ b. $f(x)=x^{27}$
   c. $f(x)=2x^6$ d. $f(x)=10x^{15}$
   e. $f(x)=4x^{-7}$ f. $f(x)=3x^{-20}$
   g. $f(x)=4x^{-\frac{3}{2}}$ h. $f(x)=-10x^{-\frac{2}{5}}$
2. Let $f(x)=x^{-9}$, find
   a. $f'(1)$ b. $f'\left(\frac{1}{3}\right)$ c. $f'(k)=k, k\neq 0.$
3. Based on Theorem 2.1, show the proof of corollary 2.1 and corollary 2.2.

ii. Derivatives of combinations of functions

a. Derivatives of a sum and difference of a functions

Activity 2.8

Given the functions $f(x)=x^2$, and $g(x)=x^3+2x$, then find

a. $f'(x)+g'(x)$ b. $(f+g)'(x)$ c. $f'(x)-g'(x)$ d. $(f-g)'(x)$

Theorem 2.2

If f and g are differentiable functions at a, then $f+g$ and $f-g$ are also differentiable at a and are given by $(f+g)'(a)=f'(a)+g'(a)$, and $(f-g)'(a)=f'(a)-g'(a)$

Proof

From the definition of derivative, we have:

$$\frac{(f+g)(x)-(f+g)(a)}{x-a} = \frac{f(x)+g(x)-f(a)-g(a)}{x-a}$$
$$= \frac{f(x)-f(a)+g(x)-g(a)}{x-a}$$
$$= \frac{f(x)-f(a)}{x-a} + \frac{g(x)-g(a)}{x-a} \qquad (i)$$

By definition of derivatives, as x gets close to a,

$$\frac{(f+g)(x)-(f+g)(a)}{x-a} = (f+g)'(a),$$

$$\frac{f(x)-f(a)}{x-a} = f'(a) \text{ and } \frac{g(x)-g(a)}{x-a} = g'(a) \qquad \text{(ii)}$$

From Eq. (i) and Eq. (ii), we obtain

$$(f+g)'(a) = f'(a) + g'(a).$$

This theorem, implies for all x at which both f and g are differentiable,

$$(f+g)'(x) = f'(x) + g'(x).$$

Corollary 2.3

If $f_1, f_2, f_3, \ldots, f_n$ are differentiable at a number a, then

$$(f_1+f_2+f_3+\ldots+f_n)'(a) = f_1'(a)+f_2'(a)+f_3'(a)+\ldots+f_n'(a) \text{ and}$$

$$(f_1-f_2-f_3-\ldots-f_n)'(a) = f_1'(a)-f_2'(a)-f_3'(a)-\ldots-f_n'(a)$$

Example 1

If $f(x) = x^3 + 3x^2$, find a formula for $f'(x)$ and compute $f'(2)$.

Solution

$$f'(x) = 3x^2 + 6x \Rightarrow f'(2) = 3\times 2^2 + 6\times 2 = 24.$$

Example 2

Find the derivative of the function $f(x)=3x^4-2x^3+10x+\sqrt{x}+\dfrac{1}{\sqrt{x}}$

Solution

$$f'(x)=12x^3-6x^2+10+\frac{1}{2\sqrt{x}}-\frac{1}{2x\sqrt{x}}$$

Exercise 2.9

1. If $f(x)=x^4-3x^2+2$, find a formula for $f'(x)$ and compute $f'(3)$.
2. Find the derivative of each of the following functions.
   a. $f(x)=5x^4+\sqrt{x}$ b. $f(x)=x+\frac{1}{x^2}$ c. $f(x)=\frac{1}{6}x^6-2x-4$
   d. $f(x)=\sqrt{x}-\frac{1}{\sqrt{x}}$ e. $f(x)=x^{10}+\frac{1}{x}+6x^{\frac{5}{3}}$ f. $f(x)=6x^{\frac{4}{3}}-\frac{1}{3x^3}-\frac{1}{2}x^2$
3. Based on the proof of the sum rule of derivative, show the proof of the difference rule.

b. Derivative of a constant multiple of a function

Activity 2.9

Given the function $f(x)=x^2$, find a. $(3f)'(x)$ b. $3(f'(x))$

Theorem 2.3

If f is differentiable at a, then for any constant number k, the function kf is differentiable at a and $(kf)'(a)=k(f'(a))$.

Proof

By definition of the derivative as $x \to a$,

$$(kf)'(a)=\frac{(kf)(x)-(kf)(a)}{x-a}=\frac{k[f(x)-f(a)]}{x-a}=k\frac{[f(x)-f(a)]}{x-a}=kf'(a).$$

Thus, for all x at which f is differentiable,

$$(kf)'(x)=kf'(x).$$

Example 1

Let $f(x)=6\sqrt{x}$, find the formula for $f'(x)$ and compute $f'(4)$.

Solution

$$f'(x)=(6\sqrt{x})'=\frac{6}{2\sqrt{x}}=\frac{3}{\sqrt{x}} \text{ and } f'(4)=\frac{3}{\sqrt{4}}=\frac{3}{2}.$$

Derivative of polynomial functions

If $f(x)=a_nx^n+a_{n-1}x^{n-1}+a_{n-2}x^{n-2}+...+a_1x+a_0,$, then

$$f'(x)=na_nx^{n-1}+(n-1)a_{n-1}x^{n-2}+(n-2)a_{n-2}x^{n-3}+...+a_1.$$

Example 2

Given $y=3x^8-\sqrt{2}x^5+\frac{3}{2}x^3+20x+1$, compute $\left.\frac{dy}{dx}\right|_{x=-1}$.

Solution

Using differentiation rule

$$\frac{d}{dx}\left(3x^8-\sqrt{2}x^5+\frac{3}{2}x^3+20x+1\right)=24x^7-5\sqrt{2}x^4+\frac{9}{2}x^2+20$$

Then, we evaluate the derivative at $x=-1$.

$$\left.\frac{dy}{dx}\right|_{x=-1}=-24-5\sqrt{2}+\frac{9}{2}+20=\frac{1}{2}-5\sqrt{2}.$$

Note: $\left.\frac{dy}{dx}\right|_{x=-1}$ means value obtained by substituting $x=-1$ into $\frac{dy}{dx}$.

Exercise 2.10

1. Find the derivatives of the following functions.

a. $f(x)=\frac{1}{10}x^{10}$

b. $f(x)=3x^{\frac{4}{3}}$

c. $f(x)=\sqrt{3}x^3$

d. $f(x)=x^4-2x^3+3x-2$

2. Find the derivatives of the following functions, and compute the indicated values.

a. $y = 2\sqrt{x}$, evaluate $\left.\frac{dy}{dx}\right|_{x=1}$ and $\left.\frac{dy}{dx}\right|_{x=4}$.

b. $y = \frac{9}{x^3}$, evaluate $\left.\frac{dy}{dx}\right|_{x=-1}$ and $\left.\frac{dy}{dx}\right|_{x=3}$.

c. $y = x^3 + \frac{1}{2}x^2 - \frac{2}{\sqrt{x}}$, evaluate $\left.\frac{dy}{dx}\right|_{x=1}$ and $\left.\frac{dy}{dx}\right|_{x=4}$.

c. Derivative of a product

Activity 2.10

Given the functions $f(x) = x^2$, and $g(x) = x$, find

a. $(fg)'(x)$ b. $f'(x)g(x) + f(x)g'(x)$

Theorem 2.4

If f and g are differentiable at a, then fg is also differentiable at a, and $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$.

Proof

By definition of the derivative as x gets very close to a,

$$\frac{(fg)(x)-(fg)(a)}{x-a} = (fg)'(a). \qquad (i)$$

and

$$\begin{aligned}\frac{(fg)(x)-(fg)(a)}{x-a} &= \frac{f(x)g(x)-f(a)g(a)}{x-a}\\ &= \frac{f(x)g(x)-f(a)g(x)+f(a)g(x)-f(a)g(a)}{x-a}\\ &= \frac{g(x)\big(f(x)-f(a)\big)+f(a)\big(g(x)-g(a)\big)}{x-a}\\ &= \frac{g(x)\big(f(x)-f(a)\big)}{x-a}+\frac{f(a)\big(g(x)-g(a)\big)}{x-a}\\ &= g(x)\frac{\big(f(x)-f(a)\big)}{x-a}+f(a)\frac{\big(g(x)-g(a)\big)}{x-a}\end{aligned}$$

As $x \to a$, the expression $\frac{f(x)-f(a)}{x-a} = f'(a)$ and $\frac{g(x)-g(a)}{x-a} = g'(a)$. Thus,

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

From this theorem, we generalize that, for all x at which both f and g are differentiable

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Example

1. Find the derivative of the following functions.

   a. $g(x) = (x+6)(x^2+4)$ b. $f(x) = x^2(\sqrt{x}+2x)$

Solution

a. Using product property,

$$g'(x) = (x+6)'(x^2+4) + (x+6)(x^2+4)' = (x^2+4) + 2x(x+6) = 3x^2+12x+4.$$

b. Using product property,

$$f'(x) = \left(x^2\right)'(\sqrt{x}+2x) + x^2(\sqrt{x}+2x)'$$

$$= 2x(\sqrt{x}+2x) + x^2\left(\frac{1}{2\sqrt{x}}+2\right)$$

$$= 2x\sqrt{x} + 4x^2 + \frac{x^2}{2\sqrt{x}} + 2x^2$$

$$= 6x^2 + \frac{5x\sqrt{x}}{2}$$

Exercise 2.11

Find $f'(x)$ of the following functions using product rule.

a. $f(x) = (2x+1)(x^2+x-5)$ b. $f(x) = (x^2-2)(x^3+4x)$

c. $f(x) = x(x^2+\sqrt{x})$ d. $f(x) = (x+2\sqrt{x})(x^2-2)$

d. Derivative of a quotient

Theorem 2.5

If f and g are differentiable at a, and $g(a) \neq 0$, then $\frac{1}{g}, \frac{f}{g}$ are differentiable at a and

i. $\left(\frac{1}{g}\right)'(a) = \frac{(1)'g(a) - 1(g'(a))}{(g(a))^2} = -\frac{g'(a)}{(g(a))^2}$

ii. $\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{(g(a))^2}$.

Proof

i. By definition of derivative as x gets very close to a,

$$\frac{\frac{1}{g(x)} - \frac{1}{g(a)}}{x-a} = \left(\frac{1}{g}\right)'(a)$$

And $\left(\frac{1}{g}\right)'(a) = \frac{\frac{1}{g(x)} - \frac{1}{g(a)}}{x-a} = \frac{g(a)-g(x)}{(x-a)g(x)g(a)} = -\frac{g(x)-g(a)}{x-a} \cdot \frac{1}{g(x)g(a)}$

As $x \to a$, the quantity $\frac{g(x)-g(a)}{x-a} = g'(a)$

Thus $\left(\frac{1}{g}\right)'(a) = -\frac{g'(a)}{(g(a))^2}$

ii. $\left(\frac{f}{g}\right)'(a) = \left(f \cdot \frac{1}{g}\right)'(a)$

Using product rule $\left(f \cdot \frac{1}{g}\right)'(a) = f'(a)\left(\frac{1}{g}\right)(a) + f(a)\left(\frac{1}{g}\right)'(a)$

$$= f'(a)\frac{1}{g(a)} + f(a)\left(-\frac{g'(a)}{(g(a))^2}\right)$$

$$= \frac{f'(a)g(a) - f(a)g'(a)}{(g(a))^2}$$

From this theorem, we see that, for all x at which both f and g are differentiable

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

Example

Find $f'(x)$ a. $f(x)=\dfrac{1}{x^2+3x-1}$ b. $f(x)=\dfrac{8x^5}{x^3+2x}$

Solution

Using quotient rule

a. $f'(x)=-\dfrac{\left(x^2+3x-1\right)'}{\left(x^2+3x-1\right)^2}=-\dfrac{2x+3}{\left(x^2+3x-1\right)^2}$

b. $f'(x)=\dfrac{(8x^5)'(x^3+2x)-8x^5(x^3+2x)'}{(x^3+2x)^2}$

$=\dfrac{(40x^4)(x^3+2x)-8x^5(3x^2+2)}{(x^3+2x)^2}=\dfrac{40x^7+80x^5-24x^7-16x^5)}{(x^3+2x)^2}$

$=\dfrac{16x^7+64x^5}{(x^3+2x)^2}.$

Exercise 2.12

Find the derivative of each of the following functions.

a. $f(x)=\frac{1}{x^2+1}$ b. $f(x)=\dfrac{2x^3+3}{x-5}$

c. $f(x)=\frac{x^3+3x}{x^2}$ d. $f(x)=\frac{2x^4}{\sqrt{x}}$

2.1.4.4 The chain rule

Activity 2.11

1. Let $f(x)=2x+3$ and $g(x)=x^2-1$. Find each of the following functions.

   a. $f'(x)$ b. $f'(g(x))$ c. $f(g(x))$ d. $f'(g(x))\cdot g'(x)$

2. Find the derivatives of the following functions.

   a. $f(x)=\left(3x^2+2\right)^2$ b. $f(x)=\left(\sqrt{x^3+2x-7}\right)^3$

Theorem 2.6

Let g be differentiable at a, and f be differentiable at $g(a)$. Then $f \circ g$ is differentiable at a, and $(f \circ g)'(a) = f'(g(a)) \cdot g'(a)$.

The Chain rule can be interpreted as saying that $(f \circ g)'(x) = f'(g(x))g'(x)$ for all x such that g is differentiable at x and f is differentiable at $g(x)$.

Proof

For $g(x) - g(a) \neq 0$,

$$\frac{f(g(x)) - f(g(a))}{x - a} = \frac{f(g(x)) - f(g(a))}{x - a} \cdot \frac{g(x) - g(a)}{g(x) - g(a)}$$

By definition of derivative as x gets very close to a,

$$\frac{(f \circ g)(x) - (f \circ g)(a)}{x - a} = \frac{f(g(x)) - f(g(a))}{x - a}$$

$$= \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a}$$

As $x \to a$, the expressions

$$\frac{f(g(x)) - f(g(a))}{g(x) - g(a)} = f'(g(a)) \text{ and } \frac{g(x) - g(a)}{x - a} = g'(a)$$

Thus, $(f \circ g)'(a) = f'(g(a))\, g'(a)$.

Example 1

For each of the following functions find a formula for $f'(x)$.

a. $f(x) = \sqrt{x^4 + 3x^2 - 5}$ b. $f(x) = (5x + 4)^{21}$

Solution

Using the Chain rule

a. f is the composition of two simple functions $g(x) = \sqrt{x}$ and $h(x) = x^4 + 3x^2 - 5$. That is, $f(x) = g(h(x))$.

By the chain rule, $f'(x) = g'\big(h(x)\big)\,h'(x)$.

But $g'(x) = \dfrac{1}{2\sqrt{x}}$ and $h'(x) = 4x^3 + 6x$

Thus, $f'(x) = g'\big(h(x)\big)\,h'(x) = \dfrac{1}{2\sqrt{x^4+3x^2-5}} \cdot \big(4x^3+6x\big) = \dfrac{2x^3+3x}{\sqrt{x^4+3x^2-5}}$.

b. $f'(x) = 21(5x+4)^{21-1} = 21(5x+4)^{20}(5) = 105(5x+4)^{20}$.

Example 2

Let $y = \sqrt{4+x^5}$. Find a formula for $\dfrac{dy}{dx}$ and compute $\left.\dfrac{dy}{dx}\right|_{x=2}$.

Solution

$\dfrac{dy}{dx} = \dfrac{1}{2\sqrt{4+x^5}}\dfrac{d}{dx}(x^5+4) = \dfrac{5x^4}{2\sqrt{4+x^5}}$ and $\left.\dfrac{dy}{dx}\right|_{x=2} = \dfrac{20}{3}$.

Exercise 2.13

Find the derivative of the following functions using the chain rule.

a. $f(x) = (2x+1)^4$

b. $f(x) = (x^3 + 2x)^{25}$

c. $f(x) = \sqrt{x^2+3x-1}$

d. $f(x) = \frac{1}{(3x+2)^2}$

e. $f(x) = \sqrt{x^2 + \dfrac{1}{x^2}}$

f. $f(x) = \left(\frac{3x-5}{2-7x}\right)^6$

2.1.5 Maximum and Minimum Points

The differentiation of various functions renders great services in solving the problems concerned with finding out the maximum and minimum values of quantities. In the various fields of engineering and technology, we have to find the maximum or minimum values of one quantity with respect to another quantity. For example, in finding the radius and height of a cylinder that is to be manufactured with the metal sheet of a given surface area, so that the capacity of cylinder is maximum. Sometimes it is necessary to find the least cost for the transmission of given horsepower. We can find the dimensions of a plot of a given perimeter so that its area is maximum and so on.

2.1.5.1 Increasing and Decreasing Functions

Activity 2.12

1. Find the real zeros of each of the following functions.

a. $f(x) = 4x - 8$ b. $f(x) = x^2 - x - 12$ c. $f(x) = \dfrac{1-\sqrt{x}}{(x-2)^2}$

2. Consider the graph of the following function.

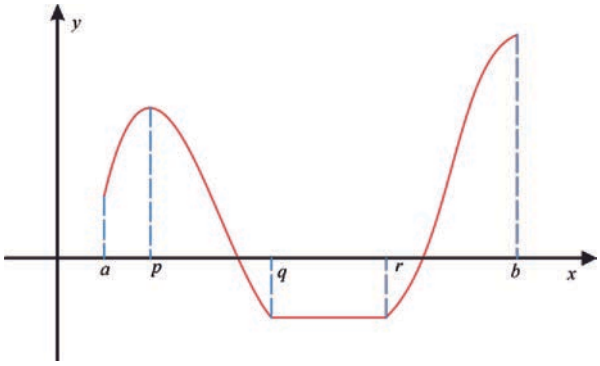


Figure 2.10

a. Identify the intervals in which the graph is rising and falling as you move from left to right starting at *a*.

b. Identify the intervals in which the graph is neither rising nor falling.

3. Solve each of the following inequalities using sign chart method:

a. $2x^2 + 3x - 2 \geq 0$ b. $x^2 + 3x - 4 < 0$

Definition 2.8

Let f be a function on an interval I. If x_1 and x_2 are in I,

i. For $x_1 < x_2$ if $f(x_1) \leq f(x_2)$, then f is said to be increasing on I.

ii. For $x_1 < x_2$ if $f(x_1) \geq f(x_2)$, then f is said to be decreasing on I.

iii. For $x_1 < x_2$ if $f(x_1) < f(x_2)$, then f is strictly increasing on I.

iv. For $x_1 < x_2$ if $f(x_1) > f(x_2)$, then f is strictly decreasing on I.

Geometrically, a function is increasing if its graph rises and decreases if its graph falls while x moves to the right.

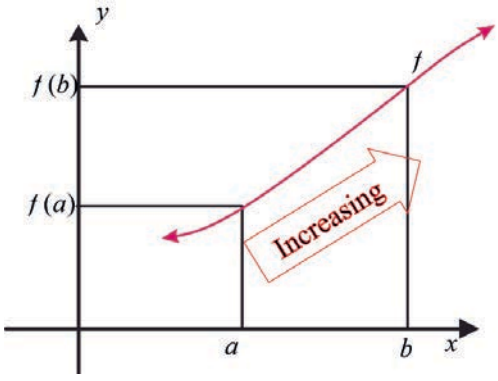


Figure 2.11a

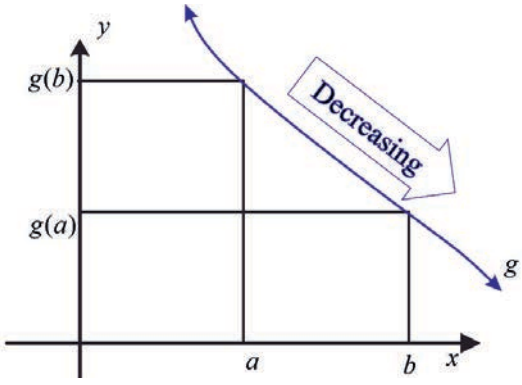


Figure 2.11b

Increasing and decreasing test

Suppose that $f(x_1) \geq f(x_2)$, f is differentiable in the interior of an interval I.

a. If $f'(x) \geq 0$, for all x in the interior of I, then f is increasing on I.

b. If $f'(x) \leq 0$, for all x in the interior of I, then f is decreasing on I.

c. If $f'(x) > 0$ and $f'(x) = 0$ on I, then f is strictly increasing on I.

d. If $f'(x) < 0$ and $f'(x) = 0$ on I, then f is strictly decreasing on I.

Note

A function that is either increasing or decreasing is known as a monotonic function.

Definition 2.9

A number c in the domain of a function f is said to be a **critical number** of f if and only if either $f'(c)=0$ or f has no derivative at c.

Example 1

Find the critical numbers of the given functions.

a. $f(x)=4x^3-5x^2-8x+20$ b. $f(x)=2\sqrt{x}(6-x)$

Solution

a. $f'(x)=12x^2-10x-8$

$$f'(x)=0 \Rightarrow 2(6x^2-5x-4)=2(2x+1)(3x-4)=0 \Rightarrow x=-\frac{1}{2} \text{ or } x=\frac{4}{3}$$

Hence, the critical numbers of f are $x=-\frac{1}{2}$ and $x=\frac{4}{3}$.

b. Applying the product rule of derivatives:

$$f'(x)=\frac{2}{2\sqrt{x}}(6-x)-2\sqrt{x}=\frac{6-x}{\sqrt{x}}-\frac{2x}{\sqrt{x}}=\frac{6-3x}{\sqrt{x}}=\frac{3(2-x)}{\sqrt{x}} \Rightarrow$$

$f'(x)=0 \Rightarrow x=2$

and $f'(x)$ does not exist when $x=0$. Hence, the critical numbers of f are $x=0$ and $x=2$.

Example 2

Find the interval on which the function $f(x)=3x^4-4x^3-12x^2+5$ is increasing and decreasing.

Solution

$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x^2 - x - 2) = 12x(x-2)(x+1)$

Hence, the critical numbers f are $x = -1, 0, 2$

		-1		0		2	
$12x$	----		----	0	++++		+++++
$x+1$	----	0	++++		++++		++++
$x-2$	----		-----		------	0	++++
$f'(x) = 12x(x-2)(x+1)$	----	0	++++	0	-----	0	++++

From the sign chart above,

$f'(x) \geq 0$ on $[-1,0] \cup [2,\infty)$ and $f'(x) \leq 0$ *on* $(-\infty,-1] \cup [0,2]$.

Therefore, the function is increasing on an interval $[-1,0] \cup [2,\infty)$ and decreasing on $(-\infty,-1] \cup [0,2]$.

Exercise 2.14

1. Find all the critical numbers of the following functions.

   a. $f(x) = x^3 + 6x^2$

   b. $f(x) = \dfrac{2x^2}{1-x^2}$

   c. $f(x) = 2x^3 + 9x^2 - 24x$

   d. $f(x) = 8x\sqrt{1-x^2}$

2. Find the interval on which the following functions are increasing and decreasing.

   a. $f(x) = x^3 + 6x^2$

   b. $f(x) = 2x^2 + 6x - 10$

   c. $f(x) = -2x^3 + 3x^2 + 12x + 6$

   d. $f(x) = x - 6\sqrt{x-1}$

   e. $f(x) = (x+3)^2(x-1)^2$

2.1.5.2 Minimum and maximum values of a function

Activity 2.13

Consider the graph of the following function.

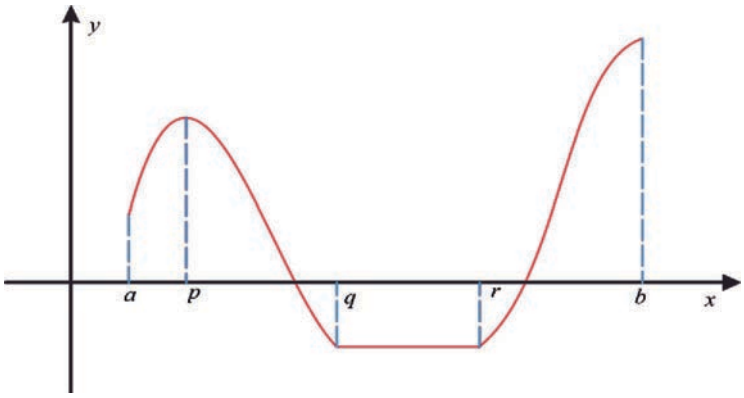


Figure 2.12

a. Where does the function attain maximum value and a minimum value?

b. What happens to the gradient at a maximum point and at a minimum point?

Definition 2.10

If there exists an open interval (a,b) containing c such that $f(x) < f(c)$ for all x in (a,b) other than c in the interval, then $f(c)$ is a relative (local) maximum value of f and a point $(c, f(c))$ is called relative maximum point. If $f(x) > f(c)$ for all x in (a,b) other than c, then $f(c)$ is a relative (local) minimum value of f and the point $(c, f(c))$ is known as relative(local) minimum point (See Figure 2.13 below). Functions may have any number of relative extrema.

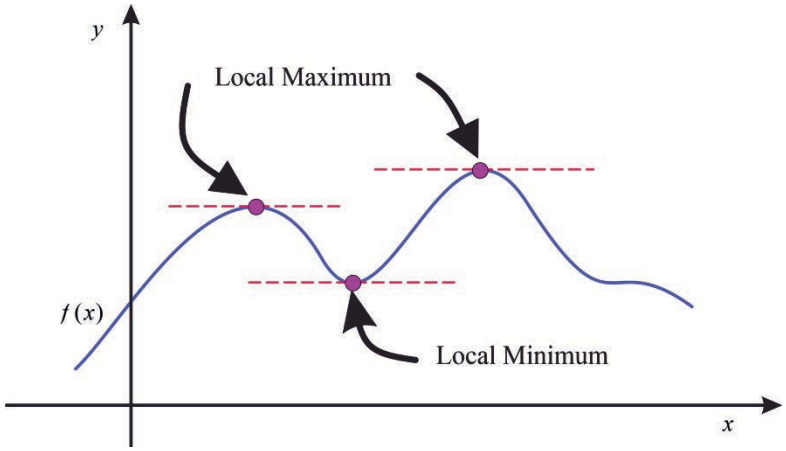


Figure 2.13

Definition 2.11

If c is in the domain of f and for all x in the domain of the function, $f(x) \leq f(c)$, then $(c, f(c))$ is an absolute maximum piont of the function f. If for all x in the domain $f(x) \geq f(c)$ then $(c, f(c))$ is an absolute minimum point of the function f. (see Figure 2.14). The absolute maximum and minimum values of f are called absolute extreme values.

Note

- Absolute extrema are not necessarily unique.
- At either local maximum or local minimum point the gradient is zero.

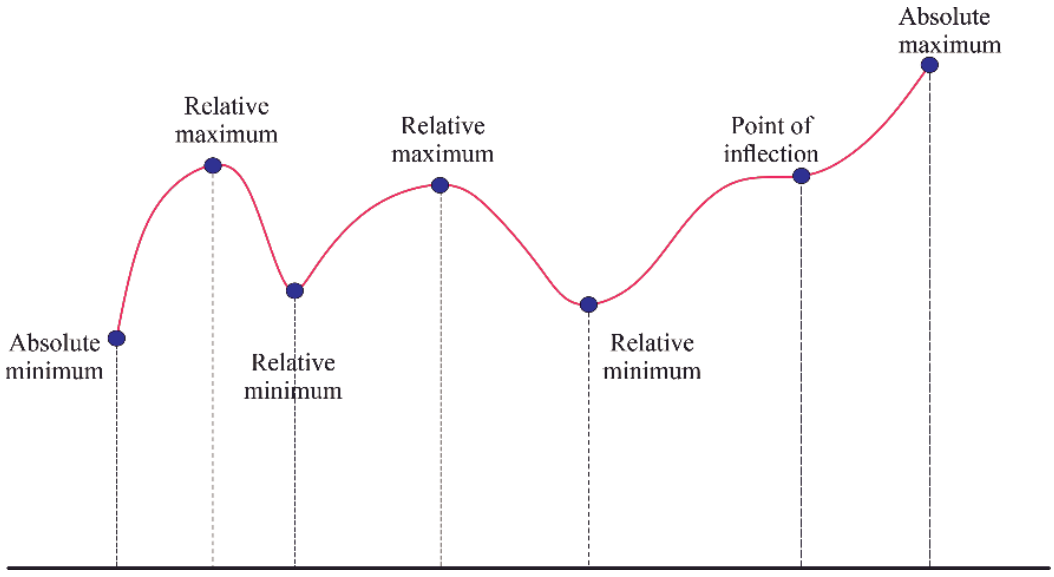


Figure 2.14

Procedure to find absolute extreme values on closed intervals

We can find the absolute maximum and minimum values of a function f on a closed interval $[a,b]$ by the following steps:

Step 1: Find all the critical numbers of a function f.

Step 2: Evaluate f at all critical numbers and at the end points a and b.

Step 3: Choose the largest and the smallest values obtained in step2. These values are the maximum and the minimum of a function f on $[a,b]$.

Example

Find the absolute maximum and minimum values of the function $f(x)=x^2-2x$ on $[-1,2]$.

Solution

$f'(x)=2x-2=2(x-1)$

$f'(x)=0 \Rightarrow x=1$.

Thus, the critical number of f is $x=1$.

Thus, to determine the extreme value of the function we need to evaluate the value of the function at the boundary and critical number.

$$f(-1)=1+2=3\ ,\ f(1)=1-2=-1 \text{ and } f(2)=4-4=0.$$

Therefore, the absolute maximum value is $f(-1)=3$ and the absolute minimum value is $f(1)=-1$. Thus, $(-1,3)$ is the maximum point and $(1,-1)$ is the minimum point of f.

Exercise 2.15

Find the absolute maximum and minimum values of the following functions on a given interval.

a. $f(x)=x^2+2x+3,\quad [-2,2]$

b. $f(x)=x^3+3x^2-9x+5,[-2,2]$

c. $f(x)=3x^4-26x^3-60x^2-11$ on $[1,5]$.

d. $f(x)=\dfrac{x^2-2x+4}{x-2},[-3,1]$

The first derivative test

Let c be a critical number of a function f on an open interval I containing c.

a. If the sign of f' changes from negative to positive at c, then f has a relative minimum at c.

b. If the sign of f' changes from positive to negative at c, then f has a relative maximum at c.

c. If the sign of f' does not change at c, then f has no relative maximum or minimum at c.

Procedure to find relative extreme value

To determine relative extreme value of a function using the first derivative test, one can follow the following steps.

Step 1: Find all the critical numbers of f.

Step 2: Form the sign chart for $f'(x)$ using the critical number as poles.

Step 3: Apply first derivative test and decide the extreme values if any.

Example 1

Find the relative maximum and minimum values of

a. $f(x)=2x^3+3x^2-12x-3$ b. $f(x)=(x-1)^2(x-3)^2$

Solution

a. $f'(x)=6x^2+6x-12=6(x^2+x-2)=6(x+2)(x-1)$

Hence, the critical numbers are $x=-2,1$

		-2		1	
$6(x+2)$	----	0	++++		+++++
$x-1$	----		----	0	+++++
$f'(x)=6(x+2)(x-1)$	+++++	0	----	0	+++++

From the sign chart above we see that, f' changes its sign from positive to negative at $x=-2$ and changes from negative to positive $x=1$, therefore, the function has a local maximum value at $x=-2$ and a local minimum value at $x=1$. This implies the point $(-2, f(-2))=(-2,17)$ is a relative maximum and $(1, f(1))=(1,-10)$ is a relative minimum point of the function.

Thus, 17 is a relative maximum value and – 10 is a relative minimum value.

b.

$$\begin{aligned} f'(x) &= 2(x-1)(x-3)^2 + 2(x-1)^2(x-3) \\ &= 2(x-1)(x-3)(x-3+x-1) \\ &= 2(x-1)(x-3)(2x-4) \\ &= 4(x-1)(x-2)(x-3) \end{aligned}$$

Hence, the critical numbers are $x = 1, 2, 3$.

		1		2		3	
$4(x-1)$	----	0	++++		++++		+++++
$x-2$	----		-----	0	++++		++++
$x-3$	----		-----		------	0	++++
$f'(x) = 4(x-1)(x-2)(x-3)$	----	0	++++	0	-----	0	++++

From the sign chart, we see that f' changes its sign from negative to positive at $x = 1$ and $x = 3$ and changes from positive to negative at $x = 2$. Therefore, $(1, f(1)) = (1, 0)$ and $(3, f(3)) = (3, 0)$ are relative minimum points and $(2, f(2)) = (2, 1)$ is the relative maximum point of the function.

Thus, 0 is a relative maximum value and 1 is a relative minimum value.

Example 2

If $f(x) = ax^3 + bx - 5$ has a relative minimum value of -6 at $x = \frac{1}{2}$, find a and b.

Solution

Given $f(\frac{1}{2}) = \frac{1}{8}a + \frac{1}{2}b - 5 = -6 \Leftrightarrow a + 4b = -8$ (i)

and $f'(\frac{1}{2}) = 0$, $f'(x) = 3ax^2 + b \Rightarrow \frac{3a}{4} + b = 0 \Leftrightarrow 3a + 4b = 0$ (ii)

(i) and (ii) imply $\begin{cases} a + 4b = -8 \\ 3a + 4b = 0 \end{cases} \Rightarrow a = 4 \text{ and } b = -3.$

Exercise 2.16

1. Find the relative extreme value of the following functions (if any).

   a. $f(x)=2x^3-9x^2-24$ b. $f(x)=\frac{1}{4}x^3-3x$ c. $f(x)=\frac{x^2+x+1}{x^2-x+1}$

2. If $f(x)=x^3+ax+b$ has a relative minimum value of 4 at $x=2$, find a and b.

2.1.6 Equation of Tangents and Normal Lines to Curves

Activity 2.14

1. Find the equation of a line containing the points $(1,3)$ and $(-2,3)$.
2. Find the equation of a line perpendicular to the line found in question 1 above containing the point $(1,3)$.
3. Write the equation of a tangent line to a curve if its slope is 3 at the point $(-2,5)$.
4. What is the relationship between the slopes of a tangent line and a normal line to a curve at a point of tangency?

Is there any relation ship between the first derivative of a function at a point of tengency and the slope of a tangent line to the curve of the function?

We have seen in section 2.1.5 that the first derivative of a function at a given point is interpreted as the slope of the tangent line at that point. This implies f' is the slope of the line tangent to the graph of f at a point $(a, f(a))$. Thus, the equation of the tangent line to the curve $y=f(x)$ at $x=a$ is given by $y=f'(a)(x-a)+f(a)$.

A line that is perpendicular to the tangent line at the point of tangency is known as the **normal line** to the curve. Since the product of the slope of two non-vertical perpendicular lines is -1, the slope of the normal line at the point of tangency $(a, f(a))$ is $\frac{-1}{f'(a)}$ and its equation is given by $y=-\frac{1}{f'(a)}(x-a)+f(a)$.

Note

1. If the first derivative of f at $x=a$ is zero, then f has a horizontal tangent line with equation $y=f(a)$ and vertical normal line given by $x=a$.
2. If the first derivative of f at $x=a$ undefined, then f has a vertical tangent line with equation $x=a$ and a horizontal normal line given by $y=f(a)$.

Example 1

Find the equation of the line tangent and the normal line to the graph of the following functions at the given point.

a. $f(x)=x^3-2x;\ (1,-1)$

b. $f(x)=\sqrt{20-x^2};(2,4)$

c. $f(x)=\begin{cases} x, & \text{if } x>3 \\ x^2\text{-}6, & \text{if } x\le 3 \end{cases}$

d. $f(x)=x\sqrt{x^2+16}\ ;(0,0)$

Solution

a. $f'(x)=3x^2-2\Rightarrow f'(1)=1$

Thus,

i. The equation of the tangent line to the graph of f at (1,-1) is

$$y=f'(1)(x-1)+f(1)=1(x-1)-1=x-2.\ \Leftrightarrow y=x-2.$$

ii. The equation of the normal line to the graph of f at (1,-1) is

$$\begin{aligned} y&=-\frac{1}{f'(1)}(x-1)+f(1)\\ &=-1(x-1)-1\\ &=-x. \end{aligned}$$

$$\Leftrightarrow y=-x$$

b. $f'(x)=\dfrac{-x}{\sqrt{20-x^2}}\Rightarrow f'(2)=\dfrac{-1}{2}$

i. The equation of the tangent line to the graph of f at $(2,4)$ is

$$y = f'(2)(x-2) + f(2) = -\frac{1}{2}(x-2) + 4 = -\frac{1}{2}x + 5$$

$$\Leftrightarrow y = -\frac{1}{2}x + 5.$$

ii. Since the slope of a normal line is negative reciprocal of that of the tangent line, the slope of the normal line in this problem is 2. Hence, the equation of the normal line to the graph of f at $(2,4)$ is

$$y = \frac{-1}{f'(2)}(x-2) + f(2) = 2(x-2) + 4 = 2x.$$

$$\Leftrightarrow y = 2x.$$

c. $f'(x) = \begin{cases} 1, & \text{if } x > 3 \\ 2x, & \text{if } x > 3 \end{cases} \Rightarrow f'(1) = 2.$

i. The equation of the tangent line to the graph of f at $(1,-5)$ is

$$y = f'(1)(x-1) + f(2) = 2(x-1) - 5 = 2x - 7.$$

$$\Leftrightarrow y = 2x - 7.$$

ii. The equation of the normal line to the graph of f at $(1,-5)$ is

$$y = -\frac{1}{f'(1)}(x-1) + f(1) = \frac{-1}{2}(x-1) - 5 = \frac{-1}{2}x - \frac{9}{2}.$$

$$\Leftrightarrow y = -\frac{1}{2}x - \frac{9}{2}.$$

d. $f'(x) = \sqrt{x^2+16} + \frac{x^2}{\sqrt{x^2+16}} \Rightarrow f'(0) = 4$

i. The equation of the tangent line to the graph of f at $(0,0)$ is

$$y = f'(0)(x-0) + f(0) = 4(x-0) + 0 = 4x.$$

$$\Leftrightarrow y = 4x.$$

ii. The equation of the normal line to the graph of f at $(0,0)$ is

$$y = \frac{-1}{f'(0)}(x-0) + f(0) = \frac{-1}{4}(x-0) = \frac{-1}{4}x.$$

$$\Leftrightarrow y = -\frac{1}{4}x.$$

Example 2

Find all points at where $f(x)=\sqrt{x^2-6x}$ has vertical tangent line.

Solution

The function will have a vertical tangent line at its first derivative is undefined in its domain and $f'(x)=\dfrac{2x-6}{2\sqrt{x^2-6x}}=\dfrac{x-3}{\sqrt{x^2-6x}}$ is undefined at the points where $x(x-6)=0\Rightarrow x=0$ or $x=6$. Hence, $f(x)$ has a vertical tangent line at $(6,0)$ and $(0,0)$.

Exercise 2.17

1. Find the equation of the line tangent and the normal line to the graph of the following functions at the given point.

   a. $f(x)=x^3+x+2;\ (0,2)$ b. $f(x)=(1-x^3)\sqrt{x+2};\ (-1,2)$.

2. Find all points at where the following functions has vertical tangent line.

   a. $f(x)=\sqrt{x^2-x}$ b. $f(x)=2\sqrt{x}$

3. If $f(x)=x^2+ax+b$ and $g(x)=x^3-c$ have the same tangent line at $(1,2)$ find the values of the constants a,b and c and the equation of their normal lines.

2.2 Application of Derivative

Activity 2.15

1. Given a set $S = \{3,4,5,6,7\}$ and $f(x) = 3x + 2$
   a. Find $K = \{f(x) \mid x \in S\}$.
   b. What is the smallest element of K?
   c. What is the largest element of K?
2. Given a set S on an open interval $(3,7)$ and $f(x) = 3x + 2$
   a. Find $T = \{f(x) \mid x \in S\}$.
   b. Can you list all the elements of T?
   c. Can you guess the smallest element of T?
   d. Can you guess the largest element of T?
3. Given a set S as a closed interval $[3,7]$ and $f(x) = 3x + 2$
   a. Find $H = \{f(x) \mid x \in S\}$.
   b. Can you list all the elements of H?
   c. Can you guess the smallest element of H?
   d. Can you guess the largest element of H?

Applications of derivatives are varied not only in mathematics but also in real life. To give an example, derivatives have various important applications in mathematics such as finding the rate of change of a quantity, finding the approximate value, finding the equation of the tangent line and the normal line to a curve, and finding the minimum and maximum values of algebraic expressions. Derivatives also has a wide range of applications in real-life usage. For instance, in business, derivatives are used to find profit and loss for the future of the investment using graphs. It is used to calculate the rate of change of distance of a moving body with respect to time. Derivatives can have different interpretations in each of the sciences. For instance; chemists who study a chemical reaction may be interested in the rate of change in the concentration of a reactant with respect to time (the rate of reaction). A biologist is interested in the rate of change of the population of a colony of bacteria with respect to time. In section 2.1, you have studied derivatives and have developed methods to find derivatives of various functions. Once you have developed the properties of the

mathematical concepts once and for all, you can then turn around and apply these results to all of the sciences. You have already investigated some of the applications of derivatives, but now that you know the differentiation rules, you are in a better position to pursue the applications of differentiation in a greater depth.

In this section, you will study problems involving variables that are changing with respect to time. If two or more such variables are related to each other, then their rates of change with respect to time are also related.

2.2.1 Applications of Derivatives in Finding Rate of Change

Example 1

If the radius of a circle is increasing at a rate of 1.5 cm/sec., find the rate at which the area is increasing at the instant when the diameter is 12 cm.

Solution

We know that area of a circle with radius r is given by $A = \pi r^2$ and we are given $\frac{dr}{dt} = 1.5\text{cm/sec}$, to find $\frac{dA}{dt}$ at diameter $d = 12\text{cm}$.

$$A = \pi r^2 \Rightarrow \frac{dA}{dt} = \pi \frac{d(r^2)}{dt} = \pi(2r\frac{dr}{dt}) = 2\pi r \frac{dr}{dt}$$

Besides when the diameter $d = 12\text{cm}$, $r = 6\text{cm}$.

Hence, $\frac{dA}{dt}(6) = 2\pi(6)(1.5) = 18\pi \text{cm}^2/\text{sec}$.

Example 2

An object is moving along the path $3y = x^2$. If the x-coordinate of the object is increasing at a rate of 2 ft/min at the instant when $x = 6$, find the rate at which the y-coordinate is increasing.

Solution

Given $3y = x^2$ and $\frac{dx}{dt} = 2\text{ft/min}$

$$3y = x^2 \Rightarrow 3\frac{dx}{dt} = 2x\frac{dx}{dt} \Leftrightarrow \frac{dy}{dt} = \frac{2x}{3}\frac{dx}{dt}$$

Thus, at $x = 6$, $\frac{dy}{dt} = \frac{2}{3}(6)(2\text{ ft/min}) = 8\text{ ft/min}$

Example 3

Suppose the side of an equilateral triangle is increasing at a rate of 2 cm/sec. Find the rate at which the area is increasing at the instant when the side is 4 cm.

Solution

The area of an equilateral triangle having one side of lenth s is given by $A = \frac{\sqrt{3}}{4}s^2$

We are given, $\frac{ds}{dt} = 2\text{cm/sec.}$

To find $\frac{dA}{dt}$ when $s = 4\text{cm}$

$$A = \frac{\sqrt{3}}{4}s^2 \Rightarrow \frac{dA}{dt} = 2s\frac{\sqrt{3}}{4}\frac{ds}{dt}$$

$$\left.\frac{dA}{dt}\right|_{s=4} = 2 \times \frac{\sqrt{3}}{4} = 4\sqrt{3}\ \text{cm}^2/\text{sec}$$

Example 4

Air is being pumped into a spherical balloon at the rate of 4 cm^3 / min. Find the rate of change of the radius when the radius is 2 cm.

Solution

Let r be the radius of the sphere. The volume V of the sphere is given by

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt} = \frac{1}{4\pi} cm/\min.$$

Example 5

The displacement of a particle at any time t is given by $S = t^3 - 3t^2 + 5t + 7$. Find its velocity and acceleration at the end of 2 hrs, where S is in kilometers and t is in hrs.

Solution

Given $S = t^3 - 3t^2 + 5t + 7$

Velocity $v = \frac{dS}{dt} = 3t^2 - 6t + 5$

Acceleration $a = \frac{dv}{dt} = 6t - 6$

Hence, the velocity at the end of 2 hrs is

$$v(2) = \left.\frac{ds}{dt}\right|_{t=2\text{hrs}} = 3(2)^2 - 6(2) + 5 = 5 \text{ km/hr.}$$

Acceleration at the end of 2 hrs is $a = \frac{dv}{dt} = 6(2) - 6 = 6 \text{ km/hr}^2$.

Exercise 2.18

1. A circular plate of metal is being expanded by heating. The radius of plate increases at the rate of 0.3 cm per second. Find the rate of increase of area when its radius becomes 15 cm.

2. The volume of a sphere is increasing at the rate of 0.2π cm^3 / sec. Find the rate of increase of radius when the radius is 10 cm.
3. A square plate is being expanded by heating. If its side is increasing at the rate of 0.2 cm per second, then at what rate the area is increasing when the side is 20 cm?
4. A man 2 m tall walks at a uniform speed of 8 m/minute from a lamp post 6 m high. Find the rate at which the length of his shadow increases.
5. A varying force is applied on a body of unit mass and if the force increases at the rate of 2 N/sec., then find the rate at which the acceleration in the body is changing.

2.2.2 Applications of Derivatives in Business and Economics

Activity 2.16

1. Where do you think we use the concept of derivative other than in mathematics?
2. Is the concept of derivative vital in the study of economics, road taxation and traffic education?

From this activity, one can see the development of multiple strategies, into which companies incorporate derivatives. Companies, both in and out of the financial industry have begun to use derivatives as a method of speculating and generating income.

Example 1

The profit function of a company can be represented by $P(x) = x - 0.00001x^2$, where x represents units sold. Find the optimal sales volume and the amount of profit to be expected at that volume.

Solution

Marginal profit: $MP = \frac{dP}{dx} = \frac{d}{dx}(x - 0.00001x^2) = 1 - 0.00002x.$

To get maximum profit now we put marginal profit equals to zero. So, $1 - 0.00002x = 0 \Rightarrow x = \frac{1}{0.00002} = 50,000$ units.

Now, to check whether this value is maximum or minimum we can use the first derivative test.

		50,000	
$P'(x) = 1 - 0.00002x$	++++	0	----

From this sign chart the sign of $P'(x)$ changes from positive to negative at $x = 50,000$. Thus, by the first derivative test, we get maximum profit when the company sold $x = 50,000$ units of production. Now, by putting the value of x in the profit function we get maximum profit.

$$P = f(50,000) = 50,000 - 0.00001(50000)^2$$
$$= 50,000 - 0.00001(2500,000,000) = 50,000 - 25,000 = 25,000$$

The optimum output for the company will be 50,000 units of x and the maximum profit at that volume will be 25, 000 birr.

Example 2

The demand equation for a certain product is $P = 6 - \frac{1}{2}x$ birr, where x represents the amount of product. Find the level of production which results in maximum revenue.

Solution

The revenue functionis $R(x)$.

Thus, $R(x) = x\left(6 - \frac{1}{2}x\right) = 6x - \frac{1}{2}x^2$ birr.

The marginal revenue is given by $R'(x) = 6 - x$

The graph of $R(x)$ is a parabola that opens downward as shown in Figure 2.15.

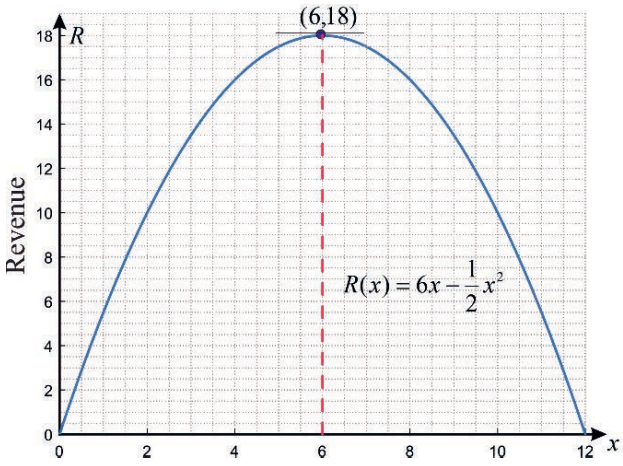


Figure 2.15

It has a horizontal tangent precisely at that x for which $R'(x) = 0$, that is, for that x at which marginal revenue is 0. The only such value of x is $x = 6$. The corresponding value of revenue is $R(6) = 6(6) - \frac{1}{2}(6)^2 = 18$ birr.

		6	
$R'(x) = 6 - x$	++++	0	----

Thus, by the first derivative test, the rate of production resulting in maximum revenue is $x = 6$, which results in total revenue of 18 birr.

Example 3

Promoters of international fund-raising concerts must walk a fine line between profit and loss, especially when determining the price to charge for admission to closed-circuit TV showings in local theatres. By keeping records, a theatre determines that at an admission price of 26 birrs, it averages 1000 people in attendance. For every drop in price of 1 birr, it gains 50 customers. Each customer spends an average of 4 birrs on concessions. What admission price should the theatre charge to maximize total revenue?

Solution

Let x be the number of birr by which the price of 26 Birr should be decreased. (If x is negative, the price is increased.) We first express the total revenue R as a function of x. Note that the increase in ticket sales is $50x$ when the price drops x Birr:

$$\begin{aligned} R(x) &= (\text{revenue from tickets}) + (\text{revenue from concessions}) \\ &= (\textit{number of people}) \cdot (\textit{Ticket price}) + (\textit{number of people}) \cdot 4 \\ &= (1000 + 50x) \cdot (26 - x) + (1000 + 50x) \cdot 4 \\ &= 26{,}000 - 1000x + 1300x - 50x^2 + 4000 + 200x \\ &= -50x^2 + 500x + 30{,}000 \end{aligned}$$

To find x such that $R(x)$ is a maximum, we first find $R'(x)$:

$$R'(x) = -100x + 500$$

This derivative exists for all real numbers x. Thus, the only critical values are where $R'(x) = 0$; so we solve that equation:

$$-100x + 500 = 0 \Rightarrow x = 5$$

This corresponds to lowering the price by 5 Birr. To determine whether $R(x)$ has a maximum value or not at this critical point we use the first derivative test.

		5	
$R'(x) = 500 - 100x$	++++	0	----

Since the sign chart changes from negative to positive at $x = 5$, by first derivative test, in order to maximize revenue, the theatre should charge 26 Birr-5 Birr or 21 Birr per ticket.

Example 4

For a company, the total revenue R and the total cost C for yard maintenance of x homes are given by $R(x) = 1000x - x^2$, and $C(x) = 3000 + 20x$.

Suppose that the company is adding 10 homes per day at the moment when the 400th customer is signed. At that moment, what is the rate of change of

a. total revenue? b. total cost? c. total profit?

Solution

a. $\dfrac{dR}{dt} = \dfrac{dR}{dx}\dfrac{dx}{dt} = (1000 - 2x)\dfrac{dx}{dt}$

$\left.\dfrac{dR}{dt}\right|_{x=400} = (1000 - 2\times 400)(10) = 2000$ birr per day.

b. $\dfrac{dC}{dt} = \dfrac{dC}{dx}\dfrac{dx}{dt} = (20)\dfrac{dx}{dt} = 20\times 10 = 200$ birr per day.

c. Since $p(x) = R(x) - C(x)$, we have

$\dfrac{dp}{dt} = \dfrac{dR}{dt} - \dfrac{dC}{dt} = 2000$ birr per day - 200 birr per day

$= 1800$ birr per day.

Exercise 2.19

Given $C(x) = 62x^2 + 27{,}500$ and $R(x) = x^3 - 12x^2 + 40x + 10$ where $C(x)$ and $R(x)$ represent, respectively, the total cost and revenue from the production and sale of x items. Find each of the following:

a. Total profit , $P(x)$.

b. Total cost, revenue, and profit from the production and sale of 50 units of the product.

c. The marginal cost, revenue, and profit when 50 units are produced and sold.

2.2.3 Application of Derivative in Road Taxation

Example

A motorist has to pay an annual road tax of 50 birrs and 110 birrs for insurance. His car does 30 miles to the gallon which costs 75 pence (per gallon). The car is serviced every 3000 miles for 20 birrs, and depreciation is calculated in pence by multiplying the square of the mileage by 0.001. Obtain an expression for the total annual cost of running the car. Hence find an expression for the average total cost per mile and calculate the annual mileage which will minimize the average cost per mile.

Solution

Suppose he covers x miles in a year.

Tax per annum = 50 birrs; Insurance per annum = 110 birrs;

Cost of petrol $= \frac{0.75x}{30}$; service charge $= \frac{20x}{3000}$; depreciation $= \frac{0.001x^2}{100}$ (0.001 is in pence and is divided by 100 to get the amount in birr).

Total cost (TC): $C = 50 + 110 + \left(\frac{0.75x}{30}\right) + \frac{20x}{3000} + 0.00001x^2$

Average total cost (TC) per mile: $\frac{C}{x} = M = \frac{160}{x} + \frac{0.75}{30} + \frac{20}{3000} + 0.00001x$

$$\frac{dM}{dx} = 0$$

$$\Rightarrow \frac{-160}{x^2} + 0.00001 = 0$$

$$\Leftrightarrow 0.00001x^2 = 160$$

$$\Leftrightarrow x^2 = \frac{160}{0.00001} = 16,000,000$$

$$\Leftrightarrow x = \pm 4,000$$

Now, to check whether this critical number $(x = 4000)$ gives a maximum or minimum cost, we can use a sign chart and apply the first derivative test.

		-4000		4000	
$x - 4000$	----		-----	0	++++
$x + 4000$	----	0	++++		++++
$x^2 - 16{,}000{,}000$	+++		-----		++++

From the sign chart, we see that the sign of the derivative changes from negative to positive at $x = 4000$. So, the motorist can cover 4000 miles in a year to minimize the average cost per mile.

Exercise 2.20

1. A rectangular area of 3200 ft^2 is to be fenced off. Two opposite sides will use fencing costing 1 dollar per foot and the remaining sides will use fencing costing 2 dollars per foot. Find the dimensions of the rectangle of the least cost.
2. A manufacturer wants to design an open box that has a square base and a surface area of 48 square unit. What dimensions will produce a box with a maximum volume?
3. A closed cylindrical can having a volume of 128π cm^3 is to be formed. Find the dimensions (the radius and height) of the cylinder that will minimize the amount of material to be used?

2.3 Introduction to Integration

Activity 2.17

1. Is it possible to determine the distance a vehicle has traveled if we know its velocity function?
2. Can we determine a company's total profit if we know its marginal-profit function?
3. An object moves with a velocity of $v(t) = \frac{1}{2}t$, where t is in minutes and v is in feet per minute.

   a) How far does the object travel during the first 30 min?

   b) How far does the object travel between the first hour and the second hour?
4. Find the areas of the region below $y = 2x$ and above the x-axis shown in the figure 2.16 (a) below.

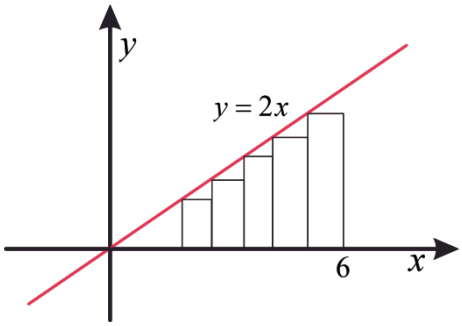


Figure 2.16 (a)

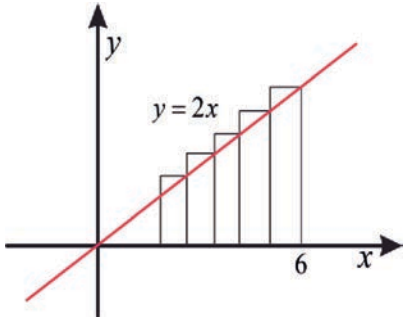


Figure 2.16 (b)

5. Find the area of the region below the line $y = 2x$ shown in the fig 2.16 (b).

   Upper sum on the interval $[0,6]$, with equal partition of $[0,6]$.
6. Find the area of the region just below the line $y = 2x$ over the interval $[0,6].$)
7. Compare the values of 4 up to 6 above.

2.3.1 Areas of Regions

Integral calculus is primarily concerned with the area below the graph of a function (specifically, the area between the graph of a function and the x-axis). In this section, we assume that all functions are non-negative; that is, $f(x) \geq 0$.

Example 1

A vehicle travels 50 mi/hr for 2 hr. How far has the vehicle traveled?

Solution

The answer is 100 miles. We treat the vehicle's velocity as a function, $v(x) = 50$. We graph this function, sketch a vertical line at $x = 2$, and obtain a rectangle. This rectangle measures 2 units horizontally and 50 units vertically. Its area is the distance the vehicle has traveled:

$v(t) = 50$

Distance $A = 2 \times 50 = 100$

$2 \text{ hr} \times \dfrac{50 \text{ miles}}{1 \text{ hr}} = 100 \text{ miles}.$

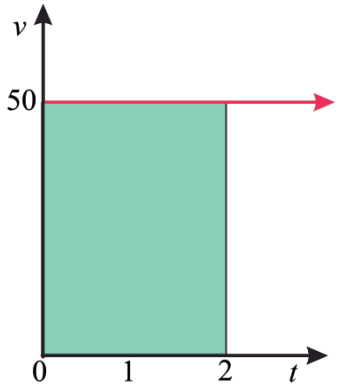


Figure 2.17

Example 2

Green Leaf Skateboards determines that for the first 50 skateboards produced, its cost is 40 birr per skateboard. What is the total cost to produce 50 skateboards?

Solution

The marginal-cost function is $C'(x) = 40,\ 0 \le x \le 50$. Its graph is a horizontal line.

If we mark off 50 units along the x-axis, we get a rectangle, as in Example 1.

The area of this rectangle is $40 \times 50 = 2000$. Therefore, the total cost to produce 50 skateboards is 2000 birr:

$$(50 \text{ skateboards} \times 40 \frac{\text{Birr}}{\text{skateboard}} = 2000 \text{ Birr}).$$

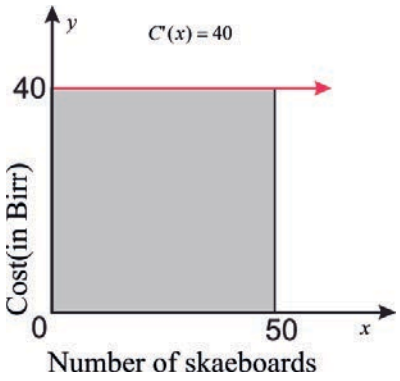


Figure 2.18

Example 3

The velocity of a moving object is given by the function $v(x) = 3x$, where x is in hours and v is in miles per hour. Use geometry to find the area under the graph, which is the distance the object has traveled during the first 3 hours $(0 \le x \le 3)$.

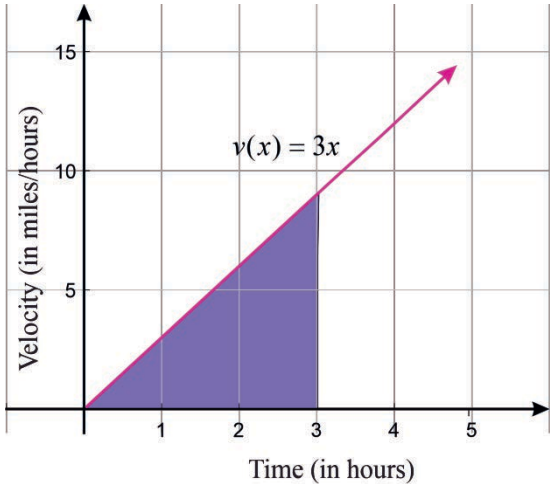


Figure 2.19

Solution

The graph of the velocity function is shown above. We see the region corresponding to the time interval $0 \le x \le 3$ is a triangle with base 3 and height 9 (since $v(3) = 9$).

Therefore, the area of this region is $A = \frac{1}{2}(3)(9) = \frac{27}{2} = 13.5$.

The object that traveled 13.5 miles during the first 3 hours.

In each of the Examples 1 through 3, the function was a rate function; its output units formed a rate (miles per hour in Examples1 and 3, Birr per skateboard in Example 2). The units of the area were derived by multiplying input units by output units.

Exercise 2.21

1. The velocity of a moving object is given by the function $v(x) = 3x$, where x is in hours and v is in miles per hour. Use geometry to find the area under the graph, which is the distance the object has traveled between the third hour and the fifth hour $(3 \le x \le 5)$.
2. A company has a marginal profit function modeled by the linear function $P'(x) = 0.15x$, where x is in months and P' is in thousands of dollars per month. Sketch this graph and use it to determine the total profit earned by the company in a year $(0 \le x \le 12)$.

Riemann summation

Activity 2.18

1. Examples 1 through 3 in this section suggest a pattern:

✓ If $f(x)=k$, where k is a non-negative constant, its graph is a horizontal line of height k. The region under this graph over the interval $[0,x]$ is a rectangle, and its area is $A=k.x$ (height times base).

✓ If $f(x)=mx$, its graph is a line of slope $m\geq 0$, passing through the origin. The region under this graph over an interval $[0,x]$ is a triangle, and its area is

$$A=\frac{1}{2}(x)(mx)=\frac{1}{2}mx^2.$$

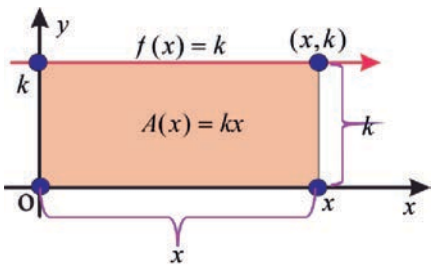


Figure 2.20

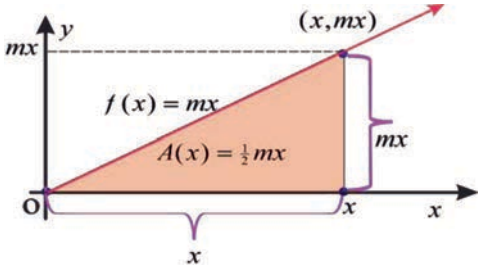


Figure 2.21

In these two cases, the area function is an anti-derivative of the function that generated the graph.

a. Is this always true?

b. Is the formula for the area under the graph of any function that function's anti-derivative?

c. How do we handle curved graphs for which area formulas may not be known?

2. Write summation notation for

i) $5+10+15+20+25$ ii) $g(x_1)\Delta x+g(x_2)\Delta x+...+g(x_{19})\Delta x.$

Example 1

A Skateboard has the following marginal-cost function for producing skateboards: For up to 50 skateboards, the cost is 40 ETB per skateboard. For quantities from 51 through 125 skateboards, the cost drops to 30 ETB per skateboard. After 125 skateboards, it drops to 25 ETB per skateboard. If x represents the number of skateboards produced, the marginal cost function C' is

$$C'(x) = \begin{cases} 40, & \text{for } 0 \le x \le 50, \\ 30, & \text{for } 50 < x \le 125, \\ 25, & \text{for } 125 < x \le 150. \end{cases}$$

Find the total cost to produce 150 skateboards.

Solution

We are extending Example 2.We calculate the areas of the rectangles formed by the horizontal lines of the graph of the marginal-cost function:

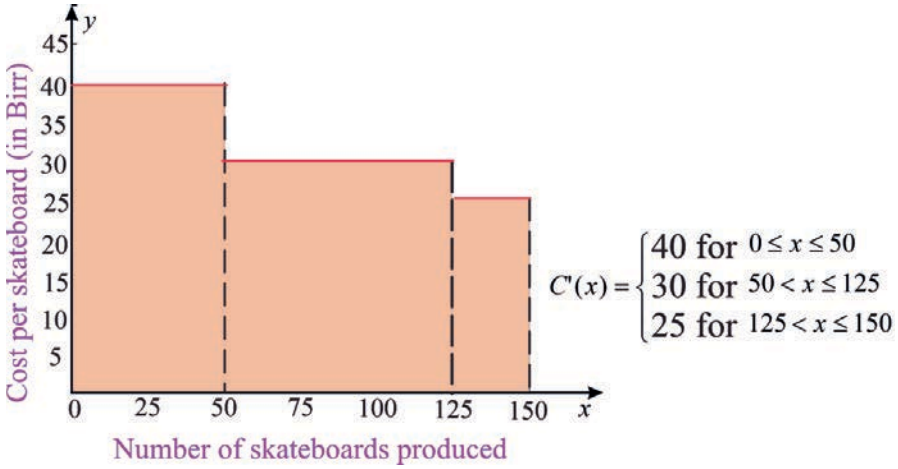


Figure 2.22

The total cost to produce 150 skateboards is found by summing those areas:

$$\text{Total cost} = (40)(50) + (30)(75) + (25)(25) = 4875 \text{ Birr.}$$

Example 1 illustrates the first steps of a Riemann summation, a method that allows us to determine the area under a curved graph. We use rectangles to approximate the area under a curve given by $y = f(x)$, a continuous function, over an interval $[a, b]$. Riemann summation is accomplished with the use of summation notation, introduced below. In Figure 2.23, $[a, b]$ is divided into four subintervals, each having width $\Delta x = \frac{(b-a)}{4}$.

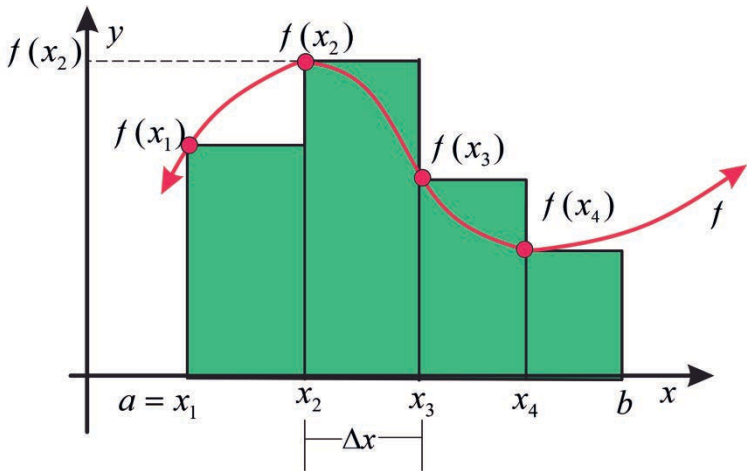


Figure 2.23

The heights of the rectangles shown are

$$f(x_1),\ f(x_2),\ f(x_3) \text{ and } f(x_4).$$

The area of the region under the curve is approximately the sum of the areas of the four rectangles:

$$f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x + f(x_4)\Delta x$$

We can denote this sum with summation notation, or sigma notation, which uses the Greek capital letter sigma, Σ:

$$\sum_{i=1}^{4} f(x_i)\Delta x$$

This is read "the sum of the product $f(x_i)\Delta x$ from $i = 1$ to $i = 4$."

Before we continue, let's consider some examples involving summation notation.

Example 2

Find the sum of expression $\sum_{i=1}^{4} 3^i$.

Solution

$$\sum_{i=1}^{4} 3^i = 3^1 + 3^2 + 3^3 + 3^4 = 120.$$

Example 3

Express $\sum_{i=1}^{30} h(x_i)\Delta x$ without using summation notation.

Solution

$$\sum_{i=1}^{30} h(x_i)\Delta x = h(x_1)\Delta x + h(x_2)\Delta x + \ldots + h(x_{30})\Delta x.$$

Approximation of area by rectangles becomes more accurate as we use smaller subintervals and hence more rectangles, as shown in the following figures.

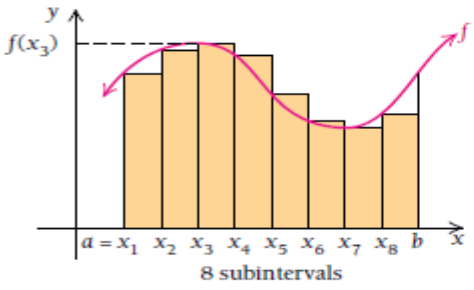


Figure 2.24a

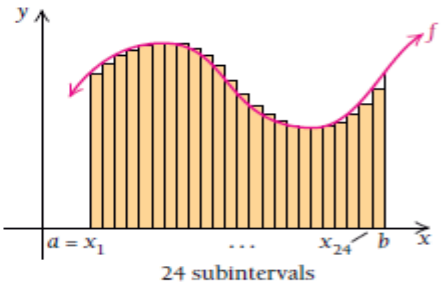


Figure 2.24b

In general, suppose that the interval $[a, b]$ is divided into n equally sized subintervals, each of width $\Delta x = \frac{(b-a)}{n}$ and we then construct rectangles with heights $f(x_1), f(x_2), \ldots, f(x_n)$.

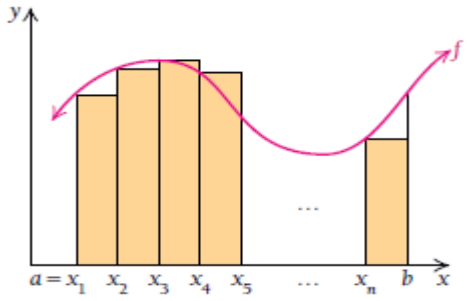


Figure 2.25

The width of each rectangle is Δx, so the first rectangle has an area of $f(x_1)\Delta x$, the second rectangle has an area of $f(x_2)\Delta x$, and so on. The area of the region under the curve is approximated by the sum of the areas of the rectangles A_n:

$$\text{Area}(A_n) = \sum_{i=1}^{n} f(x_i)\Delta x.$$

Definition 2.12

Let $p = \{x_0, x_1, x_2, ..., x_n\}$ be any partition on $[a, b]$ of a given function f. For each k between 1 and n, let t_k be an arbitrary number in $[x_{k-1}, x_k]$. Then the sum

$$f(t_1)\Delta x_1 + f(t_2)\Delta x_2 + f(t_3)\Delta x_3 + ... + f(t_n)\Delta x_n$$

is called **a Riemann sum for** f on $[a, b]$ and is denoted

$$\sum_{k=1}^{n} f(t_k)\Delta x_k.$$

Thus, $\sum_{k=1}^{n} f(t_k)\Delta x_k = f(t_1)\Delta x_1 + f(t_2)\Delta x_2 + f(t_3)\Delta x_3 + ... + f(t_n)\Delta x_n$

Riemann sum (pronounced "Ree-mahn") was developed by the great German mathematician G. F. Bernhard Riemann (1826–1866). Riemann sums can be calculated using any x-value within each subinterval. For simplicity, in this text, we will always use the left endpoint of each subinterval.

Steps for the Process of Riemann Summation

1. Draw the graph of $f(x)$.
2. Subdivide the interval $[a, b]$ into n subintervals of equal width. Calculate the width of each rectangle by using the formula $\Delta x = \frac{b-a}{n}$.
3. Construct rectangles above the subintervals such that the top left corner of each rectangle touches the graph.
4. Determine the area of each rectangle.
5. Sum these areas to arrive at an approximation for the total area under the curve.

Exercise 2.22

1. Find the approximate area of the region enclosed by the graph of function $f(x) = x^2 + 1$, the x-axis, $x = 1$ and $x = 5$.

   a. Use the following steps to find it.

   Step 1: Interval $[1, 5]$ is divided into four subintervals, each having width $\Delta x = \frac{5-1}{4} = 1$

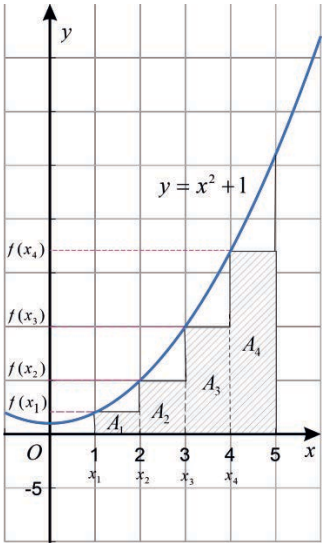


Figure 2.26

Step 2: Find the values of $f(x_1)$, $f(x_2)$, $f(x_3)$ and $f(x_4)$, when $x_1 = 1$, $x_2 = 2$, $x_3 = 3$ and $x_4 = 4$.

Step 3: Since the area of the region is approximatily the sum of the areas of four rectangles A_1, A_2, A_3 and A_4, calculate $A_1 + A_2 + A_3 + A_4 = \sum_{i=1}^{4} f(x_i)\Delta x$.

b. Find the approximate area of the same region in the same way as a, when the interval $[1, 5]$ is divided into eight subintervals, each having width $\Delta x = \frac{5-1}{8} = 0.5$.

2. Express $\sum_{i=1}^{6} (i^2 + i)$ without using summation notation.

2.3.2 Definite Integrals

Activity 2.19

Let $f(x) = 3x^2$ defined on the interval $[0,5]$. Then, find the Riemann sum of f on $[0,5]$

a. By dividing the interval $[0,5]$ in to 5 sub-intervals of equal width.

b. By dividing the interval $[0,5]$ in to 10 sub-intervals of equal width.

The key concept developed in this section is that the more subintervals we use, the more accurate the approximation of area becomes. As the number of subdivisions $dx = \frac{b-a}{n}$ increases, the width of each rectangle Δx decreases. If n is allowed to approach infinity, then Δx approaches 0; these are limits, and the approximations of the area becomes closer and closer to the true area under the graph. The exact area underneath the graph of a continuous function $y = f(x)$ over an interval $[a,b]$ is, by definition, given by a definite integral.

Definition 2.13

Let a given piecewice contionuous function $y = f(x)$ be non-negative, $f(x) \geq 0$, over an interval $[a,b]$. **A definite integral** is the Riemann sum of the areas of the rectangles under the graph of the function $y = f(x)$ over the interval $[a,b]$ as the value of n increases indefinitely (equivalently, $dx = \frac{b-a}{n}$ approaches very close to 0). That is:

$$\text{Exact area} = \sum_{i=1}^{n} f(x_i)\Delta x = \int_a^b f(x)dx, \text{ as } \Delta x \text{ gets very close to 0.}$$

Notice that the summation symbol becomes an integral sign (the elongated "s" is Leibniz notation representing "sum") and Δx becomes dx. If $f(x) \geq 0$ over an interval $[a,b]$, the definite integral represents the area.

Example

Determine the value of $\int_0^2 (3x+2)dx$.

Solution

This definite integral is a command to calculate the exact area underneath the graph of the function $f(x) = 3x + 2$ over the interval $[0,2]$.

We sketch the graph and note that the region is a trapezoid. Thus, we can use geometry to determine this area (a Riemann sum is not needed here).

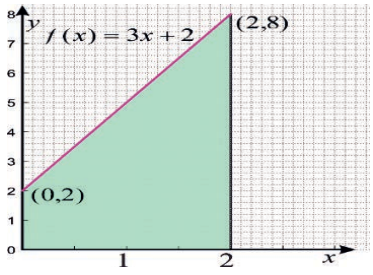


Figure 2.27

We find that the area is 10. Therefore, $\int_0^2 (3x+2)dx = 10$.

Exercise 2.23

Derermine the values of the following definite integrals:

a. $\int_0^2 (2x+1)dx$ b. $\int_0^4 2xdx$ c. $\int_1^5 xdx$ d. $\int_{-2}^3 (x+2)dx$

Anti-derivative

Activity 2.20

1. Find at least three different functions which have the derivative $2x$. Describe the similarities (and differences) between the functions.
2. Check that all of these functions:

   $f(x)=x^2+3x+1$, $g(x)=x^2+3x$, $h(x)=x^2+3x+12$, and $k(x)=x^2+3x+\pi$

   have the same derivative; $2x+3$.

Definition 2.14

The process of finding $f(x)$ from its derivative $f'(x)$ is known as anti - differentiation or integration. $f(x)$ is said to be the **anti-derivative** of $f'(x)$.

$$f(x) \xrightarrow{\text{differentiation}} f'(x)$$
$$f(x) \xleftarrow{\text{Anti-differentiation}} f'(x)$$

Integration is the reverse operation of differentiation.

- The set of all anti-derivatives of a function $f(x)$ is called the **indefinite integral** of $f(x)$.
- The indefinite integral of $f(x)$ is denoted by $\int f(x)dx$ read as "the integral of $f(x)$ with respect to x".
  - ✓ The symbol $\int$ is said to be the **integral sign**.
  - ✓ dx denotes that the variable of integration is x.
  - ✓ If a function has an integral, then it is said to be integrable.
  - ✓ If $F'(x) = f(x)$, then $\int f(x)dx = F(x) + c$.
  - ✓ c is known as the constant of integration.

In this section, you will see how to find the integrals of constant and power functions.

i. Integration of a power function

Example 1

$\int x\,dx = \frac{1}{2}x^2 + c$ Because $\frac{d}{dx}\left(\frac{1}{2}x^2 + c\right) = \frac{1}{2}\times 2x + 0 = x$

$\int x^2\,dx = \frac{1}{3}x^3 + c$ Because $\frac{d}{dx}\left(\frac{1}{3}x^3 + c\right) = \frac{1}{3}\times 3x^2 + 0 = x^2$

$\int x^3\,dx = \frac{1}{4}x^4 + c$ Because $\frac{d}{dx}\left(\frac{1}{4}x^4 + c\right) = \frac{1}{4}\times 4x^3 + 0 = x^3$

$\int 1\,dx = x + c$ Because $\frac{d}{dx}(x + c) = 1 + 0 = 1$

In general, the following formula holds. However, α is a real number and

$\alpha \neq -1$, $\int x^\alpha\,dx = \frac{1}{\alpha+1}x^{\alpha+1} + c$ (c is the constant of integration)

Note: Since $1 = x^0$, $\int 1\,dx = \int x^0\,dx = \frac{1}{0+1}x^{0+1} + c = x + c$

Example 2

Find the following indefinite integrals:

a. $\int x^7\,dx$ b. $\int x^{-3}\,dx$ c. $\int x^{\frac{1}{2}}\,dx$ d. $\int x^{-\frac{4}{3}}\,dx$

Solution

a. $\int x^7\,dx = \frac{1}{7+1}x^{7+1} + c = \frac{1}{8}x^8 + c.$

b. $\int x^{-3}\,dx = \frac{1}{-3+1}x^{-3+1} + c = -\frac{1}{2}x^{-2} + c = -\frac{1}{2x^2} + c.$

c. $\int x^{\frac{1}{2}}\,dx = \frac{1}{\frac{1}{2}+1}x^{\frac{1}{2}+1} + c = \frac{2}{3}x^{\frac{3}{2}} + c = \frac{2}{3}x\sqrt{x} + c.$

d. $\int x^{-\frac{4}{3}}\,dx = \frac{1}{-\frac{4}{3}+1}x^{-\frac{4}{3}+1} + c = \frac{5}{2}x^{-\frac{1}{3}} + c = \frac{5}{2\sqrt[3]{x}} + c.$

Note: c is the constant of integration.

Exercise 2.24

Find the following indefinite integrals:

a. $\int x^5\,dx$ b. $\int x^{-4}\,dx$ c. $\int x^{\frac{1}{3}}\,dx$ d. $\int x^{-\frac{3}{2}}\,dx$

e. $\int x^{10}\,dx$ f. $\int x^{-2}\,dx$ g. $\int x^{\frac{3}{2}}\,dx$ h. $\int x^{-\frac{3}{5}}\,dx$

ii. Properties of the indefinite integral

1. Let k be a constant, $\int kf(x)dx = k\int f(x)dx$
2. $\int\big(f(x)+g(x)\big)dx = \int f(x)dx + \int g(x)dx$
3. $\int\big(f(x)-g(x)\big)dx = \int f(x)dx - \int g(x)dx$

Note

$\int k\,dx = kx + c$, where k is a given constant and c is the constant of integration.

Example

Find the following indefinite integrals:

a. $\int 10\,dx$ b. $\int (2x^5 - x)\,dx$ c. $\int \left(x^2 + \sqrt{x}\right) dx$

Solution

$a. \int 10\,dx = 10x + c$

$b. \int (2x^5 - x)\,dx = 2 \cdot \frac{1}{6}x^6 - \frac{1}{2}x^2 + c = \frac{1}{3}x^6 - \frac{1}{2}x^2 + c$

$c. \int \left(x^2 + \sqrt{x}\right) dx = \frac{1}{3}x^3 + \frac{2}{3}x^{\frac{3}{2}} + c$

Note: Add only one constant of integration c at the end.

Exercise 2.25

Find the following indefinite integrals.

a. $\int 99\,dx$ b. $\int 10x^4\,dx$ c. $\int \frac{3}{x} dx$ d. $\int (x^2 - x)\,dx$

e. $\int (3x^2 - 2x + 5)\,dx$ f. $\int (-2x^2 - x - 6)\,dx$ g. $\int \left(x^{-2} - 4x^3 + \sqrt{x}\right) dx$

2.3.3 Area and Definite Integrals

In Sections 2.3.1 and 2.3.2, we considered the area under the graph of a function f. We have yet to establish the general rule to find the exact area under the graph of f. As we will see, we can use the anti-derivative of a function to determine the exact area under the graph of the function. This process is called integration.

Definition 2.15

Let f be a given function over the interval $[a,b]$ and F be any anti-derivative of f. Then the definite integral of f from a to b is: $\int_a^b f(x)dx = F(b) - F(a)$.

Evaluating definite integrals is called *integration*. The numbers a and b are known as *the limits of integration*.

Theorem 2.7 (The Fundamental Theorem of Calculus)

Let f be a given non-negative function over an interval $[0,b]$, and let $A(x)$ be the area between the graph of f and the x-axis over the interval $[0,x]$, with $0<x<b$. Then $A(x)$ is a differentiable function of x and $A'(x)=f(x)$.

Note

To find the area under the graph of a nonnegative function f over the interval $[a,b]$:

1. Find any anti-derivative $F(x)$ of $f(x)$. Let $C=0$ for simplicity.
2. Evaluate $F(x)$ at $x=b$ and $x=a$, and compute $F(b)-F(a)$.

   The result is the area under the graph over the interval $[a,b]$.

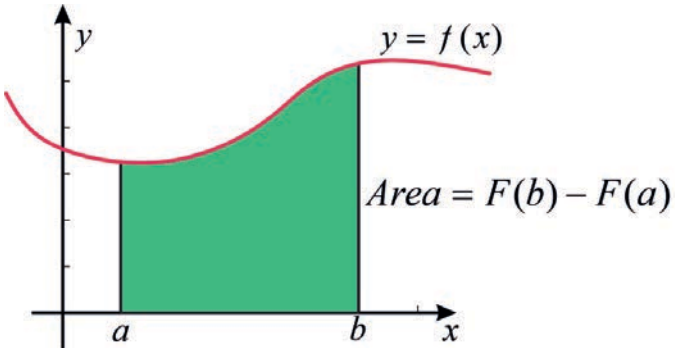


Figure 2.28

Example 1

Find the area under the graph of $y=x^2+1$ over the interval $[-1,2]$.

Solution

In this case, $f(x)=x^2+1$, with $a=-1$ and $b=2$.

1. Find any anti-derivative of $x=b$ of $f(x)$. We choose the simplest one:

$$F(x)=\frac{x^3}{3}+x.$$

2. Substitute 2 and -1, and find the difference $F(2)-F(-1)$:

$$\int_{-1}^{2}(x^2+1)dx=\left[\frac{x^3}{3}+x\right]_{-1}^{2}$$

$$=\left[\frac{2^3}{3}+2\right]-\left[\frac{(-1)^3}{3}+(-1)\right]$$

$$=6$$

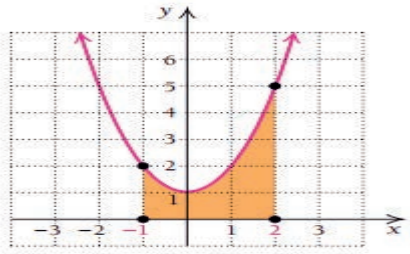


Figure 2.29

We can make a partial check by counting the squares and parts of squares shaded on the graph in Figure 2.29.

Exercise 2.26

1. Refer to the function and graph in Example 1.
   a. Calculate the area over the interval $[0,5]$.
   b. Calculate the area over the interval $[-2,2]$.
   c. Can you suggest a shortcut for part (b)?
2. Evaluate each of these integrals.

a. $\int_{1}^{3}(3x^2+2x)dx$ b. $\int_{-1}^{0}(x^3-3x+1)dx$ c. $\int_{4}^{15}(0.002x^4-0.3x^2+4x-7)dx$

Example 2

Let $y = x^3$ represent the number of kilowatts (kW) generated by a new power plant each day, x days after going online. Find the area under the graph of $y = x^3$ over the interval $[0, 5]$ and interpret the significance of the area.

Solution

In this case, $f(x) = x^3, a = 0,$ and $b = 5.$

1. Find any anti-derivative $F(x)$ of $f(x)$. We choose the simplest one:

$$F(x) = \frac{x^4}{4}$$

2. Substitute 5 and 0, and find the difference $F(5) - F(0)$:

$$F(5) - F(0) = \frac{5^4}{4} - \frac{0^4}{4} = \frac{625}{4} = 156\frac{1}{4}.$$

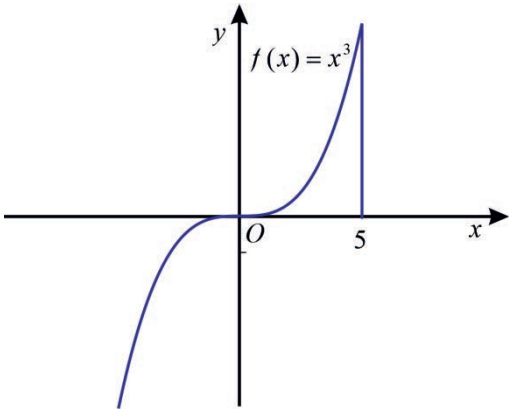


Figure 2.30

The area represents the total number of kilowatts generated during the first 5 days.

The difference $F(b) - F(a)$ has the same value for all anti-derivatives F of a function $f(x)$ whether the function is nonnegative or not. It is called the definite integral of $f(x)$ from a to b.

Properties of definite integrals

If f and g are given function on $[a,b], k \in \Re$, and $c \in [a,b]$, then

a. $\int_a^a f(x)dx = 0$

b. $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

c. $\int_a^b f(x)dx = -\int_b^a f(x)dx$

d. $\int_a^b kf(x)dx = k\int_a^b f(x)dx$

e. $\int_a^b (f(x) \pm g(x))dx = \int_b^a f(x)dx \pm \int_b^a g(x)dx$

Example 3

Evaluate $\int_a^b x^2 dx$.

Solution

Using the anti-derivative $F(x) = \frac{x^3}{3}$, we have $\int_a^b x^2 dx = \frac{b^3}{3} - \frac{a^3}{3}$.

It is convenient to use an intermediate notation: $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$,

where $F(x)$ is an anti-derivative of $f(x)$.

Example 4

Evaluate $\int_{-1}^4 (x^2 - x)dx$

Solution

$$a) \quad \int_{-1}^4 (x^2 - x)dx = \left[\frac{x^3}{3} - \frac{x^2}{2}\right]_{-1}^4$$

$$= \left(\frac{4^3}{3} - \frac{4^2}{2}\right) - \left(\frac{(-1)^3}{3} - \frac{(-1)^2}{2}\right)$$

$$= \frac{64}{3} - 8 + \frac{1}{3} + \frac{1}{2} = 14\frac{1}{6}.$$

Exercise 2.27

Evaluate the following definite integrals:

a. $\int_1^3 x^2\,dx$ b. $\int_0^2 (x^2-2x+8)\,dx$ c. $\int_{-2}^3 (-3x^2+4x-5)\,dx$ d. $\int_1^4 \left(2x+\frac{1}{x^2}\right)dx$

2.3.4 More on Area

When we evaluate the definite integral of a nonnegative function, we get the area under the graph over an interval.

Activity 2.21

Find the areas of a region bounded by the function $f(x)$ and the interval $[a,b]$, where a. $f(x)=k,\ k>0$. b. $f(x)=x$. c. $f(x)=x^2$.

Example 1

Find the area under $y=\frac{1}{x^2}$ over the interval $[1,b]$.

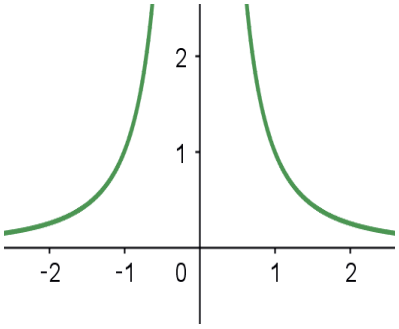


Figure 2.31

Solution

$$\int_1^b \frac{dx}{x^2}=\int_1^b x^{-2}dx=\left[\frac{x^{-2+1}}{-2+1}\right]_1^b=\left[\frac{x^{-1}}{-1}\right]_1^b=\left[-\frac{1}{x}\right]_1^b=\left(-\frac{1}{b}\right)-\left(-\frac{1}{1}\right)=1-\frac{1}{b}.$$

The definite integral is also defined for $f(x)<0$. We can use geometry to determine the value of some definite integrals, as the following example suggests.

Example 2

Compare the definite integrals of the functions $y = x^2$ and $y = -x^2$

Solution

$$\int_0^2 x^2 dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3}.$$

and

$$\int_0^2 -x^2 dx = \left[-\frac{x^3}{3}\right]_0^2 = -\frac{2^3}{3} - \frac{0^3}{3} = -\frac{8}{3}.$$

The graphs of the functions $y = x^2$ and $y = -x^2$ are reflections of each other across the x-axis (see Figure2.32a and Figure2.32b). Thus, the shaded areas are the same, $\frac{8}{3}$. The evaluation procedure for $y = -x^2$ gave us $-\frac{8}{3}$. This illustrates that for negative-valued functions, the definite integral gives us the opposite of the area between the curve and the x-axis.

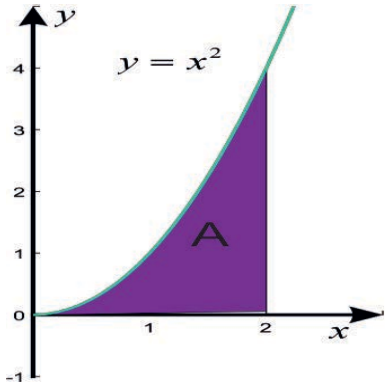


Figure 2.32a

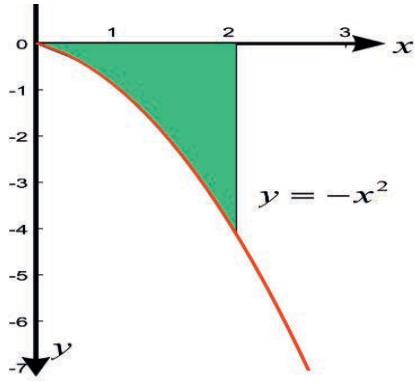


Figure 2.32b

When $f(x) \geq 0$ in the interval $[a,b]$, the area S which is enclosed by the graph of $f(x)$, the two straight lines $x = a, x = b$ and the x-axis is $S = \int_a^b f(x)dx$

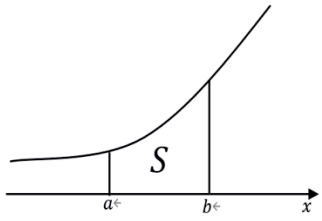


Figure 2.33

When $f(x) \leq 0$ in the interval $[a, b]$, the areas S which is enclosed by the graph of $f(x)$, the two straight lines $x = a, x = b,$ and the x-axis is

$$S = -\int_a^b f(x)dx$$

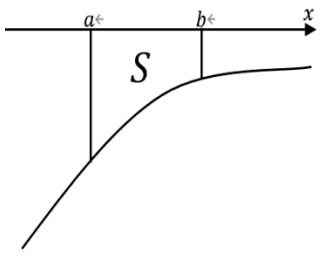


Figure 2.34

Example 3

Find the area S which is enclosed by the graph of $f(x)$and the x-axis.

a. $f(x) = -x^2 + 1$ b. $f(x) = x^2 - 1$

Solution

a. From the figure on the right, the area S is

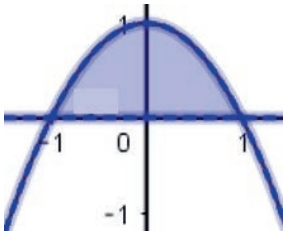


Figure 2.35

$$S = \int_{-1}^{1}(-x^2 + 1)\,dx$$

$$= \left[-\frac{x^3}{3} + x\right]_{-1}^{1}$$

$$= \left(-\frac{1^3}{3} + 1\right) - \left\{-\frac{(-1)^3}{3} + (-1)\right\}$$

$$= \frac{2}{3} + \frac{2}{3} = \frac{4}{3}$$

a. From the figure on the right, the area S is

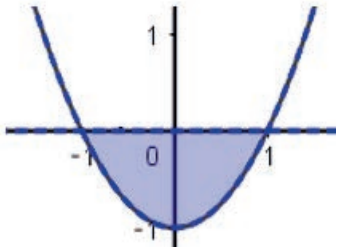


Figure 2.36

$$S = -\int_{-1}^{1}(x^2 - 1)\,dx$$

$$= -\left[\frac{x^3}{3} - x\right]_{-1}^{1}$$

$$= -\left[\left(\frac{1^3}{3} - 1\right) - \left\{\frac{(-1)^3}{3} - (-1)\right\}\right] = -\left(-\frac{2}{3} - \frac{2}{3}\right) = \frac{4}{3}$$

When the area S is divided into the upper side and the lower side of the x-axis, the upper area and the lower side area are obtained respectively and added them.

In the figure below, the area S of the area enclosed by the graph of $f(x)$ and the x-axis can be found by $S = A + B$.

In the interval $[a, b]$, $f(x) \leq 0$. In the interval $[b, c]$, $f(x) \geq 0$.

So, we can find the area S,

$$S = -\int_a^b f(x)\,dx + \int_b^c f(x)\,dx$$

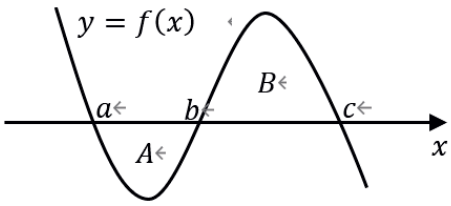


Figure 2.37

Example 4

Find the area S of the region enclosed between the graph of $f(x) = x^2 - 1$, the x-axis, and the y-axis in the interval $[0, 2]$.

Solution

Note that 1 is the x-intercept of the function $f(x) = x^2 - 1$ in $[0, 2]$.

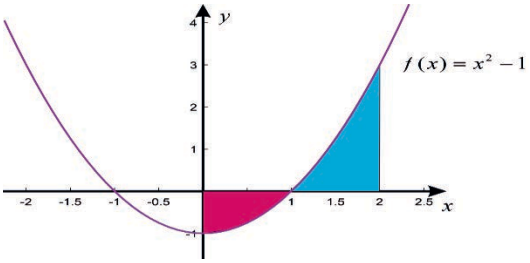


Figure 2.38

$$S = \int_0^2 (x^2 - 1)dx = -\int_0^1 (x^2 - 1)dx + \int_1^2 (x^2 - 1)dx$$

$$= -\left[\frac{x^3}{3} - x\right]_0^1 + \left[\frac{x^3}{3} - x\right]_1^2$$

$$= -\left[\left(\frac{1^3}{3} - 1\right) - \left(\frac{0^3}{3} - 0\right)\right] + \left[\left(\frac{2^3}{3} - 2\right) - \left(\frac{1^3}{3} - 1\right)\right]$$

$$= -\left(-\frac{2}{3}\right) + \frac{4}{3} = \frac{2}{3} + \frac{4}{3} = 2$$

This shows that the area above the x-axis exceeds the area below the x-axis by 2 unit. So, the area of the region is 2 units.

The definite integral of a continuous function over an interval is the sum of the areas above the x-axis minus the sum of the areas below the x-axis.

Example 5

Consider $\int_{-1}^{2}(-x^3+3x-1)dx$. Predict the sign of the result by examining the graph, and then evaluate the integral.

Solution

From the graph, it appears that there is considerably more area below the x-axis than above. Thus, we expect that

$$\int_{-1}^{2}(-x^3+3x-1)dx<0.$$

Evaluating the integral, we have

$$\int_{-1}^{2}(-x^3+3x-1)dx=\left[-\frac{x^4}{4}+\frac{3}{2}x^2-x\right]_{-1}^{2}$$

$$=\left(-\frac{2^4}{4}+\frac{3}{2}.2^2-2\right)-\left(-\frac{(-1)^4}{4}+\frac{3}{2}.(-1)^2-(-1)\right)$$

$$=(-4+6-2)-\left(-\frac{1}{4}+\frac{3}{2}+1\right)$$

$$=0-2\frac{1}{4}=-2\frac{1}{4}.$$

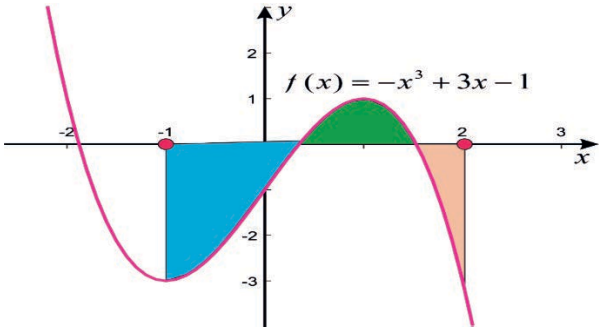


Figure 2.39

As a partial check, we note that the result is negative, as predicted.

Exercise 2.28

1. Find the area S which is enclosed by the graph of $f(x)$ and the x-axis.
   a. $f(x) = x^2 - 4$ b. $f(x) = -x^2 + 4$
   c. $f(x) = x^2 - 2x$ d. $f(x) = -x^2 - 2x$
2. Find the area S which is enclosed by the graph of $f(x)$, the two straight lines and the x-axis.
   a. $f(x) = x^2$, the two straight lines $x = 0$ and $x = 4$
   b. $f(x) = x^2 + 3$, the two straight lines $x = -1$ and $x = 3$
3. Find the area S which is enclosed between the graph of $f(x) = x(x + 2)(x - 1)$ and the x-axis.

2.3.5 Application of Integration

Activity 2.22

Do you think that integration has any other applications than finding the area under the curve of the function? Write down what you think these applications might be.

In this section, we will look at some applications of integrals. Integration is used in different fields of study such as business, physical sciences, economics, etc.

Example 1

A certain Airlines determines that the marginal profit resulting from the sale of x seats on a plane traveling from Addis Ababa to New York City, in hundreds of dollars, is given by $P'(x) = \sqrt{x} - 6$.

Find the total profit when 60 seats are sold.

Solution

We integrate to find $P(60)$:

$$P(60)=\int_0^{60}P'(x)dx=\int_0^{60}(\sqrt{x}-6)dx=\left[\frac{2}{3}x^{\frac{3}{2}}-6x\right]_0^{60}\approx -50.1613.$$

When 60 seats are sold, the airline's profit is -$5016.13. That is, the airline will lose $5016.13 on the flight.

Example 2

Suppose that the velocity function of a car is $v(t)=5t^4$ and that the displacement at t = 0 is $s(0)=9$. Find $s(t)$. Assume that $s(t)$ is in meters and $v(t)$ is in meter per second.

Solution

First $s(t)=\int v(t)dt=\int 5t^4dt=t^5+C.$

Next, we determine C by using the initial condition $s(0)=9$, which is the displacement of the car at time $t=0$: $s(0)=0^5+C=9\Rightarrow C=9$ Thus, $s(t)=t^5=9.$

Example 3

The driver of a vehicle traveling at 40 mi/ hr (58.67 ft /sec) applies the brakes, softly at first, then harder, coming to a complete stop after 7 seconds. The velocity as a function of time is modeled by $v(t)=-1.197t^2+58.67$, where v is in feet per second, t is in seconds, and $0\le t\le 7$. How far did the vehicle travel while the driver was braking?

Solution

The distance traveled $s(t)$ is given by the definite integral of $v(t)$:

$$s(t) = \int_0^7 (-1.197t^2 + 58.67)dt = \left[-\frac{1.197}{3}t^3 + 58.67t \right]_0^7$$

$$= -\frac{1.197}{3}(7)^3 + 58.67(7) - 0$$

$$\approx 273.83 \text{ ft}$$

Exercise 2.29

1. Pure Water Enterprises finds that the marginal profit, in dollars, from drilling a well that is x feet deep is given by $P'(t) = \sqrt[5]{x}$. Find the profit when a well 250 ft deep is drilled.
2. A company estimates that its sales will grow continuously at a rate given by the function $S'(t) = 20e^t$, where $S'(t)$ is the rate at which sales are increasing, in dollars per day, on day t.
   a. Find the accumulated sales for the first 5 days.
   b. Find the sales from the 2^{nd} day through the 5^{th} day.

Summary

- **Rate of change, gradient, and derivative of functions**
  - A **rate of change** is a rate that describes how one quantity changes in relation to another quantity.
  - If $y = f(x)$ then the **average rate of change** of y with respect to x on the interval $[a, b]$ is
    $\frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a}$.
  - A **secant line** is a line that intersects the graph of the function $y = f(x)$ at exactly two points.
  - A **tangent line** to a curve or a graph of a function $y = f(x)$ is a line that touches the curve exactly at one point. The point where the tangent line touches the graph is said to be the point of tangency.
  - A **gradient** is a measure of steepness and direction of a line.
    $\text{Gradient} = \frac{\text{vertical change}}{\text{horizontal change}} = \frac{\Delta y}{\Delta x}$.
  - The derivative of a function $f(x)$ at a number a in the domain of f, denoted by $f'(a)$ is the gradient of a tangent line to the graph of $f(x)$ at $(a, f(a))$. That is as h gets very close to zero from both sides, $f'(a) = \frac{f(a+h) - f(a)}{h}$.

- **The derivatives of some functions**
  1. If $f(x) = k$, then $f'(x) = 0$.
  2. If $f(x) = x^n$, then $f'(x) = nx^{n-1}$.
  3. If $f(x) = \sqrt{x}$, then $f'(x) = \frac{1}{2\sqrt{x}}$.

- **Derivatives of combinations of functions**

If f and g are differentiable functions, then $f+g$, kf, fg and $\frac{f}{g}$ are also differentiable and are given as follows:

i. $(f+g)'(x)=f'(x)+g'(x)$.

ii. $(kf)'(x)=kf'(x)$, where k is a constant number.

iii. $(fg)'(x)=f'(x)g(x)+f(x)g'(x)$.

iv. $\left(\frac{f}{g}\right)'(x)=\frac{f'(x)g(x)-f(x)g'(x)}{(g(x))^2}$

- **Differentiations of the composition of functions**

✓ Let g be differentiable at x, and f be differentiable at $g(x)$. Then fog is differentiable at x, and $(\text{fog})'(x)=f'(g(x))g'(x)$ for all x such that g is differentiable at x and f is differentiable at $g(x)$.

✓ If y is a function of u, and u is a function of x, then

$$\frac{dy}{dx}=\frac{dy}{du}\frac{du}{dx}$$

- **Equation of tangent line and normal line to a curve**

a. The equation of the tangent line to the curve $y=f(x)$ at $x=a$ is given by

$$y=f'(a)(x-a)+f(a).$$

b. The equation of the normal line to the curve $y=f(x)$ at $x=a$ and $f'(x)\neq 0$ is given by

$$y=\frac{-1}{f'(a)}(x-a)+f(a).$$

- A **critical value** of a function f is any number c in the domain of f for which the tangent line at $(c,f(c))$ is horizontal or for which the derivative does not exist. That is, c is a critical value if $f(c)$ exists and $f'(c)=0$ or $f'(c)$ does not exist.

- Let I be the domain of f. $f(c)$ is a **relative minimum** if there exists within I an open interval I_1 containing c such that $f(c) \le f(x)$, for all x in I_1 and is a **relative maximum** if there exists within I an open interval I_2 containing c such that $f(c) \ge f(x)$, for all x in I_2.
- **The First-Derivative Test for Relative Extrema**

  Let f be a function that is differentiable on an interval I. Then

  i. If f' changes sign from positive to negative at c then f has **a local maximum** value at c for some critical value c.

  ii. If f' changes sign from negative to positive at c then f has **a local minimum** value at c for some critical value c.
- The **anti-derivative** of $f(x)$ is the set of functions $F(x)+C$ such that $\frac{d}{dx}\left(F(x)+C\right)=f(x)$, where the constant C is called the **constant of integration.**
- Let f be continuous on $[a,b]$, and let $p=\{x_0,x_1,x_2,...,x_n\}$ be any portion of $[a,b]$. Then the sum $\sum_{k=1}^{n} f(t_k)\Delta x_k = f(t_1)\Delta x_1 + f(t_2)\Delta x_2 + f(t_3)\Delta x_3 + ... + f(t_n)\Delta x_n$ is called **a Riemann sum for** f on $[a,b]$.
- Fundamental Theorem of Calculus:

  If $f(x)=F'(x)$, then $\int_a^b f(x)dx = F(b)-F(a)$.
- If a function has areas both below and above the $x-$axis, the definite integral gives the net total area or the difference between the sum of the areas above the x-axis and the sum of the areas below the x-axis.
  - ✓ If there is more area above the x-axis than below, the definite integral will be positive.

- ✓ If there is more area below the x-axis than above, the definite integral will be negative.
- ✓ If the areas above and below the x-axis are the same, the definite integral will be 0.

- Properties of definite integrals

a. $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

b. $\int_a^b \big(f(x) \pm g(x)\big)dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$

c. $\int_a^b \big(f(x) \pm g(x)\big)dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$

d. $\int_a^b f(x)dx = -\int_b^a f(x)dx$

e. $\int_a^b \big(f(x) \pm g(x)\big)dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$

Review Exercise

1. Find the derivative of each of the following functions at the given number.

   a. $f(x)=\dfrac{x^3+3x}{x^2}$ at $x=2$

   b. $f(x)=\dfrac{5}{x^3+2}$ at $x=-4$

   c. $f(x)=\dfrac{x^4}{x^3+2}$ at $x=2$

   d. $f(x)=\dfrac{4\sqrt{x}}{2x+1}$ at $x=4$

2. The diagram in Figure 2.40 shows the graph of the function $y=\dfrac{1}{x}$ between $x=\dfrac{1}{2}$ and $x=4$.

   i. Find the slope of the secant line between

      a. $x=3$ and $x=3.1$.

      b. $x=3$ and $x=3.01$.

      c. $x=3$ and $x=3.001$.

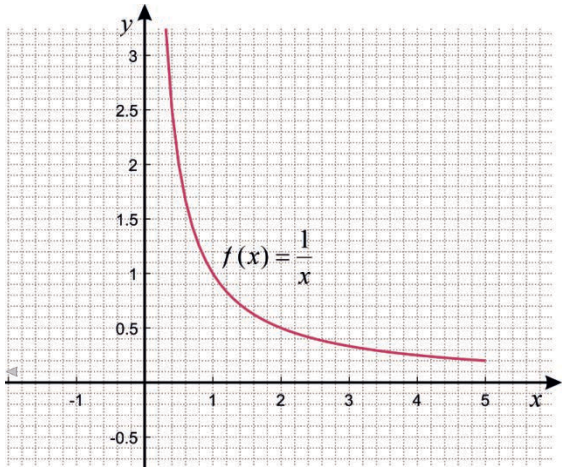


Figure 2.40

   ii. Find a formula for the slope of the secant line between $(3, f(3))$ and $(3+\Delta x, f(3+\Delta x))$ for a function f.

   iii. Determine what happens when Δx approaches 0.

3. If an object is dropped from an 80 m high window, its height y above the ground at the time t seconds is given by the formula $y=f(t)=80-4.9t^2$ (neglect air resistance).

   i. Find the average velocity of the falling object between

      a. $t=1$ sec and $t=1.1$ sec.

      b. $t=1$ sec and $t=1.01$ sec.

      c. $t=1$ sec and $t=1.001$ sec.

ii. Find a simple formula for the average velocity of the falling object between $t = 1$ sec and $t = (1 + \Delta t)$ sec.

iii. Determine what happens to this average velocity as Δt approaches 0.

4. Let $y = f(t) = t^2$ where t is the time in seconds and y is the distance in meters that an object falls on a certain airless planet. Draw a graph of this function between $t = 0$ and $t = 3$.

i. Make a table of the average speed of the falling object between

a. $t = 2$ sec and $t = 3$ sec.

b. $t = 2$ sec and $t = 2.1$ sec.

c. $t = 2$ sec and $t = 2.01$ sec.

ii. Find a simple formula for the average speed between time $t = 2$ sec and $t = (2 + \Delta t)$ sec.

iii. In your formula for average speed determine what happens as Δt approaches zero.

iv. Draw the straight line through the point $(2, 4)$ whose slope is the instantaneous velocity you just computed.

5. Find the equation of the tangent and the normal lines $f(x) = (1 - x^3)\sqrt{x + 2}$.

6. Find the derivative of each of the following functions.

a. $f(x) = x^2(2x + 5)$

b. $f(x) = \dfrac{x^2 + x}{x^3 - x + 2}$

c. $f(x) = \dfrac{x^2}{x + 1}$

d. $f(x) = x^3\left(1 - \frac{1}{x}\right)\left(x^2 - \frac{1}{x^2}\right)$

7. Consider the function $f(x) = 6x^5 - 50x^3 - 120$.

a. Find the interval on which f is increasing.

b. Find the interval on which f is decreasing.

c. Find all values of x for which f have a relative maximum.

d. Find all values of x for which f have a relative minimum.

8. A particle moves along the curve $2y = x^2 + 4$. Find the point(s) on the curve at which the y-coordinate changes 4 times as fast as the x-coordinate.

9. Let $f(x) = x^2 + px + q$. Find the values of p and q such that $f(1) = 3$ is an extreme value of f on the interval [0, 2]. Is this value maximum or minimum?

10. Find the value of a, b, c, d so that $h(x) = ax^3 + bx^2 + cx + d$ will be extreme values at $(1, 2)$ and $(2, 3)$.

11. Find the absolute maximum and the minimum values of the following function on the interval [0,2].

$$f(x) = \begin{cases} x^3 - \dfrac{x}{3}; & 0 \le x \le 1 \\ x^2 + x - \dfrac{4}{3}; & 1 < x \le 2 \end{cases}$$

12. A board 5 ft long slides down a wall. At the instant when the bottom end is 4 ft from the wall, the other end is moving down at a rate of 2 ft/s. At that moment
    a. How fast is the bottom end sliding along the ground?
    b. How fast is the area of the region between the board, the ground, and the wall changing?

13. A farmer has 240 m of fencing material and wants to fence a rectangular field that borders a straight river. (No fence is needed along the river). What are the dimensions of the field that has the largest area?

14. The radius of the balloon is increasing at the rate of 0.5 cm/sec. At what rate the surface area of the balloon is increasing when the radius is 4 cm?

15. A cyclist decelerates at a constant rate from 30 km/hr to a standstill in 45 sec.
    a. How fast is the bicyclist traveling after 20 sec?
    b. How far has the bicyclist traveled after 45 sec?

16. A factory is polluting a lake in such a way that the rate of pollutants entering the lake at time t, in months, is given by $N'(t) = 280t^{\frac{3}{2}}$, where N is the total number of pounds of pollutants in the lake at time t.

Figure 2.41

a. How many pounds of pollutants enter the lake in 16 months?

b. An environmental board tells the factory that it must begin to clean up procedures after 50,000 pounds of pollutants has entered the lake. After what length of time will this occur?

UNIT 3

STATISTICS

Unit Outcomes

By the end of this unit, you will be able to:

- Describe absolute and relative dispersion and their interpretation.
- Conceptualize specific facts about measurement in statistical data.
- Grasp basic concepts about sampling techniques.
- Appreciate the value of statistics in real life.

Unit Contents

3.1 Measures of Absolute Dispersions

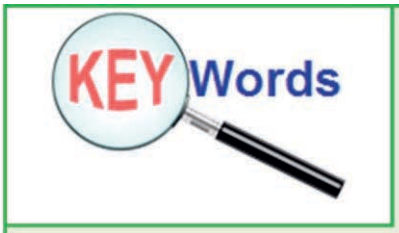
3.2 Relative Dispersions

3.3 Use of Frequency Curves

3.4 Sampling Techniques

Summary

Review Exercise



- Cluster sampling
- Coefficient of mean deviation
- Coefficient of quartile deviation
- Coefficient of range
- Coefficient of variation
- Coefficient of variation
- Inter-quartile
- Lower class boundary of the lowest class
- Mean deviation
- Multistage sampling
- Non-random sampling technique
- Quartile deviation
- Random sampling technique
- Simple random sampling
- Skewness
- Standard deviation
- Stratified sampling
- Symmetrical distribution
- Systematic sampling
- Upper class boundary
- Variation

Introduction

Statistics as a science deals with the proper collection, organization, presentation, analysis and interpretation of numerical data. Since statistics allows us to draw conclusions based on data it is applicable in almost all sciences. Some of the organizations where statistics is applicable are business, meteorology, schools, economic, social and political activities. In previous grades, you should have studied about statistics and its basic concepts such as collection of data, presentation of data, and measures of central tendencies like mean, median and mode and measures of dispersion such as range, variance, and standard deviation. In this unit, we shall revise those concepts and discuss additional measures of dispersion; namely, inter-quartile range, mean deviation, quartile deviation and standard deviations. In addition, we shall discuss some relative measures of dispersion such as the coefficient of dispersions. Finally, sampling techniques will be discussed.

3.1 Measures of Absolute Dispersions

In Grade 9, you learned about the different measures of dispersion. In this section, we shall revise measures of dispersion and discuss measure of absolute dispersion.

Activity 3.1

1. Why do you study dispersion?
2. What is the difference between measures of dispersion and measures of central tendency?

Measures of Dispersion

Recall that a measure of dispersion can be defined in either of the following ways:

a. The degree to which numerical data tends to spread about an average.

b. The scatter or variation of variables about a central value.

Types of measures of dispersion:

There are two types of measures of dispersion. These are:

i. Absolute measure of dispersion.

ii. Relative measure of dispersion.

i. Absolute measure of dispersion

The following are absolute measures of dispersions

- Range
- Inter-quartile range
- Mean deviation
- Standard deviation

Note

An absolute measure of dispersion measures the variability in terms of the same units of the data, for example if the units of the data are meters, liters, kg, etc. The units of the measures of dispersion will also be meters, liters, kg, etc.

ii. Relative measure of dispersion

The common relative measures of dispersion are:

- Coefficient of range
- Coefficient of quartile deviation
- Coefficient of mean deviation
- Coefficient of standard deviation or coefficient of variation

Note

A relative measure of dispersion compares the variability of two or more data that are independent of the units of measurement. It is used to compare the dispersion of a data set with the dispersion of another data set.

The major difference between absolute and relative measures of dispersion is that the absolute measure of dispersion measures only the variability of the data. Furthermore, it has the same unit of measurement. However, relative measure of dispersion is used to compare the variation of two or more distributions, and does not have units.

3.1.1 Range and Inter-quartile Range

Range

Activity 3.2

Consider the following data set which gives the price (in Birr) of a cup of coffee in randomly selected 20 coffee houses:

10, 5, 9, 12, 10, 12, 13, 8, 15, 12 , 12, 13, 10, 12, 5, 14, 6, 14, 10, 8.

a. Identify the highest price and lowest price of a cup of coffee.

b. Determine the difference between the highest price and the lowest price of a cup of coffee.

Range for ungrouped data

Definition 3.1

The range of an ungrouped set of data is defined as the difference between the largest value and smallest value in a set of data, that is, $R = L - S$, where L= largest value and S = smallest value.

Example 1

A certain supermarket has registered the following data about daily sales (in Birr 100) for twelve consecutive days:

$$115, 134, 120, 111, 122, 103, 80, 155, 122, 104, 110, 98.$$

Calculate the range of the data.

Solution

L = 155, S = 80, then R = L-S = $155 - 80 = 75$.

Range for grouped data

Definition 3.2

The range of a grouped set of data R is defined as the difference between the upper class boundary of the highest class $B_U(H)$, and the lower class boundary of the lowest class $B_L(L)$, that is, $R = B_U(H) - B_L(L)$.

Example 2

Consider the following data. What is the range of this distribution?

x	5-9	10-14	15-19	20-24
f	4	8	3	2

Solution

From the grouped frequency distribution, the range is calculated using

$$B_U(H) = 24.5, \qquad B_L(L) = 4.5.$$

Therefore, $R = B_U(H) - B_L(L) = 24.5 - 4.5 = 20.$

Advantages and Limitations of range

Advantage of range

- It is simple to compute.

Limitations of range

- It only depends on extreme values.
- It does not consider variations of values in between.
- It is highly affected by extreme values.

Exercise 3.1

1. Calculate the range of each of the following data sets:
   a. 25, 21, 22, 20, 19, 16
   b. 14, 17, 18, 16, 10, 18, 19, 22.
2. Calculate the range for these data sets.
   a.

x	5	7	8	9	11
f	13	10	11	12	13

   b.

x	0-20	20-40	40-60	60-80	80-100
f	3	1	2	4	1

Inter-Quartile Range

Activity 3.3

1. The following are the average daily temperatures (in°C) of ten days for two cities A and B:

   City A: 28, 24, 25, 26, 26, 20, 27, 30, 28, 27

   City B: 22, 24, 23, 26, 25, 25, 24, 26, 27, 28.

   a. Calculate the first quartile (Q_1) and the third quartile (Q_3) for each city.

   b. Determine $Q_3 - Q_1$ for each city.

2. Calculate Q_1 and Q_3 for the following data set:

x	0-5	5-10	10-15	15-20
f	1	3	3	2

In the above section, we mentioned range as the difference between the highest and the lowest values in a data set. Sometimes, it may not be possible to find the range, especially in open ended data where the highest or lowest value may be unknown. It may be the case that you know that the range is highly affected by extreme values. Under such circumstances, it may be of interest to measure the difference between the third quartile (Q_3) and the first quartile (Q_1) which is called the inter-quartile range. Inter-quartile range is a measure of variation which could possibly overcomes the limitations of range. It is defined as follows:

Definition 3.3

Inter-quartile range (IQR) is a difference between the upper and the lower quartiles; that is, $IQR = Q_3 - Q_1$.

Recall that the $k^{th}(k = 1, 2, 3)$ quartiles for ungrouped data are

a. odd $Q_k = \left(\frac{k(n+1)}{4}\right)^{th}$ item if n is odd,

b. even $Q_k = \left(\frac{\left(\frac{kn}{4}\right)+\left(\frac{kn}{4}+1\right)}{2}\right)^{th}$ item if n is even, where n is the number of observations of the data.

The $k^{th}(k = 1, 2, 3)$ quartiles for a grouped frequency distribution is

$$Q_k = L + \left(\frac{\frac{k(n)}{4} - cf}{f}\right)w,$$

where, L = lower boundary of the class containing the median or median class, cf = the cumulative frequency in the class preceding the class containing the median, f = the number of observations (frequency) in the class containing the median and w = the size of the class interval; that is, the width of the median class.

Example 3

Consider the following two sets of data:

A: 20, 70, 70, 70, 70, 70, 70,70, 100

B: 20, 30, 50, 80, 90, 100.

a. Find the ranges of A and B.

b. Find the inter-quartile ranges of A and B.

Solution

a. The ranges of A and B are R = 100 – 20 = 80 and R = 100 – 20 = 80, respectively, from which you see that they have the same range. However, if you observe the two sets of data, you can see that B is more variable than A.

b.

For A:

$Q_1 = \frac{1(n+1)}{4} = \frac{9+1}{4} = \frac{10^{th}}{4}$ item=2.5^{th} .

Therefore, Q_1 =2^{nd} item+ 0.5(3^{nd} item-2^{nd} item) $= 70 + 0.5(70 - 70) = 70.$

$Q_3 = \frac{3(n+1)}{4} = \frac{3(9+1)}{4} = \frac{30^{th}}{4}$ item=7.5^{th}.

Therefore, Q_3 =7^{th} item+ 0.5(8^{th} item-7^{th} item) $= 70 + 0.5(70 - 70) = 70.$

Thus, $IQR = 70 - 70 = 0.$

For B:

$$Q_1 = \left(\frac{\left(\frac{n}{4}\right)+\left(\frac{n}{4}+1\right)}{2}\right)^{th} = \left(\frac{\left(\frac{6}{4}\right)+\left(\frac{6}{4}+1\right)}{2}\right)^{th} = 2^{nd}\text{item}=30.$$

$$Q_3 = \left(\frac{\left(\frac{kn}{4}\right)+\left(\frac{kn}{4}+1\right)}{2}\right)^{th} = \left(\frac{\left(\frac{3\times6}{4}\right)+\left(\frac{3\times6}{4}+1\right)}{2}\right)^{th} \text{item}=5^{th}\text{item} = 90.$$

Thus, $IQR = 90 - 30 = 60.$

From the above computation, you clearly see that B possesses higher variability than A. The greater the measure of variation, the higher the dispersion of the data set.

Limitation of Inter-Quartile Range

- It only depends on two values Q_1 and Q_3. It does not consider the variability of each item in the data set.
- It ignores 50% of the data (the top 25% above $Q_3$and the bottom 25% below Q_1. It only considers the middle 50% of the values between Q_1 and Q_3.

Exercise 3.2

1. Calculate the inter-quartile range of each of the following data sets:
   a. 25, 21, 22, 20, 19, 16 b. 14, 17, 18, 16, 10, 18, 19, 22.
2. Calculate the inter-quartile range for these data sets:

a.

x	5	7	8	9	11
f	13	10	11	12	13

b.

x	22	23	24	25
f	1	2	3	1

c.

x	0-20	20-40	40-60	60-80	80-100
f	3	1	2	4	1

d.

x	0-9	10-19	20-29	30-39
f	4	2	3	1

3.1.2 Mean Deviation and Quartile Deviation

Mean deviation for ungrouped data (1)

Activity 3.4

1. The following data set represents mathematics examination scores of fifteen students graded out of 15:

   10, 11, 12, 10, 9, 10, 12, 13, 8, 15, 14, 10, 9, 15, 10

   a. Find the mean of the data.

   b. Find the deviation of each value from the mean.

   c. Determine the mean of these deviations.

   d. Determine the mean of these deviations by considering the absolute values of each deviation.

   e. Compare the results you obtained in parts (c) and (d).

2. Find the mean, median and mode for the following data set.

x	10	11	12	13
f	3	1	2	4

3. Find the mean, median and mode for the following data describing the height of students in a certain class and given in continuous grouped data form as:

Height(cm)	150-154	155-159	160-164	165-169
f	3	1	2	4

Definition 3.4

Mean deviation of data is the sum of all deviations (in absolute value) of each item from the average value divided by the total number of items.

i. Mean deviation about the mean $MD(\bar{x})$

Steps to find or compute mean deviation from the mean:

Step 1: Find the mean of the data set.

Step 2: Find the deviation of each item from the mean regardless of sign (since mean deviation assumes absolute value).

Step 3: Find the sum of the deviations.

Step 4: Divide the sum by the total number of items in the data set.

That is,

$$MD(\bar{x}) = \frac{|x_1 - \bar{x}| + |x_2 - \bar{x}| + |x_3 - \bar{x}| + \ldots + |x_n - \bar{x}|}{n} = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}$$

Example 1

Find the mean deviation about the mean of the following data.

50, 80, 80, 110, 110, 110, 140, 160.

Solution

Mean deviation about the mean

Step 1: Calculate the mean of the data set.

$$\bar{x} = \frac{50 + 80 + 80 + 110 + 110 + 110 + 140 + 160}{8} = \frac{840}{8} = 105$$

Step 2: Find the absolute deviation of each data item from the mean.

x	50	80	80	110	110	110	140	160
$\lvert x_i - \bar{x} \rvert$	55	25	25	5	5	5	35	55

Step 3: Find the sum of the deviations.

$$\sum |x_i - \bar{x}| = |50 - 105| + |80 - 105| + |80 - 105| + |110 - 105| + |110 - 105| + |110 - 105| + |140 - 105| + |160 - 105|$$

$$= 210.$$

Step 4: Divide the sum of the deviation by the total number of items in the data set.

$$MD(\bar{x}) = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n} = \frac{210}{8} = 26.25$$

Exercise 3.3

Calculate the mean deviation about the mean of the following data.

a. 15, 11, 12, 20, 19, 6, 11

b. 4, 7, 8, 9, 10, 10, 11, 12, 13, 17

Mean deviation for ungrouped data (2)

ii. Mean deviation about the median $MD(m_d)$

Here, we simply need to replace the role of the mean by the median and follow each step as above. This will give us the mean deviation about the median to be:

$$MD(m_d) = \frac{|x_1 - m_d| + |x_2 - m_d| + |x_3 - m_d| + \ldots + |x_n - m_d|}{n}$$

$$= \frac{\sum_{i=1}^{n} |x_i - m_d|}{n}$$

Example 2

Find the mean deviation about the median of the following data.

50, 80, 80, 110, 110, 110, 140, 160.

Solution

Step 1: Calculate the median of the data set.

$$m_d = \frac{\left(\frac{8}{2}\right)^{th} \text{item} + \left(\frac{8}{2}+1\right)^{th} \text{item}}{2} = \frac{110 + 110}{2} = 110$$

Step 2: Find the absolute deviation of each data item from the median.

x	50	80	80	110	110	110	140	160
$\lvert x_i - m_d \rvert$	60	30	30	0	0	0	30	50

Step 3: Find the sum of the deviations.

$$\sum_{i=1}^{n} |x_i - m_d| = |50 - 110| + |80 - 110| + |80 - 110| + |110 - 110| + |110 - 110| + |110 - 110| + |140 - 110| + |160 - 110| = 200.$$

Step 4: Divide the sum of the deviations by the total number of items in the data set.

$$MD(m_d) = \frac{\sum_{i=1}^{n} |x_i - m_d|}{8} = \frac{200}{8} = 25$$

Exercise 3.4

Calculate the mean deviation about the median of the following data.

a. 15, 11, 12, 20, 19, 6, 11

b. 4, 7, 8, 9, 10, 10, 11, 12, 13, 17

Mean deviation of ungrouped data (3)

iii. Mean deviation about the mode $MD(m_o)$

Again, proceed in a similar way as in the case of i and ii above. We get mean deviation about the mode to be:

$$MD(m_o) = \frac{|x_1 - m_o| + |x_2 - m_o| + |x_3 - m_o| + \ldots + |x_n - m_o|}{n}$$

$$= \frac{\sum_{i=1}^{n} |x_i - m_o|}{n}$$

Example 3

Find the mean deviation about the mode of the following data.

50, 80, 80, 110, 110, 110, 140, 160.

Solution

Mean deviation about the mode

Step 1: Identify the mode of the data set; mode = 110.

Step 2: Find the absolute deviation of each data item from the mode.

x	50	80	80	110	110	110	140	160
$\lvert x_i - m_o \rvert$	60	30	30	0	0	0	30	50

Step 3: Find the sum of the deviations.

$$\sum_{i=1}^{n} |x_i - m_o| = |50 - 110| + |80 - 110| + |80 - 110| + |110 - 110| + |110 - 110| + |110 - 110| + |140 - 110| + |160 - 110| = 200.$$

Step 4: Divide the sum of the deviations by the total number of items in the data set.

$$MD(m_o) = \frac{\sum_{i=1}^{n} |x_i - m_o|}{8} = 25$$

Exercise 3.5

Calculate the mean deviation about the mode of the following data.

a. 15, 11, 12, 20, 19, 6, 11 b. 4, 7, 8, 9, 10, 10, 11, 12, 13, 17

Mean deviation for discrete frequency distributions (1)

To calculate the mean deviation for a discrete frequency distribution about the mean, the median and the mode, you take similar steps as in the process for discrete data.

If $x_1, x_2, \ldots, x_n$ are values with corresponding frequencies $f_1, f_2, \ldots, f_n$, then the mean deviation is computed as follows:

i. Mean deviation from the mean

To calculate the mean deviation from the mean, take the following steps:

Step 1: Find the mean of the data set.

Step 2: Find the deviation of each item from the mean regardless of sign (since mean deviation assumes absolute value).

Step 3: Multiply each deviation by its corresponding frequency.

Step 4: Find the sum of these deviations multiplied by their frequencies.

Step 5: Divide the sum by the sum of the frequencies in the data set.

$$MD(\bar{x}) = \frac{f_1|x_1 - \bar{x}| + f_2|x_2 - \bar{x}| + f_3|x_3 - \bar{x}| + \ldots + f_n|x_n - \bar{x}|}{f_1 + f_2 + f_3 + \cdots + f_n}$$

$$= \frac{\sum_{i=1}^{n} f_i|x_i - \bar{x}|}{\sum_{i=1}^{n} f_i}$$

Example 4

Find the mean deviation of the following data about the mean.

x	f
6	2
10	6
12	11
20	13
25	8
26	2
30	4

Solution

Mean deviation of the data about the mean

Step 1: Calculating the mean:

$$\bar{x} = \frac{2\times 6 + 6\times 10 + 11\times 12 + 13\times 20 + 8\times 25 + 2\times 26 + 4\times 30}{2+6+11+13+8+2+4} = 18.2.$$

Step 2: Calculate the deviations from the mean:

x	f	$\lvert x-\bar{x}\rvert$	$f\lvert x-\bar{x}\rvert$
6	2	12.2	24.4
10	6	8.2	49.2
12	11	6.2	68.2
20	13	1.8	23.4
25	8	6.8	54.4
26	2	7.8	15.6
30	4	11.8	47.2
Total	$\sum f = 46$		$\sum f\lvert x-\bar{x}\rvert = 282.4$

Step 3: Thus, mean deviation about the mean will be

$$MD(\bar{x}) = \frac{\sum_{i=1}^{n} f_i|x_i - \bar{x}|}{\sum_{i=1}^{n} f_i} = \frac{282.4}{46} = 6.1.$$

Exercise 3.6

Calculate the mean deviation about the mean of the following data.

x	f
10	2
20	4
30	7
40	6
50	1

Mean deviation for discrete frequency distributions (2)

ii. Mean deviation about the median

Here, we simply need to replace the role of the mean by the median and follow each step as above. This will give us the mean deviation about the median to be:

$$MD(m_d) = \frac{f_1|x_1 - m_d| + f_2|x_2 - m_d| + f_3|x_3 - m_d| + \ldots + f_n|x_n - m_d|}{f_1 + f_2 + f_3 + \cdots + f_n}$$

$$= \frac{\sum_{i=1}^{n} f_i|x_i - m_d|}{\sum_{i=1}^{n} f_i}$$

Note that: Median for ungrouped data is:

a. $m_d = \left(\frac{(n+1)}{2}\right)^{th}$ item if n is odd

b. $m_d = \left(\frac{\left(\frac{n}{2}\right)+\left(\frac{n}{2}+1\right)}{2}\right)^{th}$ item if n is even.

Example 5

Find the mean deviation of the following data about the median.

x	f
6	2
10	6
12	11
20	13
25	8
26	2
30	4

Solution

Step 1: Calculating the median:

The median

$$m_d = \frac{\left(\frac{46}{2}\right)^{th} \text{ item} + \left(\frac{46}{2}+1\right)^{th} \text{ item}}{2} = \frac{20+20}{2} = 20$$

Step 2: Calculate the deviations from the median:

Step 3: Thus, the mean deviation about the median will be

$$MD(m_d) = \frac{\sum_{i=1}^{n} f_i|x_i - m_d|}{\sum_{i=1}^{n} f_i} = \frac{268}{46}$$

$$= 5.8.$$

x	f	$\lvert x-m_d\rvert$	$f\lvert x-m_d\rvert$
6	2	14	28
10	6	10	60
12	11	8	88
20	13	0	0
25	8	5	40
26	2	6	12
30	4	10	40
Total	$\sum f$ =46		268

Exercise 3.7

Calculate the mean deviation about the median of the following data.

x	f
10	2
20	4
30	7
40	6
50	1

Mean deviation for Discrete Frequency Distributions (3)

iii. Mean deviation about the mode

The steps that we need to follow here are also the same, but we shall use the mode instead of the mean or the median. Following the steps, we will get the mean deviation about the mode to be:

$$MD(m_o) = \frac{f_1|x_1 - m_o| + f_2|x_2 - m_o| + f_3|x_3 - m_o| + \ldots + f_n|x_n - m_o|}{f_1 + f_2 + f_3 + \cdots + f_n}$$

$$= \frac{\sum_{i=1}^{n} f_i|x_i - m_o|}{\sum_{i=1}^{n} f_i}$$

Example 6

Find the mean deviation of the following data set about the mode.

x	f
6	2
10	6
12	11
20	13
25	8
26	2
30	4

Solution

Step 1: Calculating the mode:

The mode $m_o = 20$.

Step 2: Calculate the deviations from the mode:

x	f	$\lvert x\text{-}m_o\rvert$	$f\lvert x\text{-}m_o\rvert$
6	2	14	28
10	6	10	60
12	11	8	88
20	13	0	0
25	8	5	40
26	2	6	12
30	4	10	40
Total	$\sum f$ =46		$\sum f\lvert x - m_o\rvert = 268$

Step 3: Thus, the mean deviation about the mode will be

$$MD(m_o) = \frac{\sum_{i=1}^{n} f_i|x_i - m_o|}{\sum_{i=1}^{n} f_i} = \frac{268}{46} = 5.8.$$

Exercise 3.8

Calculate the mean deviation about the mode of the following data.

x	f
10	2
20	4
30	7
40	6
50	1

Mean deviation for grouped frequency distributions (1)

a. Mean deviation of the data about the mean

For continuous grouped frequency distributions, mean deviation is calculated in the same way as the above except that each x_i is substituted by the midpoint (m_i) of each class.

This will give us the mean deviation about the mean to be:

$$MD(\bar{x}) = \frac{f_1|m_1 - \bar{x}| + f_2|m_2 - \bar{x}| + f_3|m_3 - \bar{x}| + \ldots + f_n|m_n - \bar{x}|}{f_1 + f_2 + f_3 + \cdots + f_n}$$

$$= \frac{\sum_{i=1}^{n} f_i|m_i - \bar{x}|}{\sum_{i=1}^{n} f_i}$$

Example 7

Find the mean deviation about the mean for the following.

x	0-5	6-11	12-17	18-23	24-29
f	5	8	7	10	3

Solution

First, you have to find the mean of the distribution

x	f	m	fm	Cf
0-5	5	2.5	12.5	5
6-11	8	8.5	68	13
12-17	7	14.5	101.5	20
18-23	10	20.5	205	30
24-29	3	26.5	79.5	33
Total	$\sum f = 33$		$\sum fm = 466.5$	

Step 1: Calculating the mean.

$$\text{The mean} = \bar{x} = \frac{\sum fm}{\sum f} = \frac{466.5}{33} = 14.14$$

Step 2: Calculate the deviations from the mean:

x	f	m	$f\lvert m-\bar{x}\rvert$
0-5	5	2.5	58.20
6-11	8	8.5	45.12
12-17	7	14.5	2.52
18-23	10	20.5	63.6
24-29	3	26.5	37.08
Total	$\sum f = 33$		$\sum f\lvert m - \bar{x}\rvert = 206.52$

Step 3: Thus, mean deviation about the mean will be

$$\frac{\sum_{i=1}^{n} f_i |m_i - \bar{x}|}{\sum_{i=1}^{n} f_i} = \frac{206.52}{33} = 6.26.$$

Exercise 3.9

Calculate the mean deviation about the mean of the following data set.

x	0-9	10-19	20-29	30-39	40-49
f	4	3	6	5	2

Mean deviation for grouped frequency distributions (2)

b. Mean deviation of the data about the median

The mean deviation about the median is given by:

$$MD(m_d) = \frac{f_1|m_1 - m_d| + f_2|m_2 - m_d| + f_3|m_3 - m_d| + \ldots + f_n|m_n - m_d|}{f_1 + f_2 + f_3 + \cdots + f_n}$$

$$= \frac{\sum_{i=1}^{n} f_i|m_i - m_d|}{\sum_{i=1}^{n} f_i}$$

Note that the median of grouped data is given by

$$m_d = L + \left(\frac{\frac{n}{2} - cf}{f}\right) w$$

where, L = lower boundary of the class containing the median or median class, cf = the cumulative frequency in the class preceding the class containing the median, f = the number of observations (frequency) in the class containing the median and w = the size of the class interval that is, the width of the median class.

Example 8

Find the mean deviation about the median for the following.

x	0-5	6-11	12-17	18-23	24-29
f	5	8	7	10	3

Solution

Step 1: Calculating the median:

$$m_d = L + \frac{(\frac{n}{2} - cf)}{f} = 11.5 + 5\left(\frac{16.5 - 13}{7}\right) = 14$$

Step 2: Calculate the deviations from the median:

x	f	m	Cf	$f\|m-m_d\|$
0-5	5	2.5	5	57.5
6-11	8	8.5	13	44
12-17	7	14.5	20	3.5
18-23	10	20.5	30	65
24-29	3	26.5	33	37.5
Total	$\sum f = 33$			$\sum f\|m - m_d\| = 207.5$

Step 3: Thus, mean deviation about the median will be

$$\frac{\sum_{i=1}^{n} f_i|m_i - m_d|}{\sum_{i=1}^{n} f_i} = \frac{207.5}{33} = 6.29.$$

Exercise 3.10

Calculate the mean deviation about the median of the following data.

x	0-9	10-19	20-29	30-39	40-49
f	4	3	6	5	2

Mean deviation for grouped frequency distributions (3)

c. Mean deviation of the data about the mode

The mean deviation about the mode is:

$$MD(m_o) = \frac{f_1|m_1 - m_o| + f_2|m_2 - m_o| + f_3|m_3 - m_o| + \ldots + f_n|m_n - m_o|}{f_1 + f_2 + f_3 + \cdots + f_n}$$

$$= \frac{\sum_{i=1}^{n} f_i|m_i - m_o|}{\sum_{i=1}^{n} f_i}$$

Note that the mode of grouped data given by

$$m_o = L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) w$$

where, L = lower class boundary of the modal class, Δ_1 = the difference between the frequency of the modal class (f_{m_o}) and the frequency of the preceding class (pre-modal class) = $f_{m_o} - f_1$

Δ_2 = the difference between the frequency of the modal class and the frequency of the subsequent class (next class) = $f_{m_o} - f_2$ and w= size of the class interval.

Example 9

Find the mean deviation about the mode for the following.

x	0-5	6-11	12-17	18-23	24-29
f	5	8	7	10	3

Solution

Step 1: Calculating the mode:

$$\text{The mode } m_o = L + \left(\frac{\triangle_1}{\triangle_1+\triangle_2}\right)w = 17.5 + \frac{3}{3+7} = 19.3.$$

Step 2: Calculate the deviations from the mode.

X	f	m	$f\|m-m_o\|$
0-5	5	2.5	82.5
6-11	8	8.5	84
12-17	7	14.5	31.5
18-23	10	20.5	15
24-29	3	26.5	22.5
Total	$\sum f = 33$		$\sum f\|m - m_o\| = 235.5$

Step 3: Thus, mean deviation about the mode will be

$$\frac{\sum_{i=1}^{n} f_i|m_i - m_o|}{\sum_{i=1}^{n} f_i} = \frac{235.5}{33} = 7.1.$$

Exercise 3.11

Calculate the mean deviation about the mode of the following data.

x	0-9	10-19	20-29	30-39	40-49
f	4	3	6	5	2

Advantage and Limitation of Mean Deviation

From the above discussions on mean deviation, it can be noticed that mean deviation can be useful for applications. If our average is "Arithmetic mean", you take the deviation about the mean. Similarly, if our average is "Median", then you take the deviation about the median and if our average is the "Mode", you take mean deviation about the mode. To decide which one of the mean deviations to use in a given situation, consider the following points:

i. If the degree of variability in a set of data is not very high, use of the mean deviation about the mean since it is comparatively the best for interpretation.

ii. Whenever there is an extreme value that can affect the mean, however, the mean deviation about the median is preferable.

Though mean deviation has some advantages, it is not commonly used for interpretation. Rather, it is the standard deviation that is commonly used and which tends to be the best measure of variation.

Advantages of mean deviation

- Compared to range, mean deviation has some advantages. Range considers only two values, whereas mean deviation takes each value into consideration.

Limitations of mean deviation

- It ignores signs of deviation by taking the absolute value of a deviation which violates the rules of algebra.

Exercise 3.12

1. Calculate the mean deviation about the mean, median and mode of each of the following data sets:

   a. 12, 9, 15, 12, 7, 10, 12. b. 4, 7, 8, 9, 10, 10, 11, 12, 13, 17

2. Calculate the mean deviation about the mean, median and mode for the following data sets:

   a.

x	12	13	14	15
f	10	12	9	11

   b.

x	0-4	5-9	10-14	15-19	20-24
f	3	1	2	4	1

Quartile deviation of ungrouped data

Activity 3.5

A certain shop has registered the following data on daily sales (in 100 Birr) for one week:

$300, 450, 540, 600, 250, 350, 420$

1. Calculate the range and inter-quartile range of the data.
2. Compare the values you obtained in question 1.

Definition 3.5

Quartile Deviation (QD) is half of the difference between the upper and lower quartiles, that is, $QD = \frac{Q_3 - Q_1}{2}$.

Note

Steps to calculate quartile deviation from ungrouped data:

Step 1: Arrange the data in ascending order.

Step 2: Assign serial numbers to each item. The first serial number is assigned to the lowest test score, while the last serial number is assigned to the highest item.

Step 3: Determine the first quartile Q_1 and third quartile Q_3.

Step 4: Use the obtained values in locating the serial numbers of the item that falls under Q_1 and Q_3.

Step 5: Subtract Q_1 from Q_3 and divide the difference by 2.

Example 10

Find the quartile deviation of the following ungrouped data.

$96, 70, 100, 89, 78, 56, 45, 78, 68, 42, 66, 89, 90, 54, 44, 67, 87, 97, 98$

Solution

Step 1: First arrange the data in ascending order:

$$42, 44, 45, 54, 56, 66, 67, 68, 70, 78, 78, 87, 89, 89, 90, 96, 97, 98, 100.$$

Step 2: Assign serial numbers to each score

x	42	44	45	54	56	66	67	68	70	78	87	87	89	89	90	96	97	98	100
S No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

Step 3: Determine the first quartile (Q_1) and third quartile (Q_3).

$$Q_1 = \frac{1(n+1)}{4} = \frac{19+1}{4} = \frac{20^{th}}{4} \text{ item} = 5^{th} \text{item and}$$

$$Q_3 = \frac{3(n+1)}{4} = \frac{3(19+1)}{4} = \frac{60^{th}}{4} \text{ item} = 15^{th} \text{item.}$$

Step 4: Use the obtained values in locating the serial numbers of the score that falls under (Q_1) and (Q_3); that is,

5^{th}item $= 56$ and 15^{th}item $= 90$.

Step 5: Subtract (Q_1) from (Q_3) and divide the difference by 2.

$$QD = \frac{Q_3 - Q_1}{2} = \frac{90 - 56}{2} = \frac{34}{2} = 17.$$

Exercise 3.13

Calculate the quartile deviation of each of the following data sets.

a. 105, 101, 102, 109, 60, 100, 120

b. 40, 70, 60, 90, 110, 100, 111, 120, 130, 170

Quartile deviation of grouped data

The formula for i^{th} quartile is $Q_i = L + \left(\frac{\frac{i(n)}{4} - cf}{f} \right) \times h, \; i = 1, 2, 3,$

where

L: The lower limit of the i^{th} quartile class

$n = \sum f$: Total number of observations

f : Frequency of the i^{th} quartile class

cf : Cumulative frequency of the class previous to i^{th} quartile class

h : The class width

Example 11

The following table gives the amount of time (in minutes) spent on the internet each evening by a group of 56 students. Calculate the inter-quartile range for the following frequency distribution.

Time spent on internet (x)	10-12	13-15	16-18	19-21	22-24
Number of students (f)	3	12	15	24	2

Solution

Class interval	*Class Boundary*	*f*	*cf*
10-12	9.5-12.5	3	3
13-15	12.5-15.5	12	15
16-18	15.5-18.5	15	30
19-21	18.5-21.5	24	54
22-24	21.5-24.5	2	56
Total		56	

In order to evaluate the Quartile Deviation of grouped data, first we find

$$Q_i = \left(\frac{i(n)}{4}\right)^{th} \text{ value}, \quad i = 1, 2, 3,$$

where n is the total number of observations.

The first Quartile Q_1 is calculated as

$$Q_1 = \left(\frac{i(n)}{4}\right)^{th} \text{ value} = \left(\frac{1(56)}{4}\right)^{th} \text{ value} = (14)^{th} \text{ value}.$$

The cumulative frequency just greater than or equals to 14 is 15. The corresponding class, 12.5-15.5 is the first quartile class. Thus,

$L = 12.5$, the lower limit of the 1^{st} quartile class

$n = 56$, total number of observations

$f = 12$, frequency of the 1^{st} quartile class

$cf = 3$, cumulative frequency of the class previous to the 1^{st} quartile class

$h = 3$, the class width

The first quartile Q_1 can be computed as follows:

$$Q_1 = L + \left(\frac{\frac{i(n)}{4} - cf}{f} \right) \times h$$

$$= 12.5 + \left(\frac{\frac{1(56)}{4} - 3}{12} \right) \times 3 = 12.5 + \left(\frac{14-3}{12} \right) \times 3$$

$$= 12.5 + 2.75$$

$$= 15.25 \text{ minutes}$$

Thus, 25% of the students spent less than or equal to 15.25 minutes on the internet.

The third Quartile Q_3 is calculated as

$$Q_3 = \left(\frac{i(n)}{4} \right)^{th} \text{ value} = \left(\frac{3(56)}{4} \right)^{th} \text{ value} = (42)^{th} \text{ value.}$$

The cumulative frequency just greater than or equals to 42 is 54. The corresponding class is 18.5-21.5 is the third quartile class. Thus,

$L = 18.5$, the lower limit of the 3^{rd} quartile class

$n = 56$, total number of observations

$f = 24$, frequency of the 3^{rd} quartile class

$cf = 30$, cumulative frequency of the class previous to the 3^{rd} quartile class

$h = 3$, the class width

The third quartile Q_3 can be computed as follows:

$$Q_3 = L + \left(\frac{\frac{i(n)}{4} - cf}{f} \right) \times h$$

$$= 18.5 + \left(\frac{\frac{3(56)}{4} - 30}{24} \right) \times 3 = 18.5 + \left(\frac{42 - 30}{24} \right) \times 3$$

$$= 18.5 + 1.5$$

$$= 20 \text{ minutes}$$

Thus, 75% of the students spent less than or equal to 20 minutes on the internet.

Therefore, the Quartile Deviation (QD) is calculated as

$$QD = \frac{Q_3 - Q_1}{2} = \frac{20 - 15.25}{2} = 2.375 \text{ minutes.}$$

Exercise 3.14

Calculate the quartile deviation for the following data sets.

a.

x	110	130	140	150
f	2	3	2	1

b.

x	0-4	5-9	10-14	15-19	20-24
f	3	1	2	4	1

3.1.3 Variance and Standard Deviation

Variance for ungrouped data

Activity 3.6

The following are the average daily temperatures (in $°C$) for one week in a certain city: **5, 16, 16, 10, 17, 20, 14**

1. Calculate the range, inter-quartile range and mean deviation about the mean of the data.
2. Compare the results you obtained in (1).

In the previous section, we discussed measures of absolute dispersions or variations. In this section, we will discuss about dispersions or variation of items/values in a data set in terms of the average/mean of the data. That is, in order to measure the extent by which the distributions of the data values are dispersed or spread from the mean, such variability is defined in definition 3.6.

Definition 3.6

The variance of a data set is the average of the squared difference of each value from the mean of the data set.

Raw data is a list of values which are given in raw form.

For Example: 6, 5, 9, 13, 11, 8, 9, 9

If $x_1, x_2, \ldots, x_n$ are n observed values, then variance σ^2 for the sample data is given by

$$\sigma^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 + \cdots . + (x_n - \bar{x})^2}{n} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}$$

where, n = number of values and $\bar{x}$ = mean.

Note

Steps to find variance for ungrouped data:

Step 1: Find the mean of the data.

Step 2: Find the deviation of each value from the mean and square it.

Step 3: Add the squared deviations.

Step 4: Divide the sum obtained in step 3 by n.

Example 1

Find the variance of the following data.

$$16, 12, 8, 18, 5, 9, 24, 20.$$

Solution

Calculate the mean. The mean $(\overline{x})$ is:

$$\bar{x} = \frac{16 + 12 + 8 + 18 + 5 + 9 + 24 + 20}{8} = 14$$

Calculate the deviation of each value from the mean and square it.

x	x-$\bar{x}$	$(x - \bar{x})^2$
16	2	4
12	-2	4
8	-6	36
18	4	16
5	-9	81
9	-5	25
24	10	100
20	6	36
Total		$\sum(x_i - \bar{x})^2 = 302$

Thus, $\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n} = \frac{302}{8} = 37.75.$

Exercise 3.15

Find the variance of the following data set:

5, 10, 4, 7, 9

Variance for ungrouped frequency distribution

Ungrouped frequency distribution is a table that describes values a certain data with their respective frequencies and listed in ascending or descending order. For example, given value and frequency like the following table.

Value(v)	x_1	x_2	x_3	...	x_n
Frequency(f)	f_1	f_2	f_3	...	f_n

If $x_1, x_2, \ldots, x_n$ are values with corresponding frequencies $f_1, f_2, \ldots, f_n$, the variance is given by

$$\sigma^2 = \frac{f_1(x_1 - \bar{x})^2 + f_2(x_2 - \bar{x})^2 + f_3(x_3 - \bar{x})^2 + \cdots . + f_n(x_n - \bar{x})^2}{f_1 + f_2 + f_3 + \cdots + f_n}$$

$$= \frac{\sum_{i=1}^{n} f_i(x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}$$

where, n = number of values and $\bar{x}$ = mean.

Note

Steps to compute/find variance from ungrouped frequency:

Step 1: Find the mean of the distribution.

Step 2: Find the deviation of each value from the mean and square it.

Step 3: Multiply the squared deviations by their corresponding frequencies and add them.

Step 4: Divide the sum obtained in step 3 by $\sum f_i$.

Example 2

Find the variance of the following data set.

x	2	3	4	6
f	5	4	2	1

Solution

Calculate the mean of the distribution.

x	f	$x - \bar{x}$	$f(x - \bar{x})^2$
2	5	-1	5
3	4	0	0
4	2	1	2
6	1	3	9
Total	$\sum f_i = 12$		$\sum f(x_i - \bar{x})^2 = 16$

The mean

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{36}{12} = 3$$

Calculate the deviation of each value from the mean and square it.

The required variance

$$\sigma^2 = \frac{\sum_{i=1}^{n} f_i(x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i} = \frac{16}{12} = 1.33.$$

Exercise 3.16

Find the variance of the following data set:

x	10	20	30	40	50
f	4	6	5	3	2

Variance for grouped data

Before we look at finding variance for grouped data, let us state the basic steps that are similar to variance for ungrouped data as follows:

Note

Steps to find variance from a grouped frequency distribution:

Step 1: Find the class mark for each class.

Step 2: Find the mean of the grouped data.

Step 3: Find the deviation of each class mark from the mean and square it.

Step 4: Find the sum of the squared deviations multiplied by their respective frequencies.

Step 5: Divide the sum obtained in step 4 by $\sum f$

Therefore, based on the above steps, the variance for grouped data is obtained by using the formula:

$$\sigma^2 = \frac{\sum_{i=1}^{n} f_i(m_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}$$

where m_i is the midpoint of each class or class mark.

Example 3

Find the variance of the following data set.

x	0-4	4-8	8-12	12-16
f	4	8	2	1

Solution

Calculate the mean of the distribution.

$$\bar{x} = \frac{\sum f_i m_i}{\sum f_i} = \frac{90}{15} = 6$$

Calculate the deviation of each value from the mean and square it.

x	f	m_i	$f(m-\bar{x})^2$
0-4	4	2	64
4-8	8	6	0
8-12	2	10	32
12-16	1	14	64
Total	$\sum f = 15$		$\sum f(m_i - \bar{x})^2 = 160$

The required variance is, therefore,

$$\sigma^2 = \frac{\sum_{i=1}^{n} f_i(m_i - \bar{x})^2}{\sum_{i=1}^{n} f_i} = \frac{160}{15} = 10.67.$$

Exercise 3.17

Find the variance of the following data.

x	0-10	10-20	20-30	30-40	40-50
f	2	8	5	4	1

Standard deviation

Activity 3.7

The following data represent the daily sales (in 1000 Birr) of certain supermarket for ten days. **30, 45, 54, 60, 25, 35, 42, 80, 70, 40**

a. Compute the different measures of dispersion (mean, range, inter-quartile range, quartile deviation).

b. Discuss similarities and differences between the different measures of dispersion.

From the previous discussion, you know that mean deviation considers all the data values. However, the mean deviation assumes only the absolute deviations of each data value from the central value (mean, median or mode). Hence, it misses algebraic considerations. To overcome the limitation of mean deviation, a better measure of variation which is known as standard deviation can be used.

The standard deviation is the most commonly used measure of dispersion. The value of the standard deviation tells us how closely the values of a data set are clustered around the mean. In general, a lower value of the standard deviation for a data set indicates that the values of the data set are spread over a relatively small range around the mean. On the other hand, a large value of the standard deviation for a data set indicates that the values of that data set are spread over a relatively large range around the mean.

Definition 3.7

The standard deviation of a data set is the positive square root of a variance. The standard deviation is represented by the Greek σ (sigma).

If $x_1,\ x_2,\ \ldots, x_n$ are n observed values, then standard deviation for the ungrouped data is given by $\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n}}$ where, n = number of values and $\bar{x}$ = mean.

The standard deviation for grouped data is given by $\sigma = \sqrt{\frac{\sum_{i=1}^{n} f_i(m_i-\bar{x})^2}{n}}$, where m_i is the midpoint of each class or class mark and f_i is the frequency of group i.

Example 4

Find the standard deviation of the data in Examples 1, 2 and 3 above.

Solution

Standard deviation of the data in Example 1 is $\sigma = \sqrt{37.75} = 6.14$, standard deviation of the data in Example 2 is $\sigma = \sqrt{1.33} = 1.15$ and standard deviation of the data in Example 3 is $\sigma = \sqrt{10.67} = 3.27$.

Advantages and limitations of standard deviation

Advantages of standard deviation

- It is rigidly defined.
- It is based on all observations.

Limitations of standard deviation

- The process of squaring deviations and then taking the square root of their mean is complicated.
- It attaches great weight to extreme values since the squared deviations are used.

Exercise 3.18

1. Determine the variance and standard deviation of the following data set:

   $6, 7, 4, 11, 10, 11, 12, 13, 17$

2. Calculate the variance and standard deviation for these data sets:

a.

x	1	2	3	5	6	7
f	2	3	2	1	2	2

b.

x	10-19	20-29	30-39	40-49
f	2	3	1	2

3.2 Interpretation of Relative Dispersions

Activity 3.8

1. Why do we study relative dispersion?
2. Two basketball teams scored the following points in ten different games:

   Team A: 30, 91, 0, 64, 42, 80, 30, 5, 117, 71

   Team B: 53, 46, 48, 50, 53, 53, 58, 60, 57, 52

   a. Calculate the mean of each team's scores.

   b. Using the results obtained in part (a), compare and discuss the performance of the two teams.

In statistics, measures of dispersions help us to interpret the variability of the values in a data set; that is, they enable us to measure how homogeneous or heterogeneous the values in a data set are. In simple terms, they show how squeezed or scattered the values in a data set are.

3.2.1 Coefficient of Range

Activity 3.9

Consider the following two sets of data:

Data A: $2, 7, 7, 7, 7, 7, 7, 10$

Data B: $2, 3, 5, 8, 9, 10$

1. Calculate the range of each data set.
2. Based on the value you obtained in question 1, what is your interpretation of the two data?

Coefficient of range is a relative measure based on the value of range and defined as follows:

Definition 3.8

Coefficient of Range $CR = \frac{L-S}{L+S}$, where L = largest value and S = smallest value.

Example 1

Let us take two sets of observations of marks of students. Set A contains the marks of seven students in Physics out of 25 and set B contains marks of the same number of students in Mathematics out of 50 as shown below:

Physics: 10, 15, 18, 20, 20, 21, 21

Mathematics: 20, 25, 26, 27, 30, 35, 35

a. Find the ranges of the sets A and B.

b. Calculate the coefficients of ranges of the sets A and B.

Solution

Physics: 10, 15, 18, 20, 20, 21, 21

Mathematics: 20, 25, 26, 27, 30, 35, 35

The range and coefficient of range are calculated as:

Subjects	Range	CR
Physics	21-10=11	$\frac{21-10}{21+10}=0.35$
Mathematics	35-20=15	$\frac{35-20}{35+20}=0.27$

From the above computation, we can observe that the range is 11 in Physics and that the range is 15 in Mathematics. This is based on absolute measures of dispersion but not relative measures of dispersion. However, the reality is that the two subjects cannot be compared directly as their base is not the same. When we convert these two values into coefficients of range, we see that the coefficient of range for Physics is greater than that of Mathematics. Thus, there is greater dispersion or variation in Physics than Mathematics. This implies that the marks of students in Mathematics are more stable than their marks in Physics. We will learn how to make interpretations like this using relative measures of dispersion.

Example 2

a. Find the coefficient of range of the following data set:

$22, 28, 31, 23, 24.$

b. From the following data, calculate the coefficient of range.

x	5	15	25	35	45	55
f	10	10	30	50	40	30

c. From the following data, calculate coefficient of range.

Mark	0-10	10-20	20-30	30-40	40-50	50-60
No. students	1	5	3	4	5	2

Solution

a. $L = 31$ and $S = 22$, then $CR = \frac{L-S}{L+S} = \frac{31-22}{31+22} = \frac{9}{53} = 0.17$.

b. $L = 55$ and $S = 5$, then $CR = \frac{L-S}{L+S} = \frac{55-5}{55+5} = \frac{50}{60} = 0.83.$

c. Upper limit of highest class $L = 60$ and lower limit of lower class $S = 0$, then

$$CR = \frac{L-S}{L+S} = \frac{60-0}{60+0} = 1.$$

Exercise 3.19

1. Consider the following data which shows the hourly wages earned (in Birr) by ten workers

$$42, 17, 83, 59, 72, 76, 64, 45, 40, 32.$$

Find the CR of the data.

2. Find the CR of the following distribution.

x	10	15	17	18	20
f	3	4	9	3	6

3. The large L and small S values of gross incomes of two companies (in Birr) are given below:

Company	S	L
A	1000	43200
B	1200	36000

a. Calculate the CR of each company.

b. Which company has the more variable income?

3.2.2 Coefficient of Quartile Deviation (CQD)

Coefficient of quartile deviation (1)

Activity 3.10

Consider the following two data sets.

A: 150, 180, 190, 30, 20, 70, 100

B: 130, 160, 150, 150, 140, 160, 170

1. Find the first and the third quartiles for each data set.
2. Determine the range and inter-quartile range for each data set.
3. Compare which set of data has higher variation.

The coefficient of quartile deviation is a measure of relative dispersion and is based on the upper quartile Q_3 and the lower quartile Q_1 is defined as follows:

Definition 3.9

Coefficient of Quartile Deviation: $CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1}$

Example 1

For the data given in Example 1 in the previous section, calculate the coefficient of quartile deviation of both subjects.

Solution

Physics: 10, 15, 18, 20, 20, 21, 21

Mathematics:20, 25, 26, 27, 30, 35, 35

For mathematics:

$Q_1 = \frac{1(n+1)}{4} = \frac{7+1}{4} = \left(\frac{8}{4}\right)^{th}$ item=2^{nd}item $= 25$ and

$Q_3 = \frac{3(n+1)}{4} = \frac{3(7+1)}{4} = \left(\frac{24}{4}\right)^{th}$ item=6^{th} item $= 35$

Thus, coefficient of quartile deviation of mathematics is

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{35 - 25}{35 + 25} = \frac{10}{60} = 0.17.$$

For physics:

$Q_1 = \frac{1(n+1)}{4} = \frac{7+1}{4} = \left(\frac{8}{4}\right)^{th}$ item=2^{nd}item $= 15$ and

$Q_3 = \frac{3(n+1)}{4} = \frac{3(7+1)}{4} = \left(\frac{24}{4}\right)^{th}$ item=6^{th} item $= 21$

Thus, coefficient of quartile deviation of physics is

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{21 - 15}{21 + 15} = \frac{6}{36} = 0.17.$$

We see from the above computation that coefficient of quartile deviation for both Physics and Mathematics is 0.17 for both subjects. Therefore, the inference is that the marks or scores of students in both subjects indicate the same median performance.

Example 2

The number of complaints received by the manager of Geda supermarket was recorded for each of the last 10 working days as it is given below:

21, 15, 18, 10, 5, 17, 21, 19, 28, 25.

Find coefficient of quartile deviation of the data.

Solution

Sorted data: 5, 10, 15, 17, 18, 19, 21, 21, 25, 28.

$$Q_1 = \left(\frac{\left(\frac{n}{4}\right) + \left(\frac{n}{4}+1\right)}{2}\right)^{th} = \left(\frac{\left(\frac{10}{4}\right) + \left(\frac{10}{4}+1\right)}{2}\right)^{th} = 3^{rd}\ \text{item} = 15.$$

$$Q_3 = \left(\frac{\left(\frac{kn}{4}\right) + \left(\frac{kn}{4}+1\right)}{2}\right)^{th} = \left(\frac{\left(\frac{3\times10}{4}\right) + \left(\frac{3\times10}{4}+1\right)}{2}\right)^{th}\ \text{item}$$

$$= 8^{th}\ \text{item} = 21.$$

Thus, coefficient of quartile deviation is: $CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{21-15}{21+15} = \frac{6}{36} = 0.17.$

Exercise 3.20

1. The marks obtained, in English, are by 9 students given below. Calculate the coefficient of quartile deviation for data.

   45, 32, 37, 46, 39, 36, 41, 48, 36.

2. Calculate the coefficient of quartile deviation for data set: 2, 4, 6, 8, 10, 5, 6, 9, 4, 6.

Coefficient of quartile deviation (2)

Example 3

From the following data, calculate the coefficient of quartile deviation.

Mark	0-10	10-20	20-30	30-40	40-50	50-60
No. students	10	20	30	50	40	30

Solution

Calculation of cumulative frequency

Mark	No. students	*cf*
0-10	10	10
10-20	20	30
20-30	30	60
30-40	50	110
40-50	40	150
50-60	30	180

$$Q_1 = \left(\frac{n}{4}\right)^{th} = \left(\frac{180}{4}\right)^{th} = 45^{th}\text{item.}$$

Thus, Q_1 lies in the class 20-30 and hence,

$$Q_1 = L + \frac{(\frac{n}{4} - cf)w}{f} = 20 + \frac{(\frac{180}{4} - 30)10}{30} = 20 + \frac{150}{30} = 25.$$

$$Q_3 = \left(\frac{3n}{4}\right)^{th} = \left(\frac{540}{4}\right)^{th} = 135^{th}\text{item}$$

Thus, Q_3 lies in the class 40-50 and hence

$$Q_3 = L + \frac{(\frac{3n}{4} - cf)w}{f} = 40 + \frac{(\frac{3(180)}{4} - 110)10}{40} = 40 + \frac{250}{40} = 46.25.$$

Therefore, coefficient of quartile deviation of the data is

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{46.25 - 25}{46.25 + 25} = \frac{21.25}{71.25} = 0.3.$$

Exercise 3.21

1. Calculate the coefficient of quartile deviation for the continuous grouped data sets given below:

Class boundaries	*f*
29.5-39.5	8
39.5-49.5	87
49.5-59.5	190
59.5-69.5	304
69.5-79.5	211
79.5-89.5	85
89.5-99.5	20

2. The Q_1 and Q_3 of gross incomes of two companies are given below:

Company	Q_1	Q_3
A	5000	54300
B	1000	45320

a. Calculate the *CQD* of each company.

b. Which company has more variable income?

3.2.3 Coefficient of Mean Deviation (CMD)

Activity 3.11

Consider the following data which shows the amount of sugar, in kilograms, sold by a small shop over for ten days:

$$4, 5, 6, 4, 6, 3, 7, 6, 3, 5$$

1. Determine the value of the mean deviation of the data about the mean, median and mode.
2. Compare the values you obtained in question 1.

The coefficient of quartile deviation is a measure of relative dispersion that is based on the mean deviation. It may be taken from the mean, median or mode and is defined as follows:

Definition 3.10

The coefficient of Quartile Deviation from the mean is $CMD = \frac{MD}{\bar{x}}$, where MD = mean deviation from the mean and $\bar{x}$ = mean.

Definition 3.11

The coefficient of Quartile Deviation from the median is $CMD = \frac{MD}{m_d}$, where MD = mean deviation from the median and m_d = median.

Definition 3.12

The coefficient of Quartile Deviation from the mode is $CMD = \frac{MD}{m_0}$, where MD = mean deviation from the mode and m_o = mode.

Example

Find the coefficient of mean deviation of the following data set about the mean.

x	0	1	2	3	4
f	1	9	7	3	4

Solution

First we calculate mean

$$\bar{x} = \frac{0\times 1 + 1\times 9 + 2\times 7 + 3\times 3 + 4\times 4}{1+9+7+3+4} = \frac{48}{24} = 2.$$

x	f	$\lvert x-\bar{x}\rvert$	$f\lvert x-\bar{x}\rvert$
0	1	2	2
1	9	1	9
2	7	0	0
3	3	1	3
4	4	2	8
Total	$\sum f = 24$		$\sum f\lvert x-\bar{x}\rvert = 22$

Thus, $MD = \frac{\sum f|x-\bar{x}|}{\sum f} = \frac{22}{24} = 0.92.$

Therefore, $CMD = \frac{MD}{\bar{x}} = \frac{0.92}{2} = 0.46.$

Exercise 3.22

1. Calculate the coefficient of quartile deviation from the mean of the following set of examination marks: $45, 32, 37, 46, 39, 36, 41, 48, 36.$
2. Find the coefficient of quartile deviation from the mode of the continuous grouped data given below:

Income	35-39	40-44	45-49	50-54	55-59	60-64	65-69
f	13	15	17	28	12	10	5

3. The median and mean deviation from the median of gross incomes of two companies are given below:

Company	m_d	$MD(m_d)$
A	50000	4430
B	20000	2200

a. Calculate the CMD of each company from the median.

b. Which company has more consistent income?

3.2.4 Coefficient of Variation (CV)

Activity 3.12

Find the variance σ^2 and the standard deviation σ for the following data set:

$$50, 30, 80, 110, 130.$$

The coefficient of variation is a measure of relative dispersion that is based on standard deviation. It is used to compare the variations or the performance of two sets of data. A large value indicates that there is greater variability while a small value indicates less variability.

Definition 3.13

The coefficient of Variation is a unit-less relative measure that is used to measure the degree of consistency given as a ratio of the standard deviation to the mean, that is, $CV = \frac{\sigma}{\bar{x}} \times 100$

Example 1

Calculate the variance, standard deviation and coefficient of variation from the following marks obtained by nine students.

$$45, 32, 37, 46, 39, 36, 41, 48, 36.$$

Solution

First we compute mean of the data.

$$\bar{x} = \frac{45 + 32 + 37 + 46 + 39 + 36 + 41 + 48 + 36}{9} = 40$$

Calculate the deviation for each value from the mean and square it.

x	$x - \bar{x}$	$(x - \bar{x})^2$
45	5	25
32	-8	64
37	-3	9
46	6	36
39	-1	1
36	-4	16
41	1	1
48	8	64
36	-4	16
Total		$\sum(x - \bar{x})^2 = 232$

Thus, the variance is $\sigma^2 = \frac{\sum(x-\bar{x})^2}{n} = \frac{232}{9} = 25.78.$

and the standard deviation $\sigma = \sqrt{\frac{\sum(x-\bar{x})^2}{9}} = \sqrt{25.78} = 5.08.$

Therefore, $CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{5.08}{40} \times 100 = 12.7.$

Example 2

Calculate the coefficient of variation from the following mass (in grams) of 60 mangos:

Wight	65-84	85-104	105-124	125-144	145-164	165-184	185-204
f	9	10	17	10	5	4	5

Solution

x	m	f	$f(m-\bar{x})^2$
65-84	74.5	9	20736
85-104	94.5	10	7840
105-124	114.5	17	1088
125-144	134.5	10	1440
145-164	154.5	5	5120
165-184	174.5	4	10816
185-204	194.5	5	25920
Total		$\sum f = 60$	$\sum f(m-\bar{x})^2 = 72960$

First compute the mean of the data.

$$\bar{x} = \frac{\sum fm}{\sum f} = \frac{7350}{60} = 122.5.$$

Calculate the deviation for each mid-point from the mean and square it.

Calculate the variance: $\sigma^2 = \frac{\sum f(m-\bar{x})^2}{\sum f} = \frac{72960}{60} = 1216.$

Thus, $\sigma = \sqrt{\frac{\sum f(m-\bar{x})^2}{60}} = \sqrt{1216} = 34.87.$

Therefore, $CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{34.87}{122.5} \times 100 = 28.46.$

Exercise 3.23

1. Find the CV of the following distribution.

x	2	5	7	8	10
f	3	4	9	3	6

2. Consider the following data set which show the amount of milk sold, in liters, by two farmers over ten days.

   Farmer A: 42, 17, 83, 59, 72, 76, 64, 45, 40 ,32

   Farmer B: 28, 70, 31, 0, 59, 108 ,82, 14, 3, 95

   a. Calculate the CV for each farmer.

   b. Which farmer consistently sold more milk?

3. The mean and standard deviation of the gross incomes of two companies are given below:

Company	$\bar{x}$	σ
A	12000	2400
B	20000	4400

a. Calculate the CV of each company.

b. Which company has more variable income?

3.3 Use of Frequency Curves

The shape of a frequency curve describes the distribution of a data set. Such a description is made possible after the frequency curve of a frequency distribution is drawn. In this section, you will see how the measures of central tendency (mean, mode and median) determine the skewness of a distribution.

3.3.1 Mean, Median and Mode on the Frequency Curve

Activity 3.13

Consider the following data:

Data A: 20, 30, 40, 50, 50, 60, 50, 70, 80

Data B: 20, 30, 10, 40, 80, 50, 80, 60, 80

a. Calculate and compare the mean, median and mode for each data set.

b. Construct frequency curves for each data set and discuss your observations.

A measure of central tendency or a measure of dispersion alone does not tell us whether or not the distribution is symmetrical. It is the relationship between the mean, median and mode that tells us whether the distribution is symmetrical or skewed.

Example

Consider the following frequency distribution:

x	20-21	22-23	24-25	26-27	28-29
f	4	12	24	12	4

a. Draw the bar graph and frequency curve.

b. Calculate the mean, median and mode.

c. Describe the relationship between the mean, median and mode, and the skewness of the distribution.

Solution

a. The bar graph and frequency curve of this frequency distribution are as follows.

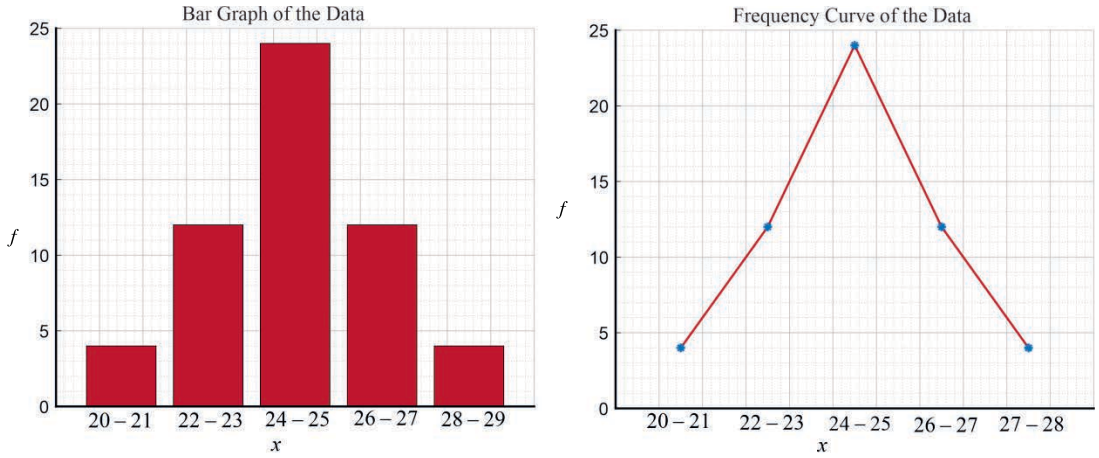


Figure 3.1

b. Mean = median = mode = 24.5.

c. From (a) and (b), you see that whenever mean = median = mode, then the distribution is perfectly symmetrical.

Investigate what happens to the skewness of a distribution, if mean > median > mode.

From the discussions outlined above, you can make the following generalizations:

a. If the mean is smallest in value, and the median is larger than the mean but smaller than the mode, then the distribution is negatively skewed. That is, if

Mean < Median < Mode then the distribution is negatively skewed (skewed to the left).

b. For a unimodal distribution in which the values of mean, median and mode coincide (that is, Mean = Median = Mode), the distribution is said to be perfectly symmetrical.

c. If the mean is the largest in value, and the median is larger than the mode but smaller than the mean, then the distribution is positively skewed. That is, if Mean> Median > Mode, then the distribution is positively skewed (skewed to the right).

Observe the following figure for a, b and c above.

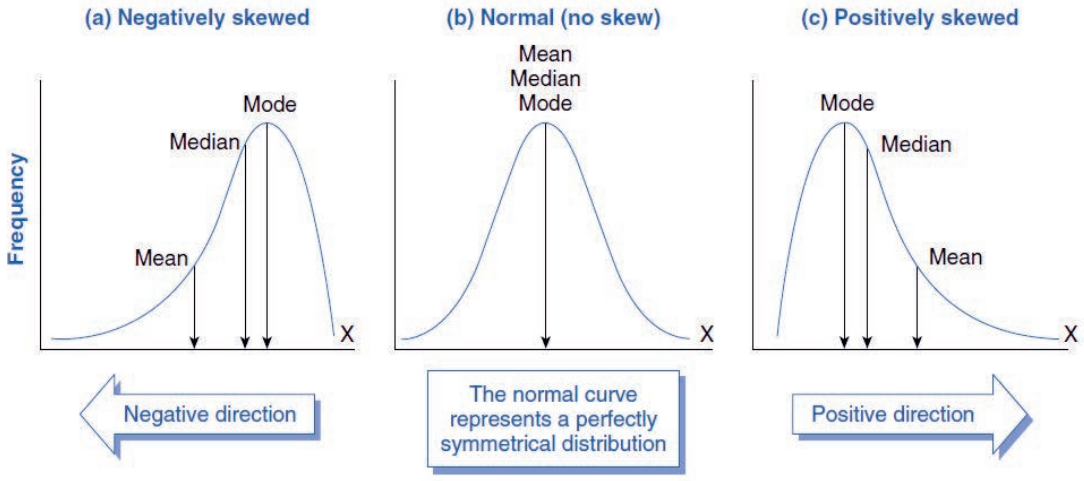


Figure 3.2: Skewed distribution figure

Exercise 3.24

Consider the following frequency distribution:

x	0-20	20-40	40-60	60-80	80-100
f	8	12	30	14	6

a. Draw the bar graph and frequency curve.

b. Calculate mean, median and mode.

c. Describe relationships between the mean, median and mode, and the skewness of the distribution

3.3.2 Measurements of Skewness

Pearson's coefficient of skewness

Activity 3.14

Consider the following data sets:

Data A: 2, 3, 4, 5, 5, 6, 5, 7, 8

Data B: 2, 3, 1, 4, 8, 5, 8, 6, 8

1. Calculate the mean, median and standard deviation for each data set.
2. Compare the values obtained in question 1.

In the discussion above, we used the relationships between the measures of central tendency to determine the skewness of a distribution. With the help of central tendencies and standard deviation, it is also possible to determine the skewness of a distribution. This is sometimes called a mathematical measure of skewness. Mathematically, skewness can be measured by calculating:

i. Pearson's coefficient of skewness.

ii. Bowley's coefficient of skewness.

Pearson's coefficient of skewness is obtained by expressing the difference between the mean and the median relative to the standard deviation. It is usually denoted by α. Pearson's coefficient of skewness is

$$\alpha = \frac{3(\text{mean} - \text{median})}{\text{standard deviation}}.$$

The interpretation of skewness by this approach follows our prior knowledge. From the previous discussion, if the mean = the median, we can see that the distribution is symmetrical. Looking at Pearson's coefficient of skewness, if the mean is equal to the median then, $\alpha = 0$ so the distribution is symmetrical. Following the same approach, we can state the following interpretation on skewness using Pearson's coefficient of skewness.

Interpretation

1. If Pearson's coefficient of skewness $\alpha = 0$, then the distribution is symmetrical.
2. If Pearson's coefficient of skewness $\alpha > 0$ (positive), then the distribution is skewed positively (skewed to the right).
3. If Pearson's coefficient of skewness $\alpha < 0$ (negative), then the distribution is negatively skewed (skewed to the left).

Example 1

Calculate Pearson's coefficient of skewness for the data given below and determine the skewness of the distribution.

x	100	105	110	115	120
f	3	4	5	2	1

Solution

$\bar{x} = 108$, $m_d = 110$ and $SD = 5.7$. Thus,

$$\alpha = \frac{3(\text{mean} - \text{median})}{\text{standard deviation}} = \frac{3(108 - 110)}{5.7} = -1.05 < 0.$$

Therefore, the distribution is negatively skewed.

Exercise 3.25

Calculate Pearson's coefficient of skewness from the data below and determine the skewness of the distribution.

x	10	20	30	40	50
f	4	6	5	3	2

Bowley's coefficient of skewness

Previously, you saw how to determine skewness by using relationships between mean, median and standard deviation. It is also possible to determine skewness by using positional measures of central tendency, the quartiles.

A coefficient of skewness that uses quartiles is Bowley's coefficient of skewness, which is usually denoted by β, and is given by:

$$\beta = \frac{Q_3 + Q_1 - 2\,(\text{median})}{Q_3 - Q_1}$$

The interpretation for skewness based on Bowley's coefficient of skewness is also the same as that of Pearson's.

Interpretation

1. If Bowley's coefficient of skewness $\beta = 0$, then the distribution is symmetrical.
2. If Bowley's coefficient of skewness $\beta > 0$ (positive), then the distribution is skewed positively (skewed to the right).
3. If Bowley's coefficient of skewness $\beta < 0$ (negative), then the distribution is negatively skewed (skewed to the left).

Example 2

Find Bowley's coefficient of skewness for the following data and determine the skewness of the distribution.

$11, 8, 7, 13, 12, 14, 15, 17, 20, 16, 19$

Solution

Arranging the data in ascending order, you get:

$7, 8, 11, 12, 13, 14, 15, 16, 17, 19, 20.$

From this arranged data, you can determine the quartiles as:

$Q_1 = 11$, $Q_2 = \text{median} = 14$ and $Q_3 = 17$. Thus,

Bowley's coefficient of skewness is

$$\beta = \frac{Q_3 + Q_1 - 2(\text{median})}{Q_3 - Q_1} = \frac{17 + 11 - 2(14)}{17 - 11} = 0.$$

Therefore, the distribution is symmetrical.

Exercise 3.26

1. Calculate Bowley's coefficient of skewness for the data below and determine the skewness of the distribution.

x	10	20	30	40	50
f	4	6	5	3	2

2. Compare Bowley's coefficient of skewness calculated in question 1 with Pearson's coefficient calculated in Exercise 3.24. Interpret the result.

3.4 Sampling Techniques

Sampling

In Grade 9 and Grade 11, you did some work in statistics including the collection and tabulation of statistical data, frequency distributions and histograms, measures of location (mean, median and mode(s), quartiles, deciles and percentiles), measures of dispersion for both ungrouped and grouped data, and some ideas of probability. In this section, you will study sampling techniques.

Activity 3.15

If you were asked to study something, perhaps you would start by collecting data. Discuss the following questions.

1. Why do you need to collect data?
2. How would you collect the data?
3. From where would you collect the data?

Statistics as a science deals with the proper collection, organization, presentation, analysis and interpretation of numerical data. Since statistics is useful for making decisions or forecasting future events, it is applicable in almost all sciences. It is useful in social, economic and political activities. It is also useful in scientific investigations. Some examples of applications of statistics are given below.

1. Statistics in schools

In schools, teachers rank their students at the end of a semester based on information collected through different methods (exams, tests, quizzes, etc.) which gives an indication of the students' performance.

2. Statistics in business

Statistics is widely used in business to make business forecasts. A successful business must keep a proper record of information in order to predict the future course of the business, and should be accurate in its statistical and business forecasting. Statistics can also be used to help in formulating economic policies and evaluating their effect.

3. Weather forecasting

Statistics is used heavily in the field of weather forecasting. In particular, probability is used by weather forecasters to assess how likely it is that there will be rain, snow, clouds, etc. on a given day in a certain area.

Forecasters will regularly say things like “there is a 90% chance of rain today after 5 pm” to indicate that there’s a high likelihood of rain during certain hours.

4. Sales tracking

Retail companies often use descriptive statistics like the mean, median, mode, standard deviation, and interquartile range to track the sales behavior of certain products.

This gives companies an idea of how many products they can expect to sell during different time periods and allows them to know how much they should keep in stock.

5. Health insurance

Health insurance companies often use statistics and probability to determine how likely it is that certain individuals will spend a certain amount on healthcare each year.

For example, an actuary at a health insurance company might use factors like age, existing medical conditions, current health status, etc. to determine that there’s an

80% probability that a certain individual will spend $10,000 or more on healthcare in a given year.

6. Traffic

Traffic engineers regularly use statistics to monitor the traffic in different areas of a city, which allows them to decide whether or not they should add or remove roads to optimize traffic flow.

Also, traffic engineers often use time series analysis to monitor how traffic changes throughout the day so they can optimize the behavior of traffic lights.

7. Investing

Investors use statistics and probability to assess how likely it is that a certain investment will pay off.

For example, a given investor might determine that there is a 5% chance that the stock of company A will increase 100 times during the upcoming year. Based on this probability, they'll decide how much of their portfolio to invest in the stock.

8. Medical studies

Statistics is regularly used in medical studies to understand how different factors are related. For example, medical professions often use correlation to analyze how factors like weight, height, smoking habits, exercise habits, and diet are related.

If a certain diet and overall weight is found to be negatively correlated, a medical professional may recommend a diet to an individual who needs to lose weight.

9. Manufacturing

Statistics is often used in manufacturing to monitor the efficiency of different processes. For example, manufacturing engineers may collect a random sample of widgets from a certain assembly line and track how many of the widgets are defective.

They may then perform a test to determine if the proportion of widgets that are defective is lower than a certain value that is considered acceptable.

10. Urban planning

Statistics is regularly used by urban planners to decide how many apartments, shops, stores, etc. should be built in a certain area based on population growth patterns.

For example, if an urban planner sees that population growth in a certain part of the city is increasing at high rate compared to other parts of the city then they may decide to prioritize building new apartment complexes in that part of the city.

Note

1. Collection of data is the basis for any statistical analysis. Great care must be taken at this stage to get accurate data. Inaccurate and inadequate data may lead to wrong or misleading conclusions and cause poor decisions to be made.
2. Recall that a population in statistics means the complete collection of items (individuals) under consideration.

It is often impractical and too costly to collect data from the whole population or to make census survey. Consequently, it is frequently necessary to use the process of sampling, from which conclusions are drawn about a whole population. This leads you into an essential statistical concept called sampling which is important for practical purposes.

Definition 3.14

A sample is a limited number of items taken from a population which is being studied or investigated.

A sample needs to be taken in such a way that it is a true representation of the population. It should not be biased so as to cause a wrong conclusion. Avoiding bias requires the use of proper sampling techniques.

Before examining sampling techniques, you need to note the following.

1. **Size of a sample**: There is no single rule for determining the size of a sample of a given population. However, the size should be adequate in order to represent the population.

i. **Homogeneity** or **heterogeneity** of the population: If the population has a homogeneous nature, a smaller size sample is sufficient. (For example, a drop of blood is sufficient to take a blood test from someone).

ii. **Availability of resources**: If sufficient resources are available, it is advisable to increase the size of the sample.

2. **Independence:** Each item or individual in the population should have an equal chance of being selected as a member of the sample.

Exercise 3.27

Your school has 4 sections for grade 12 and the numbers of students are shown in the table below.

Section	Boys	Girls	Total
A	20	30	50
B	20	20	40
C	10	30	40
D	20	10	30
Total	70	90	160

Suppose you want to conduct a survey about the grade 12 students of your school. Since time is limited, you have to select 20 students only.

a. What is the size of the population?

b. What is the size of the sample?

c. Are the following methods of selection acceptable? If not, explain why it is not acceptable.

1) Select 20 girls.

2) Select 20 students from Section A.

3) Select 10 boys and 10 girls.

4) Select 5 students each from the four sections.

Random sampling

There are various techniques of sampling, but they can be broadly grouped into two:

i. Random or probability sampling. **ii.** Non-random or non-probability sampling.

Random or Probability Sampling

Simple random sampling

Simple random sampling is the selection of individuals for which every individual has equal chance of being selected. To apply this method, you can either use the lottery method or a table of random numbers (attached at the end of the textbook).

The lottery method

In this method you need to take the following steps.

Step 1: Prepare slips of paper which are identical in size and color.

Step 2: Write names or code numbers for each member of the population.

Step 3: Fold the slips and put them in a container and mix them up.

Step 4: A blindfold selection are made while blindfolded until a sample of the required size is obtained.

Example 1

A mathematics teacher in a school wants to determine the average height of grade 12 students. There are 8 sections of grade 12 in the school. Assuming that there are 45 students in each class and that she require a sample size of 48 (6 from each section), how can she use the lottery method to select her sample?

Solution

- ✓ Prepare 45 cards of same size and color with number 0 written on 39 of them and the number 1 written on 6 of them.
- ✓ Put the cards on a table with the numbers facing down.
- ✓ Invite the students (one at a time) to come and pick a card.
- ✓ Those who pick cards with the number one on them will be members of the sample.
- ✓ Repeat the same process for each group.

Using a table of random numbers

For this method, you need to use a table of random numbers, and you need to take the following steps.

Step 1: Each member of the population is given a unique number from a set of consecutive numbers.

Step 2: Select arbitrarily one random number from the table of random numbers.

Step 3: Starting with the selected random number, read the consecutive list of random numbers and match these with the members of the population in their consecutive number order.

Step 4: Sort the selected random numbers into either ascending or descending order.

Step 5: If you need a sample of size "n", then select the sample that corresponds with the first "n" random numbers.

Example 2

For the problem in Example 1 above, use the random numbers table attached at the end of the textbook to select a sample of 48 students (6 from each section).

Solution

- ✓ Give each student a roll number from 1 to 45 in alphabetical order.
- ✓ Select arbitrarily one random number from the table of random numbers.
- ✓ From the selected random number, read 45 consecutive random numbers and attach each to the consecutive numbers given to each member of the population.
- ✓ Sort the selected random numbers (together with the numbers from 1 – 45) into ascending or descending order.
- ✓ Take the first 6 random numbers and the corresponding roll numbers. The students whose roll numbers are selected will be part of the sample.

Advantages of simple random sampling

- Lack of bias
- Simplicity
- Less knowledge required

Limitations of simple random sampling

- Difficult to access lists of the full population
- Time consuming
- Cost

Exercise 3.28

1. In your class, there are 20 students whose roll numbers are from 1 to 20. You want to select five students using the random numbers table attached at the end of this textbook. Following the steps explained below, select five students.

1) Select one random number from the table in a random way.
2) Read 20 consecutive numbers from the table and attach them to each of the roll numbers as listed.
3) Sort the random numbers (together with the roll numbers) in ascending order.
4) Take the first 5 random numbers and the corresponding roll numbers.

Roll Number	Random Number
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Random Number in Ascending Order	Roll Number Rearranged

11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

2. Which roll numbers did you select? Compare the results with others.

Systematic sampling

Systematic sampling is another random sampling technique used for selecting a sample from a population. In order to apply this method, you take the following steps:

If N = size of the population and n = size of the sample, then we use $k = \frac{N}{n}$ for a sampling interval. After this, we arbitrarily select one number between 1 and k, and then every next sample member is selected by considering the k^{th} member after the selected one.

Example 3

In a class, there are 110 students with class list numbers written from 1-110.

You need to select a sample of 10 students. How can you apply the systematic sampling technique?

Solution

You apply the systematic sampling technique as follows:

$N = 110,\ n = 10$ which implies

$$k = \frac{N}{n} = \frac{110}{10} = 11.$$

First, sort the list in ascending order and choose one number at random from the first 11 numbers. If the selected number is 6, then the sample numbers that you obtain by taking every 11^{th} number until you get the tenth sample. The selected sample will be $6, 17, 28, 39, 50, 61, 72, 83, 94, 105$.

Note that in systematic sampling, you use $S_n = S_1 + (n-1)k$ where S_1 is the first randomly selected sample, $S_n$stands for n^{th} member of a sample and k is the sampling interval.

Advantages of systematic sampling

- Easy to execute and understand
- Control and sense of process
- Clustered selection eliminated
- Low risk factor

Limitations of systematic sampling

- Assumes the size of the population can be determined
- Need for a natural degree of randomness
- Great risk of data manipulation

Exercise 3.29

You want to know how many words on average are listed on one page of a dictionary. The dictionary has 1200 pages in total. You have decided to select 20 pages as a sample for your survey. Supposing you apply the systematic sampling technique, answer the following questions.

a. What is the sampling interval?

b. If you have chosen 13 as the first number, list all the page numbers to be selected as samples.

Stratified sampling

Stratified sampling is useful whenever the population under consideration has some identifiable stratum or categorical difference where, in each stratum, the data values or items are supposed to be homogeneous. In this method, the population is divided into homogeneous groups or classes called strata and a sample is drawn from each stratum. Once you identify the strata, you select a sample from each stratum either by simple random sampling or systematic sampling.

Example 4

If you consider students in a group, you may consider intervals of age as strata. In such a case, you could take the age groups 12-14, 15-17 and 18–20 as stratification of the students. Samples are taken from each stratum proportionally. In this case, strata are age groups from 12-14, 15-17 and 18-20.

Advantages of stratified sampling

- More representative sample
- Great precision
- Administrative convenience

Limitations of stratified sampling

- The success of stratified random sampling depends on effective stratification and appropriate size of the sample to be drawn from each of the stratum. If stratification is faulty, the result will be biased.
- Disproportional stratified sampling requires the assignment of weights to different strata and if the weights assigned are faulty, the resulting sample will not be representative and might give biased results.
- Time consuming and tedious.

Exercise 3.30

In City A, there are 60 secondary schools with 250 classes in total. Suppose that all the sections have 25 boys and 25 girls each. Among the classes, 100 are Social Science stream and 150 are Natural Science stream. You want to select 30 classes from them to conduct a survey about the desired future occupation of the students.

a. In this case, what are the strata?

b. If you have to choose 30 classes as your sample, how many classes you should choose from Social Science stream and Natural Science stream, respectively?

Cluster sampling

Cluster sampling divides the population into subgroups, but each subgroup has similar characteristics to the whole sample. Now, instead of sampling individuals from each subgroup, you randomly select entire subgroups.

Example 5

The list of all agricultural farmers in a village or district may not be easily available but the list of villages or districts is generally available. In this case, every farmer in a sampling unit and every village or district is the cluster.

Advantages of cluster sampling

- Requires fewer resources
- More feasible

Limitations of cluster sampling

- Biased samples
- High sampling error

Figure 3.3 represents the above four sampling techniques.

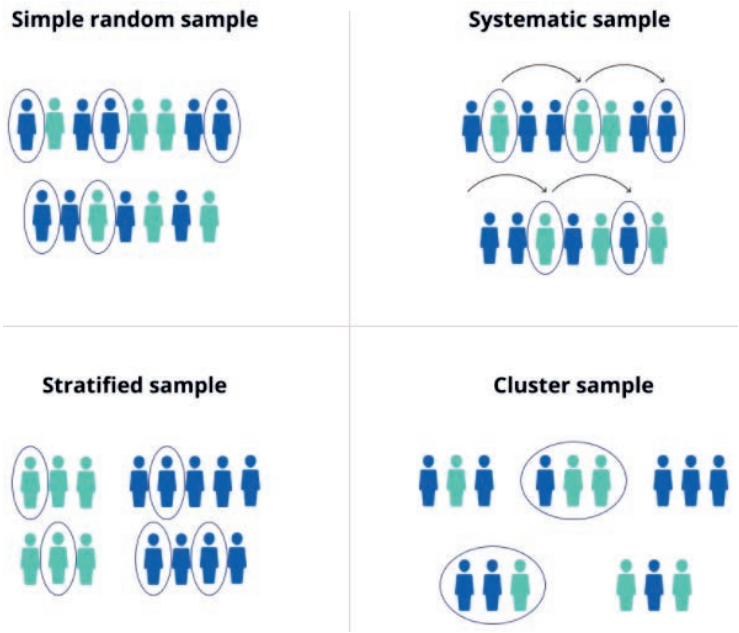


Figure 3.3: Pictorial description of the four sampling techniques

Exercise 3.31

In City B, there are 40 secondary schools with 60,000 students in total. Suppose the City Education Bureau (CEB) has selected 1,000 students from them for survey purposes applying one of the following methods. Name each of the sampling methods.

a. CEB collected students' rosters from all the schools and combined them into one. After assigning a new roll number to each student, CEB selected those numbers of students: 10, 70, 130, 190,. ...

b. CEB collected students' rosters from all the schools and combined them into one. After assigning a new roll number and a random number to each student, CEB rearranged the random numbers in ascending order. CEB selected the first 1,000 students from the list.

c. CEB classified the schools into urban and rural schools. There are 25 urban schools (45,000 students in 900 sections) and 15 rural schools (15,000 students in 500 sections). CEB selected 15 sections from the urban schools and 8 sections from the rural schools to cover about 1,000 students in total.

d. CEB first selected 5 urban schools and 3 rural schools. In each school, CBE further selected Section B of all grades. Since there were 1,600 students in the selected 32 sections, CBE further selected 1,000 students randomly from them.

Multistage Sampling

Multistage sampling is a more complex form of cluster sampling, in which smaller groups are successively selected from large populations to form the sample population used in your study. Due to this multi-step nature, the sampling method is sometimes referred to as phase sampling.

Steps for Carrying out Multistage Sampling Technique

For this method, you need to take the following steps.

Step 1: Define the first sampling frame, assign a unique number to each group in the population, and select a small sample of discrete clusters to form your primary sampling units.

Step 2: Define a second sampling frame for the primary sampling units selected in step 1 and then select random samples from these.

Step 3: If necessary, repeat the previous step.

Step 4: Select the final sample group from the sub-groups using a form of probability sampling, such as simple random sampling or systematic sampling.

Example 6

A research firm in Ethiopia conducted a survey in which it divided the country into its counties and randomly selected some of these counties as a cluster sample (the first stage of sampling).

Each county was then divided into its towns, and areas were chosen at random from each town (the second stage of sampling).

Finally, each town was divided into small areas and households were selected at random from each area. These households formed the sample population for the research study (third stage of sampling).

Advantages of multistage sampling

- Practical for primary data collection for large populations that are geographically dispersed.
- Reduces the costs and time associated with data collection.
- Provides flexibility, as researchers can break down the population as often as necessary to create the sample population they need.
- Allows each stage to use its own sampling method, whether it be stratified sampling, cluster sampling or simple random sampling.

Limitations of multistage sampling

- It introduces a considerable degree of subjectivity, based on the sampling design that surrounds the formation of the sub-groups and their selection.
- To form the sampling frames for multistage random sampling, group-level information is required, sometimes at a national level depending on the target population.
- Typically not as accurate as using simple random sample with the same sample size.

So far, five different sampling techniques have been discussed. It is important to remember that, no one technique is better than the others. Each has its own advantages and limitations.

Exercise 3.32

Suppose, as a statistics officer, you have to survey 1000 households of your Region or City. To select the sample households, you use the multistage sampling method. As the first stage of sampling, you randomly select four Zones or Sub-cities. From each Zone or Sub-city, you further select two Woreda randomly as the second stage sampling.

a. How many households do you have to select from each selected Woreda, if an equal number of households should be selected from each Woreda?

b. Suppose the total numbers of households of the eight selected Woreda are as follows:

100, 300, 300, 400, 500, 600, 800, 1000

If you want to select households proportional to the total number of households in each Woreda, determine the number of sample households to be selected from each Woreda.

Some Statistical Application Software

How to compute the measures of dispersion using microsoft Excel

Range

Excel does not offer a function to compute range. However, we can easily compute it by subtracting the minimum value from the maximum value. The formula would be =MAX( )-MIN( ) where the dataset would be the referenced in both the parentheses. The =MAX( ) and =MIN( ) functions would find the maximum and the minimum points in the data. The difference between the two is the range. The higher the value of the range, the greater is the spread of the data.

Max. Score = 100	Arun	John
Math	100	45
Physics	40	65
Chemistry	20	70
Programming	100	80
Average	**=AVERAGE(C3:C6)**	**=AVERAGE(D3:D6)**
Range	**=MAX(C3:C6)-MIN(C3:C6)**	**=MAX(D3:D6)-MIN(D3:D6)**

Variance

The calculation of variance differs slightly depending on whether the data set describes a sample or the entire population. We have already seen that variance is nothing but the average of the squared deviations. When we are computing the variance for a population, we divide the sum of squared deviations by *n*. However, when we compute the variance for a sample, we divide the sum of squared deviations by (*n* - 1).

This change is taken care of by Excel with two different functions: =VAR.P( ) for the population variance, and =VAR.S( ) for the sample variance.

Max. Score = 100	Arun	John
Math	100	45
Physics	40	65
Chemistry	20	70
Programming	100	80
Average	**=AVERAGE(C3:C6)**	**=AVERAGE(D3:D6)**
Population Variance	**=VAR.P(C3:C6)**	**=VAR.P(D3:D6)**
Sample Variance	**=VAR.S(C4:C7)**	**=VAR.S(D4:D7)**

If we treat our data set as the population, then the variance for Arun is 1275, and the variance for John is 162.5. If we treat our data as a sample, the variance for Arun is 1189.58, and the variance for John is 50.

Older versions of Excel used =VARP( ) and =VARS( ) to calculate population variance, and sample variance, respectively.

Microsoft Excel also supports two other functions that calculate variance, =VARA( ) for sample variance, and =VARPA() for population variance. These differ from the other variance functions in how they treat certain text strings within the data.

=VARA( ) and =VARPA( ) can handle the following text strings that =VAR.S( ) and =VAR.P( ) ignore:

1. Logical values such as TRUE and FALSE are counted, and treated as 1, and 0, respectively.
2. Any text value is counted, and is treated as 0.

Standard Deviation

We already know that the standard deviation is nothing but the square root of variance. Naturally, if the variance computation is different for a sample and for a population, the standard deviation would be different as well. Similar to variance, Excel offers two functions, =STDEV.S( ) for sample standard deviation, and =STDEV.P( ) for population standard deviation.

Max. Score = 100	Arun	John
Math	100	45
Physics	40	65
Chemistry	20	70
Programming	100	80
Average	**=AVERAGE(C3:C6)**	**=AVERAGE(D3:D6)**
Population SD	**=STDEV.P(C3:C6)**	**=STDEV.P(D3:D6)**
Sample SD	**=STDEV.S(C3:C6)**	**=STDEV.S(D3:D6)**

Older versions of Excel support =STDEV() for sample standard deviation, and =STDEVP() for population standard deviation.

Standard deviation can also be computed on logical strings, and text, just like variance. The function for sample variance is =STDEVA( ) and the function for population variance is =STDEVPA( ), if the text values are to be counted. The treatment of the text values remain the same as with =VARA( ) and =VARPA( ) functions.

Average Absolute Deviation

To avoid the problems associated with squaring the quantities with dimensions, we may want to check dispersion using the average of absolute values of the deviations. Excel includes a function for it as well. It can be computed with =AVEDEV( ).

Max. Score = 100	Arun	John
Math	100	45
Physics	40	65
Chemistry	20	70
Programming	100	80
Average	**=AVERAGE(C3:C6)**	**=AVERAGE(D3:D6)**
Average Dev.	**=AVEDEV(C3:C6)**	**=AVEDEV(D3:D6)**

=AVEDEV( ) would ignore any text values, or logical values. However, cells with 0 are counted, and treated as numbers.

Inter-Quartile Range (IQR)

Microsoft Excel has two functions to compute quartiles. The inter-quartile range has to be calculated as the difference between the quartile 3 and quartile 1 values.

Quartiles can be calculated using =QUARTILE.INC( ) or =QUARTILE.EXC( ). Both functions calculate the quartiles by calculating the percentiles on the data. However, the =QUARTILE.EXC calculates exclusive quartiles, and cannot calculate quartile 0 or quartile 4 (the extreme values are excluded). The inclusive function =QUARTILE.INC( ) can be used to calculate quartiles including quartile 0 and quartile 4.

Both functions have the following syntax: =QUARTILE.INC(range, quartile number) where the quartile number can be between 0 and 4. Any quartile number outside these values would return an error.

Max. Score = 100	Arun	John
Math	100	45
Physics	40	65
Chemistry	20	70
Programming	100	80
Quartile 3	**=QUARTILE.INC(C3:C6,3)**	**=QUARTILE.INC(D3:D6,3)**
Quartile 1	**=QUARTILE.INC(C3:C6,1)**	**=QUARTILE.INC(D3:D6,1)**
IQR	**=C7-C8**	**=D7-D8**

Max. Score = 100	Arun	John
Math	100	45
Physics	40	65
Chemistry	20	70
Programming	100	80
Quartile 3	**=QUARTILE.EXC(C3:C6,3)**	**=QUARTILE.EXC(D3:D6,3)**
Quartile 1	**=QUARTILE.EXC(C3:C6,1)**	**=QUARTILE.EXC(D3:D6,1)**
IQR	**=C7-C8**	**=D7-D8**

In both the above examples, Excel would calculate the quartile values by extrapolation because there are not enough data points. The quartile boundaries would lie between two values in our data set.

Older versions of Excel had a single function for quartile, =QUARTILE() and that was identical to the =QUARTILE.INC() function in the current versions. There was no equivalent of =QUARTILE.EXC() function in the earlier versions.

Summary

1. Measures of dispersion help to decide the degree of variability.
2. There are two types of measure of dispersion.
   i. Absolute and ii. Relative.
3. Range $= L - S$.
4. Inter-quartile range $= IR = Q_3 - Q_1$.
5. Mean deviation about the mean $= MD(\bar{x}) = \frac{\sum |x-\bar{x}|}{n}$.
6. Mean deviation about the median $= MD(m_d) = \frac{\sum |x-m_d|}{n}$.
7. Mean deviation about the mode $= MD(m_o) = \frac{\sum |x-m_o|}{n}$.
8. Quartile deviation $= QD = \frac{Q_3-Q_1}{2}$.
9. Variance $= \sigma^2 = \frac{\sum f(x-\bar{x})^2}{\sum f}$.
10. Standard deviation $= \sigma = \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}}$.
11. Coefficient of range $= CR = \frac{L-S}{L+S}$.
12. Coefficient of quartile deviation $= CQD = \frac{Q_3-Q_1}{Q_3+Q_1}$.
13. Coefficient of mean deviation $= CMD = \frac{MD}{\bar{x}}$.
14. Coefficient of variation $= CV = \frac{\sigma}{\bar{x}} \times 100$.
15. For a given distribution:
    - If $\bar{x} > m_d > m_o$, then the distribution is **positively skewed.**
    - If $\bar{x} < m_d < m_o$, then the distribution is **negatively skewed.**
    - If $\bar{x} = m_d = m_o$, then the distribution is **symmetrical**.
16. Pearson's coefficient of skewness $= \alpha = \frac{3(\text{mean}-\text{median})}{\text{standard deviation}}$
    - If $\alpha > 0$, then the distribution is **positively skewed**.
    - If $\alpha < 0$, then the distribution is **negatively skewed**.
    - If $\alpha = 0$, then the distribution is **symmetrical.**

17. Bowley's coefficient of skewness $= \beta = \frac{Q_3+Q_1-2(\text{median})}{Q_3-Q_1}$

- If $\beta > 0$, then the distribution is **positively skewed**.
- If $\beta < 0$, then the distribution is **negatively skewed**.
- If $\beta = 0$, then the distribution is **symmetrical.**

18. A **population** is the complete set of items which are of interest in any particular situation.

19. It is not possible to collect information from the whole population because it is costly in terms of time, energy and resources. To overcome these problems, we take only a certain part of the population called a **sample.**

20. A sample serves as representative of the population, so that we can draw conclusions about the entire population based on the results obtained from the sample.

21. There are two methods of sampling:

i. The random (probability) sampling method.

ii. The non-random (non-probability) sampling method.

22. In random sampling, every member of the population has an equal chance of being selected.

Review Exercise

1. Find the range, inter-quartile range, coefficient of range and coefficient of quartile deviation of the following data set.

Class	15-20	20-25	25-30	30-35
f	8	21	15	4

2. Find the mean deviation about the mean of mass (in kg) and the coefficient of mean deviation about the mean of 460 infants born in hospital in one year from the following table.

Mass	2-2.5	2.5-3	3-3.5	3.5-4	4-4.5	4.5-5
f	17	97	187	135	18	6

3. Find variance and standard deviation of the data which represents the marks scored by ten students in a class test:

   $30, 29, 25, 33, 35, 38, 37, 48, 40, 44.$

4. Consider the data given below:

   $$4, 8, 7, 10, 11$$

   a. Find the standard deviation for the data.

   b. Add 10 to all the values then find the standard deviation of the new values.

   c. Multiply all the values by 5 then find the standard deviation of the new data.

   d. Compare the results you obtained in parts b and c with the results you obtained in part a and write a conclusion.

5. Calculate the coefficient of variance of the amount of money (in Birr) a group of children spent on food during a school trip .The amounts spent are:

   $$10, 15, 5, 20, 25, 30, 35, 40.$$

6. 53 students were asked to write the total number of hours per week they spent to studying mathematics. With this information find the mean deviation about the mode, the standard deviation and the coefficient of variation.

x	4	6	8	10	12
f	8	21	15	4	5

7. The mean and standard deviation of 15 observations are found to be 10 and 5, respectively. On rechecking it was found that one of the observations with the value 8 was incorrect. Calculate the mean and standard deviation if the value of the corrected observation was 21.
8. Suppose that each day laboratory technician A completes 40 analyses with standard deviation of 5. Technician B completes 160 analyses with standard deviation 15. Which employee shows less variation?
9. Find the range, mean, variance and standard deviation of the first n natural numbers.
10. A wall clock strikes the bell once at 1 o'clock, 2 at 2 o'clock and 3 at 3 o'clock and so on. How many times will it strike in a particular day? Find variance and standard deviation of the number of strikes the bell makes a day on the hour.
11. 1500 visitors turned up to the city library during the first month of the academic year. They who borrowed 6750 books altogether. Find and interpret the coefficient of variation if the square of the sum of the borrowed books is 33315.
12. The following data obtained from the police records of the accident cases in Dire Dawa city police station from February 2009-August 2013 G.C.

X(per month)	*variance*	*Standard Deviation*
Death	21.20	3.83
Heavy injury	47.20	12.85
Light injury	88.40	12.70
No. of accident	276.40	12.80

   a. Determine the coefficient of variation of death, heavy injury, light injury and number of accidents.
   b. Based on the values obtained in part (a) give your interpretation of the data.
13. The mean and coefficient of variation of seven observations are 8 and 50, respectively. If five of those are 12, 14, 10, 4 and 2, then find the remaining two observations.

14. For a group of 100 students the mean and coefficient of variation of their marks were found to be 60 and 25, respectively. Later on it was found that the scores 45 and 70 were wrongly entered as 40 and 27. Find the corrected mean and coefficient of variation.

15. If in the following distribution the mean deviation about the mean and coefficient of mean deviation about the mean are 180 and 10, respectively, then find the missing frequency f.

x	11-13	13-15	15-17	17-19	19-21	21-23	23-25
f	7	6	9	13	F	5	4

16. The mean deviation about the median and coefficient of mean deviation about the median of the following data are 440 and 8, respectively. Find the values of s and t if the total frequency is 82.

x	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
f	5	17	2	12	s	2	7	t	9	4

UNIT 4

INTRODUCTION TO LINEAR PROGRAMMING

Unit Outcomes

By the end of this unit, you will be able to:

- Deduce how to find regions of inequality graphs.
- Solve systems of linear inequality.
- Construct linear programing problems.
- Solve real life problems of linear programming problems.
- Use spreadsheet to solve Linear Programming Problems.

Unit Contents

4.1 Graphical Solutions of System of Linear Inequalities

4.2 Maximum and Minimum Values

4.3 Applications

Summary

Review Exercise

- Bounded solution region
- Constraints
- Decision variables
- Equation of a line
- Feasible region
- Half planes solution
- Minimum value
- Maximum value
- Objective function
- Optimal value
- Slope of a line
- Vertex (corner point)

Introduction

In the previous grades, you discussed systems of linear equations and their applications in solving the day-to-day problems. In Grade 9, you studied about linear inequalities and system of linear inequalities in two variables and their solutions by graphical method.

In this unit, you shall study some linear programming problems and their solutions only by graphical method though there are many other methods to solve such kinds of problems. As part of its components, the unit introduces different real life application problems of Linear Programming Problems. Finally, you will practice how to solve linear programing problems by the help of spreadsheet.

Application of the Unit as a career in Mathematics - software engineering: A software engineer creates, writes, develops and interprets different computer programs. Software engineers use mathematics to write a program and to solve various computing problems. If you are interested in a career in software engineering, therefore, you should study these mathematical subjects: algebra, linear programming, statistics and calculus.

4.1 Graphical Solutions of System of Linear Inequalities

Activity 4.1

1. Plot the linear equation of the form $y = mx + b$ for different values of m and b. For instance, $m < 0, m > 0$ and $m = 0$.
2. Find slope and y-intercept of the following:
   a. $y = 3x + 2$
   b. $y = \frac{1}{3}x + 2$
   c. $y = -4x + 5$
   d. $x + 2y = 2$
3. Draw the graph of the following:
   a. $y = 2$
   b. $x = -4$
   c. $y = -3$

In Unit 4 of Grade 9 Textbook, you studied about inequalities and how their solutions are determined. In this section, you will study graphical solutions of linear inequalities in one and two variables. You will also learn about how to translate word problems into mathematical expression. Here we get certain statements involving a sign "<" (less than), ">" (greater than), "≤" (less than or equal) and" ≥" (greater than or equal) which are known as inequalities.

In the process, you recognize that the study of inequalities is very useful in solving problems in the field of science, mathematics, statistics, economics, finance, psychology, and so on.

4.1.1 Inequalities

Let us consider the following situations:

Alem went to market with Birr 500 to buy wheat flour which was available in packets of 5kg. The price of one packet of wheat flour was Birr 130. If x denotes the number of packets of wheat flour she bought, then the total amount of Birr she spent

would be 130x. Since, she had to buy wheat flour in packets, she would not be able to spend the total amount of Birr 500. Explain.

Clearly the statement is not an equation as it does not involve the sign of equality.

Hence, $130\,x < 500$ (1)

Consider the inequality (1), $130\,x < 500$ clearly, where x can never be fraction or negative. Left hand side of this inequality is 130 x and the right hand side is 500.

When $x = 0$, i.e.$130 \times 0 = 0 < 500$, the statement is true.

When $x = 1$, i.e.$130 \times 1 = 130 < 500$, the statement is true.

When $x = 2$, i.e.$130 \times 2 = 260 < 500$, the statement is true.

When $x = 3$, i.e.$130 \times 3 = 390 < 500$, the statement is true.

When $x = 4$, i.e.$130 \times 4 = 520 < 500$, the statement is false.

In the above situation, you find that the values of x, which make the above inequality a true statement, are 0, 1, 2, 3. These values of x, which make the inequality a true statement, are called solutions of inequality and the collection of these numbers, {0, 1, 2, 3} is called the solution set.

Thus, any solution of an inequality in one variable is a value of the variable that makes it a true statement. However, the solutions of the inequality shown above is by *trial and error* which is not very efficient because, it is time consuming and sometimes not feasible. Thus, you must have more efficient systematic techniques for solving inequalities. Before looking at these techniques, you should go through some more properties of numerical inequalities and follow them as rules, while solving the problems in inequalities.

You will recall the following rules while solving linear equations:

Rule 1 Equal numbers may be added to (or subtracted from) both sides of an equation.

Rule 2 Both sides of an equation, may be multiplied (or divided) by the same non-zero number.

Nevertheless, Rule 2 has an exception when the operation involves negative number. The sign of inequality is reversed whenever you multiply (or divide) both sides of an inequality by a negative number.

Consider the following rules for solving an inequality:

Rule 1 Equal numbers may be added to (or subtracted from) both sides of an inequality without affecting the sign of inequality.

Rule 2 Both sides of an inequality can be multiplied (or divided) by the same positive number. But when both sides are multiplied or divided by a negative number, then the sign of the inequality is reversed.

Example 1

Solve $7x + 3 < 5x + 9$ and, then, show the graph of the solution on a number line.

Solution

By Rule 1, we subtracted 3 and $5x$ from both sides of the inequality.

$$7x + 3 < 5x + 9,$$
$$7x - 5x < 9 - 3,$$
$$2x < 6,$$
$$x < 3.$$

Which is equivalent to $(-\infty, 3)$, the graphical representation of the solution is given in Figure 4.1.

Figure 4.1

Example 2

Solve the following problem and then, show the graph of the solution on a number line.

$$\begin{cases} 2x - 3 < 1 \\ 4x - 1 \geq 1 \end{cases}$$

Solution

From the first, $2x < 4$, $x < 2$. From the second, $4x \geq 2, x \geq \frac{2}{4}$, $x \geq \frac{1}{2}$. Then, the answer is $\frac{1}{2} \leq x < 2$.

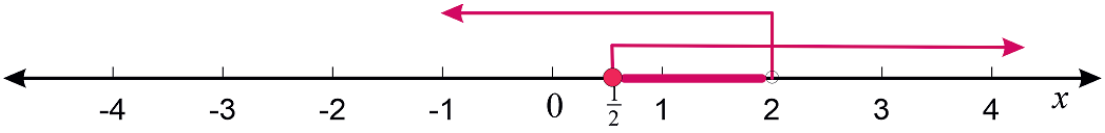


Figure 4.2

Example 3

Solve $\frac{3x-5}{2} \geq \frac{x+1}{4} - 2$ and show the graph of the solution on a number line.

Solution

By Rule 2

$$\frac{3x-5}{2} \geq \frac{x+1}{4} - 2$$

$$\frac{3x-5}{2} \times 4 \geq \frac{x+1}{4} \times 4 - 2 \times 4$$

$$2(3x-5) \geq x + 1 - 8$$

$$6x - 10 \geq x - 7$$

$$5x \geq 3$$

$$x \geq \frac{3}{5}$$

That is, the interval $[\frac{3}{5}, \infty)$ whose graphical representation of the solution is given in Figure 4.3.

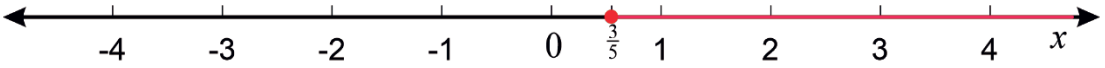


Figure 4.3

Exercise 4.1

Solve the following inequalities and show their solutions on number lines.

a. $5x + 1 < 2x + 7$

b. $\begin{cases} 3x + 1 < 16 \\ -2x + 5 \leq 13 \end{cases}$

c. $-1 < -2x + 3 \leq 2$

d. $\frac{2x}{3} - 3 > \frac{16x}{21} - \frac{13}{3} - \frac{2x}{15}$

e. $\frac{x-4}{2} < \frac{7x}{2} - (3x + 2)$

f. $5(x - 4) + 17 - 14x > 13 - 4x$

g. $12x - 1 \leq 3(4x - 3)$

h. $11(2x - 15) < x + 3$

i. $\frac{x}{x-3} < 2, x \neq 3$

j. $\frac{x+2}{x} - 1 \geq 0,\ x \neq 0.$

4.1.2 Word Problem of Linear Inequality

To solve the word problem related to real world situations you can use the following procedures:

Step 1: Understand the problem

Step 2: Setting up the model

Step 3: Solving the problem

Example 1

A student has Birr 120 and wants to buy some exercise books and pens. The cost of one exercise book is Birr 20 and that of a pen is Birr 10. Write the mathematical expression.

Solution

Let x denotes the number of exercise books and y denotes the number of pens that the student will buy, then the total amount that will be spent by the student is Birr $(20\,x + 10\,y)$. Then, we have:

$$20\,x + 10\,y \leq 120\,. \qquad (2)$$

In this case the total amount spent may be up to Birr 120. Note that expression (2) consists of two statements

$$20\,x + 10\,y < 120, \quad (3)$$

$$20\,x + 10\,y = 120. \quad (4)$$

Accordingly, expression (3) is not an equation, i.e., it is an inequality while statement (4) is an equation. Statement, $130\,x < 500$ is an example of linear inequalities in one variable and (2) is an example of linear inequalities in two variables.

Some more examples of inequalities are:

$$ax + b < 0, \quad (5)$$

$$ax + b > 0, \quad (6)$$

$$ax + b \geq 0, \quad (7)$$

$$ax + by \leq 0, \quad (8)$$

$$ax + by \geq 0. \quad (9)$$

Inequalities (5) and (6) are strict inequalities while inequalities (7), (8) and (9) are slack inequalities. Besides, inequalities from (5) to (7) are linear inequalities in one variable x when $a \neq 0$, while inequalities (8) and (9) are linear inequalities in two variables x and y when $a \neq 0$, and $b \neq 0$.

Example 2

A taxi charges a flat rate of Birr 1.75, plus an additional Birr 0.75 per km. If Selam has at most Birr 10 to spend on the car ride, how far could she travel?

Solution

The total cost to travel x distance is T, $T = 0.75x + 1.75$. She has only Birr 10.00. Therefore, $T = 0.75x + 1.75 \leq 10$. Solving for x, you obtain $x \leq 11$. Selam would travel a maximum distance of 11km.

Example 3

Find all pairs of consecutive odd natural numbers both of which are larger than 20, and their sum is less than 80.

Solution

Let n be any natural number, then $2n-1$ and $2n+1$ are two consecutive odd natural numbers. From the given information

$$2n-1>20, \tag{1}$$

$$(2n-1)+(2n+1)<80. \tag{2}$$

Simplifying (2), we get $4n<80$,

$$\text{i.e. } n<20. \tag{3}$$

From (1), $2n>21$,

$n>\frac{21}{2}$.

Combining (1) and (3), we obtain, $10.5<n<20$, thus, n takes the values, 11, 12… 18, 19. So, the required possible pairs will be (21, 23), (23, 25), (25, 27), (27, 29), (29, 31), (31, 33), (33, 35), (35, 37), (37, 39).

Exercise 4.2

1. To get an "A" grade in a course, one must obtain an average of 85 marks or more in five examinations (each out of 100 marks). If Semira's marks in the first four examinations are 87, 82, 81 and 80, then, find the minimum marks that Semira must receive in the fifth examination to obtain "A" grade in the course.
2. Find all pairs of consecutive odd positive integers both of which are smaller than 20 and their sum is more than 21.

3. Show that the following word problems are expressed by inequalities not by equation.
   a. Lenssa practices her bicycle for at least 12 hours per week. She practices for $3/4$ of an hour each session. If Lenssa has already practiced 3 hours this week, how many more sessions remain for her to meet or exceed her weekly practice goal?
   b. The base of a rectangular prism has a length of 13 cm and a width of $1/2$ cm. The volume of the prism is less than or equal to 65cm^3. Find all possible heights of the prism.

4.1.3 Linear Inequalities in Two Variables and Their Graphical Solutions

In the previous section, you have seen that an inequality in one variable is a visual representation and is a convenient way to represent the solutions of the inequality. Now, you will discuss the graph of a linear inequality in two variables.

Activity 4.2

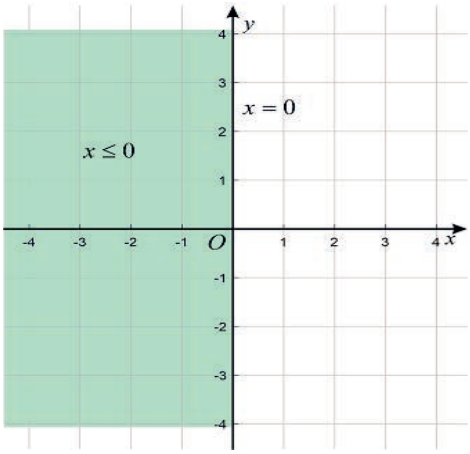
Choose any three points and show in which side of the line $y = x$, in the xy-coordinate plane.

You know that a line divides the Cartesian plane into two parts. Each part is known as a half plane. A vertical line will divide the plane into left and right half planes and a non-vertical line will divide the plane into lower and upper half planes.

Definition 4.1

Half-Plane: The region on a side of a line in the xy-plane.

Vertical Line



Non-vertical Line

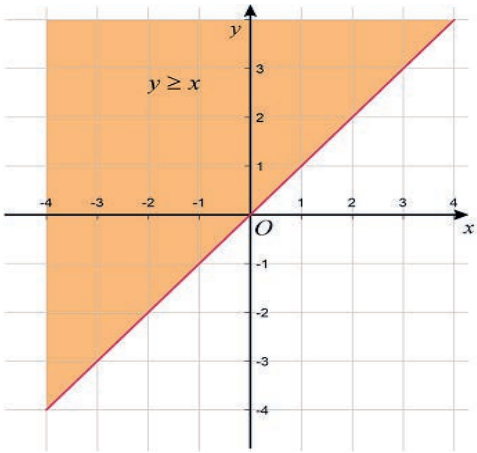


Figure 4.4

A point in the Cartesian plane will either lie on a line or on either of the half planes I or II.

Steps to draw the graph of inequalities

1. Change the inequality sign to an equal sign, and then plot the line.
   a. If an inequality is of the form $y > mx + c$ or $y < mx + c$, then the points on the line $y = mx + c$ are not included in the solution region. So, draw a broken or dotted line,
   b. If an inequality is of the type $y \geq mx + c$ or $y \leq mx + c$, then the points on the line $y = mx + c$ are also included in the solution region. So, draw a solid line.
2. Test a point created in one half-plane.
   a. If the point in the half-plane satisfies the inequality, then the entire half-plane satisfies the inequality.
   b. If the point does not satisfy the inequality, then the entire half-plane does not satisfy the inequality.
3. If a point from one half plain fails to satisfy, test a point from the other half-plane.
4. Shade in any half-planes that satisfy the inequality.

Example 1

Graph the solution set for the following inequalities:

a) $y < 5x + 7$ b) $y \geq 2x + 3$

Solution

a) First change the inequality to the equation, $y = 5x + 7$. It is the line that passes through $(0, 7)$ and its slope is 5. Note that the line is dashed (broken) since the inequality is <, the points on the line are not included in the region. Secondly, test the point (0, 0).

$$(LHS) = 0,$$
$$(RHS) = 5 \times 0 + 7 = 7.$$
$$(LHS) < (RHS)$$

Thus, this point satisfies the inequality. Therefore, every point in the lower side of the line $y = 5x + 7$ satisfies the inequality. Finally, shade the region in the half-plane to the right of the line.

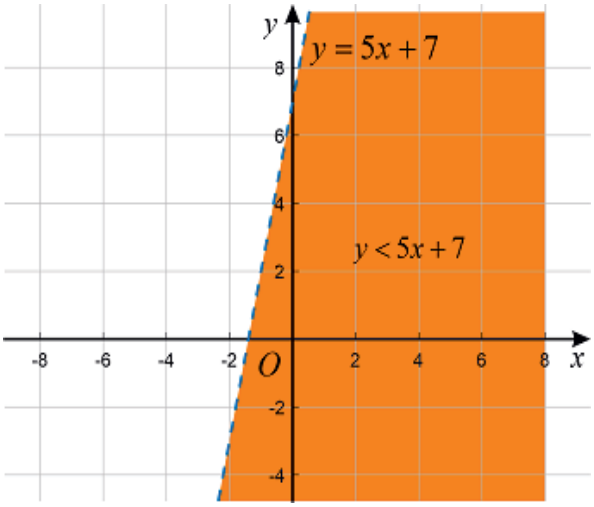


Figure 4.5

b) To plot the region of the solution, first draw the graph of $y = 2x + 3$, whose slope is 2 and y-intercept is $(0, 3)$.

To identify the region test the point $(0, 0)$,

$$(LHS) = 0,$$

$$(RHS) = 2 \times 0 + 3 = 3$$

$$(LHS) < (RHS).$$

Thus, this point does not satisfy the inequality. So the region should be on the other side as shown in the Figure 4.6. Note that the line is solid since the inequality sign is ≥. This means that any point on the line satisfy the given inequality.

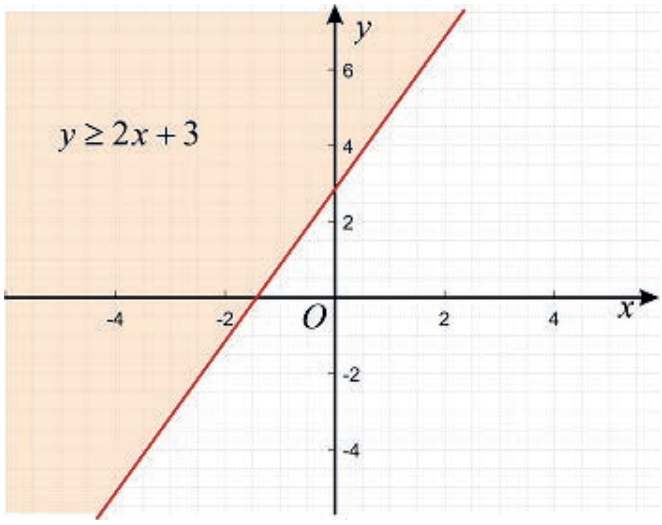


Figure 4.6

Example 2

Write and graph an inequality that describes all ordered pairs whose *y*-coordinate is at least 5.

Solution

For this problem the value of x is all real numbers. But the y value is any number greater than or equal to 5, i.e., $y \geq 5$ is the required inequality. The graphical representation is shown in Figure 4.7.

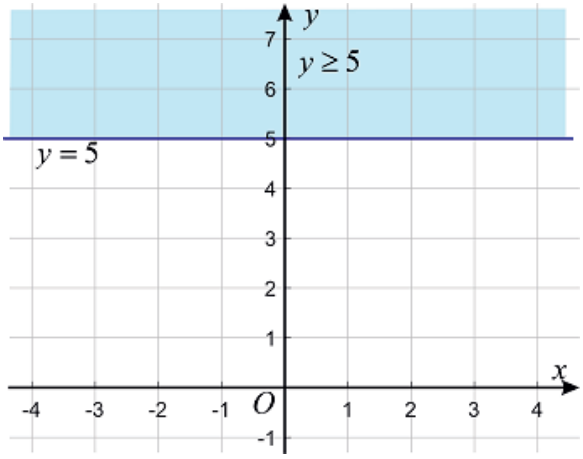


Figure 4.7

Exercise 4.3

1. Graph the solution set for the following inequality:

   a. $y > 5x + 3$ b. $y \leq -\frac{1}{2}x + 7$

   c. $x > 4$ d. $5x + 2y > 6$

2. Is $(2, 5)$ a solution of the inequality $y + 5x \geq 1$?

3. Determine whether or not $(2, \frac{1}{2})$ is a solution to $5x - 2y < 10$.

4. Write an inequality that describes all points in:

   a. the upper half-plane above x-axis;

   b. the lower half-plane below x-axis;

   c. the half-plane left of y-axis;

   d. the half-plane right of y-axis.

4.1.4 Systems of Linear Inequalities in Two Variables and Graphical Solutions

A system of linear inequalities is a collection of two or more linear inequalities to be solved simultaneously. A graphical solution of a system of linear inequalities is the graph of all ordered pairs (x, y) that satisfy all the inequalities. Such a graph is called the solution region (or the feasible region).

Activity 4.3

A student wants to spend at most Birr 15.00 for his breakfast milk and bread. Each cup of milk costs Birr 2.40 and each loaf of bread costs Birr 1.20. Write an inequality that represents the number of cups of milk (x) and loaf of bread (y) a student can buy on his budget.

How to Solve a System of Linear Inequalities in Two Variables

1. Using the technique of graphing inequalities, graph all of the given inequalities.
2. If the inequality is"$y \geq$ ...." or " $>$ ....", then you shadow the upper side of the line. Similarly, if the inequality is"$y \leq$ ....", or " $y <$ ....", then you shadow the lower side of the line.
3. The overlapped shaded region is the solution of the system.
4. Regarding the intersection of the lines,
   a. it is part of the solution if all given inequalities include the lines (i.e., $\leq$ or $\geq$)
   b. it is not part of the solution if any one of the given inequalities does not include the line (i.e., $<$ or $>$).

Example 1

Find a graphical solution to the system of linear inequalities $\begin{cases} y \geq -x + 3 \\ y \leq 2x \end{cases}$

Solution

First draw the lines $y = -x + 3$ and $y = 2x$ in the same coordinate plane.

For the first inequality, shade the region above the line $y = -x + 3$ and for the second shade the region below the line $y = 2x$.

When solving the system $\begin{cases} y = -x + 3 \\ y = 2x \end{cases}$ simultaneously we arrive at the point of intersection.

Thus, the overlapped shaded region (green part) is the region that meets the given system of linear inequalities. In this example, the point $(1, 2)$ is also part of the solution.

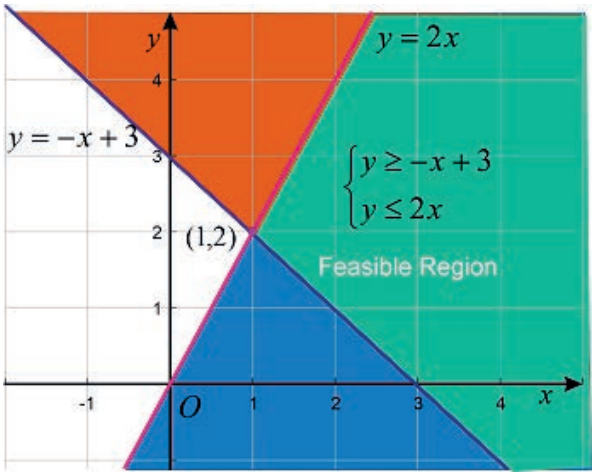


Figure 4.8

Example 2

Find a graphical solution to the system of linear inequalities.

$$\begin{cases} 5x - y > -7 \\ 2x + y \geq 14 \end{cases}$$

Solution

First solve the inequality for y.

$$\begin{cases} y < 5x + 7 \\ y \geq -2x + 14 \end{cases}$$

Then, draw the lines on the same coordinate.

$$\begin{cases} y = 5x + 7 \\ y = -2x + 14 \end{cases}$$

The first inequality is $<$, so shade the lower side of the dot line. The second one is $\geq$, hence, shade the upper side of the solid line. Thus, the overlapped shaded region (orange part) is the region that meets the given system of linear inequalities.' To find the point of intersection of the two lines, solve

$\begin{cases} y - 5x = 7 \\ y + 2x = 14 \end{cases}$, simultaneously, and obtain the point (1, 12). Which is not part of the solution region.

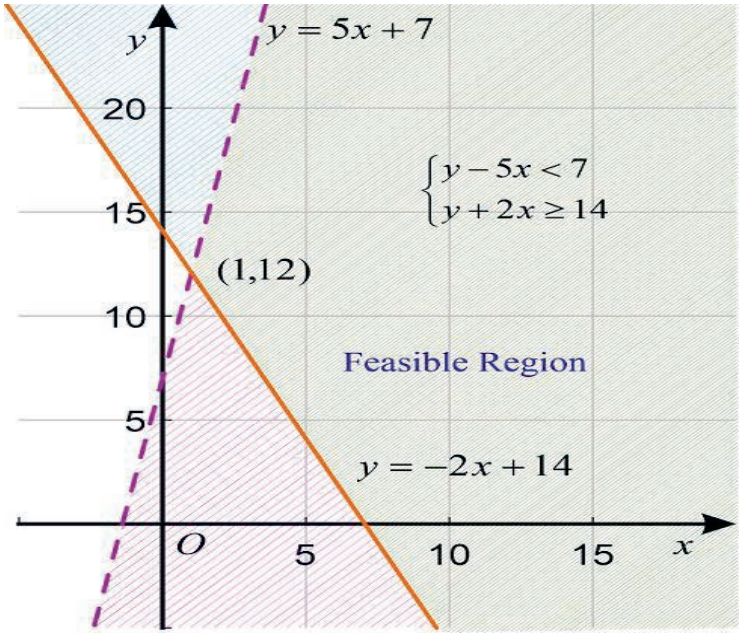


Figure 4.9

Example 3

Determine whether or not the order pairs $(0, 0), (2, 3), (-8, 2)$,and $(-4, 6)$ are in the solutions of the system

$$\begin{cases} x + 3y > 3 \\ -x + y \leq 6 \end{cases}$$

Solution

To test whether or not each point is in the solution region of the system, we use the following table:

Order pairs	Satisfies $x + 3y > 3$	Satisfies $-x + y \leq 6$	Satisfies both inequalities
(0, 0)	No	Yes	No
(2, 3)	Yes	Yes	Yes
(-8, 2)	No	No	No
(-4, 6)	Yes	No	No

Therefore, the only point in the solution region is (2, 3).

Exercise 4.4

1. Solve the following systems of inequalities graphically.

a. $\begin{cases} y \leq 2x + 2 \\ y \geq -2x + 6 \end{cases}$ b. $\begin{cases} 2x + y < 8 \\ x + y \geq 1 \end{cases}$ c. $\begin{cases} y - 2x \leq 4 \\ 2x + y \geq 4 \end{cases}$

d. $\begin{cases} x - y < 0 \\ x + 2y > 3 \end{cases}$ e. $\begin{cases} 2x + 3y \geq 6 \\ x + 5y \leq 4 \end{cases}$ f. $\begin{cases} y - x \geq 0 \\ 2x - 3y \leq 1 \end{cases}$

g. $\begin{cases} y + x > 0 \\ y - 2x < 5 \end{cases}$

2. Determine whether or not the ordered pair is a solution to the system $\begin{cases} x + 4y \geq 10 \\ 3x - 2y < 12 \end{cases}$

a. $(-2, 4)$ b. $(3, 1)$ c. $(1, 3)$ d. $(0, 0)$

4.1.5 Further on the System of Inequalities

Definition 4.2

A point of intersection of two or more boundary lines of a solution region is called a vertex (or a corner point) of the region.

Example 1

Solve the following system of inequalities graphically.

$$\begin{cases} 5x + 4y \leq 40 \\ x \geq 1 \\ y \geq 2 \end{cases} \quad (1)$$

Solution

We first rewrite the first inequality, $y \leq -\frac{5}{4}x + 10$

Draw the graphs of the lines,

$$y = -\frac{5}{4}x + 10 \quad (2)$$
$$x = 1 \quad (3)$$
$$y = 2 \quad (4)$$

Then, we know that the inequality $y \leq -\frac{5}{4}x + 10$ represents the shaded region below the line $y = -\frac{5}{4}x + 10$ and the second inequality represents the shaded region right of the line $x = 1$ but third inequality represents the shaded region above the line $y = 2$.

To find the vertices of the region, solve the equation (2) and (3), (2) and (4) simultaneously.

For (2) and (3), substituting $x = 1$ into (4), $y = -\frac{5}{4} \times 1 + 10 = \frac{35}{4}$

Therefore, $(x, y) = (1, \frac{35}{4})$

Similarly, for (2) and (4), substituting $y = 2$ into (2),

$$2 = -\frac{5}{4}x + 10,$$
$$\frac{5}{4}x = 8,$$
$$x = \frac{32}{5}.$$

Therefore, $(x, y) = (\frac{32}{5}, 2)$

Hence, the shaded region including all the points on the lines are also the solution of the given system of the linear inequalities.

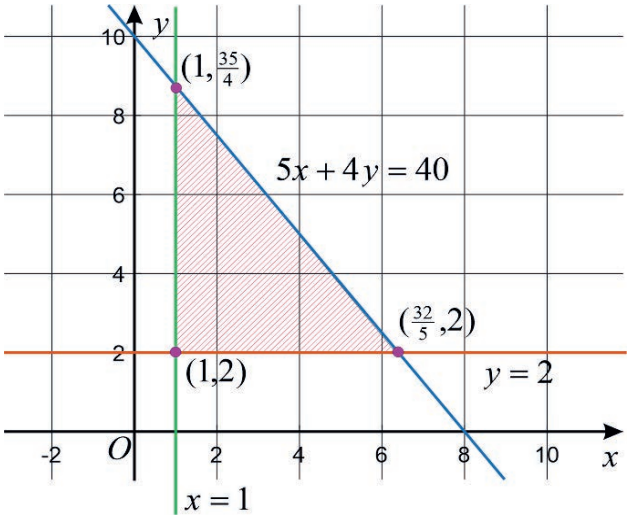


Figure 4.10

In many practical situations involving system of inequalities, the variables x and y often represent quantities that cannot have negative values. For example, number of units produced, number of articles purchased, and number of hours worked, etc. Clearly, in such cases, $x \geq 0, y \geq 0$ and the solution region lies only in the first quadrant.

Example 2

Ficho needs to purchase supplies of answer sheets and pencils for a standardized test to be given to the juniors at her high school. The number of the answer sheets needed is at least 40 more than the number of pencils. The pencils cost Birr 2.00 and the answer sheet cost Birr 1.00. Ficho's budget for these supplies allows for a maximum cost of Birr 400.

a. Write a system of inequalities to model this situation.
b. Graph the system.
c. Can Ficho purchase 100 answer sheets and 150 pencils?
d. Can Ficho purchase 150 answer sheets and 100 pencils?

Solution

a. Letting x represents the number of answer sheets and y is the number of pencils. $\begin{cases} x \geq y + 40 \\ x + 2y \leq 400 \\ x \geq 0 \\ y \geq 0. \end{cases}$

Re-write the inequalities:

$$\begin{cases} y \leq x - 40 \\ y \leq -\frac{1}{2}x + 200 \\ x \geq 0 \\ y \geq 0 \end{cases}$$

Then, draw the lines of equations:

$$\begin{cases} y = x - 40 & (1) \\ y = -\frac{1}{2}x + 200 & (2) \\ x = 0 & (3) \\ y = 0 & (4) \end{cases}$$

To find the three vertices, solve each pair of equations simultaneously.

For (1) and (2),

$$x - 40 = -\frac{1}{2}x + 200, \quad \frac{3}{2}x = 240, \quad x = 240 \times \frac{2}{3} = 160$$

$$y = 160 - 40 = 120.$$

Therefore, $(x, y) = (160,120)$. The other two vertices are also found by solving (1) and (4), and (2) and (4) simultaneously, (40, 0) and (400, 0).

b. The graph is as shown in Figure 4.11.

c. Ficho cannot purchase 100 answer sheets and 150 pencils.

d. Ficho can purchase 150 answer sheets and 100 pencils.

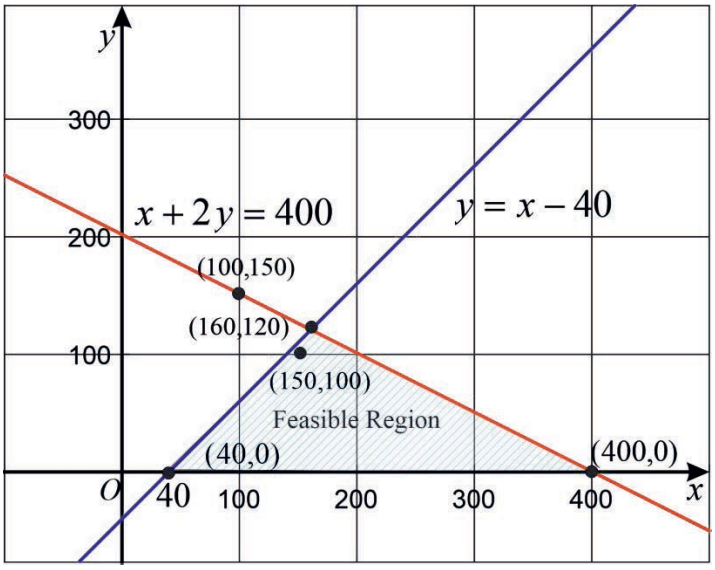


Figure 4.11

Example 3

Graph the solution set of the system

$$\begin{cases} y \leq -2x + 22 \\ y \leq -x + 13 \\ y \leq -\frac{2}{5}x + 10 \\ x \geq 0 \\ y \geq 0 \end{cases}$$

Solution

Draw the lines: $y = -2x + 22$, $y = -x + 13$ and $y = -\frac{2}{5}x + 10$. All the first three inequalities are $\leq$. Then, shade the region on the first quadrant below those lines. Finally, you get the shaded region as shown in Figure 4.12.

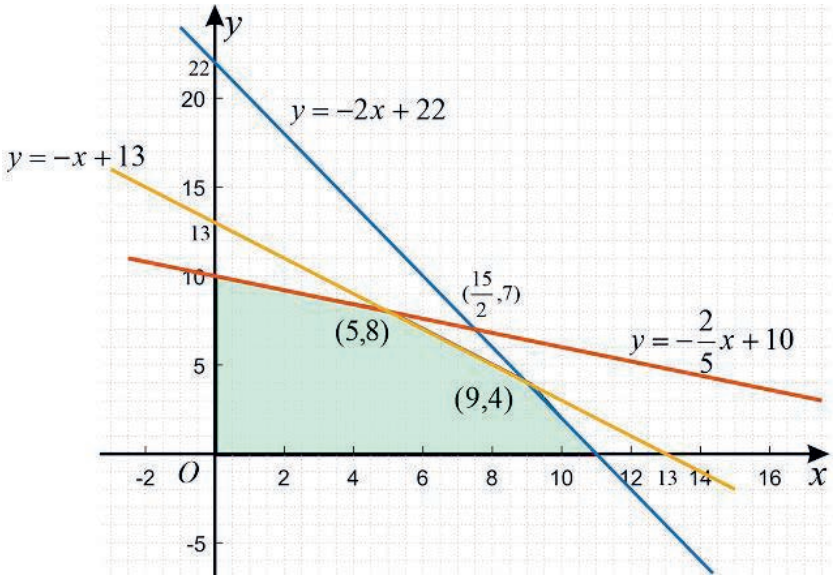


Figure 4.12

Exercise 4.5

1. Solve the system of linear inequalities $\begin{cases} y \leq -2x + 5 \\ x \geq 0 \\ y \geq 1 \end{cases}$

2. Solve the system $\begin{cases} y \leq -\frac{1}{2}x + 5 \\ y \leq -x + 8 \\ y \leq x \\ x \geq 0 \\ y \geq 0. \end{cases}$

3. A rectangular field with at most 200 meter of fencing is constructed. Write a linear inequality in terms of the length l and the width w. Sketch the graph of all possible solutions to this problem.

4. A company sells one product for Birr 8 and another for Birr 12. How many of each product must be sold so that revenues are at least Birr 2,400? Let x represent the number of products sold at Birr 8 and let y represent the number of products sold at Birr 12. Write a linear inequality in terms of x and y and sketch the graph of all possible solutions.

5. Yiseresh sells her photographs at a Gallery at a street fair. At the start of the day, she wants to have at least 25 photos to display at her Gallery. Each small photo she displays costs her Birr 40.00 and each large photo costs her Birr 80.00. She doesn't want to spend more than Birr 2000.00 on photos to be displayed.
   a. Write a system of inequalities to model this situation.
   b. Graph the system.
   c. Can she display 20 small and 10 large photos?
   d. Can she display 10 small and 30 large photos?

4.2 Maximum and Minimum Values

Activity 4.4

A student has 100m metallic wire. And he wants to enclose a land that he has by the wire. Which shape maximizes the area enclosed by the wire and which shape minimizes the area? Discuss the answer in class.

Terminologies

Optimization problem A problem which seeks to maximize or minimize a linear function (say of two variables x and y) subject to certain constraints as determined by a set of linear inequalities is called an optimization problem. A special but a very important class of optimization problems is linear programming problem.

Linear programming is defined as a field of mathematics that deals with the problem of finding the maximum and minimum values of a given linear expression (called **objective function**), where the variables are subject to certain conditions (called **linear constraints)** expressed as linear inequalities. Linear programming problems are of much interest because of their wide applicability in industry, commerce, management science and so on.

Constraints mean the situation in which we cannot just take any x and y when looking for the x and y that optimise our objective function. If you think of the

variables x and y as a point (x, y) in the xy −plane then you call the set of all points in the xy-plane that satisfy our constraints are the **feasible region**. Any point in the feasible region is called a **feasible point.**

Before we proceed further, we now formally define some terms we have used above and shall be using in the linear programming problems:

Objective function Linear function $z = ax + by$, where a, b are constants, which has to be maximized or minimized is called a linear objective function.

Feasible solutions are points within and on the boundary of the feasible region that represent feasible solutions of the constraints.

Optimal (feasible) solution: Any point in the feasible region that gives the optimal value (maximum or minimum) of the objective function is called an **optimal solution.**

Example 1

A furniture dealer deals in only two items - tables and chairs. He has Birr 50,000 to invest and has storage space of at most 60 pieces. A table costs Birr 2500 and a chair Birr 500. He estimates that he can make a profit of Birr 300 from the sale of one table and Birr 100 from the sale of one chair. He wants to know how many tables and chairs he should buy from the available money so as to maximize his total profit, assuming that he can sell all the items which he buys.

Solution

Mathematical formulation of the problem

Let x be the number of tables and y be the number of chairs that the dealer buys. Clearly, x and y must be non-negative. The dealer has two limitations - investment capital and storage capacity. Suppose he decides to buy only tables but not chairs. So, he can buy 50000 ÷ 2500, i.e., 20 tables. His profit in this case will be Birr (300 × 20) = Birr 6,000.00.

On the other hand, let's suppose he chooses to buy only chairs but not tables. With his capital of Birr 50,000, he can buy 50000 ÷ 500, i.e., 100 chairs. But he can store only 60 pieces. Therefore, he is forced to buy only 60 chairs which will give him a total profit of (60 × 100) =Birr 6,000.00. There are many other possibilities. The dealer is constrained by the maximum amount he can invest and maximum number of items he can store.

$2500x + 500y \leq 50000$ or $5x + y \leq 100$, investment limitation (1)

$x + y \leq 60$, storage limitations (2)

$x \geq 0,\ y \geq 0$, non-negativity constraints (3)

We have to maximize the linear function z subject to certain conditions determined by a set of linear inequalities with variables as non-negative.

If the dealer wants to maximize profit, $Z = 300x + 100y$ is a linear objective function. Variables x and y are called decision variables.

Definition 4.3

Suppose f is a function with domain $I = \{x : a \leq x \leq b\}$

I. A number $M = f(c)$ for some c in I is called the **maximum** value of f on I, if $M \geq f(x)$, for all x in I.

II. A number $m = f(d)$ for some d in I is called the **minimum** value of f on I, if $m \leq f(x)$, for all x in I.

III. A value which is either a maximum or a minimum is called an **optimum** (or **extremum**) value of f on I.

You will now discuss how to find solutions to a linear programming problem. In this section, you will be concerned only with the graphical method.

Graphical method of solving linear programming problems

In Section 4.1, you learnt how to graph a system of linear inequalities involving two variables x and y and to find its solutions graphically.

Let us refer to the problem of investment in tables and chairs discussed in Example 1. You will now solve this problem graphically. Let us graph the constraints stated as linear inequalities (1)- (3). The graph of this system (shaded region) consists of the point's common to all half planes determined by the inequalities (1) to (3). Each point in this region represents a **feasible choice** open to the dealer for investing in tables and chairs.

In Figure 4.13(a), the region *OABC* is the feasible region for the problem. The region other than feasible region is called an **infeasible region**.

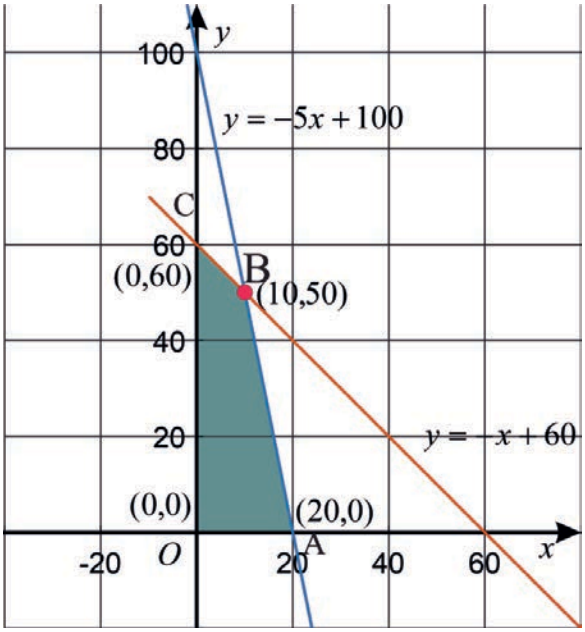


Figure 4.13(a)

In Figure 4.13(a), every point within and on the boundary of the feasible region *OABC* represents feasible solution to the problem. For example, the point (10, 50) is a feasible solution of the problem and so are the points (0, 60), (20, 0) etc. The point (25, 40) is an infeasible solution of the problem.

You see that every point in the feasible region *OABC* satisfies all the constraints given in (1) to (3). Since there are **infinitely many points**, it is not evident how we should go about finding a point that gives a maximum value of the objective function $Z = 300x + 100y$.

The possible candidate to give a maximum value are the corner points (vertices) of the bounded (feasible) region. Such points are: *O, A, B* and *C* and it is easy to find their coordinates as (0, 0), (20, 0), (10, 50) and (0, 60) respectively. Let us now compute the values of Z at these points. The Z values will be obtained by substituting the coordinates of the corners into the objective function

$$Z = 300x + 100y,$$

At O, $Z(0,0) = 300 \times 0 + 100 \times 0 = 0$.

At A, $Z(20,0) = 300 \times 20 + 100 \times 0 = 6000$.

At B, $Z(10,50) = 300 \times 10 + 100 \times 50 = 3000 + 5000 = 8000$.

At C, $Z(0,60) = 300 \times 0 + 100 \times 60 = 0 + 6000 = 6000$.

Accordingly, you observe that the maximum profit to the dealer results from the investment strategy (10, 50), i.e. buying 10 tables and 50 chairs. This method of solving linear programming problem is referred as **Corner Point Method**. You can verify the solution as follows. Solving the objective function for y,

$$Z = 300x + 100y$$

$$y = -3x + \frac{Z}{100}$$

This means, it is a line, whose slope is -3, and y-interception is $\frac{Z}{100}$. We shall move this line within the region to find the maximum value of y-interception, and the value of Z.

As shown in the Figure 4.13(b), the value of y-interception will be maximum, when the line passes through B (10, 50). Then, the value of Z will be:

$$y = -3x + \frac{Z}{100}$$

$$50 = -3 \times 10 + \frac{Z}{100}$$

$$80 = \frac{Z}{100} \Leftrightarrow Z = 8000$$

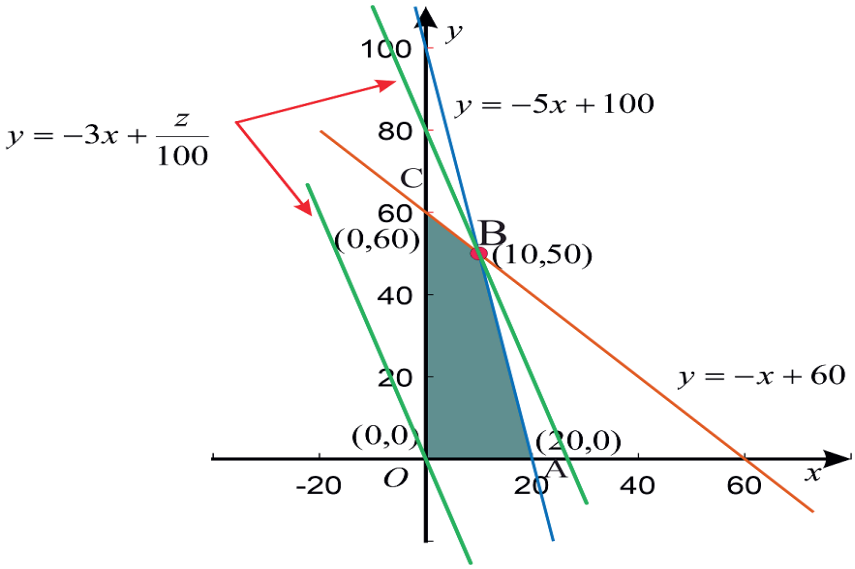


Figure 4.13(b)

Exercise 4.6

1. A furniture dealer deals in only two items - tables and chairs. He has Birr 50,000 to invest and has storage space of at most 60 pieces. A table costs Birr 2500 and a chair Birr 500. Suppose that the dealer can make profit of Birr 200 from the sale of one table and Birr 50 from that of a chair.
   a. Find the linear objective function of the profit.
   b. What is the maximum profit can the dealer make?
2. Consider an ice-cream manufacturing company that produces only two types of ice-cream A and B. Both the ice-cream require Milk and Cream only. To manufacture each unit of A and B, the following quantities are required: Each unit of A requires 1 unit of Milk and 3 units of Cream. Each unit of B requires 1 unit of Milk and 2 units of Cream. The company kitchen has a total of 5 units of Milk and 12 units of Cream. On each sale, the company makes a profit of: Birr 6 per unit A sold, Birr 5 per unit B sold. Now, the company wishes to maximize its profit. How many units of A and B should it produce respectively?

4.2.1 Steps to Solve a Linear Programming Problem

Definition 4.4

A corner point of a feasible region is a point in the region which is the intersection of two boundary lines.

Definition 4.5

A feasible region of a system of linear inequalities is said to be bounded if it can be enclosed within a line. Otherwise, it is called unbounded. Unbounded means that the feasible region extends indefinitely in any direction.

We use the following theorems which are fundamental in solving linear programming problems. However, the proofs of these theorems are beyond the scope of the textbook.

Theorem 4.1

Let R be the feasible region for a linear programming problem and let $Z = ax + by$ be the objective function. When Z has an optimal value (maximum or minimum), where the variables x and y are subject to constraints described by linear inequalities, this optimal value must occur at a corner point (vertex) of the feasible region.

Theorem 4.2

Let R be the feasible region for a linear programming problem, and let $Z = ax + by$ be the objective function. If R is **bounded**, then the objective function Z has both a **maximum** and a **minimum** value on R and each of these occurs at a corner point (vertex) of R.

Remark

If R is **unbounded**, then a maximum or a minimum value of the objective function may not exist. However, if it exists, it must occur at a corner point of R. (By Theorem 4.1).

Steps to solve a linear programming problem by corner point method

Step 1: Draw the graph of the feasible region.

Step 2: Determine the coordinates of the corner points either by inspection or by solving the two equations of the lines intersecting at that point.

Step 3: Evaluate the objective function at each corner points. Let M and m be the largest and smallest values of these points respectively.

Step 4: If the feasible region is bounded, M and m are the maximum and minimum values of Z. If the feasible region is unbounded:

i. M is the maximum value of Z, if the open half plane determined by $ax + by > M$ has no point in common with the feasible region. Otherwise, Z has no maximum value.

ii. Similarly, m is the minimum value of Z, if the open half plane determined by $ax + by < m$ has no point in common with the feasible region. Otherwise, Z has no minimum value.

Optimal solutions always exist when the feasible region is bounded, but may or may not exist when it is unbounded. You will now illustrate these steps of Corner Point Method by considering additional examples:

Example 1

Solve the linear programming problem by Corner Point Method.

Maximize $Z = x + y$, subject to the following constraints:

$$\begin{cases} x + 2y \leq 4 \\ x - y \leq 1 \\ x \geq 0 \\ y \geq 0 \end{cases}$$

Solution

The given constrains are changed to

$$\begin{cases} y \leq -\frac{1}{2}x + 2 \\ y \geq x - 1 \\ x \geq 0 \\ y \geq 0 \end{cases}$$

Draw the region of constrains by drawing the lines of equations:

$$\begin{cases} y = -\frac{1}{2}x + 2 \\ y = x - 1 \\ x = 0 \\ y = 0 \end{cases}$$

As shown in the Figure 4.14, the region is bounded. The corner points of the region are (0, 0), (1, 0), (0, 2) and (2, 1). We can evaluate Z at each point as follows:

$$Z = x + y$$

When (0, 0), then $Z = 0 + 0 = 0$

When (1, 0), then $Z = 1 + 0 = 1$

When (0, 2), then $Z = 0 + 2 = 2$

When (2, 1), then $Z = 2 + 1 = 3$

Or solve the objective function for y, $y = -x + Z$. The value Z is the y-intercept of the line. When the line passes the point (2, 1), the value Z has the maximum value of 3. Therefore, the maximum value of Z is 3, and it occurs when $x = 2, y = 1$.

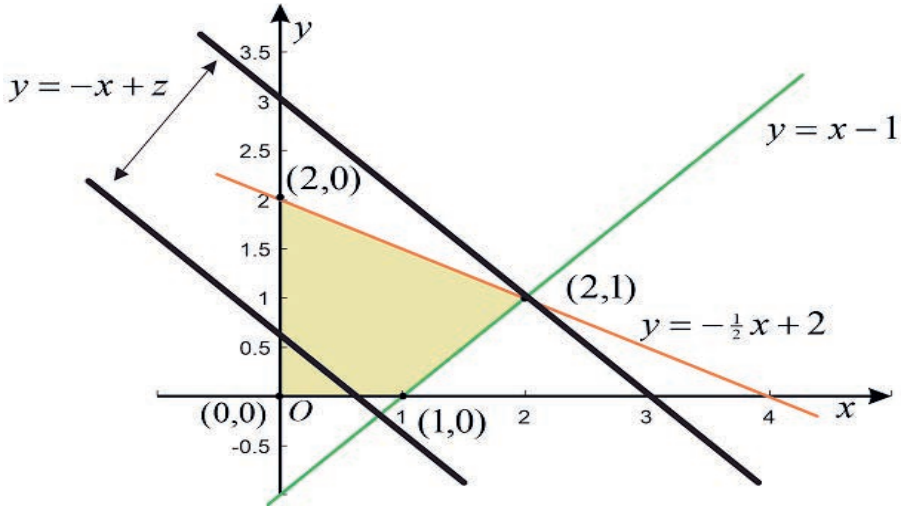


Figure 4.14

Example 2

Solve the linear programming problem by Corner Point Method.

Maximize $Z = 3x + 2y$, subject to the same constraints as Example 1:

Solution

From the constraints you sketched the feasible region shown in Figure 4.14 and observed that the feasible region is **bounded.** So, you can test each vertex to determine the maximum value of Z. The corner points of this region are (0, 0), (1, 0), (2, 1) and (0, 2) and you can evaluate Z at each corner point.

Corner points (vertices)	Corresponding Values of Z
(0, 0)	$3 \times 0 + 2 \times 0 = 0$
(1, 0)	$3 \times 1 + 2 \times 0 = 3$
(2, 1)	$3 \times 2 + 2 \times 1 = 8$
(0, 2)	$3 \times 0 + 2 \times 2 = 4$

Thus, the maximum value of Z is 8, and occurs when $x = 2$ and $y = 1$.

Exercise 4.7

Solve the following Linear Programming Problems by Corner Point Method:

a. Maximize $Z = x + y$,

$$\text{Subject to: } \begin{cases} 2x+3y \le 12 \\ 2x+y < 8 \\ x \ge 0, y \ge 0 \end{cases}$$

b. Find values of x and y that maximize,

$$Z = x + 3y,$$

$$\text{Subject to: } \begin{cases} 2x+3y \le 24 \\ x-y < 7 \\ y \le 6 \\ x \ge 0, y \ge 0 \end{cases}$$

c. Minimize: $Z = -40x + 20y$,

$$\text{Subject to: } \begin{cases} 2x-y \ge -5 \\ 3x+y \ge 3 \\ 2x-3y \le 12 \\ x \ge 0, y \ge 0 \end{cases}$$

Further on a Linear Programming Problem

Activity 4.5

Draw the region enclosed by the following lines and identify the vertices.

a. $2x + 5y \le 10, x \ge 0, y \ge 0.$

b $x + y \le 40,\ 2x + 3y \le 72, x \ge 0, y \ge 0.$

Example 1

Find the maximum value of $Z = 4x + 6y$

$$\text{Subject to } \begin{cases} y \le x+1 \\ y \le -x+27 \\ y \le -\frac{2}{5}x+18 \\ x \ge 0, y \ge 0 \end{cases}$$

Solution

The region is determined by drawing the lines,

$$\begin{cases} y = x+1 \\ y = -x+27 \\ y = -\frac{-2}{5}x+18 \end{cases}$$

The resulting bounded region coordinates are $(0,0), (27,0), (15,12), (\frac{85}{7}, \frac{92}{7})$ and $(0,1)$. Solving the objective function for y, $y = -\frac{2}{3}x + \frac{Z}{6}$.

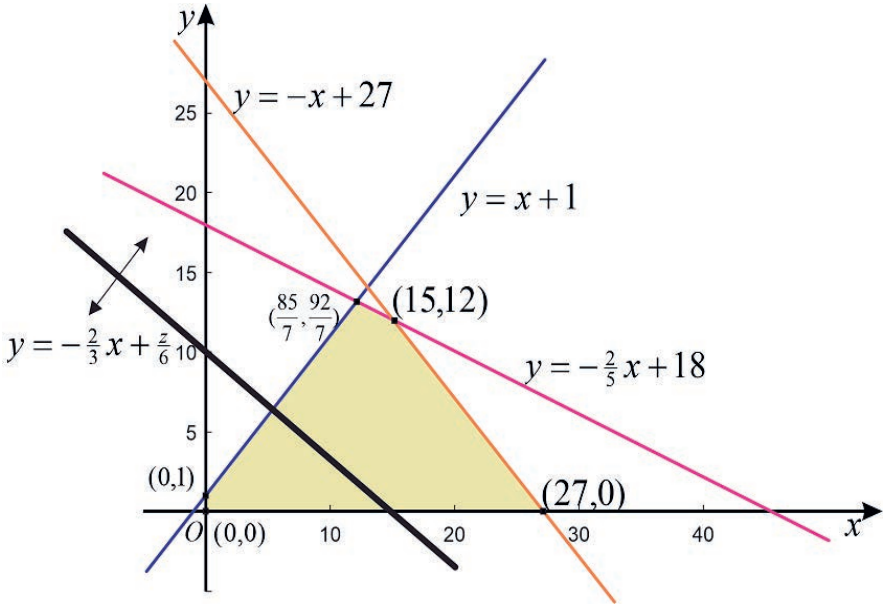


Figure 4.15

Testing the objective function at the corner points gives the following.

Corner points	Value of $Z = 4x + 6y$
(0, 0)	0
(27, 0)	108
(15, 12)	132
$(\frac{85}{7}, \frac{92}{7})$	892/7 or 127.43
(0, 1)	6

Thus, the maximum value of Z is 132 when $x = 15$ and $y = 12$.

Example 2

Find values of x and y which minimize the value of the objective function, $Z = 2x + 4y$,

Subject to $\begin{cases} y \geq -\frac{1}{2}x + 5 \\ y \geq -3x + 10 \\ x \geq 0, y \geq 0 \end{cases}$

Solution

From the given constraints the feasible region is as shown in Figure 4.16, which is unbounded. So the linear programing problem doesn't have maximum value. The corner points are at (0, 10), (2, 4) and (10, 0) with values given below.

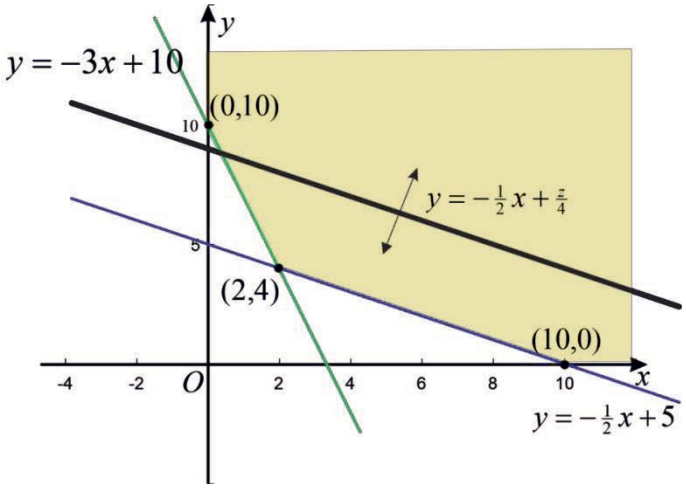


Figure 4.16

Here points (2, 4) and (10, 0) give the minimum value $Z = 20$. So, the solution is not unique. In fact, every point on the line segment through (2, 4) and (10, 0) gives the same minimum value of $Z = 20$.

Corner points	Values of Z
(0, 10)	$2\times0+4\times10=40$
(2, 4)	$2\times2+4\times4=20$
(10, 0)	$2\times10+4\times0=20$

From the above example, we can observe that

i. An optimization problem can have infinite solutions.

ii. Not all optimization problems have a solution, for instance, the problem in Example 2 doesn't have a unique set of (x, y) value corresponding to the minimum value for Z.

Exercise 4.8

Solve the following Linear Programming Problems graphically:

a. Maximize $Z = 3x + 4y$

Subject to the constraints: $\begin{cases} x + y \leq 4 \\ x \geq 0 \\ y \geq 0 \end{cases}$

b. Maximize or Minimize $Z = 5x + 6y$

Subject to $\begin{cases} 2x + y \leq 16 \\ x - y \leq 10 \\ x \geq 0 \\ y \geq 0 \end{cases}$

c. Maximize or Minimize $Z = 2x + 3y$

Subject to $\begin{cases} 3x + 7y \leq 42 \\ x + 5y \leq 42 \\ x \geq 0 \\ y \geq 0 \end{cases}$

4.3 Applications

In this section, you shall illustrate the application of linear programming to different real world problems. Some of the known linear programming problems are listed below:

4.3.1 Real Life Problems

A. Diet Problems: The basic idea behind the diet problem is choosing quantities of foods so as to satisfy nutritional requirements and ensure that the price of the resulting diet is within a reasonable range. Two versions of the problem can, at least in theory, be thought of: either we minimize the cost of the diet and ensure by way of the constraints that some nutritional constraints are satisfied, or we maximize the nutritional content of the diet subject to a budget constraint.

Example 1

World Food Program (WFP) uses at least 800 kg of special feed daily in conflict areas. A special feed is a mixture of corn and soybean meal with the composition of a kilogram of corn that contains 10% protein and 2% fiber costs Birr 30 per kg. A kilogram of soybean meal constitutes 60% protein and 6% fiber and costs Birr 90 per kg. The dietary requirements for the special feed are at least 30% protein and at most 4% fiber. WFP wishes the daily minimum cost feed mix.

Solution

The first step in solving such real life Linear Programming problems is to define the relevant decision variable. In any linear programming model, the decision variables should completely describe the decisions to be made. We define the two decision variables:

x is the number of kilogram of corn in the daily mix,

y is the number of kilogram of soybean meal in the daily mix.

The information is summarized as

	Content		Cost (Birr/kg)	Mixture in a day (kg)
	Protein (%)	Fiber (%)		
Corn	10	2	30	x
Soybean	60	6	90	y

In any linear programming problem (LPP), the decision maker wants to maximize (usually revenue or profit) or minimize (usually costs) which are some function of the decision variables. Formulation of the objective function becomes the second step. The objective function that seeks to minimize the total daily cost of the feed mix is expressed as

Minimize $Z = 30x + 90y$

The 3rd step is to construct the constraint (or restriction) on decision variables. In this example, the constraints reflect the daily amount needed and the dietary requirements.

Because WFP needs at least 800 kg of feed a day, the associated constraint is expressed as: $x + y \geq 800.$

As for the protein daily requirement constraints, the amount of protein included in x kg of corn and y kg of soybean meal is $0.1x + 0.6y$ kg. This quantity should equal at least 30% of the total feed mix; i.e. $0.1x + 0.6y \geq 0.3(x + y)$.

In a similar manner, the fiber requirement at most 4% is constructed as;

$0.02x + 0.06y \leq 0.04(x + y) = 0.04x + 0.04y.$

The complete LPP model becomes

Minimize $Z = 30x + 90y$

Subject to: $\begin{cases} x + y \geq 800 \\ 0.2x - 0.3y \leq 0 \\ 0.02x - 0.02y \geq 0 \\ x \geq 0, y \geq 0 \end{cases}$ equivalently, $\begin{cases} y \geq -x + 800 \\ y \geq \frac{2}{3}x \\ y \leq x \\ x \geq 0, y \geq 0 \end{cases}$

The above three steps are collectively known as mathematical formulation or mathematical modelling of a real world.

To solve the problem graphically, you draw the lines $y = -x + 800$, $y = \frac{2}{3}x$, and $y = x$ in the same coordinate axis. Shade the region above the lines $y = -x + 800$, $y = \frac{2}{3}x$, and below the line $y = x$.

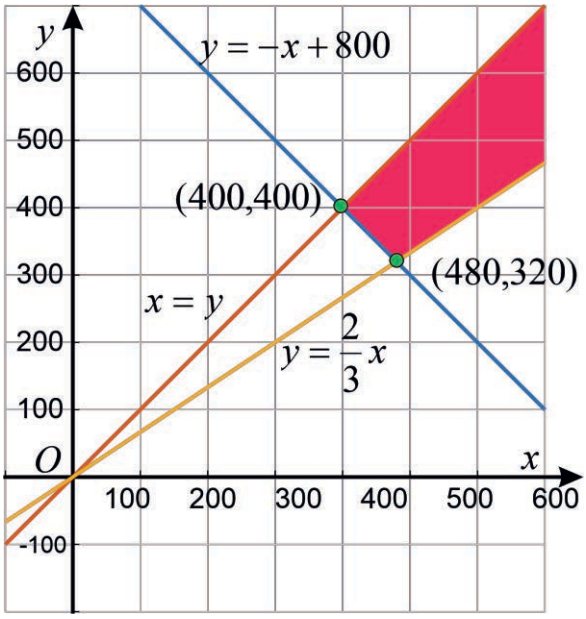


Figure 4.17

From the Figure 4.17, the possible optimum solution can be obtained at the points $(400, 400)$ and $(480, 320)$. These points are the intersections of the lines $y = -x + 800$ and $y = \frac{2}{3}x$, and $y = x$.

Testing the points at $(400, 400)$.

$$Z = 30 \times 400 + 90 \times 400 = 48{,}000$$

At $(480, 320)$, $Z = 30 \times 480 + 90 \times 320 = 43{,}200$. The associated minimum cost of the feed mix is Z= Birr 43,200 per day.

Exercise 4.9

1. Assume that there are two products, cereal and milk, for breakfast and assume that a person must consume at least 60 units of iron and 70 units of protein to stay alive. Assume that one unit of cereal costs Birr 120 and contains 30 units of iron and 5 units of protein and one unit of milk costs Birr 30 and contains 15 units of iron and 10 units of protein. Find the cheapest diet which will satisfy the minimum daily requirement.
2. A dietician has to develop a special diet using two foods M and N. Each packet of food M contains 15 units of calcium, 6 units of iron, 7 units of cholesterol and 9 units of vitamin A. Each packet of the same quantity of food N contains 5 units of calcium, 22 units of iron, 4 units of cholesterol and 3 units of vitamin A. The diet requires at least 245 units of calcium, at least 460 units of iron and at most 300 units of cholesterol. How many packets of each food should be used to minimize the amount of vitamin A in the diet? What is the minimum amount of vitamin A?

B. Manufacturing Problems: In these problems, we determine the number of units of different products which should be produced and sold by a firm when each product requires a fixed manpower, machine hours, labor hour per unit of product, warehouse space per unit of the output, etc. in order to make maximum profit.

Example 1

A furniture company makes two models A and B of office furniture. Each piece of Model A requires 9 labor hours for fabricating and 1 labor hour for finishing. Each piece of Model B requires 12 labor hours for fabricating and 3 labor hours for finishing. For fabricating and finishing, the maximum labor hours available are 180 and 30 respectively. The company makes a profit of Birr 8,000 on each piece of model A and Birr 12,000 on each piece of Model B. How many pieces of Model A

and Model B should be manufactured per week to realize a maximum profit? What is the maximum profit per week?

Solution

First construct the mathematical expression. The decision variables are x as the number of pieces of Model A and y as the number of pieces of Model B both manufactured by the company. The information is summarized in the following table.

Model	Required labor force (hour)		Profit (Birr/piece)	Piece per week
	Fabricating	Finishing		
A	9	1	8000	x
B	12	3	12000	y

Then total profit in Birr is $Z = 8000x + 12000y$

Thus, you have the following mathematical model for the given problem.

$$\text{Maximize } Z = 8000\,x + 12000\,y \quad (1)$$

Subject to the constraints:

$$9x + 12y \leq 180 \text{ (Fabricating constraint)} \quad (2)$$

$$x + 3y \leq 30 \text{ (Finishing constraint)} \quad (3)$$

$$x \geq 0, y \geq 0 \text{ (Non-negative constraints)} \quad (4)$$

The feasible region (the shaded) determined by the linear inequalities (2)-(4) is shown in the Figure 4.18. Note that the feasible region is bounded.

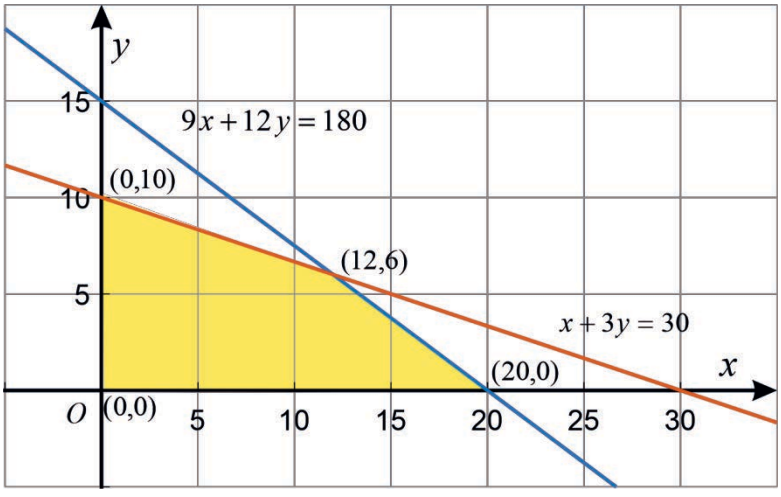


Figure 4.18

Corner point	$Z = 8000x + 12000y$
(0, 0)	0
(20, 0)	160000
(12, 6)	168000
(0, 10)	120000

The objective function (1) at each corner point is summarized in the table. From this, you find that maximum value of Z is 168,000 at B (12, 6). Hence, the company should produce 12 pieces of Model A and 6 pieces of Model B to realize a maximum profit Birr 168,000.

Example 2

XY car manufacturer produces two basic models, the M13 and M14. These cars are sold to dealers at a profit of Birr 20,000 per M13 and Birr 10,000 per M14. A M13 requires 150 hours for assembly, 50 hours for painting and finishing and 10 hours for checking and testing. The M14 requires 60 hours for assembly, 40 hours for painting and finishing and 20 hours for checking and testing. The total number of hours available per month is: 30,000 in the assembly department, 13,000 in the painting and finishing department and 5,000 in the checking and testing department. Let x be the number of M13 and y be the number of M14 models manufactured per month. From this:

a. Write down the set of constraint inequalities.
b. Use graph paper to represent the set of constraint inequalities.
c. Shade the feasible region on the graph paper.
d. Write down the profit generated in terms of x and y.
e. How many cars of each model must be produced in order to maximize the monthly profit?
f. What is the maximum monthly profit?

Solution

The information in the problem is summarized as follows:

Department	Model 13	Model 14	Hours available per month
Assembly	150	60	30,000
Painting and finishing	50	40	13,000
Checking and testing	10	20	5,000

a. Adding number of hours required for assembling each model,

$$150x + 60y \leq 30000.$$

Adding number of hours required for painting and finishing each model,

$$50x + 40y \leq 13000.$$

Adding number of hours required for checking and testing each model,

$$10x + 20y \leq 5000.$$

b. Drawing the constraints in coordinate plane we obtain the graph in Figure 4.19.

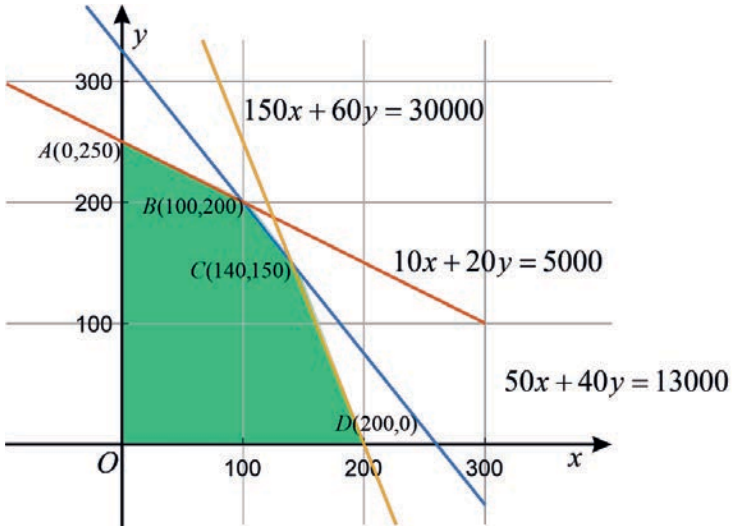


Figure 4.19

The region which satisfies all the three constraints are enclosed with vertices $O\ (0,0), A\ (0,250), B\ (100,200), C\ (140,150),$ and $D\ (200,0)$.

c. The feasible region is the shaded region with corner points OABCD.

d. The company gets a profit for a unit sell of M13 Birr 20,000 and a unit sell of M14 is Birr 10,000. This is summarized as $Profit = 20{,}000x + 10{,}000y$; the total profit the company earns per month.

e. To obtain a product mix which will give the maximum profit, we test each corner point of the shaded region.

Corner points	$Profit = 20000x + 10000y$
O (0, 0)	0
A (0, 250)	2,500,000
B (100, 200)	4,000,000
C (140, 150)	4,300,000
D (200, 0)	4,000,000

f. To maximize the objective function, we select Point C. The maximum profit can be made if 140 Model M13 cars and 150 Model M14 cars are sold. The maximum possible profit is Birr 4,300, 000.

Exercise 4.10

1. Assume that in the context of a production problem there are two products, P1 and P2, which are manufactured on three machines, M1, M2, and M3. Each product has to be processed on each of the three machines, but the order in which the products are processed on the machines is assumed to be immaterial. The two products are sold for the (fixed) unit price of Birr 12 and Birr 27, respectively, while the capacities of the three machines are 15, 14, and 12 hours per day, respectively. Assume now that it takes 9 minutes to process one unit of P1 on M1, 6 minutes to process one unit of P1 on M2, and 10 minutes to process one unit of P1 on M3. The corresponding data for P2 are 12, 14, and 18. Finally, it is required that at least 70 product units are made and sold. Formulate the production problem and determine the optimal solution.
2. A wood products firm uses left over time at the end of each week to make goods for stock. Currently, there are two products on the list of items that are produced

for stock; a chopping board and a knife holder. Both items require three operations: cutting, gluing and finishing. The manager of the firm has collected the following data on these products:

Item	Profit per unit in Birr	Time per unit (minutes)		
		Cutting	Gluing	Finishing
Chopping board	20	1.4	5	12
Knife holder	60	0.8	13	3

The manager has also determined that during each week 56 minutes of cutting, 650 minutes of gluing and 360 minutes of finishing time are available.

Required:

a) Formulate the problem as a linear programming model.

b) Determine to maximize the profit of each decision variables.

3. A company is making two products A and B. The cost of producing one unit of product A and B is Birr 60 and Birr 80 respectively. As per the agreement, the company has to supply at least 200 units of product B to its regular customers per month. One unit of product A requires one machine hours whereas product B has machine hours available abundantly within the company. Total machine hours available for product A are 400 hours. One unit of each product A and B requires one labor hour each and total of 500 labor hours are available. The company wants to minimize the cost of production by satisfying the given requirement. Formulate the problem as a linear programming problem and find the minimum cost.

C. Allocation Models: the most practical applications of linear programming problem is showed in allocation models. All allocation models in common attempt to allocate a scarce resource so as to optimize the consequence of that allocation. These allocations are to be performed so as to maximize their profit, minimize their cost, or optimize other efficiency criteria specified by the decision maker.

Example 1

A patient in a hospital is required to have at least 84 units of drug A and 120 units of drug B each day. Each gram of substance M contains 10 units of drug A and 8 units of drug B, and each gram of substance N contains 2 units of drug A and 4 units of drug B. Suppose both substances M and N contain an undesirable drug C, 3 units per gram in M and 1 unit per gram in N. How many grams of each substance M and N should be mixed to meet the minimum daily requirements and at the same time minimize the intake of drug C? How many units of drug C will be in this mixture?

Solution

Define the decision variable as x number of grams of substance M used, y number of grams of substance N used.

	Content (unit)		Amount of undesirable drug C (unit)	Amount (g)
	A	B		
M	10	8	3	x
N	2	4	1	y

The objective of this allocation problem is to minimize drug C from substances M and N. The constraints are the minimum requirements of each drug

$10x + 2y \geq 84$ . . . . . from drug A

$8x + 4y \geq 120$ . . . . . from drug B.

Combining the above, we obtain the LPP.

$$\text{Minimize } C = 3x + y$$

$$\text{Subject to: } \begin{cases} 10x + 2y \geq 84 \\ 8x + 4y \geq 120 \\ x \geq 0, y \geq 0 \end{cases}$$

The feasible region can be determined by drawing the lines $y = -5x + 42$ and $y = -2x + 30$ both inequalities are greater than or equal to $(\geq)$, you shade the region above the lines. Finally you get the region painted by green color in Figure 4.20.

The proposed optimal points are (0, 42), (4, 22) and (15, 0). The values of the objective function at these points are, C at (0, 42) is 42, C at (4, 22) is 34 and C at (15, 0) is 45. The minimum intake of drug C is 34 units and it attained by taking of 4 grams of substance M and 22 grams of substance N.

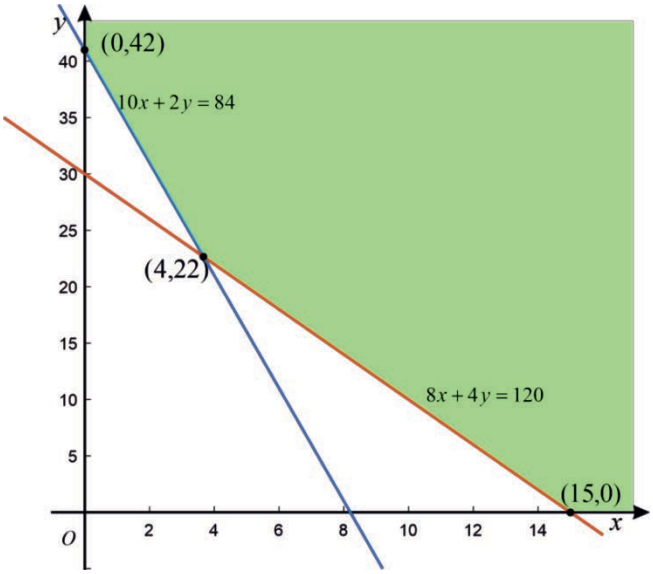


Figure 4.20

Exercise 4.11

1. One of the Ethiopian Airline airplanes can carry a maximum of 320 passengers. A profit of Birr 750 is made on each business class ticket and a profit of Birr 450 is made on each economy class ticket. The airline reserves at least 20 seats for business class. However, at least 4 times as many passengers prefer to travel by economy class than by the business class. Determine how many tickets of each type must be sold in order to maximize the profit for the airline. What is the maximum profit?

D. Transportation Problems: In these problems, we determine a transportation schedule in order to find the cheapest way of transporting a product from plants/factories situated at different locations to different markets.